Learning Axioms and Axiom Schemas from Their Instances

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Abstract

Proofs show axiom instances, not axioms. We ask when a checker can learn from them which axioms and schemas a community uses, soundly against adversarial provers. As in an earlier report, the checker is cautious: it accepts the common instances of all templates in a fixed class that cover the data. We answer three questions. (1) Instances $\varphi(t)$ determine the instance schema $\varphi(z)$, a first-order pattern; for one variable, two with different head symbols suffice. Passing to $\forall x\varphi$ is an ω-rule step: truth-safe for all φ exactly when the closed-term substructure is elementary (in $\mathbb{N}$, not in $\mathbb{R}$), and not derivable in general. (2) Every ZF schema, in each formulation considered, is a higher-order pattern; from patterns to rigid second-order templates, instances determine it iff their bodies' main symbols vary and every argument place is used. First-order anti-unification works only for some formulations and encodings; for most others no finite union of first-order schemas is sound. PA induction is not a pattern: anti-unification fails on it, determinate templates learn it. (3) For unlabelled mixtures we propose DTRC: cluster the data by refuting merged templates, then check each cluster cautiously. Under refutation separation it recovers the labels and needs no more data than a labelled learner; otherwise it is sound relative to a residue of unrefuted merges, and dependence on refutation depth is unavoidable. We prove separation for PA, check it on sampled ZF and ZFC data, and report experiments; in one, injected mistakes made DTRC accept a false sentence.

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1 Introduction

A mathematical community uses axioms: single sentences, such as Extensionality, and schemas, such as induction in Peano arithmetic or Separation in set theory, each standing for infinitely many sentences. Its proofs show only instances, often without saying which axiom they instantiate. We ask when a learner can recover from such positive instances exactly which axioms the community uses. The aim is checking, not proving: we want a verifier that accepts a candidate axiom step iff it is an instance of an axiom in use, and that stays sound when the candidates come from an untrusted, possibly adversarial, prover.

This is the setting of the report Hänni and Claude (Anthropic) (2026) (*the parent report*) on learning to check inferences. Its central tool is the *cautious* verifier: fix a class of hypotheses and accept a sentence iff every hypothesis in the class that contains the data contains it. If the target is a member of the class (more precisely, an intersection of members), this verifier is sound at all times against every prover, and it is exact once the data contain an *anchor*, a finite set of instances that every covering hypothesis extends to the whole target (Section 2.4). For rules cited by name the parent report used first-order anti-unification (Plotkin, 1970; Reynolds, 1970). PA induction does not fit: $\mathrm{Ind}(\varphi) = (\varphi(0) \wedge \forall x(\varphi \to \varphi[Sx/x])) \to \forall x\varphi$ substitutes terms into the motive φ. A follow-up analysis (*the earlier analysis*; unpublished, its record is in the accompanying repository, and we re-prove what we use) learned induction with *determinate* second-order templates. In these every metavariable has an occurrence applied to distinct bound variables, such as $P(x)$ in $T_{\mathrm{Ind}} = P(0) \wedge \forall x(P(x) \to P(Sx)) \to \forall x\, P(x)$, and arguments contain no metavariables (Remark A.3 shows why). Matching is unique and linear, and two instances are an anchor iff their motives have different main symbols and at least one has x free. Kaarel Hänni then asked three questions about this method.

1.1 The questions

Verbatim:

1. does the same thing work for learning an axiom of the form forall x phi(x) from a bunch of sentences of the form phi(x)?
2. does the same thing work for learning each axiom schema of ZFC from a bunch of instances?
3. anyway can you try to come up with a method that works for all these? ideally, your method would also work for learning many axiom(scheme)s at once, even if they do not come with labels for instances of which axiom scheme they are

"The same thing" is the method for induction. We call these Questions 1–3 (not to be confused with the axioms Q1–Q7 of Robinson arithmetic Q). We read the data of Question 1 as instances $\varphi(t)$ at terms t: since all data are read under universal closure, a datum $\varphi(x)$ with x free would already be the axiom (Proposition 4.10).

The answers compare four classes of templates (Section 2.2): first-order patterns FO, whose metavariables are 0-ary; Miller's higher-order pattern templates PAT (Miller, 1991), in which every metavariable occurrence is applied to distinct bound variables; determinate templates DT°, in which every metavariable has at least one such occurrence and the others may have any metavariable-free arguments, as $P(0)$ and $P(Sx)$; and rigid second-order templates SO° without this requirement. So FO $\subseteq$ PAT $\subseteq$ DT° $\subseteq$ SO°. Data are formulas whose free variables are parameters, read under universal closure; bound variables are de Bruijn indices, and values of

metavariables contain no free bound variable, so capture cannot occur (the λ convention). Besides data, a learner may use a *refutation oracle*: sound but one-sided evidence that a sentence is false, such as a Δ_0 counterexample in $\mathbb{N}$ or a hereditarily finite counterexample to a Π_1 sentence of set theory (Section 2.5).

1.2 The answers in brief

- *Question 1: yes for the instance schema, not for the sentence.* The instance schema $\varphi(z)$ is a first-order pattern; two instances whose terms have different head symbols are an anchor (with several variables, Plotkin's distinctness is needed too), for first-order anti-unification and the determinate learner alike. Passing from the closed instances to $\forall x\varphi$ is an ω-rule step: truth-safe for every φ exactly when the closed-term substructure is elementary (as in $\mathbb{N}$, not in $\mathbb{R}$), and not derivable in general. A cautious learner takes this step silently when parameters are admissible values; a learned closedness guard prevents it, and data with a parameter at every variable license it as ordinary generalization.
- *Question 2: yes, and more easily than for induction.* The axioms other than Separation and Replacement are single sentences, each its own anchor. Every ZF schema, in each formulation considered, is a Miller pattern, and in every class from PAT to SO$^\circ$ finitely many instances are an anchor iff their bodies' main symbols are not all equal and every argument place is used by some body; two instances can suffice. Pattern anti-unification and the determinate learner recover each schema from any anchor. Plain first-order anti-unification works only for some formulations and encodings. PA induction is not a pattern; first-order and pattern anti-unification fail on it, and the determinate learner learns it.
- *Question 3: yes, under a condition on the world that we make explicit.* The method DTRC clusters untagged data by refuting the templates that would merge them, then verifies each cluster cautiously. If every template covering data of two targets has a refuted instance (*refutation separation*), it recovers the hidden labels, is target-sound at all times and is exact once each target's data contain an anchor, so it needs no more data than a learner given the labels. Separation is proved for Q's axioms with induction and checked on sampled ZF and ZFC data. Without it DTRC is sound only relative to a residue of unrefuted merges, and no computable learner can avoid depending on the refutation depth.

The rest of this subsection gives the results behind these answers; Table 1 collects the anchor conditions.

Question 1 (Section 4). Plotkin's events are (R): the terms at each x_i do not all have the same root symbol, and (D): for $i \neq i'$ some datum puts different terms at x_i and $x_{i'}$. Closed instances satisfying them are an anchor for the instance schema $\sigma_\varphi = \varphi(z_1, \ldots, z_k)$ in FO and in DT$^\circ$, and if the language has enough closed terms, as L_A does, only these are (Theorem 4.2 and Proposition 4.3); in SO$^\circ$ the events do not suffice. For one variable the probability that N i.i.d. instances violate (R) is exactly $\sum_f p_f^N$, p_f the probability of root f (Theorem 4.4). The ω-step is truth-safe in M for every φ iff the closed-term substructure of M is elementary (Proposition 4.7, by the Tarski–Vaught criterion (Tarski and Vaught, 1957)). This holds in $\mathbb{N}$ and in every model of true arithmetic, but not in the ordered field $\mathbb{R}$, where $\neg(x \cdot x = 1 + 1)$ is true at every closed term, nor in the universe V of sets or in a model of $\mathsf{PA} + \neg\mathrm{Con}(\mathsf{PA})$ (Example 4.8). In general the step is not derivable: $\mathsf{Q} \vdash 0 + \underline{n} = \underline{n}$ for each n, but Q with all these instances does not prove $\forall x\,(0 + x = x)$ (Example 4.9). With parameters admissible, the cautious learner

target	FO (first-order lgg)	PAT (pattern lgg)	DT$^\circ$	SO$^\circ$
single axiom (Ext, AC, ...)	the axiom itself, in every class			
instance schema of $\forall \bar{x} \varphi$	(R)+(D)	(R)+(D)	(R)+(D)	(R)+(D) do not suffice
ZF schema	only some formulations; most others: no sound finite union	(R)+(N)	(R)+(N)	(R)+(N)
PA induction	unsound; no sound finite union	not in PAT; pattern lgg unsound	(R)+(N)	(R)+(B)
any $T^* \in$ DT$^\circ$			(R*)+(N)+(U)	

Table 1: When finitely many genuine instances are an anchor, by target and class (with the class's anti-unification learner where there is one). (R): the root symbols of the values (terms, bodies, motives) are not all equal; (D): Plotkin's distinctness; (N): every argument place is used by some value; (B): some motive has a free occurrence of x inside no compound term whose only variable is x; (R*), (U): root variation at every occurrence and no coincidence. "If" directions need no assumption; some "only if" directions use mild richness assumptions ((Rich), or enough closed terms). Sources, by row: Proposition 5.1; Theorem 4.2 and Proposition 4.3 (the PAT entry follows from FO $\subseteq$ PAT $\subseteq$ DT$^\circ$; the exact SO$^\circ$ condition is open); Table 11 and Theorems 5.6 and 5.9; Propositions 5.12, 5.15 and 5.17 and Theorems 5.13 and 5.16; Theorem 3.8.

in FO or DT$^\circ$ accepts $\varphi(\bar{p})$, which is $\forall \bar{x} \varphi$, after the first anchor (Proposition 4.5). For first-order schemas a closedness guard learned from closed data prevents this (DT$^\circ$ has no guards); data with a parameter at every variable make the step ordinary generalization (Proposition 4.10). If a metavariable may take a bound variable (named variables, or de Bruijn indices treated as constants), the schema can gain capture instances, such as $\exists y \neg(y = y)$ from $\exists y \neg(y = x)$, false in every structure; whenever capture instances exist, the unguarded verifier accepts some after an anchor. Learned guards repair this: closedness excludes capture as well, and a no-capture guard keeps it excluded once parameter data have removed closedness (Proposition 4.6, Table 5).

Question 2 (Section 5). The ZF schemas, in each textbook formulation considered (Table 6), are patterns because $L_\in$ has no function symbols: an instance only renames variables inside the body (Theorem 5.2). Their anchors are characterized in Theorem 5.9; pattern anti-unification (Pfenning, 1991; Baumgartner et al., 2017) and the DT$^\circ$ learner recover each schema from any anchor (Corollary 5.10), with exact failure probabilities (Theorem 5.11). First-order anti-unification works iff the metavariable is applied to literally the same variables at every occurrence, which depends on the formulation and on how bound variables are written (Theorem 5.2 and Proposition 5.3; Table 11 in Appendix D). Separation passes, with a freshness guard learned from the data; without it the lgg has Russell's instance (Proposition 5.8). In the other failing cases with textbook names or de Bruijn indices, no finite union of guarded first-order schemas covers the schema without a false member (Theorem 5.6 and Proposition 5.5). The one exception is spelled-out Replacement with textbook names: the lgg of anchor data, with the guards learned from them, over-generalizes truth-soundly, as all its guarded instances are theorems of ZF, and of a base theory exactly when that theory proves Collection; no refutation can expose it (Proposition 5.7).

PA induction is not a pattern. On raw instances first-order anti-unification can be unsound from two instances and is unsound on every anchor, no finite union of first-order schemas is truth-sound, some unsound lggs are never refuted by true closed quantifier-free sentences, and pattern anti-unification is unsound too (Propositions 5.12 and 5.15 and Theorems 5.13 and 5.14). The DT$^\circ$ learner is exact under (R)+(N) (Theorem 5.16), and T_{Ind} is the least of the 26 DT$^\circ$ templates covering any anchor (Proposition D.6). The obstruction is the substitution $\varphi[Sx/x]$, not induction: recording it, or Tarski's substitution-free form IndEq (Tarski, 1965), makes induction a first-order pattern, and Q plus the instances of IndEq axiomatizes PA (Propositions 5.18 and D.7).

The general theory (Section 3). These anchor conditions are cases of one theorem. For a target $T^* \in \mathrm{DT}^\circ$, finite data have finitely many minimal covering templates, one of them below T^* (Theorem 3.3). The cautious verifier is a polynomial-time test of "features" of the data (Theorem 3.4), although there can be 4^n minimal templates (Proposition 3.13). Root variation at every occurrence, non-vacuity and no coincidence make data an anchor; under a richness assumption (Rich), which always holds with parameters, every anchor satisfies them and its least covering template is the target (Theorem 3.8 and Corollary 3.14). For first-order targets these events are (R)+(D), for one formula metavariable (R)+(N) (Corollary 3.9). Failure probabilities decay exponentially (Theorem 3.10), and the worst-case number of escalations (queries passed to an oracle) is $\Theta(N^2)$ in the sentence size, which refutes an earlier conjecture that it is linear (Theorem 3.11 and Proposition B.6).

Question 3 (Section 6). From positive data a bound on the number of targets is necessary: without one the cautious union learner accepts only the data, whatever sound negatives it has (Proposition 6.4; Gold, 1967). With a bound, spare slots cost data diversity that no sound negative can buy, because splitting a target into sound specializations is never refuted (Theorems 6.5 and E.2). With the exact bound and negatives that refute every template covering anchor data of two targets, untagged learning is as easy as tagged learning, and the learner can find such negatives itself (Theorem 6.6). DTRC (Determinate Templates with Refutation Clustering) needs no bound. Starting from singletons, it repeatedly merges two clusters whose union has a minimal covering DT$^\circ$ template with no instance refuted within depth d, verifies each final cluster cautiously against its unrefuted minimal templates, and asserts the union. Within-target merges are never refuted; so under separation the clusters are the hidden label classes for every merge order, and the guarantees stated above follow, with the tagged learner's failure probability if separation holds for all instances (Theorem 6.8). Refutation does the work that citations do in tagged data.

Without separation DTRC is sound relative to a residue of unrefuted cross-target merges, truth-sound if these are, and exactness can fail by fragmentation (Theorem 6.10); on some data no learner with a target-blind oracle is both exact for one valid practice and target-sound for another (Proposition 6.11). Learning $\forall x\varphi$ from its instances is such a merge, truth-sound iff $\forall x\varphi$ is true (Proposition 6.13). With Δ_0 evaluation in $\mathbb{N}$ and quantifier-free φ, "refuted at no depth" means "true" (Proposition 6.14); at a fixed depth this holds only relative to the residue, and for quantified φ it can fail. No computable learner is both sound relative to the full-depth residue and eventually exact on every practice separated at some finite depth (Theorem 6.15, via the MRDP theorem (Matiyasevich, 1993)). The exact k-union verifier is polynomial for $k \leq 2$ and coNP-complete for each fixed $k \geq 3$; the refutation test of DTRC is NP-complete, and polynomial under (Rich) once the cluster contains an anchor (Theorem 6.16; not independently refereed). On natural targets, Δ_0 evaluation in $\mathbb{N}$ provably separates Q's axioms and induction; hereditarily finite counterexamples and logic, with Russell's paradox for the comprehension-shaped axioms,

separate ZF and ZFC on every sampled cross pair and in every run (Section 6.11); and Section 7 proves separation of its PA and ZF mixtures for an ideal refuter (Corollary 7.3). Weak forms of Union and Power merge into a false universal-set schema that neither logic nor hereditarily finite counterexamples refute, only coherence (Proposition 6.17). Minimum description length is no substitute for refutation: it tracks the statistics of usage, not the logical boundaries of schemas (Section 6.10).

Experiments (Section 7). An implementation of the learner (in the class DT°_F, with 0-ary term metavariables) and of DTRC matches the predicted rates on Question 1 (mean 3.27 instances to exactness, 3.25 predicted). On Separation, a Replacement variant, $\in$-induction and PA induction it is exact in most runs from 2 or 3 tagged instances, with no non-instance accepted in 927 probes (Table 9). On mixtures of the PA and of the ZF axioms it recovers the hidden labels in every run. With a pass that shares ambiguous data between clusters it matched a tagged learner given every label a datum deserves, target by target, in every run (Table 10); this is guaranteed only for targets that label a cluster of the first phase, as all did here (Proposition F.5 and Example F.6). The limits show too: the evidence for separation beyond the mixtures is narrow (Table 16); similar targets merge into valid templates (Lemma F.11); and with injected mistakes and refuter budgets of 80 and 400, DTRC accepted a false sentence, whose merge only a budget of 2000 refuted (Proposition 7.4).

1.3 Contributions

Besides the answers, the paper gives a general theory of learning one determinate template (Section 3); anchor conditions for several metavariables, which differ between PAT, DT° and SO° and are exact for the first two (Theorem D.3); the method DTRC with its theory of bounds, separation, residues, depth, noise and cost (Section 6); an implementation with baselines, a proof of separation for its mixtures with an ideal refuter, and a reported failure (Section 7); and corrections to earlier claims of this project, such as the conjecture of linear escalation bounds (Appendix G).

1.4 What is not achieved

- *Separation is assumed, not guaranteed.* It is proved for Q's axioms with induction and, for an ideal refuter, for the experimental mixtures, and only computed on sampled ZF and ZFC pairs. It fails for some similar targets (Lemma F.11 and Proposition 6.17), no computable learner can know the separating depth (Theorem 6.15), and a characterization of separation in terms of the targets is open.
- *Refutation is one-sided and incomplete.* It never certifies that a template is the target (Lemma 2.10). Truth-sound over-generalizations are invisible to it (Proposition 5.7), some unsound templates are never refuted by closed truths (Theorem 5.14), and where the ω-step is not truth-safe, a false universal can have only true closed instances, which no evaluation of closed instances refutes (Propositions 4.7 and C.5). Our oracles are incomplete and the implemented refuter has a budget, so "not refuted" certifies nothing.
- *Cost.* In the worst case there are exponentially many minimal templates and quadratically many escalations, the union verifier is coNP-complete for $k \geq 3$ and the refutation test NP-complete. The implementation uses DT°_F without the feature verifier, on small synthetic data, not a corpus from a proof library.

- *Mistakes.* The theorems assume a valid practice and correct data. Unrefutable mistakes can be absorbed into a cluster or fragment it (Proposition E.12), a frequent coherent family of mistakes cannot be told from a rule by positive data, and mistakes made the implementation accept a false sentence (Proposition 7.4).
- *Rare targets and scope.* A target seen once is never generalized, a single axiom is learned only if it occurs among the data, and an unrefuted singleton mistake looks like a single axiom (Section 7.4). We learn which instances a community uses, not the theory they generate, and from closed instances the truth of $\forall x\varphi$ is certified only as far as the oracle reaches. Open problems remain in every part; Section 8 collects them.

1.5 Reader's guide

Section 2 fixes the framework and lists the named conditions (Table 3). Section 3 is the theory of one determinate template, used by the later sections. Sections 4 to 6 answer Questions 1–3, each ending with a summary that can be read first (Sections 4.7, 5.6 and 6.12). Section 7 reports the experiments, and Section 8 discusses the answers, related work and open problems. The appendices, one per section, hold the full proofs and the secondary results the main text points to (numbered with a letter, as Proposition D.6); Appendix G says how the results were checked and what was corrected. Theorem-like statements carry a status (proved, computed, known, conjecture, or a combination) and a gray tag naming the result in the research record of the accompanying repository, which also holds the scripts and the reports of adversarial referees; refuted conjectures stay labelled as refuted.

2 Setting

This section fixes the framework of the three questions. The learning framework is that of Hänni and Claude (Anthropic) (2026), *the parent report*; the unpublished follow-up analysis of PA induction (Section 1) is *the earlier analysis*; we restate what we use from both. The rest of the paper relies on three facts: matching in DT° is unique and linear (Theorem 2.5); the cautious verifier is sound iff the target is closed in the class, and exact iff the data contain an anchor (Lemma 2.8); and refutation is one-sided (Lemma 2.10). Table 3 lists the named conditions of the paper; Appendix A has the technical definitions and longer proofs.

2.1 Data in closure-normal form

We use arithmetic $L_A = \{0, S, +, \cdot, =, <\}$ and set theory $L_\in = \{\in, =\}$, with $\neg, \wedge, \vee, \to, \leftrightarrow, \forall, \exists$; bounded quantifiers are abbreviations. The general results hold for every first-order signature, with a richness assumption where stated.

Closure-normal form. Formulas may contain *parameters* $a, b, c, p, q, w_0, w_1, \ldots$, an infinite supply of extra constants. A *datum* is a formula whose free variables are parameters, read under its universal closure. This removes the outer prefix of variable length that schema instances carry ($\forall \bar{w}$ in a Separation instance $\forall \bar{w}\, \forall z \exists y \forall x\, (x \in y \leftrightarrow x \in z \wedge \varphi(x, z, \bar{w}))$), so all instances share one frame. The frame's quantifiers stay; in particular, *leading universal quantifiers are not stripped*: $\forall x\, (x + 0 = x)$ and $p + 0 = p$ are different data with equivalent closures, and only the second is an instance of the instance schema $z + 0 = z$ (Section 4).

Bound variables. In the theory and the code, bound variables are de Bruijn indices (de Bruijn, 1972); in the text we write named variables. A *sentence* has no free index (parameters are allowed). Positions p (prefix order $p \leq q$, incomparability $p \perp q$), subtrees $u|_p$, the number $b(p)$ of binders above p, free indices $\mathrm{FV}(u)$ and shifting are defined in Appendix A.1. The *root symbol* of a term or formula is its head symbol (an index, or a hole z_m, counts as its own symbol).

2.2 Templates and four classes

Definition 2.1 (Bodies, templates, classes)**.** brief §3; single §1; cases §1 A *body* of type $\iota^n \to s$ ($s \in \{\iota, o\}$) is $\lambda z_1 \dots z_n.\beta$ with β an object term ($s = \iota$) or formula ($s = o$) that has no free indices but may contain *holes* $z_1, \dots, z_n$ (leaves of sort ι), parameters and binders. *Plugging* $\beta[u_1, \dots, u_n]$ replaces each z_m by u_m, with the free indices of u_m shifted by the binders of β above that hole.

A *metavariable* M has type $\iota^n \to o$ (*formula metavariable*: $P, Q, F, \dots$) or $\iota^n \to \iota$ (*term metavariable*: $f, g, \dots$; z if 0-ary). A *template* T is built like a sentence, except that at a position of sort s there may be an *occurrence* $M(t_1, \dots, t_n)$ of some $M : \iota^n \to s$, whose arguments t_i are *metavariable-free* terms (they may contain indices bound above the occurrence, and parameters). $\mathrm{Occ}(T)$ is the set of occurrence positions; it is an antichain. The *rigid skeleton* $\mathrm{skel}(T)$ is the set of the other positions outside argument lists. An occurrence is a *pattern occurrence* if its arguments are pairwise distinct bound variables (for $n = 0$: always). A template without metavariables is *ground*, i.e. a single sentence. The classes are: SO°, all such templates (rigid occurrences, metavariable-free arguments); DT°, the *determinate templates*, in which every metavariable has at least one pattern occurrence, while other occurrences may have any metavariable-free arguments, e.g. $P(0)$, $P(Sx)$; DT°_F, the members of DT° whose term metavariables are 0-ary (the class implemented, Section 7); PAT, the higher-order *pattern templates* (Miller, 1991), in which every occurrence is a pattern occurrence; and FO, the *first-order patterns*, in which every metavariable is 0-ary. So $\mathrm{FO} \subseteq \mathrm{PAT} \subseteq \mathrm{DT}^\circ \subseteq \mathrm{SO}^\circ$ and $\mathrm{FO} \subseteq \mathrm{DT}^\circ_F \subseteq \mathrm{DT}^\circ$.

Since bodies have no free indices and their own binders are de Bruijn binders, plugging cannot capture: in named notation, values of metavariables are closed λ-terms whose only free names are parameters. We call this the λ *convention*; it is the default.

Definition 2.2 (Instances, generality, minimal covering templates)**.** single §1 A *substitution* θ maps each metavariable of T to a body of its type, and the *instance* $T\theta$ replaces each occurrence $M(\bar{t})$ by $\theta(M)[\bar{t}]$ (one-step β-reduction; no new redex arises). $\mathrm{inst}(T) = \{T\theta : \theta\}$. A *template substitution* σ maps each M to a *template body* $\lambda\bar{z}.\gamma$, where γ may contain occurrences $N(\bar{u})$ with metavariable-free $\bar{u}$. $T \succeq T'$ (T is *at least as general as* T') iff $T' = T\sigma$ for some such σ; $T \equiv T'$ iff $T \succeq T' \succeq T$. T *covers* D if $D \subseteq \mathrm{inst}(T)$. For a class $\mathcal{C}$ and finite nonempty D, $\mathrm{Min}_{\mathcal{C}}(D)$ is the set of $\succeq$-minimal covering templates in $\mathcal{C}$, one per $\equiv$-class; $\mathrm{Min}(D) := \mathrm{Min}_{\mathrm{DT}^\circ}(D)$.

The relation $\succeq$ is transitive, and $T \succeq T'$ implies $\mathrm{inst}(T) \supseteq \mathrm{inst}(T')$. Whether the converse holds in DT° is open and not needed: minimality is syntactic, while the verifier uses only instance sets single §13. In DT°, $T \equiv T'$ iff T' is T up to renaming of metavariables and a permutation of each metavariable's argument places (Lemma A.2).

Example 2.3 (The four classes)**.** [proved] single Cor D.3; cases A2′; prior induction C1 See Table 2. (1) The instances of $z + 0 = z$ are $t + 0 = t$ for terms t over L_A and parameters. (2) T_{Sep} is Separation as in Kunen (1980) (wording recalled, not re-checked against the book). Its occurrence $P(x, z)$ is a pattern occurrence, so $T_{\mathrm{Sep}} \in \mathrm{PAT} \setminus \mathrm{FO}$. The side condition "$y$ not free in φ" is built in: a body for P has no free variables but holes and parameters, and y is not an argument. (3) In

template	example of	FO	PAT	DT$^\circ$	SO$^\circ$
$z + 0 = z$	instance schema	yes	yes	yes	yes
$T_{\text{Sep}} = \forall z \exists y \forall x (x \in y \leftrightarrow x \in z \wedge P(x, z))$	Separation	no	yes	yes	yes
$T_{\text{Ind}} = (P(0) \wedge \forall x (P(x) \to P(Sx))) \to \forall x P(x)$	PA induction	no	no	yes	yes
$f(S0) + f(0) = f(S0)$	—	no	no	no	yes

Table 2: The hierarchy FO $\subseteq$ PAT $\subseteq$ DT$^\circ$ $\subseteq$ SO$^\circ$ on the templates of Example 2.3. All ZF(C) schemas lie in PAT; PA induction lies in DT$^\circ \setminus$ PAT (Section 5).

T_{Ind} the occurrences $P(x)$ are pattern occurrences, $P(0)$ and $P(Sx)$ are not: $T_{\text{Ind}} \in \text{DT}^\circ_F \setminus \text{PAT}$. (4) $f(S0)+f(0) = f(S0)$ has no pattern occurrence. It covers $0+0 = 0$ ($f := \lambda h.0$) and $S0+0 = S0$ ($f := \lambda h.h$), but not $SS0 + 0 = SS0$: a body with $\beta[0] = 0$ is h or 0, so $\beta[S0] \in \{S0, 0\}$. These two data anchor $z + 0 = z$ in DT$^\circ$ (Proposition 4.3; anchors: Definition 2.6), but not in SO$^\circ$.

Arguments must be metavariable-free: with metavariables inside arguments, as in $M(F(x))$ (the class DT of the earlier analysis), the set of minimal covering templates can be infinite, and the earlier enumeration-based learner was unsound (Remark A.3); in DT$^\circ$ it is finite (Theorem 3.3).

2.3 Matching

Lemma 2.4 (Preservation). *[proved]* single Lemma 1.1 *Let $T \in \text{SO}^\circ$ and let θ be a substitution or a template substitution. If $p \in \text{skel}(T)$, $(T\theta)|_p$ has the root symbol of $T|_p$; if $p \in \text{Occ}(T)$ carries $M(\bar{t})$, then $(T\theta)|_p = \theta(M)[\bar{t}]$.*

Theorem 2.5 (Matching). *[proved; computed]* single Thm A; prior induction C1 *Let $T \in \text{DT}^\circ$ and let s be a sentence.*

(a) At most one substitution θ satisfies $T\theta = s$.

(b) The following decides $s \in \text{inst}(T)$ and returns θ in time $O(|T| + |s|)$ (unit-cost index arithmetic): walk skel(T) *against s; read each M off one pattern occurrence $M(\bar{y})$ at p as $\theta(M) := \lambda \bar{z}.(s|_p)[z_m/y_m]$, failing if $s|_p$ has another free index; check the other occurrences $M(\bar{t})$ at q by comparing $s|_q$ with $\theta(M)[\bar{t}]$ lazily.*

(c) For $T, T' \in \text{DT}^\circ$: $T \succeq T'$ iff freeze$(T') \in \text{inst}(T)$*, where* freeze *turns each metavariable of T' into a fresh rigid symbol of the same type. So generality is decidable in linear time.*

(d) For $T \in \text{SO}^\circ$, membership $s \in \text{inst}(T)$ is decidable in polynomial time, but there can be $2^{\Omega(|s|)}$ matchers.

Proof idea. At a pattern occurrence, $s|_p = \theta(M)[\bar{y}]$ (Lemma 2.4) forces $\theta(M)$; the rest is evaluation, linear because occurrence positions form an antichain. (d) is projection/imitation. Details: Appendix A.3. □

2.4 Protocol, cautious verifier and anchors

Targets, data, hypotheses. A *target* is a set R^* of sentences, typically $\text{inst}(\sigma)$ or $\bigcup_{i \le k'} \text{inst}(\sigma_i)$ for templates σ_i. The practice is assumed correct, i.e. all data lie in R^* (mistakes: Proposition E.12). Data come as a *text* (finite $D_1 \subseteq D_2 \subseteq \cdots \subseteq R^*$ with union R^*) or i.i.d.: each datum picks schema i with probability π_i, then an instance $\sigma_i\Theta$ with $\Theta \sim \Lambda_i$. *Tagged* data carry the index i (as a proof assistant records the axiom it uses); *untagged* data do not. The hypothesis

classes are $H_1(\mathcal{C}) = \{\mathrm{inst}(T) : T \in \mathcal{C}\}$; tagged calculi $H_k^{\mathrm{tag}}(\mathcal{C})$, k-tuples of templates, one per tag; and $H_k(\mathcal{C}) = \{\bigcup_{l \le k} \mathrm{inst}(T_l) : T_l \in \mathcal{C}\}$.

Definition 2.6 (Protocol; cautious verifier; anchors)**.** parent def:setting:protocol, def:caution:vs In rounds $t = 1, 2, \ldots$ a *prover* submits a sentence q_t and the *verifier* answers ACCEPT, REJECT or ESCALATE; on ESCALATE an oracle says whether $q_t \in R^*$, and the labelled query joins the evidence. The prover may be randomized, adaptive and computationally unbounded; it knows the target, the verifier's code, the data and the history, but not future data or coins. It models any proof search, the learner's own included. A deterministic verifier is *sound for* $S^\dagger$ if no prover ever gets a sentence outside $S^\dagger$ accepted; *target-sound* if $S^\dagger = R^*$; *truth-sound* if $S^\dagger = \mathrm{Tr}$, the set of data whose closure is true in the intended structure ($\mathbb{N}$, or the universe V of sets, assumed to satisfy ZFC). For a class $\mathcal{H}$, positive evidence D and negative evidence N, the *version space* is $\mathrm{VS}(D, N) = \{h \in \mathcal{H} : D \subseteq h,\ h \cap N = \emptyset\}$. The *version-space verifier* accepts q iff $q \in \bigcap \mathrm{VS}(D, N)$, rejects iff $\mathrm{VS}(D, N) \neq \emptyset$ and $q \notin \bigcup \mathrm{VS}(D, N)$, and escalates otherwise. Its acceptance set $\mathrm{Acc}_{\mathcal{H}}(D, N) = \bigcap \mathrm{VS}(D, N)$ (with $\bigcap \emptyset =$ all sentences) is the *cautious verifier*; $\mathrm{Acc}_{\mathcal{H}}(D) = \mathrm{Acc}_{\mathcal{H}}(D, \emptyset)$, $\mathrm{Acc}(D) = \mathrm{Acc}_{H_1(\mathrm{DT}^\circ)}(D)$, and $\mathrm{Acc}_k(D, N) = \mathrm{Acc}_{H_k(\mathrm{DT}^\circ)}(D, N)$, with version space $\mathrm{VS}_k(D, N)$. A finite $A \subseteq h^*$ is an *anchor* for h^* in $\mathcal{H}$ if every $h \in \mathcal{H}$ with $A \subseteq h$ contains h^*. The *closure* of h^* is $\mathrm{cl}_{\mathcal{H}}(h^*) = \bigcap\{h \in \mathcal{H} : h \supseteq h^*\}$; h^* is *$\mathcal{H}$-closed* if $\mathrm{cl}_{\mathcal{H}}(h^*) = h^*$.

For nonempty positive data every covering template lies above one of finitely many minimal ones (Theorem 3.3(c)), so $\mathrm{Acc}(D) = \bigcap_{T \in \mathrm{Min}(D)} \mathrm{inst}(T)$, decidable in polynomial time (Theorem 3.4 and Corollary 3.6). $\mathrm{Acc}(\emptyset) = \emptyset$ (distinct ground templates have disjoint instance sets), so $\emptyset$ is never an anchor. As the 0-ary formula metavariable P has every sentence as an instance, positive data reject nothing outright; the cost is the number of escalations (Section 3.6).

Theorem 2.7 (Caution is sound)**.** *[known]* parent thm:caution:vs (a) *If $R^* \in \mathcal{H}$ and the data lie in R^*, then $R^* \in \mathrm{VS}(D, N)$ at all times (positive labels lie in R^*, negative ones outside), so the version-space verifier is target-sound against every prover at every time.* (b) *A deterministic verifier that is sound for every target in $\mathcal{H}$ accepts, at each history it produces against a target in $\mathcal{H}$, only sentences in that history's $\bigcap \mathrm{VS}$: the cautious verifier is the most permissive sound one.*

The cautious verifier is reliable learning (Rivest and Sloan, 1988), or perfect selective classification (El-Yaniv and Wiener, 2010), applied to sentences. Realizability can be weakened to closedness; this is the sense in which soundness is relative to the class.

Lemma 2.8 (Sound iff closed)**.** *[proved]* prior induction C6.1; parent thm:imitation:anchor *Let $\mathcal{H}$ be any class of sets of sentences, h^* a target, and $D \subseteq h^*$ finite (positive data only).*

(a) $\mathrm{Acc}_{\mathcal{H}}(D) \subseteq \mathrm{cl}_{\mathcal{H}}(h^*)$*; so if h^* is $\mathcal{H}$-closed (e.g. $h^* \in \mathcal{H}$), the cautious verifier is sound for every $D \subseteq h^*$. Also, $D \subseteq D'$ implies* $\mathrm{Acc}_{\mathcal{H}}(D) \subseteq \mathrm{Acc}_{\mathcal{H}}(D')$.

(b) If D contains an anchor, $\mathrm{Acc}_{\mathcal{H}}(D) = \mathrm{cl}_{\mathcal{H}}(h^*)$*. So once the data contain an anchor, the cautious verifier is sound iff h^* is $\mathcal{H}$-closed, and it is then exact.*

(c) If h^* is $\mathcal{H}$-closed, $\mathrm{Acc}_{\mathcal{H}}(D) = h^*$ *iff D is an anchor; on a text for h^* the verifier is eventually exact iff h^* has a finite anchor, and then it stays exact.*

The proof is in Appendix A.2. For example, PA induction is closed in the class DT°_s of templates of size at most s iff $s \ge 12$ ($|T_{\mathrm{Ind}}| = 14$); for $7 \le s \le 11$ the closure is larger, so on anchor data the verifier accepts a false sentence, whatever further data arrive (Proposition D.6). For tagged data the version space factors: (i, q) is accepted iff $q \in \mathrm{Acc}(D^{(i)})$, for $D^{(i)}$ the data of tag i. For untagged unions a bound k is unavoidable, since otherwise $\mathrm{Acc}(D) = D$ (Proposition 6.4; Gold, 1967).

2.5 Refutation oracles

Positive data never exclude a hypothesis that contains them. The second source of evidence is the world: sound but one-sided evidence that a sentence is false.

Definition 2.9 (Refutation oracle)**.** untagged §1; parent lem:twotier:onesided A *refutation oracle* is a family $\mathrm{Ref}_0 \subseteq \mathrm{Ref}_1 \subseteq \cdots$ of sets of data (refuted within budget or depth d) with (OS) $\mathrm{Ref}_d \cap \mathrm{Tr} = \emptyset$ for all d (one-sided soundness), and (Dec) membership in Ref_d, and whether $\mathrm{inst}(T) \cap \mathrm{Ref}_d \neq \emptyset$ (T is *d-refuted*), are decidable uniformly in d and (T, d). $\mathrm{Ref}_\infty = \bigcup_d \mathrm{Ref}_d$. A fixed finite set of false sentences, used at every d, is an oracle; this is how the audited prototype of Section 6.11 is read (Proposition E.18). Section 7 treats its budgeted, history-dependent refuter directly.

The oracles used are sound, *target-blind* (their answers are a fixed function of the query) and incomplete untagged §1; experiments Prop E2. *R*-Δ_0 at depth d instantiates parameters and leading universal quantifiers of an arithmetic sentence of size $\leq d$ by numerals $\leq d$, refutes $\forall$ (verifies $\exists$) by a numeral $\leq d$, and verifies $\forall$ (refutes $\exists$) only by equality reasoning from the true closed equations of $\mathbb{N}$. *R-HF* gives hereditarily finite counterexamples to Π_1-shaped set-theoretic sentences, sound by Δ_0-absoluteness. *R-logic* certifies logical falsity by a propositionally unsatisfiable set of ground instances of the Skolem form; *R-coh* (coherence) applies it to the sentence together with designated true sentences. The implementation's evaluators are sound in the same sense (Proposition 7.1).

Lemma 2.10 (Refutation is one-sided)**.** *[proved]* untagged Lemma 1.1; parent lem:twotier:onesided *Let the oracle satisfy (OS) and $R^* \subseteq \mathrm{Tr}$.* (a) *A template T with $\mathrm{inst}(T) \subseteq \mathrm{Tr}$ is never refuted, at any depth, as $\mathrm{Ref}_d \cap \mathrm{Tr} = \emptyset$; in particular no target instance is. So refutation cannot tell the target from a truth-sound over-generalization, and never certifies that a template is the target.* (b) *If $R^* \in \mathcal{H}$, $D \subseteq R^*$ and $N \subseteq \mathrm{Ref}_\infty$, then $R^* \in \mathrm{VS}(D, N)$ (as $R^* \subseteq \mathrm{Tr}$ avoids N), so $\mathrm{Acc}_{\mathcal{H}}(D, N) \subseteq R^*$: refuted sentences are safe negatives. $\mathrm{Acc}_{\mathcal{H}}(D, N)$ is monotone in D and N.*

Two consequences recur: a false $\forall x \varphi$ with only true closed instances is never refuted by evaluating closed instances (Proposition C.5(a)), and a truth-sound over-generalization, such as the named lgg of anchor data of spelled-out Replacement, is never exposed (Proposition 5.7).

First-order baselines. The baseline of Sections 4 and 5 is first-order anti-unification: bound variables are encoded by names (Nm) or de Bruijn indices (dB), instances may capture bound variables, side conditions become guards, and Plotkin's lgg recovers a schema iff the events (R) and (D) hold. Appendix A.6 gives the definitions and Lemma A.4.

2.6 Running examples

Arithmetic. The axioms of Robinson's arithmetic Q (Tarski et al., 1953), numbered as throughout the paper, are Q1 $\forall x\, \neg(Sx = 0)$; Q2 $\forall x \forall y\, (Sx = Sy \to x = y)$; Q3 $\forall x\, (\neg x = 0 \to \exists y\, x = Sy)$; Q4 $\forall x\, (x + 0 = x)$; Q5 $\forall x \forall y\, (x + Sy = S(x + y))$; Q6 $\forall x\, (x \cdot 0 = 0)$; Q7 $\forall x \forall y\, (x \cdot Sy = x \cdot y + x)$. Each is a ground template. In closure-normal form an axiom may also be presented with parameters (Q4 as $a + 0 = a$) or only through closed instances (e.g. $\underline{3} + 0 = \underline{3}$, instances of $z + 0 = z$); these are different data sets, and Sections 4, 6 and 7 say which they use (the instance schema of a universal axiom is defined in Definition 4.1). PA is Q plus

$$\mathrm{Ind}(\varphi) = (\varphi(0) \wedge \forall x\, (\varphi \to \varphi[Sx/x])) \to \forall x\, \varphi,$$

where the motive φ has at most x free besides parameters; $\mathrm{inst}(T_{\mathrm{Ind}})$ is the set of these instances ($P := \lambda x.\varphi$). It is our main schema outside PAT (Section 5).

name	meaning	defined in
(R), (D)	Plotkin's events: the values' roots vary; values of distinct metavariables differ	Lem. A.4
(R^*), (N), (U)	root variation per occurrence; non-vacuity; no coincidence	Def. 3.7
(D^+), (D^{PAT}), ...	(U) for formula metavariables, resp. for several metavariables	Cor. 3.9, Thm. D.3
(B)	some induction motive has an unshielded free x	Prop. 5.17
(Rich), (Rich_c)	enough root symbols (true with parameters); enough closed terms	Def. 3.7, §4.1
(S1), (S2)	saturation: argument places used; roots not all equal	§3.2
(F)	$\text{Min}(X)$ is finite; every covering template lies above a member	Thm. 3.3, App. A.7
anchor, closure, sound	anchors, closed targets, target- and truth-soundness	Def. 2.6
(OS), (Dec)	the oracle is one-sided sound; refutation is decidable	Def. 2.9
$\text{RS}_d(D)$, global sep.	templates covering data (instances) of two targets are refuted	Def. 6.7
cross-separation, (X)	negatives meet templates covering two targets' data (anchor data)	Thms. 6.5 and 6.6
anchor-pair separation	the minimal templates of cross pairs of anchor data are refuted	§6.3
(Abs)	avoiding templates covering data of i and $j \neq i$ contain $\text{inst}(\sigma_i)$	Thm. E.2
noise-separated	templates covering a mistake and a clean datum are refuted	Prop. E.12
(H1)–(H5), (H-ref)	implemented-learner hypotheses; history-independent verdicts	Thm. 7.2, Prop. F.5

Table 3: Named conditions used in the paper.

Set theory. The ZFC axioms other than the two schemas (Ext, Pair, Union, Pow, Inf, Found, AC) are single sentences, hence ground templates, each its own anchor. Separation, Replacement and the theorem schemas Collection and $\in$-induction are pattern templates, e.g. T_{Sep} and $T_{\text{EInd}} = \forall x\,(\forall y\,(y \in x \to P(y)) \to P(x)) \to \forall x\,P(x)$; Section 5 lists the formulations.

2.7 Parameters in theory and implementation

The theory treats parameters as constants: $\text{inst}(T)$ is the set of *literal* instances, and a template that mentions a parameter rigidly matches only that parameter. This is sound but not invariant under renaming of parameters single §13. The implementation canonicalizes each datum (parameters renamed $w_0, w_1, \ldots$ by first occurrence) and matches up to injective renaming of the template's parameters, which agrees with the theory for parameter-free templates. Otherwise the minimal covering templates must be computed over all parameter alignments of the data: Appendix A.7 defines this set $\text{Min}^{\text{al}}(D)$ and proves that it has the properties of $\text{Min}(D)$ for templates all of whose parameters occur in their skeleton, in particular for parameter-free targets (Proposition A.5), while the literal $\text{Min}(D)$ can miss them (Example A.6). A parameter that occurs only inside an argument of a non-pattern occurrence is not covered (Remark A.7). The experiments' targets are parameter-free, and they use Min^{al} (written Min_F in Section 7), or the literal set when some datum is parameter-free (Appendix A.7).

3 Learning one schema

This section is the general theory of learning a *single* template of the class DT° from positive data. Fix a target $T^* \in \text{DT}^\circ$ and a finite set $D \subseteq \text{inst}(T^*)$ of its instances. The cautious verifier of Definition 2.6 accepts

$$\text{Acc}(D) = \bigcap\{\text{inst}(T) :\ T \in \text{DT}^\circ,\ D \subseteq \text{inst}(T)\}.$$

Since T^* is one of the templates intersected, $\text{Acc}(D) \subseteq \text{inst}(T^*)$: the verifier is sound at all times, against every prover. The remaining questions are about completeness and cost: when does

$\mathrm{Acc}(D)$ reach all of $\mathrm{inst}(T^*)$, how many data does that take, and what does verification cost? We show:

- the version space is finitary, and some minimal covering template lies below the target (Section 3.2);
- $\mathrm{Acc}(D)$ is a polynomial-time test of "features" of the data (Section 3.3);
- under a richness assumption that always holds with parameters, D is an anchor iff three witness events hold; for first-order targets they are Plotkin's, for PA induction and every ZF schema root variation plus non-vacuity (Section 3.4);
- exponential rates for random and for tagged data (Section 3.5);
- a tight quadratic escalation bound, refuting an earlier linear conjecture (Section 3.6);
- finite elasticity, so unions have anchors (Section 3.7), and the complexity of the version space (Section 3.8).

The union verifier and the refutation test of DTRC are analysed in Section 6.

Gray tags name the source of each result in the research notes on single templates or in the earlier analysis of PA induction (Section 2; Appendix G). Full proofs, and the secondary results that the text points to, are in Appendix B. Every result was also checked by computation, most by an independent implementation (Appendix B.9). Parameters are constants (Section 2.7); bound variables are de Bruijn indices in definitions and proofs, named in examples.

3.1 Notation and data notions

Notation is that of Section 2. In addition, $\mathrm{b}(p)$ is the number of binders strictly above p (so the bound variables in scope at p are the indices $0, \ldots, \mathrm{b}(p)-1$), $|s|$ is the number of positions, and for a map $\bar{u}$ on free indices, $s[\bar{u}]$ substitutes u_y for each free index y of s. If T covers D, θ_d is the unique matcher of $d \in D$ (Theorem 2.5) and $\beta^d_M := \theta_d(M)$.

Definition 3.1 (Common prefix, slots, scopes, features)**.** single §1 Let D be a nonempty finite set of sentences.

- The *common prefix* $C(D)$: the root is in $C(D)$ iff all data have the same root symbol, and $p \cdot i \in C(D)$ iff $p \in C(D)$ and all $d|_{p \cdot i}$ have the same root symbol. The *slots* $\mathrm{slots}(D)$ are the positions outside $C(D)$ whose parent is in $C(D)$ (or the root, if it is outside). $A(D) := C(D) \cup \mathrm{slots}(D)$ (*admissible* positions). The *scope* of a slot is $Y_\sigma(D) := \bigcup_{d \in D} \mathrm{FV}(d|_\sigma)$.
- A sentence q has the *feature* $\mathsf{Sym}(p)$ ($p \in C(D)$) if $q|_p$ exists and has the root symbol common to the data at p; $\mathsf{Scope}(\sigma)$ (σ a slot) if $\mathrm{FV}(q|_\sigma) \subseteq Y_\sigma(D)$; and $\mathsf{Eq}(\sigma, r, \bar{u})$ (σ a slot, $r \in A(D)$, $\sigma \perp r$ of the same sort, $\bar{u}$ a map from $Y_\sigma(D)$ to metavariable-free terms in scope at r) if $\mathrm{FV}(q|_\sigma) \subseteq Y_\sigma(D)$ and $q|_r = (q|_\sigma)[\bar{u}]$.
- A *D-feature* is a feature that every datum has; $\mathrm{Feat}(D)$ is the set of sentences with every D-feature.

Lemma 3.2 (Basic facts)**.** *[proved]* single §§1, 3

(a), (b) Preservation *and the* substitution facts *are Lemmas 2.4 and A.1.*

(c) Rigid prefix. *If T covers D, then* $\mathrm{skel}(T) \subseteq C(D)$ *with the same symbols, and* $\mathrm{Occ}(T) \subseteq A(D)$.

(d) Forcing. *Let $\sigma \perp r$ be positions of every datum and $Y = \bigcup_d \mathrm{FV}(d|_\sigma)$. At most one $\bar{u}$ on Y satisfies $d|_r = (d|_\sigma)[\bar{u}]$ for all $d \in D$; one exists iff one exists for every pair $\{d, d'\} \subseteq D$; and it is found in time $O(\sum_d |d|)$.*

(e) Equivalence is renaming: *Lemma A.2.*

Proofs in Appendix B.1. Part (d) is the second-order form of the fact behind Plotkin's lgg: an equation between two positions is either violated by the data or determined by them.

3.2 Minimal covering templates

$\mathrm{Min}(D)$ is the set of $\succeq$-minimal covering templates in DT°. A covering $T \in \mathrm{DT}^\circ$ is *saturated* if (S1) every argument place is used (z_m occurs in some β^d_M, for every M and m), and (S2) for every M the roots of the β^d_M, $d \in D$, are not all equal (a hole z_m counting as the root z_m).

Theorem 3.3 (Minimal covering templates)**.** *[proved; computed]*[single Thm B] *Let D be nonempty and finite.*

(a) Every covering $T \in \mathrm{DT}^\circ$ satisfies $T \succeq T'$ for some saturated covering T'.

(b) In a saturated covering template, skel $\subseteq C(D)$*; every pattern occurrence of a metavariable sits at a slot σ and has argument* set $Y_\sigma(D)$*; every occurrence sits in $A(D)$; and the arguments of each occurrence are determined by D, its position and the pattern slot of its metavariable. So there are at most $2^{|A(D)|}\,|\mathrm{slots}(D)|^{|A(D)|}$ saturated covering templates up to renaming.*

(c) Every minimal covering template is saturated. So $\mathrm{Min}(D)$ is finite up to renaming, and every covering template is $\succeq$ some member of $\mathrm{Min}(D)$.

(d) If $D \subseteq \mathrm{inst}(T^*)$, $T^* \in \mathrm{DT}^\circ$, some $T_m \in \mathrm{Min}(D)$ has $T^* \succeq T_m$, hence $\mathrm{inst}(T_m) \subseteq \mathrm{inst}(T^*)$.

Proof idea. (a) Drop an argument place that no value uses, or make rigid the common root of all values of M (with fresh metavariables below it); both steps keep coverage and determinacy, and $(|C(D)| - |\mathrm{skel}|, \sum_M \mathrm{arity}(M))$ decreases lexicographically. (b) At a pattern occurrence $d|_\sigma = \beta^d_M[\bar{y}]$, so (S2) puts σ outside $C(D)$ and (S1) gives the argument set; other arguments are forced by Lemma A.1(ii). (c) uses Lemma A.2. Appendix B.2. □

Part (d) is the finiteness hypothesis of the clustering method (Theorem 6.8): data of one schema always have a minimal covering template below it, hence a sound one. Metavariable-free arguments are essential: they make the arguments forced, and with metavariables inside arguments finiteness fails (Remark A.3), contrary to a conjecture of the earlier analysis[prior induction C8.3]. Before an anchor $\mathrm{Min}(D)$ need not have a least element: $\{\mathrm{Ind}(x = x), \mathrm{Ind}(0 = x)\}$ has exactly four minimal covering templates, one of them below T_{Ind} (Example B.1, which corrects the earlier count of eight).

3.3 The cautious verifier is a feature test

Let $T_0(D)$ be the skeleton $C(D)$ with a fresh metavariable $M_\sigma(Y_\sigma)$ at every slot (arguments in increasing order). For an Eq-feature $\varphi = \mathsf{Eq}(\sigma_0, r, \bar{u})$ of D, let T_φ be $C(D)$ cut at r, with $M_\sigma(Y_\sigma)$ at every slot not below r and the derived occurrence $M_{\sigma_0}(\bar{u})$ at r. Both are in DT° and cover D.

Theorem 3.4 (The cautious verifier checks features)**.** *[proved; computed]*[single Thm C] *For every nonempty finite set D of sentences,*

$$\mathrm{Acc}(D) = \mathrm{Feat}(D) = \mathrm{inst}(T_0(D)) \cap \bigcap \{\mathrm{inst}(T_\varphi) : \varphi \text{ an Eq-feature of } D\}.$$

Proof idea. $\mathrm{inst}(T_0(D))$ consists of the sentences with all Sym and Scope features, and T_φ adds φ; these templates cover D, so $\mathrm{Acc}(D) \subseteq \mathrm{Feat}(D)$. Conversely, a covering T is $\succeq$ a saturated T' (Theorem 3.3(a)), and by Theorem 3.3(b) $\mathrm{inst}(T')$ is cut out by D-features: Sym on its skeleton, Scope at one pattern slot per metavariable, and an Eq feature for every other occurrence. So $\mathrm{Feat}(D) \subseteq \mathrm{inst}(T)$. Appendix B.3. □

Example 3.5 (Features)**.** [proved; computed][single Thm C] For $D = \{0 + 0 = 0,\ S0 + 0 = S0\}$ the common prefix is $\cdot + 0 = \cdot$, with two slots: the left summand σ and the right-hand side r. The D-features are Sym at the three common positions, Scope at both slots (vacuous here, as no binder is in scope), and $\mathsf{Eq}(\sigma, r, \emptyset)$ with its converse $\mathsf{Eq}(r, \sigma, \emptyset)$: in both data the right-hand side equals the left summand. So $T_0(D) = (z_1 + 0 = z_2)$ and $T_\varphi = (z_1 + 0 = z_1)$ for either Eq feature, and $\mathrm{Feat}(D) = \mathrm{Acc}(D) = \mathrm{inst}(z + 0 = z)$: D is an anchor for $z + 0 = z$.

Corollary 3.6 (A polynomial verifier)**.** *[proved]*[single Cor C.1] *With $m = |A(D)| \le \min_d |d|$ and $n = \sum_d |d|$, the D-features are listed in time $O(m^2 n)$, there are $O(m^2)$ of them, and a query q is decided in time $O(m^2|q|)$. No enumeration of* $\mathrm{Min}(D)$ *is needed.*

Reading. The verifier accepts q iff q has the data's common skeleton, no slot of q mentions a bound variable that no datum mentions there, and every equation "content at r = content at slot σ with variables replaced by $\bar{u}$" that holds in all data holds in q: the second-order analogue of Plotkin's "equal columns get equal variables" (Plotkin, 1970; Reynolds, 1970). $\mathrm{Min}(D)$ can be exponentially large (Proposition 3.13), but its intersection is cut out by polynomially many equations, so the feature test replaces the earlier learner's enumeration of $\mathrm{Min}(D)$. The relation to pattern anti-unification and to other second-order generalization is discussed in Appendix B.3.

3.4 Anchors

Let $T^* \in \mathrm{DT}^\circ$ and $D = \{s_1, \ldots, s_N\}$, $s_i = T^*\theta_i$, $N \ge 1$, $\beta^i_M := \theta_i(M)$. For $\sigma \in \mathrm{Occ}(T^*)$ carrying $M(\bar{t}_\sigma)$ let $Y^*_\sigma := \bigcup_m \mathrm{FV}(t_{\sigma,m})$. Consider pairs (σ, r) with $\sigma \in \mathrm{Occ}(T^*)$, $r \in \mathrm{skel}(T^*) \cup \mathrm{Occ}(T^*)$, $\sigma \perp r$, of the same sort; such a pair is *valid* if r carries an occurrence $M(\bar{t}_r)$ of the *same* metavariable with $\bar{t}_r = \bar{t}_\sigma[\bar{u}_0]$ for some $\bar{u}_0$. A valid pair gives an equation that holds on all of $\mathrm{inst}(T^*)$.

Definition 3.7 (Witness events; richness)**.** [single §6]

- (R*) (root variation at every occurrence): for every $\sigma \in \mathrm{Occ}(T^*)$ the roots of $s_1|_\sigma, \ldots, s_N|_\sigma$ are not all equal. (At a pattern occurrence this is Plotkin's (R) for the values; at a derived occurrence $M(\bar{t})$ it concerns the plugged values $\beta^i_M[\bar{t}]$.)
- (N) (non-vacuity): for every metavariable M and argument place m, some β^i_M contains z_m.
- (U) (no coincidence): for no non-valid pair (σ, r) is there a $\bar{u}$ on Y^*_σ with $s_i|_r = (s_i|_\sigma)[\bar{u}]$ for all i.
- (Rich): the signature has a relation symbol of arity ≥ 1, and, if T^* has a term-valued metavariable, closed terms with two different root symbols exist (automatic with parameters). (Rich) holds for arithmetic, and for set theory with parameters (always available

in closure-normal form); in set theory without parameters it fails for targets with a term-valued metavariable (Example B.3), but holds for the ZF schemas, whose metavariables are formula-valued.

For example, $T^* = \forall x(0 = f(x))$ with data $\forall x(0 = 0)$, $\forall x(0 = x)$ satisfies (R*) and (N), but (U) fails: in both data the content at the rigid 0 is the content at the slot of $f(x)$ under $x \mapsto 0$, a coincidence at a non-valid pair. Indeed $\forall x(f(0) = f(x))$ covers D and misses $\forall x(0 = Sx)$.

Theorem 3.8 (Anchor theorem)**.** *[proved; computed]*[single Thm D] *If* (R*)*,* (N) *and* (U) *hold, D is an anchor for* inst(T^*) *in* DT$^\circ$*; this needs no assumption. Under* (Rich)*, every anchor satisfies* (R*)*,* (N) *and* (U)*.*

Proof idea. ($\Leftarrow$) (R*) and (N) make the common prefix, slots and scopes of D those of T^*, so instances of T^* have every Sym and Scope feature; by (U) and forcing every Eq-feature comes from a valid pair, and so holds on inst(T^*). Hence inst$(T^*) \subseteq$ Feat$(D) =$ Acc(D). ($\Rightarrow$) When an event fails, the corresponding Sym, Scope or Eq property is a D-feature, and (Rich) supplies an instance of T^* violating it. Appendix B.4. □

So the events one would first try (root variation and distinctness, as for first-order patterns) are right in spirit but must be read occurrence-wise: root variation must hold at *derived* occurrences too, and Plotkin's distinctness (D) is only the simplest coincidence.

Corollary 3.9 (Specializations)**.** *[proved; computed]*[single Cor D.1–D.3]

(a) First-order targets *(all metavariables* 0*-ary, possibly under binders):* (R*)$\wedge$(N)$\wedge$(U) *is equivalent to Plotkin's* (R)$\wedge$(D)*. So* DT$^\circ$ *has the same anchors on first-order targets as the class of first-order patterns (Plotkin, 1970; Reynolds, 1970): the larger class costs no data. For the instance schema of $\forall x\varphi(x)$ (Theorem 4.2), two instances whose terms have different roots are an anchor (with several variables,* (D) *is needed as well).*

(b) Formula metavariables only: (R*) *is equivalent to* (R)*, and under* (R)$\wedge$(N)*,* (U) *is equivalent to (D^+): for $M \neq M'$ and occurrences σ of M, r of M', no $\bar{u}$ has $\beta^i_{M'}[\bar{t}_r] = (\beta^i_M[\bar{t}_\sigma])[\bar{u}]$ for all i.* With a single formula metavariable, D is an anchor iff (R) and (N) hold.

(c) PA induction and ZF. *In closure-normal form, T_{Ind} is anchored iff the motives' main symbols are not all equal and some motive has x free (the earlier anchor theorem for induction); each ZF schema of Table 6 iff the bodies' main symbols are not all equal and every argument place of P is used by some body (freshness is built in). Single axioms are ground templates, each its own anchor. One datum is never an anchor of a schema; two can be.*

In (a)–(c) the "if" directions need no assumption and the "only if" directions use (Rich)*.*

Section 5 treats the ZF schemas in all formulations, encodings and classes. The events cannot be weakened: (R)+(N)+(D), and (U) checked only at pattern occurrences, both admit non-anchors, even for one metavariable (Proposition B.2); and (Rich) cannot be dropped (Example B.3). The exact anchor condition without (Rich) is open.

3.5 Rates

Let Λ be a law with countable support on ground substitutions for T^*, $\theta_1, \theta_2, \ldots$ i.i.d. $\sim \Lambda$, and $D_N = \{T^*\theta_1, \ldots, T^*\theta_N\}$. For $\sigma \in \mathrm{Occ}(T^*)$ let $\rho_\sigma := 1 - \max_h \Lambda(\mathrm{root}((T^*\theta)|_\sigma) = h)$, the probability that the root at σ is not the most likely one, and let $\nu_{M,m} := \Lambda(z_m \in \theta(M))$, the

probability that the value of M uses the hole z_m. For a non-valid pair (σ, r) (their set is $U(T^*)$) and a map $\bar{u}$, let $A_{\sigma,r,\bar{u}} := \{\theta : (T^*\theta)|_r = ((T^*\theta)|_\sigma)[\bar{u}]\}$ be the coincidence of r with σ via $\bar{u}$; let $\kappa_{\sigma,r} := \Lambda^{\otimes 2}\{(\theta, \theta') : \text{no } \bar{u} \text{ has } \theta, \theta' \in A_{\sigma,r,\bar{u}}\}$, the probability that two independent draws share no such coincidence, and $\chi_{\sigma,r} := \sup_{\bar{u}} \Lambda(A_{\sigma,r,\bar{u}})$, the probability of the most likely single one. Let $c(T^*) := |\mathrm{Occ}(T^*)| + \sum_M \mathrm{arity}(M) + |U(T^*)|$ (the number of events) and λ the minimum of all $\rho_\sigma, \nu_{M,m}, \kappa_{\sigma,r}$. For a *ground* target (e.g. a single ZFC axiom) $c(T^*) = 0$; we set $\lambda := 1$, and in general $\bar{c} := \max(c(T^*), 1)$. A ground T^* has $\mathrm{Acc}(\{T^*\}) = \{T^*\}$, so D_N is an anchor iff $N \geq 1$.

Theorem 3.10 (Sample complexity)**.** *[proved; computed]*[single Thm E] *The upper bounds need no assumption; the lower bounds use* (Rich)*.*

(a) For $N \geq 1$,

$$\Pr[D_N \text{ is not an anchor}] \;\leq\; \sum_\sigma (1-\rho_\sigma)^{N-1} + \sum_{M,m} (1-\nu_{M,m})^N + \sum_{(\sigma,r)\in U(T^*)} (1-\kappa_{\sigma,r})^{\lfloor N/2 \rfloor} \;\leq\; c(T^*)(1-\lambda)^{\lfloor N/2 \rfloor},$$

and this probability is at least each of $(1-\rho_\sigma)^N$, $(1-\nu_{M,m})^N$ and $\chi_{\sigma,r}^N$, for every σ, (M,m) and $(\sigma,r) \in U(T^*)$. ($D_0 = \emptyset$ is never an anchor: $\mathrm{Acc}(\emptyset) = \emptyset$.)

(b) $\lambda > 0$ iff the set $\{T^*\theta : \theta \in \mathrm{supp}\,\Lambda\}$ satisfies (R*), (N), (U)*. Then the cautious verifier is exact with probability $\geq 1-\delta$ once $N \geq 1 + (2/\lambda)\ln(\bar{c}/\delta)$; for a ground T^*, once $N \geq 1$.*

(c) Tagged data. *For k targets $T_i^* \in \mathrm{DT}^\circ$ (ground ones allowed) cited by tags of probability π_i, with parameters $\bar{c}_i, \lambda_i$, the per-tag cautious verifier (the version space factors over tags, Hänni and Claude (Anthropic), 2026) fails to be exact after N tagged data with probability at most $\sqrt{e}\sum_i \bar{c}_i \exp(-3N\pi_i\lambda_i/8)$. Order $1/(\pi_i\lambda_i)$ data are necessary in general: the failure probability is at least $(1-\pi_i)^N$ (no tag-i datum; exact for a ground tag), and at least $(1-\pi_i\rho)^N$ if some head at a pattern occurrence of T_i^* has probability $1-\rho$.*

Proof idea. By the "if" half of Theorem 3.8, failure lies in the union of the failures of the events. All roots at σ agree with probability $\sum_h p_h^N \leq (1-\rho_\sigma)^{N-1}$; a coincidence on D_N implies one on each of $\lfloor N/2 \rfloor$ disjoint, independent pairs. A union bound gives (a); (c) averages over the binomial number of data per tag. Appendix B.5. □

For one formula metavariable (induction, the ZF schemas) the coincidence terms vanish (Corollary 3.9(b)) and the bound is $(1-\rho)^{N-1} + \sum_m (1-\nu_m)^N$. For induction, with $p_{\max} := 1-\rho$ and $q := \nu$, this is exactly the earlier bound $p_{\max}^{N-1} + (1-q)^N$ prior induction C5, which dominates the exact inclusion–exclusion value (Theorem 5.11). The coincidence terms need not be tight: their exact rate is $\chi_{\sigma,r}$, which can be smaller than the rate $\sqrt{1-\kappa_{\sigma,r}}$ of the upper bound (Proposition B.4).

3.6 Escalations

In the protocol of Definition 2.6, a query the verifier does not accept is *escalated*: an oracle labels it, and a valid query joins the data. Against an honest prover (all queries are target instances) the cautious verifier's escalated queries form a chain $q_1, \ldots, q_m$ with $q_j \notin \mathrm{Acc}(\{q_1, \ldots, q_{j-1}\})$, and every such chain is forced by the honest prover that asks it in order (Hänni and Claude (Anthropic), 2026). $\mathrm{Esc}(\mathcal{H}; N)$ is the supremum of the lengths of such chains inside members of $\mathcal{H}$, from no data, on sentences of size $\leq N$; the first query is always escalated, as $\mathrm{Acc}(\emptyset) = \emptyset$. For first-order schemas over a term signature, $\mathrm{Esc}(H_1(\mathrm{FO}); N) \leq N + 1$ (Hänni and Claude (Anthropic), 2026). Every chain is honest for the universal template (a 0-ary formula metavariable).

Theorem 3.11 (Escalation bound)**.** *[proved; computed]*[single Thm F] *Let d have n positions and let $q_j \notin \mathrm{Acc}(\{d, q_2, \ldots, q_{j-1}\})$ for $j = 2, \ldots, m$ (no honesty needed). Then*

$$m - 1 \;\le\; n + \sum_p \mathrm{b}(p) + \#\{(\sigma, r) : \sigma \perp r\} \;\le\; n + n\,\mathrm{b}_{\max} + n(n-1),$$

with p, σ, r ranging over positions of d. Hence $\mathrm{Esc}(\mathrm{DT}^\circ; N) \le 1 + N + 2N(N-1) \le 2N^2 + 1$.

Proof idea. Each escalation shrinks the common prefix, enlarges a scope, or removes an equation for good (Appendix B.6).

The bound is tight: $\mathrm{Esc}(\mathrm{DT}^\circ; N) = \Theta(N^2)$, by an explicit chain of scope-growing data, which has a version inside $\mathrm{inst}(T_{\mathrm{Ind}})$ (Proposition B.6). This refutes, asymptotically, the earlier conjecture that escalations are at most linear in the size of the first escalated datum[prior induction C11], for DT°, for the class with nested arguments, for the universal template and for the induction target. Without binders at most $2N$ escalations occur (Proposition B.7); whether equation kills alone are linear is open (Conjecture B.9).

3.7 Finite elasticity and unions

A class has *finite elasticity* if there are no infinite $w_0, w_1, \ldots$ and members $L_1, L_2, \ldots$ with $\{w_0, \ldots, w_{n-1}\} \subseteq L_n \not\ni w_n$ for all n (Wright, 1989; Motoki et al., 1991). Let $H_k(\mathrm{DT}^\circ) = \{\bigcup_{l \le k} \mathrm{inst}(T_l) : T_l \in \mathrm{DT}^\circ\}$ and $\mathrm{Acc}_k(D) = \bigcap\{U \in H_k(\mathrm{DT}^\circ) : D \subseteq U\}$.

Corollary 3.12 (Elasticity and unions)**.** *[proved; known]*[single Cor F.2–F.4, Prop F.3] *(i) $\{\mathrm{inst}(T) : T \in \mathrm{DT}^\circ\}$ has finite elasticity (an elastic sequence is an escalation chain, bounded by Theorem 3.11), but not finite thickness: $\forall x(0 = 0) \wedge (0 = 0)$ is an instance of $\forall x P(x) \wedge P(t)$ for every closed t, and these instance sets differ. (ii) Finite elasticity survives unions of at most k members (Wright, 1989; Motoki et al., 1991). So $H_k(\mathrm{DT}^\circ)$ has finite elasticity, every target in it has a finite anchor (Hänni and Claude (Anthropic), 2026), and the cautious verifier over $H_k(\mathrm{DT}^\circ)$ is sound at all times and exact from some finite time on, on every text. (iii) For fixed k, $\mathrm{Esc}(H_k(\mathrm{DT}^\circ); N) = \Omega(N^k)$, already for one target over $=, +, S, 0$ (Proposition B.10).*

Explicit anchors for unions, via the failure sets of the three events, are given by Proposition B.11; a ground target (a single ZFC axiom) is learned in a union only once it is itself a datum. The union verifier reduces to Acc: $q \in \mathrm{Acc}_k(D)$ iff every partition of D into at most k nonempty blocks has a block B with $q \in \mathrm{Acc}(B)$ (Proposition B.12); this is polynomial for $k \le 2$ and coNP-complete for fixed $k \ge 3$ (Theorem 6.16).

3.8 Complexity of the version space

Matching and generality are linear (Theorem 2.5), $\mathrm{Acc}(D)$ is polynomial (Corollary 3.6), and, under (Rich), whether D is an anchor for a *given* T^* is polynomial (at most $|\mathrm{Occ}(T^*)|\,|T^*|$ pairs, each decided by forcing). But $\mathrm{Min}(D)$ itself can be huge:

Proposition 3.13 (Exponentially many minimal templates)**.** *[proved; computed]*[single Prop G.2] *Let C_n be the conjunction of n copies of $0 = 0$ and $D_n = \{\forall x(x = x) \wedge C_n,\ \forall x(0 = 0) \wedge C_n\}$, of total size $8n + 8$. Then $|\mathrm{Min}(D_n)| = 4^n$ (each of the $2n$ zeros of C_n may be rigid or a derived occurrence $f(0)$), while $\mathrm{Acc}(D_n) = D_n$.*

So $\mathrm{Min}(D)$ cannot be listed in polynomial time, and the verifier must not enumerate it. But whether D has a least covering template is decidable, and the lgg computable, in polynomial time (Proposition B.13), and, under (Rich), an anchor always has one:

topic	result
version space	every covering template lies above a member of the finite set $\mathrm{Min}(D)$, and one member lies below the target (Theorem 3.3; the earlier finiteness conjecture is true for DT°, false with nested arguments); 4 members, not 8, for $\{\mathrm{Ind}(x = x), \mathrm{Ind}(0 = x)\}$ (Example B.1)
verifier	$\mathrm{Acc}(D) = \mathrm{Feat}(D)$, decided in polynomial time (Theorem 3.4, Corollary 3.6)
anchors	iff (R*), (N), (U), "only if" under (Rich) (Theorem 3.8); first-order: (R)+(D); one formula metavariable: (R)+(N) (Corollary 3.9); weaker events fail (Proposition B.2); (Rich) cannot be dropped (Example B.3)
rates	exponential tails for one template and for tagged data, ground targets included (Theorem 3.10); exact rate of the coincidence terms (Proposition B.4)
escalations	$\mathrm{Esc}(\mathrm{DT}^\circ; N) = \Theta(N^2)$ (Theorem 3.11 and Proposition B.6; the earlier linear conjecture refuted); $\leq 2N$ without binders (Proposition B.7); equation kills linear (Conjecture B.9, conjecture)
unions	finite elasticity, infinite thickness, anchors, $\Omega(N^k)$ escalations (Corollary 3.12); explicit anchors (Proposition B.11); verifier via partitions (Proposition B.12)
complexity	$\lvert\mathrm{Min}(D)\rvert$ up to 4^n (Proposition 3.13); lgg existence polynomial (Proposition B.13); an anchor's lgg is the target, under (Rich) (Corollary 3.14)

Table 4: Results of Section 3. All are proved except Conjecture B.9 (conjecture) and Example B.1 (computed); Corollary 3.12(ii) uses known results; results added after the referee round are listed in Appendix G.3. The "only if" half of Theorem 3.8 needs (Rich), which always holds with parameters.

Corollary 3.14 (An anchor determines its template)**.** *[proved; computed]* single Cor G.4 *Under* (Rich)*, if D is an anchor for* $\mathrm{inst}(T^*)$ *in* DT°*, then T^* is the least covering template of D (up to renaming), and Proposition B.13 computes it in polynomial time.*

This extends the earlier result "T_{Ind} is the lgg of every induction anchor" to all of DT° (proofs in Appendix B.8).

3.9 Summary

For one target in DT°, positive data suffice and checking is cheap: the verifier is sound from the first datum, a polynomial feature test, exact after two well-chosen instances for PA induction and every ZF schema, and fails with exponentially small probability on random data. What one template does not give cheaply is bounded escalation (quadratic in the data size) and, for unions, more than the existence of anchors. Table 4 collects the results; open problems are in Section 8.3.

4 Universal axioms from their instances

Question 1 asks whether the method developed for PA induction (a cautious verifier over a class of templates, sound at all times and exact once the data contain an anchor; Lemma 2.8) also learns an axiom $\forall x \varphi(x)$ from sentences $\varphi(x)$. We read the data as instances $\varphi(t)$ at terms t; a datum $\varphi(x)$ with x free is, in closure-normal form, already the axiom (Proposition 4.10). The answer has four parts; proofs are in Appendix C.

(1) *The instance schema $\varphi(z)$ is learned, easily.* It is a first-order pattern. First-order anti-unification and the DT° learner both learn it, with Plotkin's events (R) and (D) as the anchor condition; for one variable, two instances such as $\varphi(0)$, $\varphi(S0)$ suffice (Theorem 4.2

and Proposition 4.3). In SO° these events do not suffice. Failure rates have exact formulas for one and two variables and an exponential bound in general (Theorem 4.4).

(2) *The instance schema is not the sentence.* The ω-step "all closed instances are true in M, hence so is $\forall\bar{x}\varphi$" is *truth-safe* in M (never leads from true closed instances to a false universal) for every φ iff the closed-term substructure of M is elementary (Proposition 4.7): in $\mathbb{N}$ and in every model of true arithmetic, but not in $\mathbb{R}$, in V, or in a model of $\mathsf{PA}+\neg\mathrm{Con}(\mathsf{PA})$ (Example 4.8). In general it is not derivable: $\mathsf{Q} \vdash 0+\underline{n} = \underline{n}$ for each n, but $\mathsf{Q} \nvdash \forall x\,(0+x = x)$ (Example 4.9).

(3) *A cautious learner takes this step silently* when parameters are admissible values (Proposition 4.5); a closedness guard prevents it, and data with a parameter at every variable license it as ordinary generalization (Proposition 4.10). In a first-order encoding, *capture* instances, possibly false in every structure, must be excluded by a guard (Proposition 4.6). Evaluating closed instances never refutes a false universal whose closed instances are true (Proposition C.5).

(4) *Without labels*, $\forall x\varphi$ is learned by a sound merge of instances (Proposition 6.13).

4.1 The target

Fix a first-order language L with at least one constant. Let $\varphi(x_1, \ldots, x_k)$, $k \geq 1$, be a parameter-free L-formula in which each x_i occurs free. A term is *closed* if it has neither variables nor parameters.

Definition 4.1 (Instance schema; witness events)**.** cases §1.1 The *instance schema* $\sigma_\varphi := \varphi[z_1/x_1, \ldots, z_k/x_k]$ has a 0-ary term metavariable z_i at each *free* occurrence of x_i. Its *closed instances* are $\mathrm{inst}_c(\varphi) = \{\varphi(\bar{t}) : \bar{t} \text{ closed}\}$, its *open instances* $\mathrm{inst}_o(\varphi) = \{\varphi(\bar{t}) : \bar{t} \text{ terms over } L \cup \mathrm{Par}\}$, Par the parameters. For data $D = \{\varphi(\bar{t}_1), \ldots, \varphi(\bar{t}_N)\}$, $\bar{t}_j = (t_{j,1}, \ldots, t_{j,k})$, Plotkin's events (Lemma A.4) read: (R_i): the roots of $t_{1,i}, \ldots, t_{N,i}$ are not all equal; ($\mathrm{D}_{ii'}$) ($i < i'$): some j has $t_{j,i} \neq t_{j,i'}$. (R) and (D) are their conjunctions.

Under the λ convention (Section 2.2), the default, $\mathrm{inst}(\sigma_\varphi)$ is $\mathrm{inst}_o(\varphi)$ if parameters are admissible values and $\mathrm{inst}_c(\varphi)$ if not. In the first-order encodings (Nm) and (dB) (Appendix A.6) a metavariable may also take a bound-variable name or index, which adds *capture instances* (Section 4.4). The question has content for languages with closed terms, such as L_A; in $L_\in$ every instance is at parameters, and a datum $\varphi(\bar{p})$ with distinct parameters already is the axiom. Several results assume ($\mathrm{Rich_c}$): *for every k there are pairwise distinct closed $t_1, \ldots, t_k$ and pairwise distinct closed $t'_1, \ldots, t'_k$ with* $\mathrm{root}(t_i) \neq \mathrm{root}(t'_i)$ *for all i* (in L_A: $t_i = S^i0$, $t'_i = 0 + S^i0$). ($\mathrm{Rich_c}$) concerns closed terms only and differs from (Rich) of Definition 3.7.

4.2 Anchors

We use Lemma A.4; every first-order schema covering D is $\succeq \mathrm{lgg}(D)$. $H_1(\mathrm{FO})$ denotes the single first-order schemas of the syntax at hand (λ, Nm or dB). Under ($\mathrm{Rich_c}$) instance sets determine schemas: for a first-order σ with 0-ary term metavariables and any first-order τ, $\mathrm{inst}(\sigma) \subseteq \mathrm{inst}(\tau)$ iff $\mathrm{inst}(\tau)$ contains the closed instances of σ iff $\tau \succeq \sigma$, in each syntax (Proposition C.1 in Appendix C.1).

Theorem 4.2 (Anchors for the instance schema)**.** *[proved; computed]* cases Thm A2 *Assume* ($\mathrm{Rich_c}$)*, and let $D = \{\varphi(\bar{t}_1), \ldots, \varphi(\bar{t}_N)\} \subseteq \mathrm{inst}_c(\varphi)$, $N \geq 1$. In each syntax the following are equivalent:*

(i) *D is an anchor for* $\mathrm{inst}_c(\varphi)$ *in* $H_1(\mathrm{FO})$*, i.e. every first-order schema covering D contains* $\mathrm{inst}_c(\varphi)$*;* (ii) $\mathrm{lgg}(D) \equiv \sigma_\varphi$*;* (iii) (R) *and* (D) *hold. Without* ($\mathrm{Rich_c}$)*, (iii)⇔(ii)⇒(i).*

Proof in Appendix C.1. One datum is never an anchor; two with componentwise different roots and pairwise distinct components are. From numeral data every anchor of $z + 0 = z$ contains $0 + 0 = 0$, which slows the rates and matters for unlabelled mixtures (Remark E.5). Further corner cases and the computational checks: Appendices C.1 and C.6.

The much larger class DT° of the induction method needs no more data for first-order targets; without determinacy it does.

Proposition 4.3 (Anchors in DT° and in SO°)**.** *[proved; computed]*[cases A2′; single D.1; untagged 5.5] *Work under the λ convention.*

(a) Let $\sigma \in \mathrm{FO}$ (0-ary metavariables of term or formula sort, possibly under binders) and $D = \{\sigma\theta_1, \ldots, \sigma\theta_N\}$ with values that have no free bound variables (parameters allowed; a formula value may contain its own binders). If (R) *and* (D) *hold, every $T \in \mathrm{DT}^\circ$ covering D satisfies $T \succeq \sigma$. In particular, data $D \subseteq \mathrm{inst}_o(\varphi)$ with* (R) *and* (D) *are an anchor for* $\mathrm{inst}(\sigma_\varphi)$ *in* $H_1(\mathrm{DT}^\circ)$*, and* $\mathrm{Min}(D) = \{\sigma_\varphi\}$*.*

(b) Under ($\mathrm{Rich_c}$)*, if $D \subseteq \mathrm{inst}_c(\varphi)$ violates* (R) *or* (D)*, then* $\mathrm{lgg}(D) \in \mathrm{FO} \subseteq \mathrm{DT}^\circ$ *covers D and misses a closed instance. So the anchors for* $\mathrm{inst}_c(\varphi)$ *in* $H_1(\mathrm{DT}^\circ)$ *are exactly those of Theorem 4.2.*

(c) In SO° *this fails, e.g. for $x + 0 = x$ and the data $0 + 0 = 0$, $S0 + 0 = S0$ (Example 2.3(4)): a metavariable without a pattern occurrence can use different arguments on different data.*

Proof in Appendix C.1, with binder analogues of (c); the key to (a) is an identity lemma (Lemma C.2). For one variable, (a) is the two-point lemma (Lemma 6.12); in general it is the "if" half of Corollary 3.9(a), proved directly.

4.3 Rates

Let $\bar{t} \sim \Lambda$ i.i.d., $p_{i,f} := \Lambda(\mathrm{root}(t_i) = f)$, $c_{ii'} := \Lambda(t_i = t_{i'})$, and let P_N be the probability that N instances violate (R) or (D), i.e. (under ($\mathrm{Rich_c}$)) are not an anchor in $H_1(\mathrm{FO})$ or $H_1(\mathrm{DT}^\circ)$.

Theorem 4.4 (Failure probabilities)**.** *[proved; computed]*[cases Thm A3] (a) *For $k = 1$, $P_N = \sum_f p_f^N$.* (b) *For $k = 2$, an exact inclusion–exclusion formula in the root and coincidence probabilities of (t_1, t_2) holds (Appendix C.2).* (c) *In general $P_N \le \sum_{i,f} p_{i,f}^N + \sum_{i<i'} c_{ii'}^N \le (2k + \binom{k}{2})(1-\rho)^N \le (2k + \binom{k}{2})e^{-N\rho}$, where $\rho = \min(\min_i \rho_i, \min_{i<i'} \Lambda(t_i \neq t_{i'}))$ and $\rho_i = \max_G \min(\Lambda(\mathrm{root}(t_i) \in G), \Lambda(\mathrm{root}(t_i) \notin G))$ is the best split of the roots, as in the coupon-collector bound of Hänni and Claude (Anthropic) (2026).*

Proof (inclusion–exclusion, union bound) in Appendix C.2. For a Galton–Watson term law ($\rho = 0.5$) the exact failure probability is below 0.01 from $N = 6$ ($k = 1, 2$), while (c) needs $N = 11$ and 13 in its exponential form, 8 and 9 in its $(1-\rho)^N$ form [cases q1_lgg (4); referee r1 (5)]. *Experiment E1* (Section 7.3, Figure 1) confirms (a): the implemented DT°_F verifier and the first-order lgg became exact at the same N in every run, with mean close to the prediction of (a) [experiments E1, Prop E7].

4.4 Parameters, guards and capture

Once the data contain an anchor, the cautious verifier over $\mathcal{H}$ accepts exactly the closure $\mathrm{cl}_\mathcal{H}(h^*) = \bigcap\{h \in \mathcal{H} : h \supseteq h^*\}$ of the target, so it is sound iff the target is $\mathcal{H}$-closed (Lemma 2.8).

Proposition 4.5 (Open instances and the hidden ω-step). *[proved; computed]*[cases Prop A4] (a) *For distinct parameters $\bar{p}$, $\varphi(\bar{p}) \in \mathrm{inst}_o(\varphi)$ has universal closure $\forall\bar{x}\varphi$, and $\mathrm{inst}_o(\varphi)$ and $\forall\bar{x}\varphi$ are interderivable in first-order logic.* (b) *Assume* ($\mathrm{Rich_c}$) *and the λ convention with parameters admissible. In $H_1(\mathrm{FO})$ and in $H_1(\mathrm{DT}^\circ)$, $\mathrm{cl}(\mathrm{inst}_c(\varphi)) = \mathrm{inst}(\sigma_\varphi) = \mathrm{inst}_o(\varphi) \supsetneq \mathrm{inst}_c(\varphi)$. So once closed data contain an anchor, the cautious verifier accepts $\varphi(\bar{p})$: it takes the ω-rule step "all closed instances, hence $\forall\bar{x}\varphi$", which no datum licenses.* (c) *With the guard* $\mathrm{closed}(z_i)$ *("the value contains no variable and no parameter"), $\mathrm{inst}_c(\varphi)$ is realizable in every syntax. By the most-specific-guard rule (Appendix A.6), closed data teach the guard, and the learner accepts neither $\varphi(\bar{p})$ nor any capture instance; one datum with a parameter at z_i deletes* $\mathrm{closed}(z_i)$.

Proof in Appendix C.3. The class DT° has no guards, so the DT° learner on closed data takes the ω-step after the first anchor whenever parameters are admissible: in E1, from $0 + 0 = 0$ and $1 + 0 = 1$ it accepts $w_0 + 0 = w_0$, the axiom in parameter form.

In a first-order encoding the step can be worse [cases referee F1]. A value $\theta(z_i)$ is *captured* if it contains a variable bound at some free occurrence of x_i (in (Nm), a name in the scope of a binder of that name; in (dB), an index that refers to an enclosing binder). $\mathrm{Cap}(\varphi)$ is the set of instances $\sigma_\varphi\theta$ with a captured value.

Proposition 4.6 (Capture). *[proved; computed]*[cases Prop A4′] *Work in syntax (Nm) or (dB).*

(a) $\mathrm{inst}(\sigma_\varphi) = \mathrm{inst}_c(\varphi) \cup \mathrm{inst}_o(\varphi) \cup \mathrm{Cap}(\varphi)$ *(the middle term if parameters are admissible), and* $\mathrm{Cap}(\varphi) \cap \mathrm{inst}_o(\varphi) = \emptyset$.

(b) Lemma C.3 says for each syntax exactly when $\mathrm{Cap}(\varphi) \neq \emptyset$*; in (dB), iff every free occurrence of some x_i lies under a binder.*

(c) Capture instances can be false in every structure while all genuine instances hold in every structure with two elements ($\exists y\,\neg(y = x)$ gives $\exists y\,\neg(y = y)$), or false in $\mathbb{N}$ while $\forall x\varphi$ is logically valid ($\exists y\,(y = Sx)$ gives $\exists y\,(y = Sy)$).

(d) Hence, if $\mathrm{Cap}(\varphi) \neq \emptyset$*, the unguarded cautious verifier accepts after an anchor sentences outside* $\mathrm{inst}_o(\varphi)$*, for suitable φ some false in every structure: it is unsound outright, not merely ω-incomplete.*

(e) With guards $\mathrm{closed}(z_i)$ *and* $\mathrm{nocap}(z_i)$ *(no captured value: the free-for condition, i.e. the λ convention as a guard), closed data teach both, and after an anchor the accepted set is* $\mathrm{inst}_c(\varphi)$*. A datum whose value at z_i contains a parameter deletes* $\mathrm{closed}(z_i)$ *but not* $\mathrm{nocap}(z_i)$*; once this has happened for every i (for $k = 1$, after one such datum), the accepted set after an anchor is* $\mathrm{inst}_o(\varphi)$*. Capture instances are never accepted.*

Proof and computed checks in Appendix C.3. Under the λ convention $\mathrm{Cap}(\varphi)$ does not arise.

4.5 The ω-gap

For an L-structure M let $M_0 := \{t^M : t \text{ closed}\}$ be its *closed-term substructure*. The criterion of Tarski and Vaught (1957) gives (proof in Appendix C.4):

Proposition 4.7 (When closed instances decide universals). *[proved; known]*[cases Prop A5] *For an L-structure M the following are equivalent:* (i) *for every L-formula $\varphi(\bar{x})$, $M \models \forall\bar{x}\varphi$ iff $M \models \varphi(\bar{t})$ for all closed $\bar{t}$;* (ii) *$M_0 \preceq M$. In particular (i) holds for term-generated M ($M_0 = M$), e.g. $\mathbb{N}$, and for every model of true arithmetic.*

syntax, guards and data	after an anchor the cautious verifier accepts
λ convention, no parameters admissible	$\mathrm{inst}_c(\varphi)$: exact
λ convention with parameters (FO or DT°), closed data	$\mathrm{inst}_o(\varphi) \ni \varphi(\bar{p})$: the ω-step; truth-sound iff $\forall\bar{x}\varphi$ is true
first-order schemas with guard $\mathrm{closed}(z_i)$, closed data	$\mathrm{inst}_c(\varphi)$: exact, no ω-step, no capture
data with a parameter at every z_i (λ, or (Nm)/(dB) with nocap)	$\mathrm{inst}_o(\varphi)$: exact for the target $\mathrm{inst}_o(\varphi)$; $\forall\bar{x}\varphi$ by generalization
(Nm) or (dB) without guards, $\mathrm{Cap}(\varphi) \neq \emptyset$	$\mathrm{inst}(\sigma_\varphi) \supseteq \mathrm{Cap}(\varphi)$: unsound; capture instances can be false in every structure

Table 5: What the cautious single-schema verifier accepts once genuine data contain an anchor ($(\mathrm{Rich_c})$ assumed). Rows 2–5: Proposition 4.5(b), (c), Proposition 4.6(e), (d).

Example 4.8 (Where the step fails)**.** [proved]cases Ex. A6, A7, A7′ (a) In $\mathbb{R}$ as an ordered field (0, 1, +, −, ·, <) closed terms denote integers, so every closed instance of $\neg(x \cdot x = 1 + 1)$ is true, but $\forall x\, \neg(x \cdot x = 1 + 1)$ is false. The theory RCF of real closed fields refutes it; the ordered-field axioms OF do not ($\mathbb{Q}$). (b) In V with $\emptyset$, $p(a, b) = \{a, b\}$, $u(a, b) = a \cup b$, closed terms denote the hereditarily finite sets, so every closed instance of $\neg\mathrm{Ind}(x)$, $\mathrm{Ind}(x) := \emptyset \in x \wedge \forall y{\in}x\; u(y, p(y, y)) \in x$ (Δ_0), is true; Infinity refutes $\forall x\, \neg\mathrm{Ind}(x)$. (c) In a model M of $\mathsf{PA} + \neg\mathrm{Con}(\mathsf{PA})$ (one exists if PA is consistent), every closed instance of $\neg\mathrm{Prf}(x, \ulcorner\bot\urcorner)$, Prf a Δ_0 proof predicate, is a true Δ_0 sentence and holds in M, but $M \models \exists x\, \mathrm{Prf}(x, \ulcorner\bot\urcorner)$.

Example 4.9 (Not derivable)**.** [proved; computed]cases Ex. A8; q1_qmodel; referee r1 (4) $\mathsf{Q} \vdash 0 + \underline{n} = \underline{n}$ for every n (Q4, then Q5 and induction on n outside Q). But Q plus all closed instances of $0 + x = x$ does not prove $\forall x\, (0 + x = x)$: there is a model of Q on $\mathbb{N} \cup \{a, b\}$, standard on $\mathbb{N}$, with $Sa = a$, $Sb = b$ and $0 + a = b$ (Appendix C.4).

Proposition 4.10 (Parameters and generalization)**.** *[proved]*cases Prop A9 *With data at parameters the gap disappears. In closure-normal form a datum $\varphi(\bar{p})$ with distinct parameters (none occurs in φ)* is *the sentence $\forall\bar{x}\varphi$, its universal closure; with trusted generalization $\forall\bar{x}\varphi$ is derived from it in one step.*

4.6 Consequences for a learner

Table 5 collects what the cautious verifier accepts, and Proposition C.4 in Appendix C.5 what is soundly learnable. Beyond the table: in (Nm) and (dB) a learner sound for $\mathrm{inst}_c(\varphi)$ must exclude capture instances whenever $\mathrm{Cap}(\varphi) \neq \emptyset$, even without parameters.

Refutation by the world (Section 2.5) helps only partly (Proposition C.5 in Appendix C.5). A false $\forall\bar{x}\varphi$ whose closed instances are true is condemned by no evaluation of closed instances, and by a coherence refutation from designated true premises Γ iff $\Gamma \cup \{\forall\bar{x}\varphi\}$ is inconsistent (RCF refutes Example 4.8(a), OF does not; ZF refutes (b)); where the ω-step is truth-safe, e.g. in $\mathbb{N}$, every false universal has a false closed instance instead. Capture instances false only in M need a designated theory that refutes them (PA refutes $\exists y\, (y = Sy)$, Q does not): against capture the world channel alone does not protect a first-order learner.

Without labels. In the untagged setting (Section 6), learning $\forall x\varphi$ means merging instances read as separate ground targets. Two closed instances of a one-variable φ with different heads have

the single minimal template σ_φ (Proposition 4.3(a)), so DTRC merges them into a cluster accepting $\varphi(a)$ for a fresh parameter a, i.e. $\forall x\varphi$, unless an instance of σ_φ is refuted (Proposition 6.13, with its caveats). With R-Δ_0, "never refuted" means "true" for quantifier-free φ (Proposition 6.14), but not in general.

4.7 Summary

The answer to Question 1 is yes for the instance schema, which first-order anti-unification and the DT$^\circ$ learner alike learn from two instances with different roots (plus (D) for several variables), and "only where $M_0 \preceq M$" for the axiom: the passage to $\forall\bar{x}\varphi$ is an ω-rule step, which a cautious learner takes silently when parameters are admissible, a closedness guard stops, and data with a parameter at every variable license; first-order encodings also need a guard against capture (Table 5). Without labels, $\forall x\varphi$ is a sound merge (Section 6.6). Open problems: Section 8.3.

5 The ZF(C) schemas, and PA induction as the contrast

Question 2 asks whether the method developed for PA induction also learns each axiom schema of ZFC from a collection of its instances. The answer is yes, and more easily than for induction. Every ZFC axiom other than Separation and Replacement is a single sentence, its own anchor (Proposition 5.1). Every ZF schema, in each textbook formulation considered here, is a higher-order pattern (Miller, 1991): its one metavariable is applied only to distinct bound variables. Two instances are an anchor iff their bodies have different main symbols and jointly use every argument place, in every class between PAT and SO$^\circ$ (Theorem 5.9), and pattern anti-unification recovers each schema from any anchor (Corollary 5.10). Plain first-order anti-unification works iff the metavariable is applied to literally the same argument tuple at every occurrence, which depends on the formulation and on how bound variables are written (Theorem 5.2). Where it fails with textbook names or de Bruijn indices, no finite union of guarded first-order schemas covers the schema without a false member (Theorem 5.6), with one exception: spelled-out Replacement in the named encoding with textbook names, whose first-order lgg of anchor data is a truth-sound over-generalization (Proposition 5.7). PA induction is the contrast. It is not a pattern, and first-order and pattern anti-unification are both unsound on it. The determinate class DT$^\circ$ learns it, and there the anchor condition is again root variation plus non-vacuity (Theorem 5.16); without determinacy a stronger condition is needed (Proposition 5.17). The difference is structural: $L_\in$ has no function symbols, so an instance of a ZF schema only renames variables inside the body, while induction substitutes the terms 0 and Sx.

Section 5.1 fixes formulations and encodings, Sections 5.2 and 5.3 treat first-order learners, Section 5.4 pattern targets, Section 5.5 induction, and Section 5.6 collects the answer. Proofs, several metavariables and the computational evidence are in Appendix D; untagged mixtures in Section 6.

Conventions. Data are instances in closure-normal form (Section 2.1): the textbook outer prefix $\forall\bar{w}$ becomes parameters, while the frame's own variables stay bound. Anchors, closures and the cautious verifier are as in Definition 2.6; first-order lggs, guards and the most-specific-guard rule as in Appendix A.6. The ZF targets lie in PAT and T_{Ind} in DT$^\circ$. Over a class H that contains the target, the cautious verifier is sound on all data and exact once they contain an anchor; over a class that does not, an anchor can make it accept the larger closure $\mathrm{cl}_H(\mathrm{inst}(T^*))$ (Lemma 2.8, Proposition D.6(b)). Every first-order schema covering a finite set P is at least as

name	template T^* (closure-normal form)	arguments of P
Sep (Kunen)	$\forall z\,\exists y\,\forall x\,(x \in y \leftrightarrow x \in z \wedge P(x,z))$	(x,z)
$\mathrm{Sep}_{\mathrm{J}}$ (Jech)	$\forall X\,\exists Y\,\forall u\,(u \in Y \leftrightarrow u \in X \wedge P(u))$	(u)
EInd	$\forall x\,(\forall y\,(y \in x \to P(y)) \to P(x)) \to \forall x\,P(x)$	(y), (x), (x)
Coll	$\forall A\,(\forall x\,(x \in A \to \exists y\,P(x,y,A)) \to \exists Y\,\forall x\,(x \in A \to \exists y\,(y \in Y \wedge P(x,y,A))))$	(x,y,A) twice
$\mathrm{Repl}_{\exists!}$	as Coll, with $\exists! y$ in place of the first $\exists y$	(x,y,A) twice
$\mathrm{Repl}_{\mathrm{sp}}$	$\forall A\,(\forall x\,(x \in A \to \exists y\,(P(x,y,A) \wedge \forall u\,(P(x,u,A) \to u = y))) \to \exists Y\,\forall x\,(x \in A \to \exists y\,(y \in Y \wedge P(x,y,A))))$	(x,y,A), (x,u,A), (x,y,A)
$\mathrm{Repl}_{\mathrm{J}}$ (Jech)	$\forall x\,\forall y\,\forall u\,(P(x,y) \wedge P(x,u) \to y = u) \to \forall X\,\exists Y\,\forall y\,(y \in Y \leftrightarrow \exists x\,(x \in X \wedge P(x,y)))$	(x,y), (x,u), (x,y)
Found	$\forall S\,(\exists x\,x \in S \to \exists x\,(x \in S \wedge \forall y\,(y \in x \to \neg y \in S)))$	ground

Table 6: The ZF schemas as templates with one formula metavariable P (closure-normal form). Plugging a body $\lambda\bar{z}.\varphi$ whose free variables are holes and parameters gives a sentence.

general as $\mathrm{lgg}(P)$ (Plotkin, 1970; Reynolds, 1970), which is why first-order failures are stated for lggs. Let $\top := \forall w\,(w = w)$ and $\bot := \neg\top$.

5.1 Formulations and encodings

Besides Separation and Replacement we treat Collection and $\in$-induction, which are theorem schemas of ZF [known]: Collection via ranks or reflection (see e.g. Lévy, 1979; we did not check the exact location), $\in$-induction via transitive closures and Foundation. Conversely, $\in$-induction with $\varphi(x) := x \notin S$ proves Foundation for S. We include both because texts and libraries state them as schemas. Table 6 lists the templates. The Kunen- and Jech-style wordings follow Kunen (1980, ch. I) and Jech (2003, ch. 1) from memory; an independent referee's recollection agrees, but we did not re-check them against the books. As far as we know Kunen treats $\exists!$ as an abbreviation, so $\mathrm{Repl}_{\exists!}$, with $\exists!$ as a primitive binder, is our encoding choice.

Proposition 5.1 (Single axioms). *[proved]*[cases §2.1; single Cor D.3] *Each ZFC axiom other than the two schemas, Choice included, is a single sentence s. Its target $\{s\}$ lies in every class, $\{s\}$ is an anchor in every class, and matching is equality.* Proof: *every template covering $\{s\}$ contains the whole target.*

Bound variables are encoded by names (Nm) or by de Bruijn indices treated as constants (dB) (Appendix A.6), or, by default, under the λ convention (Section 2.2), where capture cannot occur and freshness is built in. Named data use the textbook names of Table 6 unless said otherwise; under *α-variation* the frame's binders get arbitrary distinct names.

5.2 Which ZF schemas are first-order patterns

Let $T^* = F[P(\bar{a}_1), \ldots, P(\bar{a}_m)]$ be a pattern template over $L_\in$ with a metavariable-free, parameter-free frame F and one metavariable P of arity n, applied at its k-th occurrence to distinct bound variables $\bar{a}_k$. Let N_k be the tuple of *names* of $\bar{a}_k$ (Nm) and I_k the tuple of their *indices* at occurrence k (dB).

Theorem 5.2 (Classification). *[proved]*[cases Thm B1] *(a) $T^* \in \mathrm{PAT} \subseteq \mathrm{DT}^\circ$. (b) With textbook names,* $\mathrm{inst}(T^*)$ *is the instance set of a single first-order schema with freshness guards iff $N_1 =$*

$\cdots = N_m$. *The schema is then* $F[Z, \ldots, Z]$, *with* $\mathrm{fresh}(v, Z)$ *for each frame-bound name* v *not in* N_1. *(c) In de Bruijn notation the same holds with* $I_1 = \cdots = I_m$ *and the guard* $\mathrm{loose}(Z) \subseteq I_1$. *Well-formedness implies this guard iff* $I_1 = \{0, \ldots, d-1\}$, *where* d *is the least binder depth of an occurrence.*

Proof idea. If the tuples are equal, every occurrence reads the same text. If two differ, the lgg of the instances with bodies $\bigwedge_i (z_i = z_i)$ and its negation decouples them; filling one with the closed $\top$ and the others with $\bot$ respects every freshness guard, but no body produces the result (Appendix D.2). □

Table 11 (in Appendix D.2) applies the criterion to every formulation and encoding. It corrects two natural expectations. $\in$-induction is a de Bruijn first-order pattern, since all its occurrences read #0. And freshness is automatic only in the λ encoding, not in the de Bruijn first-order one.

Proposition 5.3 (α-variation)**.** *[proved]*[cases Prop B1α] *Under* α*-variation (with name metavariables and guards* fresh *and* distinct*), the* α*-variants of instances of* T^* *form the instance set of a single guarded first-order schema iff all occurrences of* P *are applied to the same tuple of* binders. *This is true for* Sep *and* $\mathrm{Sep_J}$, *and false for* EInd, Coll, $\mathrm{Repl}_{\exists!}$, $\mathrm{Repl_{sp}}$ *and* $\mathrm{Repl_J}$.

Proposition 5.4 (What the first-order lgg does)**.** *[proved]*[cases Prop B2] *Suppose the bodies' main symbols are not all equal. Then the first-order lgg (either encoding, textbook names) is* F *with a metavariable at each* P*-position, and positions* k, l *share it iff no body uses an argument place at which* $\bar{a}_k$ *and* $\bar{a}_l$ *differ. If all positions share one metavariable, the lgg with its most specific guards accepts only instances of* T^*.

So on anchor data the first-order lgg decouples exactly the occurrences that Theorem 5.2 cannot tie.

5.3 Where first-order anti-unification fails

Proposition 5.5 (Two instances give a false lgg instance)**.** *[proved]*[cases §2.4] *In each of the following cases two genuine instances give an lgg with a false sentence instance that respects the most specific guards and is refuted in ZF; all witnesses are in Appendix D.3. (i)* EInd, *named. From* $\mathrm{EInd}(x = x)$ *and* $\mathrm{EInd}(\neg x \in a)$ *the lgg is* $\forall x(\forall y(y \in x \to Z_1) \to Z_2) \to \forall x\, Z_2$. *Its instance* $\forall x(\forall y(y \in x \to \neg y = y) \to \forall w \neg w \in x) \to \forall x \forall w\, \neg w \in x$ *has a valid antecedent and says that every set is empty. (ii)* $\mathrm{Repl_J}$, *named or de Bruijn: the instance asserts a universal set. (iii)* Coll, $\mathrm{Repl}_{\exists!}$, $\mathrm{Repl_{sp}}$, *de Bruijn or* α*-varied. The bodies* $y = x$ *and* $\neg x = A$ *decouple the consequent's occurrence;* $y = x$ *at the antecedent's occurrence and* $\bot$ *at the others give an instance equivalent to* $\forall A \neg \exists x\, x \in A$. *(iv) Guard violations. Without its guard, Separation's lgg on data with* (R) *has Russell's instance (Proposition 5.8). Without* $\mathrm{fresh}(Y, \cdot)$, *the named lggs of* Coll, $\mathrm{Repl}_{\exists!}$ *and* $\mathrm{Repl_{sp}}$ *have the instance with body* $y = Y$, *which forces* $Y \in Y$. *For* $\mathrm{Repl_{sp}}$ *in the named encoding with textbook names, anchor data give no such instance: the lgg is truth-sound (Proposition 5.7).*

Theorem 5.6 (No finite union of first-order schemas)**.** *[proved]*[cases Lemma C1, Thm C2] *In each case below, every finite set of guarded first-order schemas whose instances include all instances of the schema has a false instance that is refutable in ZF: (i)* EInd, *named; (ii)* $\mathrm{Repl_J}$, *both encodings; (iii)* Coll, $\mathrm{Repl}_{\exists!}$, $\mathrm{Repl_{sp}}$, *de Bruijn; (iv)* Coll, $\mathrm{Repl}_{\exists!}$, $\mathrm{Repl_{sp}}$, *named under* α*-variation.*

Proof idea. Suppose a schema covers a sentence and has fewer nodes than the depth of one of its leaves ℓ. Then it has a formula metavariable Z at a proper prefix q of ℓ. If the subterm at

q is unique, the schema also covers the sentence with $\top$ or $\bot$ at q, and closed values respect every guard (Lemma D.2). Take instances with deep bodies, such as $\mathrm{EInd}(\neg^{2n}\forall w\, \neg w \in x)$, and a leaf in the occurrence that cannot be tied. For each prefix, one of $\top, \bot$ gives a false sentence (Appendix D.3). □

With textbook names, the one non-first-order case not covered is $\mathrm{Repl}_{\mathrm{sp}}$ in the named encoding. It is a genuine exception:

Proposition 5.7 (A truth-sound over-generalization). *[proved]*[cases Props C3, C3′] *Let*

$$L_S := \forall A\,(\forall x\,(x \in A \to \exists y\,(Z_1 \wedge \forall u\,(Z_2 \to u = y))) \to \exists Y\,\forall x\,(x \in A \to \exists y\,(y \in Y \wedge Z_1))).$$

This is the named lgg of $\mathrm{Repl}_{\mathrm{sp}}$ *data on which the bodies' roots vary and some body uses* y*. Take it with the guards learned from anchor data:* $\mathrm{fresh}(Y, Z_1)$*,* $\mathrm{fresh}(u, Z_1)$*,* $\mathrm{fresh}(Y, Z_2)$*,* $\mathrm{fresh}(y, Z_2)$*. (a) Every guarded instance of* L_S *is a theorem of ZF, and* $\mathrm{inst}(L_S) \supsetneq \mathrm{inst}(\mathrm{Repl}_{\mathrm{sp}})$*. (b) For any theory* B*, all guarded instances of* L_S *are theorems of* B *iff* B *proves the Collection schema.*

Proof idea. The antecedent implies $\forall x{\in}A\,\exists y\, Z_1$, and Collection gives the consequent. Conversely, $Z_1 := \varphi$ and $Z_2 := \bot$ give a formula equivalent to $\mathrm{Coll}(\varphi)$, since $\forall u(\bot \to u = y)$ is valid. This formula is not an instance of $\mathrm{Repl}_{\mathrm{sp}}$ (Appendix D.3). □

So here the first-order learner is truth-sound but not exact, and no refutation can ever detect its over-generalization. Non-anchor data can still give false instances when freshness guards apply only to formula metavariables (Appendix D.3). With Jech's image consequent the decoupling gives false instances instead, and under α-variation the phenomenon disappears. By (b), the lgg is truth-sound over ZF without Foundation iff that theory proves Collection; we do not know whether it does.

Proposition 5.8 (Separation: the guard and Russell's instance). *[proved]*[cases Prop D1] *Let the bodies' main symbols not all be equal* (R)*. (a) The unguarded lgg of Separation data (either encoding) has Russell's instance* $\forall z \exists y \forall x (x \in y \leftrightarrow x \in z \wedge \neg x \in y)$*. It is refuted in pure logic from* $\exists z \exists x\, x \in z$*. (b) The most specific guard rule always learns* $\mathrm{fresh}(y, Z)$*, since no genuine body mentions* y*. The learned guards equal the target's iff some body uses* x *and some body uses* z*. For* Coll *and* $\mathrm{Repl}_{\exists!}$ *(named), the learned guards equal the target's,* $\{\mathrm{fresh}(Y, Z)\}$*, iff every argument place is used.*

Proof. By (R) the lgg has a formula metavariable Z at the slot. (a) $Z := \neg x \in y$; for $x_0 \in z$ and the y given by the sentence, $x_0 \in y \leftrightarrow \neg x_0 \in y$. (b) A freshness guard for v is learned iff no datum's value mentions v. □

5.4 Pattern targets: anchors, pattern anti-unification and rates

Let T^* be as in Section 5.2 and $D = \{T^*[P := \lambda \bar{z}.\varphi_j] : j \le N\}$. Let (R) say that the main symbols of the bodies φ_j are not all equal, and (N_i) ($i \le n$) that some φ_j has the hole z_i free.

Theorem 5.9 (Anchors for the ZF schemas). *[proved]*[cases Thm E; single Cor D.3] *For every class* H *with* $\mathrm{PAT} \subseteq H \subseteq \mathrm{SO}^\circ$*, a finite* $D \subseteq \mathrm{inst}(T^*)$ *is an anchor in* H *iff* (R) *and* (N_i) *hold for all* $i \le n$*. In particular one instance never suffices, and two can.*

Proof idea. (R) pins the rigid part of every covering template to a prefix of the frame, and (N_i) lets every variable bound above an occurrence be recovered as a hole of the body; so a covering template, with a suitable body for each metavariable, gives back T^*. "Only if": a root-specialized or argument-dropping pattern covers non-anchor data (Appendix D.4). □

Minimal anchors, as bodies, are: $x \in z$ and $\neg(x = a)$ for Sep; $x \in x$ and $\neg(x = x)$ for EInd; $y = x$ and $\neg(x = A)$ for Coll, $\mathrm{Repl}_{\exists!}$, $\mathrm{Repl}_{\mathrm{sp}}$; $x \in y$ and $\neg(x = a)$ for $\mathrm{Repl}_{\mathrm{J}}$. If a community never lets φ mention the bounding set z, the learner stays at the sound specialization $P(x)$, which is Jech's form. Since $L_\in$ has no function symbols, for one metavariable the "shielding" that separates SO° from DT° for induction (Proposition 5.17) and for universal axioms over arithmetic (Proposition 4.3(c)) cannot occur. For $H = \mathrm{DT}^\circ$ the theorem is also a case of Corollary 3.9(c).

Several metavariables. ZF needs one metavariable. With several, as in untagged mixtures (Section 6), a Plotkin-(D)-type event appears that differs between PAT, DT° and SO°, so that SO° and DT° anchors differ even over $L_\in$ (Theorem D.3); the exact SO° condition is open (Conjecture D.4).

Corollary 5.10 (Pattern anti-unification recovers every ZF schema)**.** *[proved]*[cases Cor F, F′] *The least general pattern generalization $L_{\mathrm{pat}}(D)$ exists and is unique (Pfenning, 1991; Baumgartner et al., 2017). For $D \subseteq \mathrm{inst}(T^*)$, $L_{\mathrm{pat}}(D) \preceq T^*$, and $L_{\mathrm{pat}}(D) \equiv T^*$ iff* (R) *and every* (N_i) *hold. With several metavariables, $L_{\mathrm{pat}}(D) \equiv T^*$ iff* (R_P)*, every* ($\mathrm{N}_{P,i}$) *and* ($\mathrm{D}^{\mathrm{PAT}}_{PQ}$) *for all $P \neq Q$ hold (Theorem D.3).*

Proof. $T^* \in \mathrm{PAT}$ covers D. Under the events, Theorem 5.9 (resp. D.3) for $H = \mathrm{PAT}$ gives $L_{\mathrm{pat}}(D) \succeq T^*$. Otherwise its PAT witnesses lie above $L_{\mathrm{pat}}(D)$ and miss an instance. □

The merge rule must identify columns that are equal up to a permutation of the arguments: Replacement's $P(x, y)$ and $P(x, u)$ columns agree only up to $y \leftrightarrow u$.

Theorem 5.11 (Rates)**.** *[proved]*[cases Thm G] *Let the bodies be i.i.d. from μ. Let $C_{F,I}$ be the set of bodies with root F (no condition if $F = *$) in which no z_i, $i \in I$, is free. Then*

$$\Pr[D_N \text{ is not an anchor}] = \sum_{(F,I)\neq(*,\emptyset)} (-1)^{[F\neq *]+|I|+1} \mu(C_{F,I})^N \;\leq\; \sum_f p_f^N + \sum_i (1-q_i)^N,$$

where $p_f = \mu(\mathrm{root} = f)$ and $q_i = \mu(z_i \text{ free})$.

Proof. Inclusion–exclusion over the events "all roots are f" (pairwise disjoint) and "z_i is never free". □

5.5 PA induction: the contrast

A *motive* is a formula φ of L_A with $\mathrm{FV}(\varphi) \subseteq \{x\}$, and $\mathrm{Ind}(\varphi) = \varphi(0) \wedge \forall x(\varphi \to \varphi[Sx/x]) \to \forall x \varphi$. The target is $\mathrm{inst}(T_{\mathrm{Ind}})$ with $T_{\mathrm{Ind}} = P(0) \wedge \forall x(P(x) \to P(Sx)) \to \forall x\, P(x)$. Since $P(0)$ and $P(Sx)$ are not pattern occurrences, $T_{\mathrm{Ind}} \in \mathrm{DT}^\circ \setminus \mathrm{PAT}$. Here (R) says that the motives' main symbols are not all equal, and (N) that some motive has x free. The results are from the earlier analysis, for parameter-free motives unless said otherwise; proofs are in Appendix D.5.

First-order learners. A first-order schema is *truth-sound* if its sentence instances are true in $\mathbb{N}$. The false instances below are closed and capture-free, so the results hold in the named and in the de Bruijn encoding, with fixed or varying bound-variable names.

Proposition 5.12 (Two raw instances suffice for unsoundness)**.** *[proved]*[prior induction B1, B3] *(a)* $\mathrm{lgg}(\mathrm{Ind}(x+0=x), \mathrm{Ind}(0+x=x)) = (0+0=0) \wedge \forall x(z_1+z_2 = x \to z_3+z_4 = Sx) \to \forall x(z_1+z_2 = x)$*. It has the false instance* $(0+0=0) \wedge \forall x(0+0=x \to 0+S0=Sx) \to \forall x(0+0=x)$*. One*

raw instance never forces unsoundness, and some pairs, such as $\mathrm{Ind}(x = x)$ *and* $\mathrm{Ind}(Sx = Sx)$*, have a sound lgg. (b) If* (R) *and* (N) *hold, then* $\mathrm{lgg}(D) = L_\infty := Z_1 \wedge \forall x(Z_2 \to Z_3) \to \forall x\, Z_2$*, which has the false instance* $(0 = 0) \wedge \forall x(0 = S0 \to 0 = S0) \to \forall x(0 = S0)$*. For i.i.d. motives,* $\Pr[\mathrm{lgg}(D_N) \neq L_\infty] \leq p_{\max}^{N-1} + (1-q)^N$*.*

So the diversity that identifies T_{Ind} among determinate templates (Theorem 5.16) destroys the first-order lgg, at the same coupon-collector rate.

Theorem 5.13 (No finite union of first-order schemas)**.** *[proved]*[prior induction B2] *Let* $g_0(t) = t$*,* $g_{n+1}(t) = 0 + g_n(t)$ *and* $F' = \{\mathrm{Ind}(g_n(x) = x) : n \geq 0\}$*. Every first-order schema covering two members of* F' *has a false instance that is refutable in* Q*. Hence* k *truth-sound first-order schemas cover at most* k *members of* F'*, and no finite union of first-order schemas covers the raw induction instances (or those of open induction) truth-soundly.*

The occurrence and freshness guards learnable from F' do not help (Appendix D.5); a guard "is an induction instance" would, which is what the Sub encoding below implements. Worse, some unsound lggs are never caught by evaluating closed sentences.

Theorem 5.14 (An unsound lgg that closed truths never refute)**.** *[proved]*[prior induction referee R1] *Let* $K_0 := (z_1 + 0 = z_1) \wedge \forall x(z_1 + x = z_2 \to z_1 + Sx = Sz_2) \to \forall x(z_1 + x = z_2)$*, the lgg of* $\mathrm{Ind}(0 + x = x)$ *and* $\mathrm{Ind}(S0 + x = Sx)$*. Its instance* $(0 + 0 = 0) \wedge \forall x(0 + x = 0 \to 0 + Sx = S0) \to \forall x(0 + x = 0)$ *is false. Yet the true closed quantifier-free sentences together with all sentence instances of* K_0 *are consistent. So no sound derivation from true closed quantifier-free premises (or* Δ_0 *ones, with* $<$ *primitive) refutes an instance of* K_0*.*

K_0 is refuted once the recursion axiom Q5, $\forall x \forall y\, (x + Sy = S(x + y))$, is a designated premise [prior induction B5(d)].

Pattern anti-unification and determinate templates.

Proposition 5.15 (Pattern anti-unification is unsound on induction)**.** *[proved]*[prior induction C9] *Under* (R) *and* (N)*, the least general pattern generalization of* $D \subseteq \mathrm{inst}(T_{\mathrm{Ind}})$ *is* $T_{\mathrm{pat}} := Z \wedge \forall x(P(x) \to Q(x)) \to \forall x\, P(x)$*. It has the false non-instance* $(0 = 0) \wedge \forall x(x = 0 \to x = 0) \to \forall x(x = 0)$*.*

For the size-bounded classes DT°_s (templates of size $\leq s$), $|T|$ counts symbols, an occurrence $M(\bar{t})$ counting $1 + \sum_i |t_i|$; so $|T_{\mathrm{Ind}}| = 14$.

Theorem 5.16 (Anchors for induction in DT°)**.** *[proved]*[prior induction C3; single Cor D.3] *A finite* $D \subseteq \mathrm{inst}(T_{\mathrm{Ind}})$ *is an anchor in* DT° *iff* (R) *and* (N)*. The same holds in the size-bounded classes* DT°_s *for* $s \geq 7$*, and, by Corollary 3.9(c), in closure-normal form with parameters. Two instances suffice; one never does.*

Up to renaming exactly 26 templates of DT° contain $\mathrm{inst}(T_{\mathrm{Ind}})$, and on every anchor they are the covering templates, so $\mathrm{lgg}_{\mathrm{DT}^\circ}(D) = T_{\mathrm{Ind}}$; T_{pat} is the least of them without non-pattern occurrences. In DT°_s, $\mathrm{inst}(T_{\mathrm{Ind}})$ is closed iff $s \geq 12$; for $7 \leq s \leq 11$ the cautious verifier accepts a false sentence on every anchor (Proposition D.6, an instance of Lemma 2.8).

Proposition 5.17 (Without determinacy)**.** *[proved]*[prior induction C7] *Call a free occurrence of* x unshielded *if it lies inside no proper term* $t \neq x$ *with* $\mathrm{FV}(t) \subseteq \{x\}$*, and let* (B) *say that some motive has one. In* SO°*,* D *is an anchor iff* (R) *and* (B)*. For example,* $\{\mathrm{Ind}(Sx = S0), \mathrm{Ind}(\neg Sx = 0)\}$ *satisfies* (R) *and* (N)*, but* $Z \to \forall x\, P(Sx)$ *covers it.*

target	first-order			PAT	DT$^\circ$	anchor condition
	named	dB	named, α			
ground axioms (incl. Found, AC)	yes	yes	yes	yes	yes	the sentence itself
Sep	g	g	g	yes	yes	(R), (N_x), (N_z)
Sep_J	g	g	g	yes	yes	(R), (N_u)
EInd	no†	yes	no	yes	yes	(R), (N_x)
Coll, $\text{Repl}_{\exists!}$	g	no†	no†	yes	yes	(R), (N_x), (N_y), (N_A)
Repl_sp	no‡	no†	no†	yes	yes	(R), (N_x), (N_y), (N_A)
Repl_J	no†	no†	no	yes	yes	(R), (N_x), (N_y)
Ind, raw	no†	no†	no†	no§	yes	DT$^\circ$: (R), (N); SO$^\circ$: (R), (B)
Ind, Sub-encoded	yes	–	–	–	–	first-order lgg: (R), (N)
IndEq	g	–	–	–	–	first-order lgg: (R)

Table 7: All targets of this section. "yes": a first-order pattern (resp. in the class). "g": a first-order pattern with guards (e.g. $\text{fresh}(y, Z)$ for Sep, $\text{fresh}(Y, Z)$ for Coll, $y \notin \text{FV}(P)$ for IndEq; all: Table 11), with false instances without them. "no†": no finite union of guarded first-order schemas covers the target without a false member (Theorems 5.6 and 5.13). "no‡": not first-order, but the first-order lgg of anchor data is truth-sound iff the base theory proves Collection (Proposition 5.7). "no§": not a pattern, and the pattern lgg is unsound (Proposition 5.15). "–": not analysed. For the ZF schemas the anchor condition is the same in every class between PAT and SO$^\circ$ (Theorem 5.9). Two instances suffice in every row except the ground axioms, where one does.

Encodings that make induction first-order. Recording each step with the decidable judgment $\text{Sub}(\varphi, x, t, \psi)$ ("$\psi = \varphi[t/x]$") makes induction a first-order pattern whose lgg recovers it under the same events (R)+(N) (Proposition D.7); so the second-order learner on raw sentences loses nothing against a first-order learner on Sub-annotated data.

Proposition 5.18 (PA is Q plus one first-order pattern)**.** *[proved]*[prior induction B6] *Let* $\text{IndEq}(P) := \forall x(x = 0 \to P) \wedge \forall y(\forall x(x = y \to P) \to \forall x(x = Sy \to P)) \to \forall x\, P$*, with P a 0-ary metavariable and x, y fixed (Tarski, 1965). (a) First-order logic proves* $\text{IndEq}(\varphi) \leftrightarrow \text{Ind}(\varphi)$ *for every motive φ, up to renaming a bound variable. With parameters, the guard $y \notin \text{FV}(P)$ is needed. (b)* $\text{lgg}(\text{IndEq}(\psi_1), \ldots, \text{IndEq}(\psi_N)) = \text{IndEq}(\text{lgg}(\psi_1, \ldots, \psi_N))$*, which is* $\text{IndEq}(P)$ *iff* (R) *holds. (c)* $\mathsf{Q} + \text{inst}(\text{IndEq}) \vdash \mathsf{PA}$.

So the obstruction in Theorem 5.13 is the meta-level substitution in $\varphi(0)$ and $\varphi(Sx)$, not induction, and it differs from the non-finite axiomatizability of PA (Ryll-Nardzewski, 1952). Proof assistants record or recompute the motive (Lean, Metamath: Appendix D.5); in DT$^\circ$ the motive is the unique matcher of T_Ind.

5.6 Summary

The answer to Question 2 is yes (Table 7). All ZF schemas lie in the pattern fragment; pattern anti-unification and the DT$^\circ$ learner learn each from two suitable instances and are sound at all times. A schema is a first-order pattern iff its metavariable's argument tuple is literally the same at every occurrence in the chosen syntax (Theorem 5.2 and Proposition 5.3); otherwise, with textbook names or de Bruijn indices, every finite union of guarded first-order schemas covering it has a false member, except for spelled-out Replacement with textbook names (Theorem 5.6 and Proposition 5.7). PA induction lies outside the pattern fragment; DT$^\circ$ handles it

with the anchor events of the first-order learner on Sub-annotated data, and without determinacy the anchors need the stronger event (B). The experiments agree: on Separation, $\in$-induction, a Replacement variant and induction the implemented DT°_F learner was exact from 2–3 tagged instances in most runs and accepted no non-instance in 927 probes, while the pattern lgg failed on induction and unguarded first-order lggs failed except on EInd in de Bruijn form (Tables 9 and 11). Open problems: Section 8.3.

6 Many axioms and schemas without labels

Question 3 asks for one method that also learns "many axiom(scheme)s at once, even if they do not come with labels for instances of which axiom scheme they are". The data now come from several targets $\sigma_1, \dots, \sigma_{k'}$, and nothing says which datum is an instance of which target. This section develops the theory of this setting and the method DTRC (Determinate Templates with Refutation Clustering); Table 8 collects the guarantees. In brief:

- From positive data alone a bound k on the number of targets is necessary, and spare slots cost data diversity, which no sound negative evidence can replace (Section 6.2).
- With the exact bound, negatives that refute every cross-target merge make untagged learning as easy as tagged learning, and the learner can find them itself (Theorem 6.6).
- DTRC needs no bound. Under *refutation separation* it recovers the hidden labels, is target-sound at all times, and is exact once every target's data contain an anchor, with the tagged failure probability (Theorem 6.8). Without separation it is sound relative to a residue of unrefuted cross merges (Theorem 6.10); learning $\forall x \varphi$ from instances is such a merge, a sound one (Proposition 6.13). No computable learner can do without the refutation depth (Theorem 6.15).
- Δ_0 evaluation in $\mathbb{N}$ separates PA at small depth (proved for Q1–Q7 and T_{Ind}). Hereditarily finite counterexamples and logic separate ZF and ZFC on every sampled pair and every run, Russell's paradox doing the work for the comprehension-shaped axioms. Weak forms of Union and Power merge into a false universal-set schema that, of these channels, only coherence refutes (Section 6.11).

6.1 Setting

Data are formulas in closure-normal form with de Bruijn bound variables (Section 2.1), and templates are in DT°. Parameters are constants, except for the head classes of Appendix E.2, which identify sentences up to renaming of parameters. Let $\mathcal{S}$ be the set of such formulas and $\mathrm{Tr} \subseteq \mathcal{S}$ the set of those whose universal closure is true in the intended structure ($\mathbb{N}$ or V).

Definition 6.1 (practice, data, covering templates)**.** untagged §1 A *practice* is a finite set $\Sigma^* = \{\sigma_1, \dots, \sigma_{k'}\} \subseteq \mathrm{DT}^\circ$ of targets (ground templates, i.e. single axioms, allowed), with $R^* := \bigcup_i \mathrm{inst}(\sigma_i)$; it is *valid* if $R^* \subseteq \mathrm{Tr}$, which we assume unless said otherwise. The data form a finite multiset $D \subseteq R^*$ with *hidden* labels $D = D_1 \sqcup \dots \sqcup D_{k'}$, $D_i \subseteq \mathrm{inst}(\sigma_i)$ (a datum in two instance sets carries one label). In the random model each datum picks target i with probability π_i and then an instance $\sigma_i\theta$, $\theta \sim \Lambda_i$. Refutation oracles $(\mathrm{Ref}_d)_d$, with (OS) and (Dec), are as in Definition 2.9; $H_k = H_k(\mathrm{DT}^\circ)$, the version space $\mathrm{VS}_k(D, N)$ and the *cautious k-union learner* $\mathrm{Acc}_k(D, N) = \bigcap \mathrm{VS}_k(D, N)$ are as in Definition 2.6. For finite nonempty X,

$\mathrm{Cov}(X) := \{T \in \mathrm{DT}^\circ : X \subseteq \mathrm{inst}(T)\}$, and $\mathrm{Min}(X)$ is the set of its $\succeq$-minimal members up to renaming.

Proposition 6.2 (minimal covering templates)**.** *[proved]*[single Thm B] *For finite nonempty $X \subseteq \mathcal{S}$: (a) $\mathrm{Min}(X)$ is finite up to renaming and computable from X; (b) every $T \in \mathrm{Cov}(X)$ satisfies $T \succeq T_0$, hence $\mathrm{inst}(T) \supseteq \mathrm{inst}(T_0)$, for some $T_0 \in \mathrm{Min}(X)$; (c) if $X \subseteq \mathrm{inst}(\sigma)$ with $\sigma \in \mathrm{DT}^\circ$, some $T_0 \in \mathrm{Min}(X)$ has $\mathrm{inst}(T_0) \subseteq \mathrm{inst}(\sigma)$. (This is Theorem 3.3(c),(d).)*

$A \subseteq \mathrm{inst}(\sigma)$ is an anchor (Definition 2.6, in $H_1(\mathrm{DT}^\circ)$) if $\mathrm{inst}(T) \supseteq \mathrm{inst}(\sigma)$ for every $T \in \mathrm{Cov}(A)$. Theorem 3.8 and Corollary 3.9 characterize them ((R)+(N) for T_{Ind} and the ZF schemas, Plotkin's (R)+(D) for first-order patterns); a ground σ has the anchor $\{\sigma\}$.

Lemma 6.3 (cautious soundness; finite form)**.** *[proved; known]*[untagged Lemmas 1.1, 1.2] *(a) If $k \geq k'$ and $N \cap R^* = \emptyset$, then $R^* \in \mathrm{VS}_k(D, N)$, so $\mathrm{Acc}_k(D, N) \subseteq R^*$, and Acc_k is monotone in D and in N (Theorem 2.7(a), Lemma 2.10(b)). (b) With $\mathrm{Min}_N(B) := \{T \in \mathrm{Min}(B) : \mathrm{inst}(T) \cap N = \emptyset\}$, $\mathrm{Acc}_k(D, N)$ is the intersection of the sets $\bigcup_{B \in \pi} \mathrm{inst}(T_B)$ over all partitions π of D into at most k nonempty blocks and all choices $T_B \in \mathrm{Min}_N(B)$. So for finite N it is decidable (in time exponential in $|D|$).*

For (b), assign each datum to the first slot covering it and apply Proposition 6.2(b) to each block (Appendix E.1).

6.2 Positive data: a bound is necessary, and spare slots cost diversity

Proposition 6.4 (a bound is necessary)**.** *[proved; known]*[untagged Prop 4.3; Gold] *Let $H_\infty := \bigcup_k H_k$. For every finite $D \subseteq R^*$ and every sound N, $\mathrm{Acc}_\infty(D, N) = D$. No learner, however it uses a fixed sound oracle, identifies H_∞ in the limit from text.*

Proof idea. D is a union of ground templates; for a true non-ground σ the oracle answers identically on $\mathrm{inst}(\sigma)$ and its finite subsets, so Gold's argument applies (Appendix E.2).

Negatives cannot replace a bound: every finite set of true data is itself a consistent hypothesis. Negatives remove over-general *lumps*; a bound forbids *splits*, which replace a target by finitely many specializations that together cover its data. Splits are sound, so one-sided refutation never removes them.

Theorem 6.5 (splits; threshold)**.** *[proved]*[untagged Thm 3.2] *(a) For every N with $N \cap R^* = \emptyset$ and $k \geq k'$, $\mathrm{VS}_k(D, N)$ contains every $h \in H_k$ with $D \subseteq h \subseteq R^*$. So if such an h misses $q \in R^*$, no sound negatives make the learner accept q. (b) Suppose moreover that every D_j is nonempty and that N* cross-separates *D: every template covering data of two targets meets N. Put $m := k - k' + 1$. If every family of at most m templates that covers D_i and avoids N has union $\supseteq \mathrm{inst}(\sigma_i)$, then $\mathrm{inst}(\sigma_i) \subseteq \mathrm{Acc}_k(D, N)$. If some such family misses a $q \in \mathrm{inst}(\sigma_i)$ lying in no other target's instance set, then $q \notin \mathrm{Acc}_k(D, N)$. With pairwise disjoint instance sets this is an exact threshold.*

Proof idea. (a) $h \cap N \subseteq R^* \cap N = \emptyset$. (b) Cross-separation makes the slots used by different targets disjoint, so target i has at most m; conversely the bad family plus the other $k' - 1$ targets is a k-slot hypothesis (Appendix E.2).

Slot accounting refines this when targets share slots (Theorem E.2). For Q plus induction at $k = k' = 8$, positive data leave induction four slots: an induction instance whose motive has x free and a main connective unseen in the data is accepted iff the induction data cannot be split

into four blocks each root-homogeneous or all-vacuous; three sound negatives leave it one slot and restore the tagged anchor condition (R)+(N) (Corollary E.3). Spare slots cost diversity even for $\forall x\varphi$: if the target is the instance schema $\varphi(z)$ and $\varphi(a)$ lies in no other target's instance set, numeral-only data never make the learner accept $\varphi(a)$ once $k > k'$, since $\{\varphi(0)\} \cup \mathrm{inst}(\varphi(Su))$ is a sound split (Theorem 6.5(a)). Exact thresholds for one-variable schemas: Remark E.5.

6.3 The pigeonhole theorem

Theorem 6.6 (pigeonhole)**.** *[proved]* untagged Thm 2.1 *Let $k = k'$, $D \subseteq R^*$, $A_i \subseteq D_i$ a nonempty anchor for σ_i, and $N \cap R^* = \emptyset$ with* (X)*: for $i \neq j$, $a \in A_i$, $b \in A_j$, every $T \in \mathrm{DT}^\circ$ with $a, b \in \mathrm{inst}(T)$ meets N. Then $\mathrm{Acc}_k(D, N) = R^*$.*

Proof. The inclusion $\subseteq$ is Lemma 6.3(a). Let $h = \bigcup_{l \leq k} \mathrm{inst}(\tau_l)$ be in $\mathrm{VS}_k(D, N)$, and let L_i be the set of slots l such that τ_l covers an element of A_i. The L_i are nonempty and, by (X), pairwise disjoint; k' of them in k' slots are singletons $\{l_i\}$, so τ_{l_i} covers A_i and $\mathrm{inst}(\tau_{l_i}) \supseteq \mathrm{inst}(\sigma_i)$. □

Only anchor data matter, and (X) forces the anchors to be disjoint. The negatives need no labels: let $N(D, d)$ contain the first d-refuted instance of every d-refuted $T \in \mathrm{Min}(\{a, b\})$, over all pairs $a, b \in D$. It is sound, and if every minimal template of every cross pair of anchor data is d-refuted (*anchor-pair separation*), it satisfies (X) by Proposition 6.2(b). Hence $\mathrm{Acc}_{k'}(D, N(D, d))$ is sound at all times, decidable, and exact as soon as D contains anchors; at most $\mu \sum_{i<j} |A_i||A_j|$ negatives are needed, μ the largest $|\mathrm{Min}(\{a, b\})|$ (Corollary E.9). For PA (Q1–Q7 in parameter form, numbered as in Section 2.6, and T_{Ind}; raw DT° encoding), every cross pair among Q1–Q7 and T_{Ind} has exactly one minimal covering template: F_0 or $F_0 \to F_1$, with 0-ary formula metavariables F_0, F_1; $z_0 = z_1$; $p + z_0 = z_1$; or $p \cdot z_0 = z_1$. Four sentences refute them all: $0 = S0$, $0 = 0 \to 0 = S0$, $p + 0 = 0$ and $p \cdot 0 = S0$ (computed; Table 13). So at $k = 8$ the untagged learner is exact after one occurrence of each Q-axiom and two induction instances with different main connectives, one of them non-vacuous.

6.4 dtrc: Determinate Templates with Refutation Clustering

Definition 6.7 (coherence, DTRC, separation)**.** untagged §§5.1–5.2 A finite $X \subseteq \mathcal{S}$ is *d-coherent* if some $T \in \mathrm{Min}(X)$ is not d-refuted (equivalently, by Proposition 6.2, some covering template is not). **Algorithm** DTRC_d:

(1) normalize the data (closure-normal form, canonical parameter names, de Bruijn indices);
(2) starting from singleton clusters, repeatedly merge two clusters whose union is d-coherent, chosen by a *merge policy* (e.g. online: each datum joins the first cluster it is coherent with, then pairwise merges to a fixpoint), until no pair has a coherent union;
(3) for each final cluster C let $\mathrm{Min}^+_d(C)$ be its unrefuted minimal templates and $\mathrm{Acc}_d(C) := \bigcap_{T \in \mathrm{Min}^+_d(C)} \mathrm{inst}(T)$ ($\emptyset$ if there are none);
(4) assert $\mathrm{Acc}_d := \bigcup_C \mathrm{Acc}_d(C)$.

The practice is *d-separated on D*, $\mathrm{RS}_d(D)$, if every template covering data of two different targets is d-refuted, and *globally* d-separated if every template covering instances of two different targets is.

$\mathrm{Acc}_d(C)$ is the cautious single-template verifier of C with negatives Ref_d. Coherence is downward closed and antitone in d (Lemma E.1). By Proposition 6.2, $\mathrm{RS}_d(D)$ holds iff every minimal template of every cross pair $(a, b) \in D_i \times D_j$ is d-refuted, a finite check that the implementation performs after each run.

Theorem 6.8 (DTRC). *[proved]*[untagged Thm 5.2] *Let the practice be valid and the oracle satisfy (OS), (Dec). (i)* Within-target merges are never refuted*: every $X \subseteq \mathrm{inst}(\sigma_i)$ is d-coherent for every d. (ii)* Separation recovers the labels*: under $\mathrm{RS}_d(D)$ the final clustering is $\{D_i : D_i \neq \emptyset\}$ for every merge policy. (iii)* Soundness at all times, exactness from anchors*: under $\mathrm{RS}_d(D)$, $\mathrm{Acc}_d \subseteq R^*$, and $\mathrm{Acc}_d = R^*$ once each D_i contains an anchor of σ_i. (iv)* Rates*: under global separation and i.i.d. data, $\Pr[\mathrm{Acc}_d \neq R^*] \leq \Pr[\text{some } D_i \text{ contains no anchor}] \leq \sum_i \Pr[D_i \text{ contains no anchor}]$; the middle term is the failure probability of the tagged learner, bounded explicitly in Theorem 3.10(c) (for T_{Ind} the exact inclusion–exclusion form is that of Proposition D.7, by Theorem 5.16; for the ZF schemas see Theorem 5.11). (v)* Tests*: with n distinct data the pairwise fixpoint makes $O(n^2)$ coherence tests; under $\mathrm{RS}_d(D)$ the online policy makes at most $nk' + k'(k'-1)/2$.*

Proof idea. (i) Some $T_0 \in \mathrm{Min}(X)$ has $\mathrm{inst}(T_0) \subseteq \mathrm{inst}(\sigma_i) \subseteq \mathrm{Tr}$ (Proposition 6.2(c)), and (OS) never refutes it. (ii) Pure clusters of different targets never merge, by separation; two final clusters of one target would merge, by (i). (iii) In the cluster D_i the template T_0 of (i) is intersected; with an anchor, every member of $\mathrm{Min}(D_i)$ contains $\mathrm{inst}(\sigma_i)$. Details in Appendix E.4. □

Example 6.9 (a small run). [proved][instance of untagged Thm 5.2] Let the practice be $\{z+0=z,\ T_{\mathrm{Ind}}\}$ and $D = \{0+0=0,\ S0+0=S0,\ \mathrm{Ind}(x+0=x),\ \mathrm{Ind}(\neg x=0)\}$. A set containing an equation and an induction instance has the single covering template F_0, a 0-ary formula metavariable (the roots $=$ and $\to$ differ, so the root is a slot with no binder in scope), and $0 = S0$ refutes it. So $\mathrm{RS}_d(D)$ holds once d reaches $0 = S0$, and by Theorem 6.8(ii) DTRC returns the clusters $\{0+0=0, S0+0=S0\}$ and $\{\mathrm{Ind}(x+0=x), \mathrm{Ind}(\neg x=0)\}$ under every merge policy. Both are anchors (Corollary 3.9(a),(c): the values 0 and $S0$ have different roots; the motives have roots $=$ and $\neg$, and x is free in $x+0=x$), so by (iii) DTRC accepts exactly $\mathrm{inst}(z+0=z) \cup \mathrm{inst}(T_{\mathrm{Ind}})$. Under (Rich), which holds in arithmetic, each cluster's only minimal template is its target (Corollary 3.14).

Under separation, refutation does the job of citations in tagged data: within each recovered cluster the learner *is* the tagged cautious learner, so tagged anchors and rates (Theorems 3.8 and 3.10) transfer verbatim, and the number of targets is discovered, not given. The inequality in (iv) can be strict, since refuted non-target templates drop out before an anchor arrives. When a bound $k \geq k'$ is known, accepting only what both DTRC and $\mathrm{Acc}_k(D, N(D,d))$ accept is target-sound at all times, with or without separation, and exact under $\mathrm{RS}_d(D)$ with $k = k'$ once every D_i contains an anchor (Remark E.10).

6.5 Without separation: residue, sound merges, fragmentation

Theorem 6.10 (soundness relative to a residue). *[proved; computed]*[untagged Thm 5.4] *Call a final cluster* mixed *if it contains data of two or more targets, and let $\mathrm{Res}_d(D) := \bigcup\{\mathrm{inst}(T) : T \in \mathrm{Min}(X)$ not d-refuted, $X \subseteq D$ d-coherent and mixed$\}$. (a) $\mathrm{Acc}_d \subseteq R^* \cup \mathrm{Res}_d(D)$. (b) If every mixed final cluster C has a truth-sound $T \in \mathrm{Min}_d^+(C)$ (a* sound merge*), then $\mathrm{Acc}_d \subseteq \mathrm{Tr}$: truth-sound, though not target-sound. (c) Otherwise Acc_d is sound only relative to $\mathrm{Res}_d(D)$, which decreases in d; $\mathrm{Res}_\infty(D)$ consists of templates with no refutable instance. (d) Exactness can fail even on clean data with an anchor in every D_i and disjoint instance sets: a sound cross merge can absorb part of an anchor (*fragmentation*), depending on the merge policy.*

For (d), let $\sigma_1 = (z+0=z)$, $g = (0+S0=S0)$ a ground target and $D = \{0+0=0,\ S0+0=S0,\ g\}$, which contains the anchor $\{0+0=0, S0+0=S0\}$ of σ_1. In the order $g, 0+0=0, S0+0=S0$, the pair $\{g, 0+0=0\}$ merges through the *true* template $0+z=z$, and $S0+0=S0$ cannot join (Appendix E.4); the output rejects $p+0=p$ and accepts the true

non-target $0 + p = p$. In the order $0 + 0 = 0, S0 + 0 = S0, g$ it is exact (computed). This failure is intrinsic:

Proposition 6.11 (ambiguous practices)**.** *[proved; computed]*[untagged Rem 5.11] *For the valid practices* $P = \{z + 0 = z,\ 0 + S0 = S0\}$ *and* $P' = \{0 + z = z,\ S0 + 0 = S0\}$*, the data* $D = \{0 + 0 = 0,\ S0 + 0 = S0,\ 0 + S0 = S0\}$ *lie in* $R^*_P \cap R^*_{P'}$ *and contain anchors for both. No learner (computable or not) with a target-blind oracle is on* D *both exact for* P *and target-sound for* P'*.*

Proof. Anchors: $\{0 + 0 = 0, S0 + 0 = S0\}$, $\{0 + S0 = S0\}$ for P; $\{0 + 0 = 0, 0 + S0 = S0\}$, $\{S0 + 0 = S0\}$ for P'. The learner's output on D is the same under both; exactness for P needs $p + 0 = p$, which is not in $R^*_{P'}$. □

Only separating data such as $SS0 + 0 = SS0 \in R^*_P \setminus R^*_{P'}$ resolve this; the k'-union learner with self-generated negatives is not exact on this D either (computed), since the true template $0 + z = z$ violates (X). Residual merges are *target-sound* (a redundant axiom that is an instance of a target schema, e.g. Empty Set of Separation), *sound* (true templates below no target, as for $\forall x\varphi$ below), or *unsound unrefuted* (K_0-shaped templates, the universal-set schema; Section 6.11).

6.6 Learning $\forall x\varphi$ from instances as a sound merge

Regard each instance $\varphi(t)$ as its own ground target. The universal sentence is learned when DTRC merges the instances.

Lemma 6.12 (two-point anchors)**.** *[proved]*[untagged Lem 5.5] *Let* $\varphi(z)$ *have a single metavariable, a 0-ary* z *(term or formula sort) occurring in it. If closed* t_1, t_2 *have different head symbols,* $\{\varphi(t_1), \varphi(t_2)\}$ *is an anchor for* $\varphi(z)$ *in* DT°*. So for every set of instances of* $\varphi(z)$ *containing two with different heads,* $\mathrm{inst}(\varphi(z))$ *is contained in the instance set of every covering template.*

It is the one-metavariable case of Proposition 4.3(a), and the case of Theorem 3.8 in which (U) holds automatically; the proof rests on a two-point identity (Appendix E.4).

Proposition 6.13 (universals from instances)**.** *[proved; computed]*[untagged Prop 5.6] *Let* z *be a term metavariable and* $D = \{\varphi(t_1), \ldots, \varphi(t_n)\}$ *closed instances, two with different heads. (a)* D *is d-coherent iff* $\varphi(z)$ *is not d-refuted. (b)* $\mathrm{inst}(\varphi(z))$ *contains* $\varphi(a)$ *for a parameter* a *fresh for* φ*, which under the closure reading is* $\forall x\varphi(x)$ *(in parameter form; the string* $\forall x\varphi$ *is a different datum, Section 2.1); so* $\varphi(z)$ *is truth-sound iff* $\forall x\varphi$ *is true. (c) If* $\forall x\varphi$ *is true, every subset of* D *is coherent,* DTRC *forms one cluster under every policy, and* $\mathrm{Acc}_d \ni \varphi(a)$*: the axiom is learned and nothing false accepted. (Target-soundness for the* n *ground "targets" fails, as it must: the instances and the axiom give the same data and oracle answers.) (d) If* $\forall x\varphi$ *is false and some instance of* $\varphi(z)$ *is d-refuted, the merge is refused, though root-homogeneous subsets may still merge into an unrefuted specialization such as* $\varphi(Sz)$*. (e) If the oracle refutes every false instance of* $\varphi(z)$ *at some depth, "never refuted" means "*$\forall x\varphi$ *true". R-*Δ_0 *does so for quantifier-free* φ *(Proposition 6.14) but not for every quantified* φ*, since it verifies universals only by equational derivations from true closed equations. At a finite depth, false universals with large least counterexamples merge (part of* Res_d*). An oracle that only evaluates closed instances certifies "true at every closed term": in* $\mathbb{N}$ *this is the* ω*-rule, sound but in general not derivable (Example 4.9); in* $\mathbb{R}$ *as an ordered field, where closed terms denote integers (Example 4.8(a)),* $\neg(z \cdot z = 1 + 1)$ *is an unsound unrefuted merge, which a decision procedure for real closed fields would refute.*

Proposition 4.7 characterizes when closed instances decide universals (the Tarski–Vaught criterion); computed examples, e.g. numeral instances of Q4–Q6 merging into their instance schemas, are in Appendix E.4.

Proposition 6.14 (quantifier-free templates)**.** *[proved]*[untagged Prop 5.13] *Let $T \in \mathrm{DT}^\circ$ be an arithmetic template with no binder in its rigid part (so all its metavariables are 0-ary; formula metavariables may still take quantified closed formulas). If T has a false instance, it has a false quantifier-free instance, which R-Δ_0 refutes at some depth. Hence every such template in Res_∞ (for R-Δ_0) is truth-sound.*

Proof in Appendix E.4: at a falsifying assignment of the parameters, replace each formula value by $0 = 0$ or $0 = S0$ according to its truth value.

So without labels Question 1 has a positive answer by a sound merge. What certifies the universal is the absence of a counterexample, not a derivation; for quantifier-free φ with R-Δ_0 this is truth-sound in $\mathbb{N}$ in the limit of unbounded depth (at a fixed depth only relative to Res_d), and otherwise only relative to the strength of the oracle. Unsound permanent residues in arithmetic need quantifiers in the template.

6.7 Depth relativization is necessary

For a practice $\mathcal{P}$ with a data law, let $\mathrm{Res}_\infty(\mathcal{P})$ be the union of the instance sets of the templates that cover instances (in the support of the data law) of two different targets of $\mathcal{P}$ and have no refutable instance.

Theorem 6.15 (depth relativization)**.** *[proved]*[untagged Thm 5.7] *There is a uniformly computable family of practices (finite sets of DT° targets in arithmetic, each with a data law, all with the oracle R-Δ_0) such that no computable learner achieves both of the following, for every valid practice $\mathcal{P}$ of the family and some $\delta < \frac{1}{2}$: (a) $\Pr[\exists t : \mathrm{Acc}_t \not\subseteq R^*_{\mathcal{P}} \cup \mathrm{Res}_\infty(\mathcal{P})] \le \delta$; (b) if $\mathcal{P}$ is separated at some finite depth and each target's data law yields an anchor with positive probability, then $\Pr[\exists t\, \forall t' \ge t : \mathrm{Acc}_{t'} = R^*_{\mathcal{P}}] \ge 1 - \delta$.*

Idea. Via MRDP (Davis et al., 1961; Matiyasevich, 1993), whether a two-instance merge of a true-looking arithmetic schema is ever refuted encodes the halting problem (Appendix E.4).

The obstruction is the Π_1-completeness of "this merge is never refuted", not lack of information. DTRC with a depth schedule $d_t \to \infty$ meets (b), is sound relative to Res_{d_t} at time t and to Res_∞ in the limit, and changes its mind finitely often on every separated practice. This is the form of soundness of the parent report (Hänni and Claude (Anthropic), 2026), and the theorem shows it cannot be improved.

6.8 Whole-cluster tests and noise

Coherence is downward closed but not transitive, so DTRC tests whole clusters: single linkage on the pairwise-coherence graph can build clusters with no unrefuted minimal template; under $\mathrm{RS}_d(D)$ single linkage, complete linkage and DTRC agree (Proposition E.11). Refutable mistakes stay incoherent singletons, and under separation of the clean data mistakes never bridge targets; an unrefutable mistake can be absorbed into a target's cluster, making it unsound, or fragment it (Proposition E.12). *Robust* DTRC asserts only clusters of multiplicity $\ge s > e$ and verifies them with a trimmed verifier that tolerates e mistakes. If the clean data are separated, every mistake is refutable or noise-separated, and every coherent set of mistakes has multiplicity $< s$, it asserts

exactly the D_i of multiplicity $\geq s$, soundly, and exactly once every subset of D_i of size $\geq |D_i| - e$ contains an anchor. Fragmentation is not prevented, and a frequent coherent family of mistakes is a systematic error that no positive-data method can tell from a rule (Example E.13).

6.9 Cost: the union verifier and the refutation test

Theorem 6.8(v) counts coherence tests. A test needs at most $|\mathrm{Min}(X)|$ refutation checks; $|\mathrm{Min}(X)|$ can be 4^n for two data of size $O(n)$ (Proposition 3.13), but it is 1 when X is quantifier-free or contains an anchor (under (Rich)), and with a single negative the test is polynomial (Proposition E.14). In every test of the PA and ZF runs below, $|\mathrm{Min}(X)| = 1$. In general both the exact union verifier and the refutation test are hard; write $\mathrm{Acc}_k(D) := \mathrm{Acc}_k(D, \emptyset)$.

Theorem 6.16 (complexity)**.** *[proved; computed]* single Thm H *(a) For $k \leq 2$, "$q \in \mathrm{Acc}_k(D)$" is decidable in polynomial time (2-SAT). (b) For every fixed $k \geq 3$ it is coNP-complete (from graph k-colourability). (c) With k in the input it is coNP-complete already for quantifier-free data over $\wedge, =, S, 0$ (from Set Cover); for binder-free data and fixed k it is polynomial. (d) The refutation test of* DTRC *step 2, "does some $T \in \mathrm{Min}(X)$ avoid a given finite set R?", is NP-complete, already for $|X| = 2$ with one binder (from CNF-SAT); it is polynomial when X has a least covering template, in particular when X contains an anchor (under* (Rich)*). $R \cap \mathrm{Acc}(X) \neq \emptyset$ is a polynomial sufficient condition for blocking a merge.*

The proofs (Appendix E.6) use polynomially many *failure types*; they were not refereed independently, and the results are worst-case. In the hardness proof of (d) one datum is false; whether the test stays hard when X is true and R false in a fixed structure, as in DTRC, is open.

6.10 Without negatives: MDL and the size principle

Could compression replace refutation? Determinacy makes a two-part code well defined: each datum is coded by a covering template and its unique matcher. For Q plus induction, the question is whether MDL keeps T_{Ind} whole or splits it by the main connective of the motive. A naive code splits, by a margin linear in the number of induction data (Proposition E.16); a well-specified code gains at most $O(\log n)$ from splitting (Proposition E.17); and under a natural usage law even depth-aware adaptive codes split for $n \geq 1.6 \cdot 10^4$, while under a law for which the code is well specified it keeps T_{Ind} whole (Table 12). **MDL tracks the statistics of usage, not the logical boundaries of schemas**: it errs towards incompleteness for schemas and towards unsoundness for rarely used ground axioms, errors that the anchor condition and the refutation test of DTRC control (Appendix E.7).

6.11 Natural examples: PA, ZF and ZFC

The computations use a separate prototype implementation of DTRC (not the package of Section 7). Each run is audited so that it *is* DTRC_d with a fixed finite oracle (Proposition E.18), and every run passed.

PA. Every cross merge among Q1–Q7 and T_{Ind}, and among the instance schemas of Q's axioms, is refuted by R-Δ_0 (Table 13); for Q1–Q7 and T_{Ind} this global separation is proved, for the instance schemas it is supported by samples. On three seeds of 40 shuffled unlabelled data, DTRC returned one pure cluster per target present (7 or 8); the induction cluster's only unrefuted minimal template was T_{Ind}; 200/200 held-out induction instances and 0 of 91 non-instances (near

misses) were accepted; separation on the data was verified on every run. Induction is never at risk of a cross merge, because every template covering a Q-axiom and an induction instance contains all of $\mathrm{inst}(T_{\mathrm{Ind}})$ and is refuted; it is learned incompletely only for lack of diversity in its own data, as in the tagged case.

ZF and ZFC. The targets are Ext, Pair, Union, Power, Inf, Found, Separation $\exists b \forall c(c \in b \leftrightarrow c \in p \wedge P(c))$, Replacement with $\exists!$ spelled out and image consequent, and $\in$-induction, in closure-normal de Bruijn form; the three schemas are Miller patterns (Section 5.4). All 36 cross pairs (one sampled instance per schema) have a unique minimal covering template, refuted by an HF counterexample (22 pairs: F_0 or $F_0 \to F_1$), by pure logic (14 pairs, e.g. $\exists x F_0(x)$), and in three of the latter by **Russell's paradox**: Union–Power, Union–Separation and Power–Separation all have the minimal template $\exists x \forall y(y \in x \leftrightarrow F_0(y))$, naive comprehension, whose instance $\exists x \forall y(y \in x \leftrightarrow \neg y \in y)$ is logically false (Example E.19). On three seeds of 45 data DTRC returned 9 pure clusters, each schema cluster exact (180/180 held-out instances), and 0/30 instances of naive comprehension accepted: Separation without its guard is not learned. The Axiom of Choice is one more ground axiom: its 12 cross templates are all refuted, and on ZFC data DTRC returned 10 pure clusters with AC accepted, its anchor being {AC} (Example E.20). So the two refutation channels of the setting separate the ZFC axioms on every sampled pair and every run. Section 7 reports an independent implementation with the same picture on PA and ZF mixtures, and proves pairwise separation of its $\forall$-closed forms for an ideal refuter (Corollary 7.3).

Where separation fails. *Sound merges*—$\forall x \varphi$ from instances, redundant targets, instances shared by two true generalizations—cannot and should not be prevented; they are why DTRC is truth-sound rather than target-sound without separation. *An unsound merge that only coherence refutes*: in bounding form, $\mathrm{Union}_W = \exists b \forall c(\exists d(d \in p \wedge c \in d) \to c \in b)$ and $\mathrm{Power}_W = \exists b \forall c(\forall d(d \in c \to d \in p) \to c \in b)$ have as unique minimal covering template the *universal-set schema* $U(F) = \exists x \forall y(F(y) \to y \in x)$, whose instance $\exists x \forall y(y = y \to y \in x)$ is false.

Proposition 6.17 (the universal-set schema)**.** *[proved; computed]*[untagged Prop 7.1] *No instance of U is refuted by R-logic, by R-HF, or by any oracle sound for the structure $M_0 = \mathrm{HF} \cup \{u\}$ with $\in$ as in* HF *on* HF *and $y \in u$ for all $y \in M_0$. The template is refuted by coherence, together with the true Π_1 sentence $\forall x \neg x \in x$ or with the closure of one Separation instance.*

Proof idea. Every instance holds in M_0 with $x := u$, and R-HF's steps are sound in M_0; a witness b of $\exists x \forall y(y = y \to y \in x)$ gives $b \in b$ (Appendix E.8).

With logic and HF only, DTRC merges Union_W and Power_W and *accepts the false universal-set sentence* (computed), an unsoundness inside the residue. The learned Separation cluster accepts Russell's instance of Separation; designating its closure as true (*bootstrapped coherence*) and rerunning gives 9 pure clusters, with the universal set rejected. Bootstrapping is sound exactly when the trusted clusters are pure, and circular otherwise. K_0, the first-order lgg of $\mathrm{Ind}(0 + x = x)$ and $\mathrm{Ind}(S0 + x = Sx)$, has a false instance J_0^* (the one shown in Theorem 5.14), and R-Δ_0 and logic refute none of its instances. In DT$^\circ$ these data also have a sound T_{Ind}-specialization, and the verifier intersects both, so J_0^* is rejected; the K_0-shaped template harms only when a mistake removes the sound alternative (Example E.13), and coherence with designated or learned Q refutes it (Appendix E.8). *Permanent residue*: templates whose false instances are all consistent with the designated truths and Δ_0-correct (e.g. families containing $\neg\mathrm{Con}(\mathsf{PA})$-like Σ_1 sentences) are never refuted, and by Theorem 6.15 no computable learner can tell in general whether a merge

will be. In arithmetic they need quantifiers in the template (Proposition 6.14); a natural merge of this kind was not found (open). Proposition 7.4 reports a finite-budget effect (a residue in Res_d, not a permanent one): two injected mistakes merged into a false template at a small refutation budget, and a larger budget refuted the merge.

6.12 Summary: the answer to Question 3

DTRC answers Question 3 (Table 8). Under refutation separation untagged learning is as easy as tagged learning, and the number of targets need not be known; PA is separated at small depth (proved for Q1–Q7 and T_{Ind}), and ZF and ZFC on every sampled pair and every run. Without separation DTRC is sound relative to a residue that no computable learner can avoid in general. Four features are forced by the results above, and a naive version without them fails: (1) test whole clusters and intersect all unrefuted minimal templates (coherence is not transitive, and a K_0-shaped template can sit next to the sound one); (2) keep the refutation depth as a parameter and re-cluster as it grows; (3) when a bound $k \geq k'$ is known, intersect with the k-union learner fed with DTRC's own negatives, which restores target-soundness without separation (Remark E.10); (4) implement refutation as a growing set of refuted sentences, audited after each run (Proposition E.18). Open problems: Section 8.3.

learner	assumption	guarantee	result
union, no bound	sound negatives	accepts only the data	Prop. 6.4
k-union, $k \geq k'$	sound negatives	target-sound; sound splits survive	Thm. 6.5(a)
k-union, $k \geq k'$	cross-separation; disjoint instance sets	exact for i iff no $k - k' + 1$ avoiding templates cover D_i and miss an instance	Thm. 6.5(b)
k'-union, $N(D, d)$	anchor-pair separation	sound; exact once anchors are present	Thm. 6.6
DTRC$_d$	$\mathrm{RS}_d(D)$	recovers labels; target-sound; exact from anchors	Thm. 6.8
DTRC$_d$	none	sound relative to $\mathrm{Res}_d(D)$; truth-sound if merges are	Thm. 6.10
any computable	—	not both residue-sound and exact if separated	Thm. 6.15
DTRC $\cap$ k-union	$k \geq k'$ known	target-sound; exact from anchors if $\mathrm{RS}_d(D)$, $k = k'$	Rem. E.10
robust DTRC	RS_d on clean data; rare, noise-separated mistakes	sound; exact if $(\lvert D_i \rvert - e)$-subsets have anchors	Prop. E.12

Table 8: Guarantees for untagged data ("exact": Acc $= R^*$, per target where stated). Fragmentation and ambiguous data limit exactness without separation (Theorem 6.10(d), Proposition 6.11).

7 Implementation and experiments

We implemented the determinate-template learner and DTRC to check the theory of Sections 3 to 6 on data it did not choose, and to see what goes wrong when its hypotheses fail: when refutation has a budget, when targets look alike, and when the data contain mistakes. An adversarial referee re-ran every experiment with independent checks; the resulting corrections and withdrawn claims are listed in Appendix G. *Computed* means that a seeded script was run; re-running reproduces every number except timings. We describe the implementation (Section 7.1), its guarantees (Section 7.2), the results (Section 7.3) and the limitations (Section 7.4); proofs, further numbers and commands are in Appendix F.

In brief: on Questions 1 and 2 the implemented learner behaves as the theory predicts, and on Question 2 it accepted no non-instance in 927 probes. On untagged mixtures of the PA and ZF axioms (Question 3), DTRC recovered the hidden labels in every run, and with a sharing pass for

ambiguous data it matched, target by target, a tagged learner given every label a datum deserves. For these mixtures we prove refutation separation for an ideal refuter, and the implemented refuter achieved it on every sampled merge. But with injected mistakes and a small refutation budget, DTRC *accepted a false sentence*; similar targets can merge into valid templates that no world-based refutation separates; and the evidence for separation beyond the mixtures is narrow.

7.1 What was implemented

Data and the class. Formulas over L_A and $L_\in$ use de Bruijn indices and named parameters; each datum is in closure-normal form, canonicalized on its own (parameters $w_0, w_1, \ldots$ by first occurrence). Leading universal quantifiers are *not* stripped: the axiom Q4 is the datum $\forall x\,(x{+}0 = x)$, and $z + 0 = z$ accepts $w_0 + 0 = w_0$ but not Q4. So the PA-mixture targets below have disjoint instance sets except for $0+0=0$, and the ZF-mixture targets are pairwise disjoint (Corollary 7.3; stripped quantifiers: Appendix F.5). Matching is up to injective renaming of the template's parameters, and inst, "covers" and $\succeq$ are read in this sense (inst$^\sim$, $\succeq^\sim$ of Appendix A.7); anchors are taken in the class $\{\mathrm{inst}(T) : T \in \mathrm{DT}^\circ_F\}$. The class is DT°_F (Definition 2.1): every target below lies in it, so the cautious verifier stays sound; positive-arity term metavariables blow up the minimal covering sets (E7); and every DT° anchor is a DT°_F anchor, so the classes differ only before a DT° anchor arrives.

Minimal covering templates. They are computed from the finitely many *normal configurations* of the data's common-prefix tree (Definition F.1 and Proposition F.2), with parameters aligned (Appendix A.7): $\mathrm{Min}_F(D) := \mathrm{Min}^{\mathrm{al}}_{\mathrm{DT}^\circ_F}(D)$ is the set of minimal elements of the union of $\mathrm{Min}^{\mathrm{lit}}_F(\alpha(D)) = \mathrm{Min}_{\mathrm{DT}^\circ_F}(\alpha(D))$ over all alignments α. By Proposition A.5 it has the properties of $\mathrm{Min}(D)$ with respect to templates all of whose parameters occur in their skeleton, such as our parameter-free targets; the literal set fails for rigid parameters (Example A.6; checks E0, E0b in Appendix F.1). On the experiment data the aligned and literal sets almost never differ (Appendix F.1).

World oracles. Two three-valued evaluators return True, False or unknown: for $\mathbb{N}$, exact on numerals, with polynomial normal forms for atoms and bounded enumeration, numeral search and case splits for quantifiers; for V, with hereditarily finite sets of rank ≤ 3 and *generic objects* standing for unknown sets outside the transitive closure of the values in scope. Both use Kleene's strong connectives with supervaluation over unknown leaves, after constant propagation (Appendix F.2).

Proposition 7.1 (The oracles are sound)**.** *[proved; computed]* experiments Prop E2; recheck R3, R4, R11 *If the PA evaluator returns True (False) on a datum, its universal closure is true (false) in* $\mathbb{N}$*. If* $V \models \mathsf{ZF}$ *and the ZF evaluator returns True (False) on a datum, its universal closure is true (false) in* V*.*

The referee's independent evaluators found no contradiction on 2197 PA and 592 ZF logged verdicts and 3600 random sentences. Neither oracle is complete: no witness is constructed from a generic or a term, so $\forall a \exists b \forall x\,(x \in b \leftrightarrow a \in x)$ is not refuted and $\forall x\,(x = 0 \vee \exists y\, x = Sy)$ stays undecided. Both are truth oracles, not provability oracles.

Template refuter and dtrc. A template is *refuted* once the oracle returns False on one of its instances. Within a budget (default 80 instances) the refuter tries $\top/\bot$ assignments to the

formula metavariables, pools and enumerations of small bodies, and bodies read off the data (Appendix F.2). It is sound, *budgeted* and *history-dependent* (it caches verdicts and accumulates data-guided tries), so it is not a fixed oracle Ref_d (Definition 2.9); the guarantees below are stated for it directly, and every "refuted" count is a lower bound for an *ideal* refuter, which refutes every template with a false instance. DTRC as implemented:

(1) canonicalize the data; discard those the oracle refutes;
(2) agglomerate from singletons, trying pairs by decreasing common-prefix size (ties in data order): merging A and B succeeds iff some member of $\mathrm{Min}_F(A \cup B)$ is unrefuted, and a failed pair stays failed for clusters containing its parts;
(3) optionally, a *sharing pass*: for each final cluster C and each kept datum $d \notin C$, in data order, add d to C if the test on $C \cup \{d\}$ succeeds (clusters may then overlap);
(4) per cluster, $\mathrm{Acc}(C) = \bigcap\{\mathrm{inst}(T) : T \in \mathrm{Min}_F(C) \text{ not refuted}\}$, or C if all are refuted;
(5) assert $\bigcup_C \mathrm{Acc}(C)$.

This is DTRC of Definition 6.7 with the class DT°_F, a fixed merge order and a budgeted refuter, with four further differences: failed pairs are inherited (lossless for an ideal refuter by Proposition F.3(a), a heuristic otherwise); refuted data are discarded; a cluster whose minimal templates are all refuted accepts its own data (Definition 6.7: nothing); and with a parameter-free datum, hence a single alignment, the literal set $\mathrm{Min}^{\mathrm{lit}}_F$ is used, which for parameter-free targets changes no guarantee below (Appendix F.3).

Baselines. (i) First-order lgg per tag (Plotkin, 1970; Reynolds, 1970), with binders named by level or as de Bruijn indices (instantiation may capture). (ii) Higher-order pattern lgg per tag, in our reading of the merge rule of Baumgartner et al. (2017) (their implementation was not run): the *genuine* version generalizes a term disagreement containing a bound variable by a term variable $f(x, \dots)$; the *formula-level* version, an earlier baseline, generalizes the whole atom. (iii) The cautious k-union verifier over DT°_F, with and without refutation negatives. (iv) *Skeleton clustering*: DTRC without refutation, stopped at the true number of targets. (v) The tagged refutation-filtered verifier with *gold labels* (the generating target) or *membership labels* (every target a datum instantiates).

Data sets and metrics. *PA-mix*: Q1–Q7; 30 induction instances with random motives; 8, 6, 4 instances of $U_{\mathrm{add0}} = (z + 0 = z)$, $U_{\mathrm{mul0}} = (z \cdot 0 = 0)$, $U_{\mathrm{0add}} = (0 + z = z)$. In the main (*numerals*) regime z is a numeral uniform in 0..7; in the *mixed* regime it is a numeral (0.6), a random closed term of depth ≤ 2 (0.25) or a parameter (0.15). *ZF-mix*: $\mathrm{Ext}, \mathrm{Pair}, \mathrm{Union}, \mathrm{Pow}, \mathrm{Inf}, \mathrm{Found}$; 12 instances of T_{Sep}, 10 of Replacement $\mathrm{Repl}_{\mathrm{E}}$ ($\exists!$ spelled out as in $\mathrm{Repl}_{\mathrm{sp}}$, with the image consequent of $\mathrm{Repl}_{\mathrm{J}}$; neither formulation of Table 6), 10 of T_{EInd}, with random small bodies. Held-out sets contain schema instances only, disjoint from the training data. Metrics: the adjusted Rand index (ARI; Hubert and Arabie, 1985) on unambiguous data; *exactness*, by a sufficient syntactic test for $\mathrm{Acc} = \mathrm{inst}(T^*)$ (every unrefuted minimal template is $\succeq T^*$ and one is $\preceq T^*$), reported separately for ground axioms, where it only means that the axiom stayed a singleton; held-out acceptance; and *probes*: sampled accepted sentences, how many are instances of no target, and how many of those the oracle refutes, a lower bound on unsoundness (Proposition 7.4).

7.2 Guarantees for the implemented learner

The theorems of Section 6 (Theorems 6.8 and 6.10) carry over to DT°_F, Min_F and the budgeted, history-dependent refuter (world $\mathbb{N}$ or V; proofs in Appendix F.3). Within-target merges always

succeed, and a cluster inside the instance set of a target with true instances always intersects a never-refuted template below that target, whatever the budget (Propositions F.3 and F.4). So a budgeted refuter harms soundness only through clustering: a mixed cluster is truth-sound if one of its unrefuted minimal templates is valid, and may accept false sentences otherwise (Proposition 7.4). Exactness does depend on the budget: more refutations enlarge $\mathrm{Acc}(C)$.

Theorem 7.2 (DTRC under refutation separation)**.** *[proved]*[experiments Thm E4] *Assume* (H1) *targets $T_1, \dots, T_k \in \mathrm{DT}^\circ_F$, all of whose parameters occur in their skeletons (e.g. parameter-free), with all instances true;* (H2) *data $D = D_1 \sqcup \cdots \sqcup D_k$, $D_i \subseteq \mathrm{inst}(T_i)$, each datum an instance of exactly one target;* (H3) *a sound refuter; and* (H4) *every merge test on $A \cup B$ with $\emptyset \neq A \subseteq D_i$, $\emptyset \neq B \subseteq D_j$, $i \neq j$, fails. Then* DTRC *without sharing discards no datum, its final clusters are the nonempty D_i, and $\mathrm{Acc}(D_i) \subseteq \mathrm{inst}(T_i)$ (target-soundness), with equality whenever D_i contains an anchor of T_i. If moreover* (H-ref) *verdicts do not depend on the call history, $\mathrm{Acc}(D_i)$ equals the tagged verifier's acceptance set on D_i.*

Exactness is stated per cluster because instance sets may overlap; the implemented refuter violates (H-ref) [referee F8]. The sharing pass handles ambiguous data (Proposition F.5). If ($\mathrm{H4}^+$) every test on a set lying in no single instance set fails, and (H5) every cluster of the agglomeration lies in exactly one instance set, then each such cluster grows to all data of its target, stays target-sound, is exact with an anchor, and under (H-ref) matches the tagged learner with membership labels. A target that labels no such cluster gains nothing and can be learned incompletely although its data contain an anchor (Example F.6). With numerals every anchor of U_{add0} or U_{0add} contains $0 + 0 = 0$ (Proposition F.8), so a partition gives this datum to at most one of them, the sharing pass to both.

For the mixtures, (H4) holds with an ideal refuter: every minimal covering template of a cross pair of mixture targets has one of five shapes, each with a false instance such as $\bot$, $\forall \bar{x}\, \bot$, Russell's instance or $1 = 0$ (Proposition F.9, Table 14); a computed check (E10) of 9100 random pairs found this in every case.

Corollary 7.3 (Separation of the mixtures)**.** *[proved]*[experiments Cors E8.2, E8.3] *With an ideal refuter,* (H4) *holds iff every member of $\mathrm{Min}_F(\{a, b\})$ is refuted for all cross pairs $a \in D_i$, $b \in D_j$. Hence* (H4)*, and with it Theorem 7.2, holds for PA-mix and ZF-mix data in which no datum instantiates two targets; $0 + 0 = 0$ is the only sentence that does.*

7.3 Results

Universal axioms (Question 1; E1). On six universal axioms (2000 runs each) the DT°_F verifier and the first-order lgg became exact at the same N in every run. For the five one-variable axioms the mean N was 3.27 (predicted 3.25, mixed regime) and 8.42 (predicted 8.11, numerals); these rows share seeds and form one sample, in which 8.42 is 1.85 standard errors off, and the referee's 200 000-run simulation gives 8.117. Figure 1 shows the curves, and Section 4.3 discusses the numbers [experiments E1].

Single schemas from tagged instances (Question 2; E2). Table 9 gives exactness; no learner is exact from one instance, since a datum covers itself as a ground template. In probes ($N = 2$ and 6, 8 seeds) the DT°_F learner accepted 0 non-instances in 927. Both pattern lggs accepted none on the ZF schemas, but on induction 224 of 229 probes at $N = 2$ were non-instances, 58 of them refuted (genuine; formula-level 213/220, 46), as Proposition 5.15 predicts [prior induction C9]. First-order lggs were unsound (exhibits in Appendix F.5) on Separation (capture of

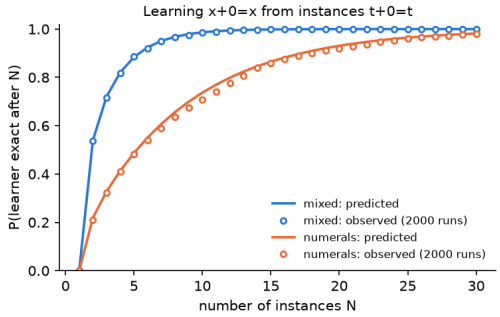


Figure 1: E1: probability that the DT°_{F} learner is exact after N instances of $x + 0 = x$, observed (2000 runs) and predicted by Theorem 4.4(a) and Proposition F.8, mixed and numerals-only regimes experiments E1.

	DT°_{F}, exact		pattern lgg, exact		first-order lgg, complete	
schema	plain	+refutation	genuine	formula-level	named	de Bruijn
T_{Sep}	23/27/30	23/27/30	23/27/30	23/27/30	25/28/30	25/28/30
$\mathrm{Repl}_{\mathrm{E}}$	23/24/30	23/24/30	23/24/30	23/24/30	28/28/30	28/28/30
T_{EInd}	27/29/30	27/29/30	24/29/30	27/29/30	24/29/30	24/29/30
T_{Ind}	25/30/30	27/30/30	0/0/0	0/0/0	25/30/30	25/30/30

Table 9: E2: runs out of 30 at $N = 2/3/8$ tagged instances. For first-order lggs only completeness (all 30 held-out instances accepted) is shown; their soundness fails except on $\in$-induction in de Bruijn indices (text) experiments E2.

the comprehension variable), Replacement and induction in both encodings, and on $\in$-induction in the named encoding; in de Bruijn indices, where $\in$-induction is a first-order pattern (Theorem 5.2), they were sound and complete on it (0 non-instances in 182 probes). On Replacement the false instances come from the image consequent of $\mathrm{Repl}_{\mathrm{E}}$; for $\mathrm{Repl}_{\mathrm{sp}}$ the named first-order lgg of anchor data is truth-sound (Proposition 5.7 and the remark after it).

The ZF schemas, Miller patterns in de Bruijn form, are learned in most runs from 2–3 instances (in every run by $N = 8$) by DT°_{F} and by the genuine pattern lgg, which is slightly later on $\in$-induction because it keeps atoms that DT°_{F} lifts to formula metavariables. (An earlier claim that the pattern lgg equals DT°_{F} on the ZF schemas rested on the formula-level baseline and is withdrawn; Appendix G.)

Untagged mixtures (Question 3; E3). Totals over seeds 0–4 for PA-mix with numerals are in Table 10. T_{Ind} is exact for DTRC, DTRC+share and both tagged learners in every seed (the pattern lgg never is), U_{mul0} in seeds 1–4 (seed 0 has no $0 \cdot 0 = 0$); these four learners differ only on U_{add0} and U_{0add}, and seed 3 has no $0 + 0 = 0$, hence no anchor of either. In the mixed regime DTRC is exact on 19/20 schema targets, the other three on 20/20. On ZF-mix, DTRC, skeleton clustering and both tagged learners have ARI 1, all 15 schema targets and 30 axioms exact, and no non-target probe; the pattern lgg is also exact, and the first-order lggs give 223 (named) and 128 (de Bruijn) non-target probes experiments E3.

Reading. DTRC recovers the labels in every run, and its exactness differs from the gold-labelled learner's only through the ambiguous $0+0 = 0$ (equal totals, different targets exact). With sharing

learner	ARI	schemas exact	axioms exact	held-out	probes / non-target / refuted
DTRC	1.000	13/20	35/35	451/500	226 / 0 / 0
DTRC+share	1.000	17/20	35/35	474/500	227 / 0 / 0
tagged, gold labels	–	13/20	35/35	–	–
tagged, membership labels	–	17/20	35/35	–	–
skeleton clustering (k given)	0.993	9/20	29/35	425/500	282 / 68 / 64
first-order lgg per tag	–	–	–	453/500	180 / 100 / 28
pattern lgg per tag	–	8/20	–	453/500	221 / 99 / 33

Table 10: E3, PA-mix, universal axioms through numerals only; totals over 5 seeds. Schemas: T_{Ind} and the three instance schemas; axioms: Q1–Q7. Both encodings, resp. both pattern versions, gave identical rows experiments E3.

it matches the membership-labelled learner target by target and seed by seed, consistently with Proposition F.5: in every run each target with data labels a cluster of the agglomeration and (H5) holds (computed), and the proposition predicts the agreement where exactness comes from an anchor (in general it needs (H-ref), which the implemented refuter lacks). With numerals the instance schemas are often not exact for any learner: n numerals uniform in 0..7 form an anchor with probability $1-(7/8)^n-(1/8)^n$ (Proposition F.8). Skeleton clustering with the true k is unsound on PA (it lumps Q5 and Q7 into $\forall x \forall y\, P(x,y)$, in one run also Q4 and Q6 into $\forall x\, P(x)$, and data of U_{add0} and U_{0add} into templates such as $z_0 + z_1 = SSz_2$) and perfect on ZF, whose targets have distinct skeletons.

Learning curves (E5). Streams are drawn i.i.d. from a mixture of all targets (weights and curves in Appendix F.5, Figure 2). DTRC+share coincides with the membership-labelled tagged learner in every stream, per target and seed, in the means and in the first N from which a target stays exact. All differences between DTRC and the gold-labelled learner concern U_{add0}, U_{0add} and come from $0+0=0$. T_{Ind}, Separation, Replacement and $\in$-induction are exact after 5–20 examples; with numerals the instance schemas are exact only after their 0-instance appears (for DTRC on U_{0add}, beyond 120 examples in four of five streams) experiments E5; referee R9.

The k-union verifier; why DT°_F (E6, E7). Without clustering, the cautious k-union verifier on Q plus induction follows the slot accounting of Theorem E.2 (counts in Appendix F.5): with refutation negatives, $k=8$ learns induction from the anchor pair $\mathrm{Ind}(x=x), \mathrm{Ind}(\neg x=0)$, and $k=9$ from three instances but not from these two. With third motive $\exists y\, x<y$ this matches the theory; with third motive $x+0=x$ it does not: the block $\{\mathrm{Ind}(x=x), \mathrm{Ind}(x+0=x)\}$ is root-homogeneous, and the DT° template $(f(0)=0 \wedge \forall x(f(x)=x \to f(Sx)=Sx)) \to \forall x\, f(x)=x$ covers it, has only true instances and misses $\mathrm{Ind}(x\cdot 0=0)$; it is not in DT°_F, so here the implemented class accepts more than DT° would. No false non-instance was accepted experiments E6; prior pa-untagged. For pairs of mixture data, full DT° gives up to 64 (PA) and 32 (ZF) minimal templates and 3 timeouts on ZF pairs, DT°_F at most 4 and 8 (cf. Proposition 3.13) experiments E7.

Separation evidence (E9). At budget 80 every cross merge on the mixtures is refuted (2200 PA and 540 ZF pair merges; all counts in Table 16, Appendix F.5), but the thousands of merges reduce to 15 (PA) and 5 (ZF) distinct templates, refuted by trivial instances (such as $1=0$, $1+0=0$, $\forall \bar{x}\, \bot$, $\top \to \bot$) or a comprehension instance. That is Proposition F.9, not evidence of

separation between *similar* targets [referee F3]. The *near-miss* and *hard* sets (Appendix F.4) contain similar targets, such as $z+0=z$ and $z+1=Sz$, or $\in$-induction and $\in$-induction along $z \in y \in x$ (EInd2). On PA, 7 of 1080 pair merges stay unrefuted, all valid, e.g. $n+z=S^n z$ from $n+0=n$ and $n+1=\underline{n+1}$ (Lemma F.11), so no world-based refuter separates them.

Two hard ZF pairs mark the limits (Proposition F.10). Union_W and Power_W have the universal-set schema (with a binary metavariable) as unique minimal covering template; the ZF oracle refutes it, but no oracle sound for HF plus a universal element does (Proposition 6.17). A merge of $\in$-induction with EInd2 is typically false and is refuted only with constant propagation (10 of 12 random merges with it, 0 without; E9b); the hard set's unrefuted merges (1 of 432 pairs, 2 of 432 sets, all of this kind) are false and are refuted at budget 400. Some other merges of that pair are valid, as is Separation with Separation from $\bigcup a$ on the body $x \in a$, so (H4) fails for these pairs in general.

Stress tests (E4). *(a) Near-miss targets* (the near-miss sets of E9; 3 seeds). On ZF all 6 targets are exact in every seed. On PA the three forms of induction are always separated, $z+0+0=z+0$ merges into $z+0=z$, and in seed 0 data of $z+0=z$ and $z+1=Sz$ merge into $5+z=S^5 z$ and $4+z=S^4 z$; every accepted template of a mixed cluster is valid (Lemma F.11).

(b) Injected mistakes. PA: 8 mistakes (induction-like sentences with a wrong step $\varphi(x) \to \varphi(x)$, a wrong base $\varphi(S0)$, or the true conclusion $\forall x\, \varphi(Sx)$; and $n+0=0$) and 3 true non-targets. ZF: 6 mistakes, 5 of them distinct (Separation with the comprehension variable free in the body, $\forall x(\varphi \to \varphi) \to \forall x \varphi$, and complement comprehension, which occurs twice). Clean targets stay exact (PA 11/11 mixed and 9–10/11 numerals, ZF 9/9). Refuted mistakes are discarded; the others become singletons or merge. Every mixed cluster was checked with a post-hoc refuter and by hand (Table 15).

Proposition 7.4 (A false sentence is accepted)**.** *[proved; computed]*[experiments Prop E10; referee F1, R15] *In the stress test with injected mistakes, PA-mix seed* 0*, both regimes, the wrong-step and the* $\forall x\, \varphi(Sx)$ *mistakes merge at refuter budgets* 80 *and* 400 *into a cluster whose only accepted template is*

$$T_F = \big(\exists x(P_0(x,0) \wedge \neg P_1(x,0)) \wedge \forall x\big(\exists y(P_0(y,x) \wedge \neg P_1(y,x)) \to \exists y(y < Sx \wedge \neg P_1(y,Sx))\big)\big) \\ \to \forall x\, \exists y\, (y < Sx \wedge \neg P_1(y,Sx)).$$

Its instance q_F *with* $P_0(y,x) := (x=0)$ *and* $P_1(y,z) := (z=2)$ *is false in* $\mathbb{N}$ *and an instance of no PA-mix target.* DTRC *accepts* q_F *at budgets* 80 *and* 400*, not at* 2000*.*

Proof. The first premise of q_F, $\exists x(0=0 \wedge \neg\, 0=2)$, is true. The second, $\forall x(\exists y(x=0 \wedge \neg\, x=2) \to \exists y(y < Sx \wedge \neg\, Sx = 2))$, is true: at $x=0$ take $y=0$, since $1 \neq 2$; at $x \geq 1$ its antecedent is false. The conclusion $\forall x\, \exists y\, (y < Sx \wedge \neg\, Sx = 2)$ fails at $x=1$. The rest is computed (E4(b))[recheck R15]. □

The current oracle refutes q_F by a case split on $x=0$ for the Π_1 premise (an earlier version returned unknown), but the refuter does not reach q_F within 400 tries, and the probes, which never sampled it, reported "0 refuted". So *with mistakes in the data, the union asserted by* DTRC *is not truth-sound even in arithmetic*, and *probe counts are lower bounds on unsoundness.* The other false templates met (two induction-step templates and five Separation-capture templates, all at budget 80; Appendix F.4) disappear at budget 400, and at 2000 no false merge remains; the remaining residual merges are valid (Lemma F.11).

7.4 Limitations

(1) Budgeted refutation can accept false sentences (Proposition 7.4). (2) No world-based method separates valid residual merges; DTRC is then truth-sound, not target-sound (Lemma 2.10(a), Theorem 6.10(b)). (3) The sharing pass needs (H4$^+$) and (H5), and helps only targets that label a cluster of the agglomeration (Example F.6). (4) The oracles construct no witnesses. (5) A one-instance target is never generalized (E4(c), with unequal frequencies: Appendix F.5), and an unrefuted singleton mistake looks like a ground axiom. (6) DT°_F lacks positive-arity term metavariables, so it can accept more than DT° (E6); the feature test of Theorem 3.4 is not implemented. (7) Without (H4) the partition can depend on the merge order. (8) Separation is proved for the mixtures only, and for an ideal refuter. (9) Scale: pure Python, about ten minutes for all experiments. Open problems: Section 8.3.

8 Discussion

Sections 4 to 6 answer the three questions and Section 7 tests the answers. This section says what they mean for the project they belong to, learning which inference rules a community uses from imitation, coherence and the world (Hänni and Claude (Anthropic), 2026). It then relates them to earlier work, collects the open problems, and states limitations and scope. It claims no new results.

8.1 What the answers mean

An instance schema is not a universal sentence. Instances $\varphi(t)$ at closed terms determine the instance schema $\varphi(z)$ (for one variable, two instances whose terms have different head symbols suffice; Theorem 4.2), but not the sentence $\forall x\varphi$. The step to $\forall x\varphi$ is an ω-rule step: truth-safe for every φ iff the closed-term substructure is elementary (Proposition 4.7), and in general not derivable, even over Q (Example 4.9). A cautious learner whose class admits parameters as values takes it silently after the first anchor (Proposition 4.5), and closed data cannot show whether a community takes it. A closedness guard blocks the step; data with a parameter at every variable license it as ordinary generalization (Table 5). Without labels, $\forall x\varphi$ is an unrefuted merge of instances (Proposition 6.13), certified by the absence of a counterexample, not by a derivation. With Δ_0 evaluation "never refuted" means "true" for quantifier-free φ, though only in the limit of unbounded depth, and not for every quantified φ (Propositions 6.13 and 6.14).

First-order learnability depends on the formulation. A ZF schema is a guarded first-order pattern iff its metavariable is applied to literally the same argument tuple at every occurrence in the chosen syntax (Theorem 5.2 and Proposition 5.3). So $\in$-induction passes with de Bruijn indices and fails with names, Collection the other way round, and α-renaming the frame changes the verdict (Table 11). No finite union of first-order schemas covers raw PA induction truth-soundly (Theorem 5.13), yet the form IndEq is a single guarded first-order pattern (Proposition 5.18). A checker should learn in a class whose verdicts do not depend on what a community regards as notation. The λ convention does: bound-variable names do not matter, freshness is built in, and capture cannot occur. In a first-order syntax capture can make an unguarded learner unsound outright, and a guard must exclude it exactly when capture instances exist (Proposition 4.6, Lemma C.3).

Why the pattern fragment matters. Every ZF schema, in every formulation considered, is a higher-order pattern (Miller, 1991): $L_\in$ has no function symbols, so an instance only renames variables inside the body. Here existing technology suffices: the pattern lgg recovers each schema from any anchor (Corollary 5.10), and for one metavariable the anchors, root variation plus non-vacuity, are the same in every class between PAT and SO° (Theorem 5.9). With several metavariables, as in untagged mixtures, a Plotkin-(D)-type event appears and differs between PAT, DT° and SO° (Theorem D.3): whether data are an anchor depends on the class.

Induction needs derived occurrences. In T_{Ind} the occurrences $P(0)$ and $P(Sx)$ apply the motive to terms, so their values are *derived* from the value at the pattern occurrence $P(x)$. First-order and pattern anti-unification are both unsound on induction (Propositions 5.12 and 5.15), and some unsound first-order lggs are never refuted by true closed quantifier-free sentences (Theorem 5.14). DT° is built for this case: a pattern occurrence for every metavariable and metavariable-free arguments keep matching unique and linear and the minimal covering templates finitely many (Theorems 2.5 and 3.3). Without determinacy anchors must be stronger (Proposition 5.17); with metavariables inside arguments minimal templates can be infinitely many (Remark A.3). The price is non-unitarity, up to 4^n minimal templates (Proposition 3.13), and caution pays it: the verifier intersects them all by a polynomial feature test (Theorem 3.4), which on the data behind K_0 rejects its false instance (Section 6.11).

Refutation is the source of labels. Without a bound on the number of schemas the cautious learner accepts only its data (Proposition 6.4); with a bound, spare slots cost data diversity that no sound negatives replace (Theorem 6.5). Refutations remove over-general *lumps*, a bound removes *splits*, which are sound and so invisible to refutation. DTRC uses no bound. Under refutation separation it recovers the hidden labels and is then the tagged learner, with tagged anchors and rates (Theorem 6.8). So the world and coherence channels, which in the parent report rule out errors, here also do the work of citations. They separate Q's axioms and T_{Ind} at small depth (proved), and ZF and ZFC on every sampled cross pair and every run (computed; Section 6.11); for the mixtures of Section 7 separation is proved for an ideal refuter (Corollary 7.3). The depth cannot be eliminated (Theorem 6.15), and compression does not replace refutation: MDL (Rissanen, 1978) and the size principle (Tenenbaum and Griffiths, 2001) follow the statistics of usage, not the logical boundaries of schemas (Section 6.10).

Truth-soundness and target-soundness come apart. "Which axioms does this community use?" asks for target-soundness, "is it safe to accept this step?" for truth-soundness, and refutation serves only the second. Merges without false instances are never refuted (Lemma 2.10): $\forall x\varphi$ from its instances, the true template $0 + z = z$ from data of two targets (Theorem 6.10), the valid residual templates of the experiments (Lemma F.11). They can leave DTRC truth-sound but not target-sound, unavoidably (Proposition 6.11); target-soundness needs labels, a known bound (Remark E.10) or the community's own judgments. Truth-soundness in turn holds only up to a residue of unrefuted unsound merges, which has natural members: the universal-set schema from weak Union and Power, when only logic and hereditarily finite counterexamples are used (Proposition 6.17); K_0-shaped templates, once a mistake removes the sound alternative (Example E.13); and, at small refutation budgets, a false sentence accepted from data with mistakes (Proposition 7.4). This is the parent report's picture in a sharp case: every signal is one-sided, and what can be proved is soundness with a declared residue.

8.2 Relation to the literature

Anti-unification and inductive logic programming. The first-order core is the least general generalization of Plotkin (1970) and Reynolds (1970), the basis of bottom-up generalization in inductive logic programming (Muggleton, 1991). Plotkin's witness events are our anchor conditions for first-order targets, in DT° too (Corollary 3.9), and the feature test (Theorem 3.4) is the second-order form of his rule that equal columns get equal variables. Metarules in meta-interpretive learning are second-order Horn clauses given in advance (Muggleton et al., 2015; Cropper and Tourret, 2020); here the second-order templates are what is learned.

Higher-order anti-unification and matching. For higher-order patterns the lgg is unique (Pfenning, 1991) and computable in linear time (Baumgartner et al., 2017). Second-order generalization beyond patterns is studied by Hirata et al. (2004) and surveyed by Cerna and Kutsia (2023); we have not checked whether DT° or the feature test appears there. DT° matching is Miller's pattern unification with one side ground, followed by evaluation (Theorem 2.5; general second-order matching: Appendix A.3). The closest relatives we know of are the deterministic second-order patterns of Yokoyama et al. (2004), which relax Miller's restriction while keeping matching deterministic, and the functions-as-constructors fragment of Libal and Miller (2016); Niederhauser and Middeldorp (2026) give a unification procedure for the former that generalizes the latter. We know these classes only from abstracts and secondary descriptions, by which, as we read them, every argument of a metavariable must contain a bound variable, which the argument of $P(0)$ does not. We have compared neither class with DT°.

Learning from positive data. By a theorem of Gold (1967), an untagged learner needs a bound or a restriction on the practice such as refutation separation (Proposition 6.4). Every anchor is a tell-tale in the sense of Angluin (1980b), and up to effectivity anchors are the $\subseteq$-tell-tales of strong-monotonic learning (Lange and Zeugmann, 1992); the cautious verifier is reliable learning (Rivest and Sloan, 1988), or perfect selective classification (El-Yaniv and Wiener, 2010), applied to sentences. First-order patterns are the tree analogue of Angluin's pattern languages (Angluin, 1980a). Finite elasticity and its preservation under bounded unions (Wright, 1989; Motoki et al., 1991) give anchors for unions of determinate templates (Corollary 3.12). Shinohara (1991) shows learnability from positive data for classes defined by length-bounded elementary formal systems, and Arimura et al. (1994) compute minimal multiple generalizations for unions of pattern languages; we have not compared their complexity results with Theorem 6.16. What the present setting adds is soundness at all times against adaptive provers, explicit anchors with rates, and refutation as negative evidence.

Axiomatizability by schemas. Vaught (1967) showed that every recursively enumerable theory that directly interprets a weak set theory is axiomatizable by a schema, a formula with a new relation symbol whose substitution instances are the axioms. Such a schema is a second-order template, with induction as the paradigm; DT° is a version with unique matching, learnable from instances. That PA is not finitely axiomatizable (Ryll-Nardzewski, 1952) concerns consequence, whereas Theorem 5.13 concerns the syntax of substitution; neither implies the other, since PA is Q plus one guarded first-order pattern (Proposition 5.18).

Proof assistants. Formal libraries record which axiom, lemma or recursor a step applies, as the tagged setting assumes, and they record or recompute motives. Lean 4's `induction` tactic

records the motive, an implicit argument of the recursor, as a λ in the kernel term (de Moura and Ullrich, 2021): the DT$^\circ$ matcher. Metamath's set.mm states induction with implicit-substitution hypotheses and distinct-variable conditions (Megill and Wheeler, 2019): a first-order pattern with freshness guards that are sound but not minimal (Section 5.5, Appendix D.5). So for formal mathematics the tagged results are the relevant ones; the untagged theory matters where citations are missing, as in informal mathematics.

8.3 Open problems

The research records leave the following problems open; gray tags name the record, pointers the nearest result. Two problems listed as open in one record are settled in another: finiteness of minimal covering templates in DT$^\circ$ (Theorem 3.3) and the complexity of the refutation test (Theorem 6.16; but see O12).

O1 The anchor condition without (Rich) (Theorem 3.8). Does $\mathrm{inst}(T) \supseteq \mathrm{inst}(T')$ imply $T \succeq T'$ in DT$^\circ$? (The verifier does not need it.) single §13

O2 A theory of parameters invariant under renaming: Proposition A.5 covers only targets whose parameters all occur in their skeleton (Remark A.7) single §13; paper review.

O3 Is the number of equation kills linear (Conjecture B.9)? Escalations are $\Theta(N^2)$ in general (Proposition B.6); the counts for the ZF schemas are not known single §13; cases §6.

O4 Size-bounded subclasses of DT$^\circ$, in which induction is closed iff $s \geq 12$ (Proposition D.6) prior induction C6.2; enumeration of $\mathrm{Min}(D)$ with polynomial delay, and counting it (Proposition 3.13); lower bounds for the coincidence terms of the rates with infinite support (Proposition B.4) single §13.

O5 Anchors in SO$^\circ$: is root-level distinctness necessary with several metavariables (conjectured not, Conjecture D.4)? What is the exact condition for instance schemas over languages with closed terms (Proposition 4.3(c))? cases §6

O6 How DT$^\circ$ relates to deterministic second-order patterns and to functions-as-constructors (Section 8.2) prior induction §3.

O7 Does ZF without Foundation prove Collection? It decides whether the named lgg of anchor data of spelled-out Replacement is truth-sound there (Proposition 5.7); unknown to us, it may be known. The lgg of non-anchor $\mathrm{Repl}_{\mathrm{sp}}$ data when guards are also learned for term metavariables (Appendix D.3). Partial or inconsistent α-renaming of binders (Proposition 5.3 covers full renaming) cases §§5–6.

O8 Spare-slot thresholds beyond one-variable schemas, e.g. for T_{Ind} with $m \geq 9$ slots (Remark E.5); usable bounds on the separation depth for natural targets, where no uniform computable bound exists (Theorem 6.15) untagged §10.

O9 A natural permanent residue, i.e. an unsound, never-refuted merge of natural axioms; in arithmetic it needs quantifiers in the template (Proposition 6.14) untagged §10.

O10 A merge rule that provably prevents fragmentation, which needs more than anchors (Propositions 6.11 and E.12); guarantees for bootstrapped coherence when only some clusters are pure (Proposition 6.17); a natural universal code that provably selects the target partition under a well-specified usage model (Section 6.10) untagged §10.

O11 Refutation separation for similar targets, characterized by the targets alone (Proposition F.10 and Lemma F.11); a sharing pass with guarantees for every target (Proposition F.5 covers only targets that label a phase-one cluster, Example F.6, and needs (H4$^+$) and (H5)) experiments §6.

O12 Is the refutation test NP-hard for true data and false negatives, as in DTRC (the proof of Theorem 6.16 uses a false datum)? And the union verifier for bounded binder depth and fixed $k \geq 3$? single §13

O13 A natural refuter that refutes every false minimal template of a cross pair within a budget polynomial in the data (Proposition 7.4); oracles that construct witnesses (Section 7.1) experiments §6.

Beyond the records, the limitations below suggest typed object languages (the motive of Lean's recursor is dependently typed), inference rules with premises and contexts, which enter here only through encodings such as Sub, and real corpora such as set.mm or a Lean library as data.

8.4 Limitations

We read Question 1's "sentences of the form $\varphi(x)$" as instances at terms; read literally, such a datum is already the axiom (Section 4). Object languages are single-sorted and first-order, data are in closure-normal form, and the theory treats parameters as constants while the implementation aligns them (Section 2.7). The practice is assumed correct outside the noise analysis (Proposition E.12), and rates assume i.i.d. data, while real usage is heterogeneous (Section 6.10). The implemented class DT°_{F} lacks term metavariables of positive arity, so it can accept more than DT° (E6), and the feature test is not implemented; the oracles are incomplete, the refuter is budgeted and history-dependent, and the data are synthetic and small (Section 7.4). Each research record was checked by an adversarial referee with independent code. A review of the paper draft found two more false propositions, now restricted (Propositions A.5 and F.5), and several overclaims (Appendix G.2). Results added after the referee round, including the material written during that review, were checked by proof and computation but not attacked by an independent referee (Appendix G.3). Bounded searches are evidence, not proof; textbook wordings of the ZF axioms are recalled, not checked against the books; and overlaps with the literature marked above as unchecked remain so.

8.5 Scope: checking, not proving

Everything here concerns checking: deciding whether a proposed sentence is licensed by the axioms and schemas a community has been seen to use. The learner never searches for proofs, and the prover of the protocol is arbitrary and may be adversarial (Section 2.4). The guarantees are soundness against every prover at all times, relative to the class and, without labels, to a residue, and exactness after an anchor, with rates; the costs are data and verification costs, not search costs. Nothing here makes a prover stronger, and an accepted sentence is one the community's practice licenses, which can differ from what is true. Possible uses, not tested here, are auditing which axioms a body of uncited proofs relies on (for instance Collection rather than Replacement, or Choice) and sending to human review the steps that no learned schema licenses, the escalations of the protocol.

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A Details for the setting

This appendix fixes the de Bruijn representation (Appendix A.1), proves the basic lemmas and the matching theorem (Appendices A.2 and A.3), states and proves that equivalence of determinate templates is renaming (Appendix A.4), explains why arguments must be metavariable-free (Appendix A.5), defines the first-order encodings and Plotkin's lgg (Appendix A.6), and treats parameter alignment, including its limits (Appendix A.7).

A.1 The de Bruijn representation

Trees. Terms, formulas, bodies and templates are finite ordered trees. A node is labelled by a function symbol or constant of the signature, a parameter, an index $\#k$ ($k \in \mathbb{N}$), a hole z_m (only in bodies), a relation symbol or $=$, a connective, a binder $\forall$ or $\exists$ (one child; it binds index 0 in its child), or a metavariable (only in templates; its children are the arguments). Every node has a sort, ι (terms) or o (formulas). Terms contain no binders. $|u|$ is the number of nodes of u; an occurrence $M(t_1, \ldots, t_n)$ counts $1 + \sum_i |t_i|$.

Positions. A position is a finite sequence of positive integers: ε is the root and $p \cdot i$ is the i-th child of p. $u|_p$ is the subtree at p; $p \leq q$ means that p is a prefix of q, and $p \perp q$ that neither is a prefix of the other. $b_u(p)$, or $b(p)$, is the number of binder nodes at proper prefixes of p.

Free indices. An occurrence of $\#k$ at position q of u is *bound in* u if $k < b_u(q)$; it is then bound by the $(k{+}1)$-th binder above q. Otherwise it is *free* and denotes the free index $k - b_u(q)$ of u, counted from the root of u. $\mathrm{FV}(u)$ is the set of free indices of u. A sentence s has $\mathrm{FV}(s) = \emptyset$, and for a position p of s, $\mathrm{FV}(s|_p) \subseteq \{0, \ldots, b_s(p) - 1\}$: the free index j of $s|_p$ refers to the $(j{+}1)$-th binder above p in s. In a template, the arguments of an occurrence at p are read relative to p: their indices refer to binders of T above p.

Shifting, substitution, abstraction, plugging.

- $\uparrow^e u$ adds e to every free index occurrence of u and leaves bound ones unchanged.
- For a term or formula u and a map $\bar{u}$ from a set $Y \supseteq \mathrm{FV}(u)$ of indices to terms, $u[\bar{u}]$ replaces each free occurrence of the index y, at a position q (where it is written $\#(y + b_u(q))$), by $\uparrow^{b_u(q)} \bar{u}_y$.

- For pairwise distinct indices $y_1, \ldots, y_n$ with $\mathrm{FV}(u) \subseteq \{y_1, \ldots, y_n\}$, the *abstraction* $u[z_m/y_m]$ replaces each free occurrence of y_m by the hole z_m. The result has no free indices, so $\lambda \bar{z}.u[z_m/y_m]$ is a body.
- For a body $\lambda \bar{z}.\beta$, plugging $\beta[u_1, \ldots, u_n]$ replaces each occurrence of z_m, at position q of β, by $\uparrow^{b_\beta(q)} u_m$. Since β has no free indices, $\mathrm{FV}(\beta[\bar{u}]) = \bigcup\{\mathrm{FV}(u_m) : z_m \text{ occurs in } \beta\}$.

Abstraction inverts plugging at distinct indices: $(u[z_m/y_m])[y_1, \ldots, y_n] = u$, since a hole at depth e is replaced by $\uparrow^e \#y_m = \#(y_m + e)$, which is the original occurrence. Index arithmetic is counted at unit cost.

Instances. For a template T and a substitution θ, $T\theta$ replaces the subtree at each $p \in \mathrm{Occ}(T)$, which is $M(\bar{t})$, by $\theta(M)[\bar{t}]$. The indices of $\bar{t}$ are shifted under the binders of $\theta(M)$, so they keep referring to the binders of T above p. For a template substitution σ, $T\sigma$ is defined in the same way; inside a template body γ, plugging also replaces the holes that occur in the arguments of occurrences $N(\bar{u})$ of γ, so these arguments stay metavariable-free.

A.2 Proofs of the basic lemmas

Proof of Lemma 2.4 (preservation). Positions strictly below an occurrence lie in its argument list, so a position $p \in \mathrm{skel}(T)$ is not below any occurrence, and no occurrence position is a prefix of p. $T\theta$ arises from T by replacing the subtrees at the occurrence positions only; hence the label at p is unchanged. If $p \in \mathrm{Occ}(T)$ carries $M(\bar{t})$, the replacement at p is $\theta(M)[\bar{t}]$, and the replacements at the other occurrences, which are incomparable with p ($\mathrm{Occ}(T)$ is an antichain), do not affect it. No further reduction takes place: bodies contain no metavariables, and arguments contain none either. For template substitutions the argument is the same. □

Lemma A.1 (Substitution facts)**.** *[proved]* single Lemma 1.2 *Let $\lambda \bar{z}.\beta$ be a body.* (i) *Composition:* $(\beta[\bar{t}])[\bar{u}] = \beta[\bar{t}[\bar{u}]]$. (ii) *Injectivity: if z_m occurs in β and $\beta[\bar{a}] = \beta[\bar{a}']$, then $a_m = a'_m$.* (iii) *If the root of β is not a hole, $\beta[\bar{a}]$ has the same root; formula bodies always have such roots.*

Proof. (i) Fix an occurrence of z_m at depth $e = b_\beta(q)$ in β. On the left, $\beta[\bar{t}]$ has $\uparrow^e t_m$ there, and $[\bar{u}]$ replaces a free index y of t_m, sitting at depth e' inside t_m and hence written $\#(y + e + e')$, by $\uparrow^{e+e'} \bar{u}_y$. On the right, $t_m[\bar{u}]$ has $\uparrow^{e'} \bar{u}_y$ at that place, and plugging shifts it by e more. The indices of β itself are all bound in β, so $[\bar{u}]$ does not touch them on the left. Both sides agree everywhere else. (ii) At an occurrence of z_m at depth e, $\beta[\bar{a}]$ has $\uparrow^e a_m$ and $\beta[\bar{a}']$ has $\uparrow^e a'_m$; shifting is injective. (iii) Plugging changes only hole positions, and the root is not a hole. Holes have sort ι, so the root of a formula body is never a hole. □

Proof of Lemma 2.8 (sound iff closed). (a) $\mathrm{VS}(D) \supseteq \{h \in \mathcal{H} : h \supseteq h^*\}$, hence $\mathrm{Acc}_\mathcal{H}(D) = \bigcap \mathrm{VS}(D)$ is contained in $\mathrm{cl}_\mathcal{H}(h^*)$; and a larger D gives a smaller version space. (b) An anchor makes $\mathrm{VS}(D) = \{h \in \mathcal{H} : h \supseteq h^*\}$. (c) If $\mathrm{Acc}_\mathcal{H}(D) = h^*$, every $h \in \mathrm{VS}(D)$ contains h^*, so D is an anchor; the converse is (b). A text eventually contains any finite subset of h^*; with monotonicity this gives the rest. □

A.3 Proof of the matching theorem

Proof of Theorem 2.5. (a) Uniqueness. Let $T\theta = s$, let $M(y_1, \ldots, y_n)$ be a pattern occurrence at p, and write $\theta(M) = \lambda \bar{z}.\beta$. By Lemma 2.4, $s|_p = \beta[y_1, \ldots, y_n]$. β has no free indices, so the free occurrences of indices in $s|_p$ are exactly the images of the holes of β, and the free index y_m

comes from z_m; since the y_m are pairwise distinct, $\beta = (s|_p)[z_m/y_m]$, and $\mathrm{FV}(s|_p) \subseteq \{y_1, \ldots, y_n\}$ is necessary. Every metavariable has a pattern occurrence, so θ is determined by s.

(b) Correctness. If a matcher exists, the skeleton walk succeeds (Lemma 2.4), step 2 computes it by (a), and the checks succeed. Conversely, suppose all steps succeed, and let θ be the computed substitution. The skeleton walk compares labels and arities at every position of $\mathrm{skel}(T)$, so s and $T\theta$ agree there, and every occurrence position of T is a position of s of the right sort. At the chosen pattern occurrence of M, $\theta(M)[\bar{y}] = ((s|_p)[z_m/y_m])[\bar{y}] = s|_p$ (Appendix A.1); at the other occurrences the check establishes $\theta(M)[\bar{t}] = s|_q$. Hence $T\theta = s$.

(b) Time. The skeleton walk costs $O(|\mathrm{skel}(T)|)$. Abstraction at the chosen pattern occurrence p_M is one pass over $s|_{p_M}$, which replaces each free y_m by z_m and fails on any other free index: $O(|s|_{p_M}|)$. The check at another occurrence $M(\bar{t})$ at q walks the body β of $\theta(M)$ and $s|_q$ in lock step, counting the binders of β passed. A non-hole node of β is compared with the current node of $s|_q$ and consumes it. A hole z_m at depth e is compared with $\uparrow^e t_m$ by walking t_m and the current subtree of $s|_q$ in lock step, shifting on the fly; on success it consumes $|t_m| \geq 1$ nodes of $s|_q$, and the walk stops at the first mismatch. So at most $|s|_q| + 1$ nodes of β are visited, and the check costs $O(|s|_q| + |\bar{t}|)$. Occurrence positions are pairwise incomparable, so the subtrees $s|_q$ at distinct occurrences are disjoint and $\sum_q |s|_q| \leq |s|$; and $\sum_q |\bar{t}_q| \leq |T|$. The total is $O(|T| + |s|)$.

(c) Generality by freezing. Let freeze replace each occurrence $N(\bar{u})$ of a metavariable of T' by $\hat{N}(\bar{u})$, where $\hat{N}$ is a fresh function symbol (if N is term-valued) or relation symbol (if N is formula-valued) of the same arity. Then $\mathrm{freeze}(T')$ is a sentence over the extended signature. If $T' = T\sigma$, let $\theta(M) := \lambda\bar{z}.\mathrm{freeze}(\gamma_M)$, where $\sigma(M) = \lambda\bar{z}.\gamma_M$. This is a ground body over the extended signature, and $T\theta = \mathrm{freeze}(T\sigma) = \mathrm{freeze}(T')$, because freezing commutes with plugging. Conversely, let $T\theta = \mathrm{freeze}(T')$, with $\theta(M) = \lambda\bar{z}.\beta_M$. In $\mathrm{freeze}(T')$ every frozen symbol is applied to arguments that contain no frozen symbol (the arguments of T' are metavariable-free). By (a), β_M is an abstraction of a subtree of $\mathrm{freeze}(T')$, so in β_M, too, every frozen symbol is applied to arguments without frozen symbols, built from holes, indices bound in β_M, and parameters. Replacing each $\hat{N}$ by N therefore turns β_M into a template body γ_M, and $\sigma(M) := \lambda\bar{z}.\gamma_M$ gives $T\sigma = T'$. The test is an instance of (b) over the extended signature, so it takes linear time.

(d) Rigid templates without determinacy (proved in the earlier analysis[prior induction C1(b)]). Let $T \in \mathrm{SO}^\circ$. Walk $\mathrm{skel}(T)$ against s as in (b). Distinct metavariables have independent bodies, so $s \in \mathrm{inst}(T)$ iff the walk succeeds and, for each metavariable M with occurrences $M(\bar{t}_1), \ldots, M(\bar{t}_k)$ at $p_1, \ldots, p_k$, there is a body β with $\beta[\bar{t}_j] = s|_{p_j}$ for all j. Decide this by recursion over a relative position r (a position of $s|_{p_1}$), with the column $c_j := (s|_{p_j})|_r$ and the number e of binders above r inside the body (the same for all j, because the body is common). A body subtree at r is either

- a hole z_m (*projection*), possible iff $c_j = \uparrow^e t_{j,m}$ for all j; or
- a node with label h and independent subtrees at the children (*imitation*), possible iff all c_j have the label h with the same arity, h is a signature symbol, a parameter, or an index $\#i$ with $i < e$ (bound inside the body), and the recursion succeeds at every child.

This is the projection/imitation analysis of second-order matching (Huet and Lang, 1978), restricted to metavariable-free arguments. The children's subproblems share no unknowns, so the recursion visits each position of $s|_{p_1}$ at most once, and each projection test is a comparison of polynomial size. Hence membership is polynomial. The number of bodies satisfies $\#(r) = \#\{m : \text{projection to } z_m \text{ works at } r\} + [\text{imitation works at } r] \cdot \prod_i \#(r \cdot i)$. For $T = P(0)$ and a sentence s with k occurrences of the constant 0, such as a conjunction of $k/2$ copies of $0 = 0$, both projection and imitation work at each 0, and only imitation works elsewhere, so there are 2^k matchers, with $k = \Theta(|s|)$. □

Evidence[single e1; single referee A]. On 1500 random DT° templates (one or two metavariables of six types, derived occurrences with random arguments) and random substitutions, a full projection/imitation count finds exactly one matcher and the algorithm of (b) returns the generating substitution in every case; on 1500 perturbed queries it agrees with the SO° test of (d); and 400/400 random pairs $T \succeq T\sigma$ are recognized by freezing. An independent matcher, written by the referee of the single-schema results and used in all of that referee's checks, produced no covering or recovery error.

Relation to unification and matching. Part (a) is the special case of Miller's pattern unification (Miller, 1991) with one side ground and each metavariable read off one occurrence. General second-order matching has finitely many matchers (Huet and Lang, 1978); its NP-hardness, usually attributed to Baxter's 1977 thesis (not checked by us), needs arguments that contain metavariables, which SO° excludes (see also Section 8.2).

A.4 Equivalence is renaming

Lemma A.2 (Equivalence is renaming)**.** *[proved]*[single Lemma R] *For $T, T' \in \mathrm{DT}^\circ$: $T \equiv T'$ iff T' arises from T by a bijective renaming of metavariables and a permutation of each metavariable's argument places.*

The key step is that a template substitution mapping T to itself fixes every pattern occurrence, hence is the identity.

Proof of Lemma A.2. "If" is clear. Conversely, let $T' = T\sigma$ and $T = T'\sigma'$. Then $T = T\tau$ with $\tau = \sigma;\sigma'$ (apply σ' to the bodies of σ).

Step 1: τ is the identity. Let $M(\bar{y})$ be a pattern occurrence of M in T, at p, and write $\tau(M) = \lambda\bar{z}.\gamma$. By Lemma 2.4, $\gamma[\bar{y}] = T|_p = M(\bar{y})$. The root of γ is not a hole (a hole plugs to an index), and not a rigid symbol (it would survive plugging). So $\gamma = N(\gamma_1, \ldots, \gamma_k)$, and plugging gives $N = M$, $k = n$ and $\gamma_j[\bar{y}] = y_j$. Each γ_j is a metavariable-free term at the root level of γ, so it contains no binder and no index (an index there would be free in the body). Since $\gamma_j[\bar{y}]$ is the index y_j, γ_j is a hole z_i with $y_i = y_j$, so $i = j$ because the y's are distinct. Hence $\tau(M) = \lambda\bar{z}.M(\bar{z})$ for every M.

Step 2: the shape of σ. Write $\sigma(M) = \lambda\bar{z}.\gamma'$. Applying σ' to γ' gives $M(\bar{z})$ by Step 1. The root of γ' is not a rigid symbol (it would survive σ') and not a hole (it would stay a hole). So it is an occurrence $N(\bar{u})$ of a metavariable N of T'; since $\bar{u}$ is metavariable-free, $\gamma' = N(\bar{u})$. Write $\sigma'(N) = \lambda\bar{w}.\delta$; then $\delta[\bar{u}] = M(\bar{z})$, and as in Step 1, $\delta = M(\delta_1, \ldots, \delta_n)$ with each δ_j a term over holes, function symbols and parameters, and $\delta_j[\bar{u}] = z_j$. A non-hole δ_j would give a non-hole root, so $\delta_j = w_{\pi(j)}$ and $u_{\pi(j)} = z_j$; π is injective because the z_j are distinct. Distinct M give distinct N, since $\sigma'(N)$ has root M. Every metavariable of $T' = T\sigma$ occurs in some $\sigma(M)$, and each $\sigma(M)$ is the single occurrence $N(\bar{u})$, so $M \mapsto N$ is a bijection.

Step 3: the arguments are permuted. N has a pattern occurrence in T'. Occurrences of T' arise only by plugging, so it is $N(\bar{u}[\bar{t}])$ for some occurrence $M(\bar{t})$ of T. Each $u_l[\bar{t}]$ is an index, and each u_l is a term over holes, function symbols and parameters, so u_l is a hole $z_{\rho(l)}$. The $u_l[\bar{t}]$ are pairwise distinct, so ρ is injective, and $u_{\pi(j)} = z_j$ gives $\rho \circ \pi = \mathrm{id}$. Hence the arity of N equals n and ρ is a permutation, and $\sigma(M) = \lambda\bar{z}.N(z_{\rho(1)}, \ldots, z_{\rho(n)})$. So T' is T with M renamed to N and its argument places permuted by ρ. □

A.5 Why arguments are metavariable-free

Remark A.3 (Nested arguments). [proved; computed][prior induction referee; single §4] The earlier analysis also considered the class DT in which arguments may contain metavariables, as in $M(F(x))$. For $D = \{\mathrm{Ind}(x = x), \mathrm{Ind}(0 = x)\}$ the templates

$$T_k = \big(0 = 0 \wedge \forall x\,(f(x) = x \to f^k(Sx) = Sx)\big) \to \forall x\, f(x) = x, \qquad k \geq 1,$$

are in DT ($f(x)$ in the conclusion is a pattern occurrence), cover D (via $f := \lambda z.z$ and $f := \lambda z.0$), are pairwise incomparable (the instance with $f := \lambda z.z + 0$ lies in T_k only), and are each minimal (a hand proof by the referee of the earlier analysis). So $\mathrm{Min}_{\mathrm{DT}}(D)$ is infinite. The earlier enumeration-based learner also failed to compute the cautious verifier of DT for targets with nested arguments, and was unsound for one such target, $\forall x(f(x) = f(x)) \wedge f(f(0)) = f(f(0))$. In DT°, $\mathrm{Min}(D)$ is finite for every D (Theorem 3.3); for this D it has exactly four elements (Example B.1).

A.6 First-order encodings, capture and guards

Classical anti-unification, the baseline of Sections 4 and 5, works on first-order trees, where a metavariable is a leaf that any subtree may replace. Under binders this differs from the λ convention, and bound variables need an encoding. In the *named* encoding (Nm), variables are names (constants of the tree signature) and binders are nodes $\mathrm{all}(x, \cdot)$; in the *de Bruijn* encoding (dB), bound variables are indices treated as constants, a metavariable at binder depth d may take a value with loose indices $< d$, and only well-formed results count. A *first-order schema* is a tree with metavariables at some positions. Unlike template instances, its instances in (Nm) and (dB) may *capture*: a value may mention a variable bound by the frame. For example, $\exists y \neg(y = z)$ has the capture instance $\exists y \neg(y = y)$, false in every structure, and the unguarded first-order form $\forall z \exists y \forall x(x \in y \leftrightarrow x \in z \wedge A)$ of Separation has Russell's instance $A := \neg(x \in y)$ (Proposition 5.8) [cases A4′, D1]. Side conditions are therefore expressed by *guards*, decidable predicates of the values from a fixed family, such as $\mathrm{fresh}(v, A)$ (v not free in the value of A), $\mathrm{loose}(A) \subseteq I$ (in (dB)), $\mathrm{closed}(z)$ and $\mathrm{nocap}(z)$. The *most-specific-guard rule* of the parent report keeps every guard of the family that holds on all data; one datum that violates a guard deletes it.

For first-order trees every finite nonempty D has a *least general generalization* $\mathrm{lgg}(D)$, unique up to renaming: the common prefix of the data, with one metavariable per distinct disagreement column (Plotkin, 1970; Reynolds, 1970).

Lemma A.4 (When the lgg recovers a schema)**.** *[known]*[parent lem:setting:recover] *Let σ be a first-order schema and $D = \{\sigma\theta_1, \ldots, \sigma\theta_N\}$. Then $\mathrm{lgg}(D) \preceq \sigma$, and $\mathrm{lgg}(D) \equiv \sigma$ iff* (R) *for each metavariable z of σ the roots of $\theta_1(z), \ldots, \theta_N(z)$ are not all equal, and* (D) *for distinct metavariables $z \neq z'$ some j has $\theta_j(z) \neq \theta_j(z')$.*

In DT°, (R) becomes root variation at every occurrence, (R*); (D) becomes the no-coincidence event (U); and a third event (N) (every argument place is used) appears (Theorem 3.8 and Corollary 3.9).

A.7 Parameter alignment

Renaming-closed instances. Let $\mathrm{par}(u)$ be the set of parameters occurring in u. A *renaming* is an injective map on parameter names, applied to every parameter occurrence. For a template T, let

$$\mathrm{inst}^{\sim}(T) = \{(\rho T)\theta : \ \rho \text{ injective on } \mathrm{par}(T),\ \theta \text{ a substitution}\}.$$

Renamings are required to be injective because a bijective renaming of parameters leaves the universal closure unchanged up to bound names, while identifying two parameters changes it. T *covers* D *up to renaming* if $D \subseteq \mathrm{inst}^{\sim}(T)$. Let $T \succeq^{\sim} T'$ iff $T' = (\rho T)\sigma$ for a renaming ρ injective on $\mathrm{par}(T)$ and a template substitution σ; equivalently, $\mathrm{freeze}(T') \in \mathrm{inst}^{\sim}(T)$. This is a preorder that contains $\succeq$, and $T \succeq^{\sim} T'$ implies $\mathrm{inst}^{\sim}(T) \supseteq \mathrm{inst}^{\sim}(T')$. (Compose: if $T' = (\rho T)\sigma$ and $T'' = (\rho' T')\sigma'$, then $T'' = ((\rho'\rho)T)(\rho'\sigma)\sigma'$. A parameter of ρT that does not occur in T' occurs only inside arguments whose holes σ discards, and the bodies of σ have all their parameters in T'. So changing ρ', on the parameters of ρT that do not occur in T', to fresh distinct names leaves T'' unchanged, and afterwards $\rho'\rho$ is injective on $\mathrm{par}(T)$. The inclusion of instance sets follows in the same way.) For parameter-free T, $\mathrm{inst}^{\sim}(T) = \mathrm{inst}(T)$.

Alignments. Let D be finite and nonempty, and let $r \in D$ be a datum with fewest parameters. An *alignment* $\alpha = (G, (\iota_d)_{d \in D \setminus \{r\}})$ consists of a set $G \subseteq \mathrm{par}(r)$ and injections $\iota_d : G \to \mathrm{par}(d)$. The aligned data $\alpha(D)$ rename, in r, each $q \in G$ to a global name g_q; in each $d \neq r$, each $\iota_d(q)$ to g_q; and every other parameter of every datum to a name of its own, used nowhere else. Write α_d for the bijective renaming applied to d. $\mathrm{Align}(D)$ is finite. For a class $\mathcal{C}$, let

$$\mathrm{Min}^{\mathrm{al}}_{\mathcal{C}}(D) = \text{the } \succeq^{\sim}\text{-minimal elements (one per } \equiv^{\sim}\text{-class) of } \bigcup_{\alpha \in \mathrm{Align}(D)} \mathrm{Min}_{\mathcal{C}}(\alpha(D)),$$

where $\mathrm{Min}_{\mathcal{C}}$ is the literal minimal set of Definition 2.2.

If some datum has no parameter, there is exactly one alignment ($G = \emptyset$). It renames all parameters apart, so no member has a parameter in its skeleton: by Lemma 2.4 such a parameter would occur in the aligned r, which has none. The aligned set $\mathrm{Min}_{\mathcal{C}}(\alpha(D))$ can then differ from the literal $\mathrm{Min}_{\mathcal{C}}(D)$ of the canonicalized data. For $D = \{(\forall x\, 0{=}0) \wedge 0{=}0,\ (\forall x\, x{=}x) \wedge w_0{=}w_0,\ (\forall x\, x{=}0) \wedge w_0{=}0\}$ in DT°_F, the literal set is $\{(\forall x\, P(x)) \wedge f_0{=}f_1,\ (\forall x\, P(x)) \wedge P(w_0)\}$ and the aligned one is $\{(\forall x\, P(x)) \wedge f_0{=}f_1\}$ (computed). The implementation uses the literal set in this case. This does no harm to Proposition A.5(b) below: a template as there is then parameter-free (its skeleton parameters would occur in the parameter-free datum), so it covers D literally, and (F) puts a member of the literal set below it.

Literal finitariness. The proof below uses one property of $\mathcal{C}$:

(F) for every finite nonempty X, $\mathrm{Min}_{\mathcal{C}}(X)$ is finite (up to $\equiv$), and every $T \in \mathcal{C}$ that covers X literally satisfies $T \succeq T_0$ for some $T_0 \in \mathrm{Min}_{\mathcal{C}}(X)$.

For $\mathcal{C} = \mathrm{DT}^{\circ}$, with parameters treated as constants, (F) is Theorem 3.3(c). For $\mathcal{C} = \mathrm{DT}^{\circ}_F$ it is Proposition F.2(c). Both classes are closed under renaming of parameters.

Proposition A.5 (Aligned minimal templates)**.** *[proved; computed]* experiments E1-al, restricted; DT° new *Let $\mathcal{C} \in \{\mathrm{DT}^{\circ}, \mathrm{DT}^{\circ}_F\}$ and let D be finite and nonempty.* (a) *Every member of* $\mathrm{Min}^{\mathrm{al}}_{\mathcal{C}}(D)$ *lies in $\mathcal{C}$ and covers D up to renaming.* (b) *Every $T \in \mathcal{C}$ that covers D up to renaming and all of whose parameters occur in* $\mathrm{skel}(T)$ *satisfies $T \succeq^{\sim} M$ for some member M.* (c) $\mathrm{Min}^{\mathrm{al}}_{\mathcal{C}}(D)$ *is finite; no template as in (b) lies strictly below a member; and if $D \subseteq \mathrm{inst}^{\sim}(T^*)$ with $T^* \in \mathcal{C}$ and every parameter of T^* in* $\mathrm{skel}(T^*)$ *(in particular if T^* is parameter-free), some member M satisfies $T^* \succeq^{\sim} M$.*

The hypothesis on the parameters in (b) and (c) cannot be dropped (Remark A.7). For DT°_F the proposition restricts the experiments' alignment result, which omits this hypothesis and whose proof has the same gap (computed check: E0b, Example A.6); the same proof gives it for DT°, with (F) supplied by Theorem 3.3 instead of Proposition F.2.

Proof. Let $\mathcal{C} \in \{\mathrm{DT}^\circ, \mathrm{DT}^\circ_F\}$; it satisfies (F).

(a) Members cover D up to renaming. Let $M \in \mathrm{Min}_{\mathcal{C}}(\alpha(D))$. It covers $\alpha(D)$ literally: $\alpha_d(d) = M\theta_d$ for every d. Each body $\theta_d(N)$ is plugged in at an occurrence of N, so its parameters occur in $\alpha_d(d)$; and α_d is a bijection from $\mathrm{par}(d)$ onto $\mathrm{par}(\alpha_d(d))$. Extend α_d^{-1} injectively to $\mathrm{par}(M)$ by sending the parameters of M that do not occur in $\alpha_d(d)$ to fresh distinct names; by Lemma 2.4 they occur in M only inside arguments that θ_d discards. Then $d = (\alpha_d^{-1}M)(\alpha_d^{-1}\theta_d)$, where $\alpha_d^{-1}\theta_d$ renames the parameters in the bodies. Hence $d \in \mathrm{inst}^\sim(M)$.

(b) Every template as in (b) lies above a member. Let $T \in \mathcal{C}$ with $d = (\rho_d T)\theta_d$ for every $d \in D$, each ρ_d injective on $\mathrm{par}(T)$. Every parameter of T lies in $\mathrm{skel}(T)$, so by Lemma 2.4 $\rho_d(\mathrm{par}(T)) \subseteq \mathrm{par}(d)$. Put $G := \rho_r(\mathrm{par}(T)) \subseteq \mathrm{par}(r)$ and $\iota_d := \rho_d \circ \rho_r^{-1} : G \to \mathrm{par}(d)$, which is injective; so $\alpha := (G, (\iota_d)) \in \mathrm{Align}(D)$. Let T' be T with each parameter p renamed to $g_{\rho_r(p)}$; then $T \succeq^\sim T' \succeq^\sim T$. For every d and $p \in \mathrm{par}(T)$, $\alpha_d(\rho_d(p)) = g_{\rho_r(p)}$: for $d = r$ by the definition of α, and for $d \neq r$ because $\rho_d(p) = \iota_d(\rho_r(p))$. Hence $\alpha_d(d) = (\alpha_d\rho_d T)(\alpha_d\theta_d) = T'(\alpha_d\theta_d)$, so T' covers $\alpha(D)$ literally. By (F), $T' \succeq M'$ for some $M' \in \mathrm{Min}_{\mathcal{C}}(\alpha(D))$, and $M' \succeq^\sim M$ for some $\succeq^\sim$-minimal element M of the finite union. So $T \succeq^\sim M \in \mathrm{Min}^{\mathrm{al}}_{\mathcal{C}}(D)$.

(c) Finiteness, minimality, and a member below the target. $\mathrm{Align}(D)$ is finite and each $\mathrm{Min}_{\mathcal{C}}(\alpha(D))$ is finite, so $\mathrm{Min}^{\mathrm{al}}_{\mathcal{C}}(D)$ is finite. If some T as in (b) satisfied $M \succ^\sim T$ for a member M, then by (b) $T \succeq^\sim M''$ for a member M'', so $M \succeq^\sim M''$, strictly (if $M'' \succeq^\sim M$ then $T \succeq^\sim M$), with both in the union, contradicting the minimality of M there. The last clause is (b) for T^*. □

Example A.6 (Literal minimal templates are not enough)**.** [proved; computed][experiments E1, E0b] Let $D = \{0 = 0 \wedge w_0 = w_0,\ w_0 = 0 \wedge w_1 = w_1\}$ (each datum canonicalized on its own). Literally, $\mathrm{Min}_{\mathrm{DT}^\circ_F}(D) = \{f_0 = 0 \wedge f_1 = f_1\}$, with 0-ary term metavariables. But $T = (f = 0 \wedge w_0 = w_0)$ covers D up to renaming (in the second datum, $f := w_0$ and the template's w_0 is renamed to w_1), and $f_0 = 0 \wedge f_1 = f_1 \succ T$ (map f_1 to w_0). So no literal minimal template lies below T, although the parameter of T lies in its skeleton. The alignment that identifies w_0 of the first datum with w_1 of the second produces T (up to renaming) as a member of $\mathrm{Min}^{\mathrm{al}}_{\mathrm{DT}^\circ_F}(D)$. This example refutes the claim, made for the literal set in an earlier version of the experiments and withdrawn (Appendix G), that a minimal template always lies below the target.

Computed (experiments, E0b): for random generating templates T^* with a rigid parameter (2–3 data each), no literal minimal template lies below T^* in 17/200 (arithmetic) and 36/200 (set theory) data sets; with $\mathrm{Min}^{\mathrm{al}}$ there are no failures, with or without a rigid parameter. The implementation caps the number of alignments at 4096 (beyond it, it falls back to the literal set) and reports truncation; none occurred. On the data of experiment E2 the two sets agree on all 720 data sets (an independent re-implementation: 360 of 360).

Remark A.7 (Parameters inside derived arguments). [proved; computed][paper review, align_cex] The hypothesis of Proposition A.5(b),(c) on the parameters is needed. A parameter that occurs in T only inside an argument of a non-pattern occurrence vanishes from every datum whose value does not use that argument, so it need not occur in the reference datum r through which alignments are built. In DT°_F, let

$$D = \{\, 0{=}0 \wedge ((\forall x\, 0{=}0) \wedge 0{=}0),\quad 0{=}0 \wedge ((\forall x\, x{=}x) \wedge w_0{=}w_0),\quad w_0{=}w_0 \wedge ((\forall x\, x{=}0) \wedge w_1{=}0) \,\}.$$

The template $T = Q \wedge ((\forall x\, P(x)) \wedge P(c))$ covers D up to renaming ($P := \lambda z.\, 0{=}0$, $\lambda z.\, z{=}z$, $\lambda z.\, z{=}0$, with c renamed to w_0, resp. w_1). The first datum is parameter-free, so there is one alignment, and $\mathrm{Min}^{\mathrm{al}}_{\mathrm{DT}^\circ_F}(D) = \{f_0{=}f_0 \wedge ((\forall x\, P_0(x)) \wedge f_1{=}f_2)\}$; T lies above no member. Replace each datum $A \wedge (B \wedge C)$ by $(A \vee \neg A) \wedge (B \to C)$: these data are covered by $T^* = (Q \vee$

$\neg Q) \wedge ((\forall x\, P(x)) \to P(c))$, all of whose instances are logically valid, and the only member is $(f_0{=}f_0 \vee \neg f_0{=}f_0) \wedge ((\forall x\, P_0(x)) \to f_1{=}f_2)$, which has false instances. So a learner that intersects the members' instance sets, aligned or literal, is unsound for this target. In DT$^\circ$ the minimality clause fails too: for $D' = \{(\forall x\, 0{=}0){\wedge}0{=}0,\ (\forall x\, x{=}x){\wedge}w_0{=}w_0,\ (\forall x\, x{=}0){\wedge}w_0{=}0\}$ the only member is $\forall x\, (f_0(x){=}f_1(x)) \wedge f_2{=}f_1(u)$. It lies strictly above $T' = \forall x\, (g(x){=}h(x)) \wedge g(c){=}h(c)$ ($u \mapsto c$, $f_2 := g(c)$), and T' covers D' but lies above no member. We checked these facts by hand and with the project's code. Alignments that identify parameters across all data, not only through r, would remove the restriction: in the proof of (b), a parameter of T that vanishes from a datum is then simply left unidentified there. Neither the theory above nor the implementation does this. The experiments are unaffected, since their targets are parameter-free.

B Proofs and further results for Section 3

This appendix gives the full proofs of Section 3, in the notation of Section 3.1, and the secondary results to which the main text only points. Bound variables are de Bruijn indices. A datum d in inst(T) has the unique matcher θ_d, and $\beta^d_M = \theta_d(M)$. All proofs were transcribed from the research record and checked again; we found no error, and some steps are spelled out in more detail than there.

B.1 Basic lemmas

Parts (a), (b) and (e) of Lemma 3.2 are Lemmas 2.4, A.1 and A.2, proved in Appendices A.2 and A.4.

Proof of Lemma 3.2(c), (d). (c) The skeleton is prefix-closed: the parent of a skeleton position is a skeleton position, because the children of an occurrence are its arguments. By Lemma 2.4, every datum carries the symbol $T|_p$ at each $p \in \mathrm{skel}(T)$; by induction on the depth of p, $p \in C(D)$. The parent of an occurrence is in $\mathrm{skel}(T) \subseteq C(D)$, so the occurrence lies in $C(D)$ or is a slot.

(d) Match $d|_\sigma$ against $d|_r$, treating the free indices of $d|_\sigma$ as first-order variables. Walk both in lock step. At a free-index occurrence y of $d|_\sigma$ under e local binders, read off the subterm of $d|_r$ at the same place and lower it by e; fail if it mentions one of those e local binders, or if two occurrences of y disagree; fail on any other symbol clash. The walk either fails or returns a partial map e_d defined exactly on $\mathrm{FV}(d|_\sigma)$, and $d|_r = (d|_\sigma)[\bar u]$ holds iff $\bar u$ extends e_d. So a common $\bar u$ on Y exists iff no e_d fails and the e_d agree on common arguments. This is a pairwise condition, and $\bar u$ is then determined on $\bigcup_d \mathrm{FV}(d|_\sigma) = Y$. The walks take time $O(\sum_d |d|)$. The argument does not use finiteness of D. □

B.2 Minimal covering templates

Proof of Theorem 3.3. (a) We use two reduction steps. Each turns a covering $T \in \mathrm{DT}^\circ$ into a covering $T' = T\sigma \in \mathrm{DT}^\circ$, so $T \succeq T'$.

Step V (vacuous place). Suppose z_m occurs in no β^d_M. Put

$$\sigma(M) := \lambda \bar z.M'(z_1, \ldots, z_{m-1}, z_{m+1}, \ldots, z_n) \qquad (M' \text{ fresh}).$$

A pattern occurrence becomes an occurrence with a sub-list of distinct indices, again a pattern occurrence. D is covered with $\theta'_d(M') := \beta^d_M$ (holes renumbered).

Step H (common root). Suppose all β^d_M ($d \in D$) have the same root.

- If the root is a hole z_m (M is term-valued), put $\sigma(M) := \lambda\bar{z}.z_m$: each occurrence $M(\bar{t})$ becomes t_m. D is still covered, since $d|_q = z_m[\bar{t}] = t_m$ at each occurrence q.
- If the root is a symbol f with children $1, \dots, k$, child j under $e_j \in \{0, 1\}$ binders of f, put $\sigma(M) := \lambda\bar{z}.f(M_1(\bar{z}, 0^{e_1}), \dots, M_k(\bar{z}, 0^{e_k}))$, with fresh M_j of arity $n + e_j$; here 0^e is the index 0 (the variable bound by f) if $e = 1$ and nothing if $e = 0$, and the holes are shifted under f by plugging. A pattern occurrence $M(\bar{y})$ becomes $f(\dots, M_j(\bar{y}\uparrow, 0), \dots)$, whose arguments are distinct indices: a pattern occurrence. Other occurrences get metavariable-free arguments. D is covered with $\theta'_d(M_j) :=$ the j-th child of β^d_M, with the variable bound by f replaced by the new hole. (If f is a constant or a parameter, $k = 0$ and M disappears.)

Let $\mu(T) = (|C(D)| - |\mathrm{skel}(T)|,\ \sum_M \mathrm{arity}(M)) \in \mathbb{N}^2$, ordered lexicographically; the first component is ≥ 0 by Lemma 3.2(c). Step H strictly increases $|\mathrm{skel}|$: the root symbol, or the index y_m, becomes rigid at least at the pattern position. Step V keeps the skeleton and lowers $\sum_M \mathrm{arity}(M)$. So the process stops, and it stops exactly when (S1) and (S2) hold, at a saturated template. Generality is transitive.

(b) $\mathrm{skel} \subseteq C(D)$ is Lemma 3.2(c). Let $M(\bar{y})$ be a pattern occurrence at σ. By Lemma 2.4, $d|_\sigma = \beta^d_M[\bar{y}]$; its root is the root of β^d_M if that is a symbol, and y_m if $\beta^d_M = z_m$ (a body has no free index, so an index root comes from a hole). Since the y_m are distinct, the roots of the $d|_\sigma$ are all equal iff the roots of the β^d_M are. By (S2) they are not, so $\sigma \notin C(D)$; its parent lies in $\mathrm{skel} \subseteq C(D)$, so σ is a slot. By (S1) each y_m is free in some $d|_\sigma$, so $\{\bar{y}\} \subseteq Y_\sigma(D)$; coverage forces $\mathrm{FV}(d|_\sigma) \subseteq \{\bar{y}\}$, so $Y_\sigma(D) = \{\bar{y}\}$. Every occurrence lies in $A(D)$ by Lemma 3.2(c). For another occurrence $M(\bar{t})$ at q, the value $\beta^d_M = (d|_\sigma)[\bar{z}/\bar{y}]$ is determined by D and the pattern slot σ; by (S1) each hole z_m occurs in some β^d_M, and by Lemma A.1(ii), t_m is determined by $d|_q = \beta^d_M[\bar{t}]$.

So, up to renaming metavariables and permuting argument places (Lemma A.2), a saturated covering template is determined by its set $Q \subseteq A(D)$ of occurrence positions and by the map that sends each $q \in Q$ to the pattern slot of the metavariable occurring there (one pattern occurrence fixed per metavariable). The skeleton is then determined too: it consists of the positions of $C(D)$ that are neither in Q nor below an element of Q. The sort of each metavariable is the sort of its positions, and its arity is $|Y_\sigma(D)|$. This gives the bound $2^{|A(D)|} \cdot |\mathrm{slots}(D)|^{|A(D)|}$.

(c) Let T be minimal. By (a), $T \succeq T'$ with T' saturated; minimality gives $T' \succeq T$, and by Lemma A.2 T is a renaming of T', hence saturated. Now let T cover D arbitrarily, and let S_T be the set of saturated covering templates T' with $T \succeq T'$. It is finite up to renaming by (b) and nonempty by (a), and $\succeq$ is a partial order on renaming classes (Lemma A.2). Take T_m $\succeq$-minimal in S_T. If U covers D and $T_m \succeq U$, then $U \succeq U'$ for some saturated U', by (a), and $T \succeq T_m \succeq U \succeq U'$; so $U' \in S_T$ and $T_m \succeq U'$, and minimality in S_T gives $U' \equiv T_m$, hence $U \equiv T_m$. So T_m is minimal among all covering templates, and $T \succeq T_m$.

(d) is (c) with $T = T^*$. □

Example B.1 (Induction data, no anchor)**.** [computed]single §4 For $D = \{\mathrm{Ind}(x = x), \mathrm{Ind}(0 = x)\}$ (both motives have root $=$, so D is no anchor), $\mathrm{Min}(D)$ consists of exactly the four templates

$$\big(\zeta_1 = \zeta_2 \wedge \forall x(f(x) = x \to f(Sx) = Sx)\big) \to \forall x\, f(x) = x, \qquad \zeta_1, \zeta_2 \in \{0, f(0)\},$$

of which only $\zeta_1 = f(0)$, $\zeta_2 = 0$ lies below T_{Ind}. The earlier analysis reported eightprior induction C8.2, from a search bounded by size 23; the template with $\zeta_1 = \zeta_2 = f(0)$ has size 24, and five of the eight lie strictly above it. Brute force at size ≤ 24 and the referee's independent enumerator confirm the four.

In detail. The D-features are: the Sym features of the common frame (including the whole antecedent $0 = 0$ and the right-hand sides x, Sx, x of the three atoms in the motive copies under $\forall x$); Scope with scope $\{x\}$ at the three slots; and the Eq features $\sigma_b \leftrightarrow \sigma_d$ (identity), $\sigma_b \to \sigma_c$ and $\sigma_d \to \sigma_c$ ($x \mapsto Sx$), and $\sigma_b, \sigma_d \to$ each of the two zeros of the antecedent ($x \mapsto 0$). (No feature leaves σ_c: its column $(Sx, 0)$ cannot be mapped onto $(x, 0)$, nor onto a rigid 0 or Sx.) A minimal template therefore merges σ_b and σ_d into $f(x)$, puts $f(Sx)$ at σ_c, and chooses at each zero of the antecedent between the rigid 0 and $f(0)$: four incomparable templates, separated by the instance $f := \lambda z.Sz$. Computed: the saturated templates number 35, of which 4 are minimal; brute force at size ≤ 24 finds the same 4, and every one of 2749 enumerated covering templates lies above one of them; at size ≤ 23 there are 8 apparent minima, as reported earlier.[single e2; referee t1, t1b]

B.3 The feature verifier

Proof of Theorem 3.4. The templates. $T_0(D)$ and every T_φ are in DT°: every M_σ has the pattern occurrence $M_\sigma(Y_\sigma)$, and in T_φ the metavariable M_{σ_0} keeps its pattern occurrence at σ_0, which is not below r and differs from r (as $\sigma_0 \perp r$). Both cover D with $\theta_d(M_\sigma) := (d|_\sigma)[\bar{z}/Y_\sigma]$, which is closed except for holes because $\mathrm{FV}(d|_\sigma) \subseteq Y_\sigma(D)$; at r, $\theta_d(M_{\sigma_0})[\bar{u}] = (d|_{\sigma_0})[\bar{u}] = d|_r$ because φ is a D-feature.

(1) $\mathrm{Acc}(D) \subseteq \mathrm{Feat}(D) = \mathrm{inst}(T_0(D)) \cap \bigcap_\varphi \mathrm{inst}(T_\varphi)$. A sentence lies in $\mathrm{inst}(T_0(D))$ iff it has every Sym and every Scope feature: such a q has the skeleton $C(D)$, and $q = T_0(D)\theta$ with $\theta(M_\sigma) := (q|_\sigma)[\bar{z}/Y_\sigma]$; conversely an instance has the skeleton, and $\mathrm{FV}(\theta(M_\sigma)[Y_\sigma]) \subseteq Y_\sigma$. If q has every D-feature, then it has the skeleton of T_φ, Scope at every slot, and $q|_r = (q|_{\sigma_0})[\bar{u}]$, so $q = T_\varphi\theta$ with the same θ. Conversely every instance $q = T_\varphi\theta$ has φ: $q|_{\sigma_0} = \theta(M_{\sigma_0})[Y_{\sigma_0}]$ and $q|_r = \theta(M_{\sigma_0})[\bar{u}] = (q|_{\sigma_0})[\bar{u}]$ by Lemma A.1(i). So $\mathrm{Feat}(D)$ equals the displayed intersection, and $\mathrm{Acc}(D) \subseteq \mathrm{Feat}(D)$ because $T_0(D)$ and the T_φ cover D.

(2) $\mathrm{Feat}(D) \subseteq \mathrm{Acc}(D)$. Let T cover D. By Theorem 3.3(a), $T \succeq T'$ with T' saturated and covering, so $\mathrm{inst}(T) \supseteq \mathrm{inst}(T')$. Fix for every metavariable M of T' a pattern occurrence $M(\bar{y})$, at the slot σ_M (Theorem 3.3(b)). A sentence q lies in $\mathrm{inst}(T')$ if it has (i) $\mathsf{Sym}(p)$ for every $p \in \mathrm{skel}(T')$, (ii) $\mathsf{Scope}(\sigma_M)$ for every M, and (iii) $\mathsf{Eq}(\sigma_M, r, \bar{u}_r)$, with $\bar{u}_r := (y_m \mapsto t_{r,m})_m$, for every other occurrence $M(\bar{t}_r)$ at r. Indeed, by (ii) $\theta(M) := (q|_{\sigma_M})[\bar{z}/\bar{y}]$ is a body, and (i), (iii) give $T'\theta = q$, since $\theta(M)[\bar{t}_r] = (q|_{\sigma_M})[\bar{u}_r] = q|_r$. Each of these is a D-feature: (i) by Lemma 3.2(c); (ii) because σ_M is a slot with $Y_{\sigma_M}(D) = \{\bar{y}\}$; (iii) because $r \in A(D)$ (Lemma 3.2(c)), $r \perp \sigma_M$ (distinct occurrences are incomparable), the sorts agree, $\bar{u}_r$ maps $Y_{\sigma_M}(D)$ to metavariable-free terms in scope at r, and T' covers D: $d|_r = \beta^d_M[\bar{t}_r] = (\beta^d_M[\bar{y}])[\bar{u}_r] = (d|_{\sigma_M})[\bar{u}_r]$. So $\mathrm{Feat}(D) \subseteq \mathrm{inst}(T') \subseteq \mathrm{inst}(T)$ for every covering T, i.e. $\mathrm{Feat}(D) \subseteq \mathrm{Acc}(D)$. □

Relation to anti-unification. $T_0(D)$ is the higher-order pattern anti-unifier without merging; the pattern lgg's merging of columns equal up to renaming (Pfenning, 1991; Baumgartner et al., 2017) is the case of an Eq feature whose $\bar{u}$ is a permutation. DT° adds *derived* occurrences, and the price is that the version space can have exponentially many minimal elements (Proposition 3.13); their intersection is still described by polynomially many equations (Corollary 3.6). Second-order generalization beyond patterns is also studied by Hirata et al. (2004) and in the frameworks surveyed by Cerna and Kutsia (2023); we have not checked whether DT° or Theorem 3.4 appears there.

Proof of Corollary 3.6. Positions in $A(D)$ are positions of every datum (a slot's parent carries the same symbol in all data, hence has the same arity), so $m = |A(D)| \leq \min_d |d|$. $C(D)$, the slots and the scopes are computed in one simultaneous walk, in time $O(n)$. For each of the at

most m^2 ordered pairs (σ, r) with σ a slot, $r \in A(D)$, $\sigma \perp r$, of the same sort, Lemma 3.2(d) decides in time $O(n)$ whether an Eq-feature exists and returns its unique $\bar{u}$. So there are at most m^2 Eq-features, plus at most m Sym and m Scope features. Each feature is checked on q in time $O(|q|)$: Sym and Scope directly; $\mathsf{Eq}(\sigma, r, \bar{u})$ by walking $q|_\sigma$ and $q|_r$ in lock step, comparing each free-index occurrence y of $q|_\sigma$ with the corresponding subterm of $q|_r$ against u_y (shifted), which consumes that subterm. Hence $O(m^2|q|)$ per query. □

B.4 Anchors

Proof of Theorem 3.8. (⇐; no assumption.) By Lemma 3.2(c), $\mathrm{skel}(T^*) \subseteq C(D)$, and by (R*) no occurrence lies in $C(D)$; since $C(D)$ is prefix-closed and the data positions are the skeleton, the occurrences and the positions below occurrences, $C(D) = \mathrm{skel}(T^*)$, $\mathrm{slots}(D) = \mathrm{Occ}(T^*)$ and $A(D) = \mathrm{skel}(T^*) \cup \mathrm{Occ}(T^*)$. For σ carrying $M(\bar{t}_\sigma)$,

$$Y_\sigma(D) = \bigcup_i \mathrm{FV}(\beta^i_M[\bar{t}_\sigma]) = \bigcup_i \bigcup_{m:\ z_m \in \beta^i_M} \mathrm{FV}(t_{\sigma,m}) = Y^*_\sigma$$

by (N). Let $q = T^*\theta$. Then q has every Sym feature (Lemma 2.4) and every Scope feature ($\mathrm{FV}(q|_\sigma) \subseteq Y^*_\sigma$). Let $\mathsf{Eq}(\sigma, r, \bar{u})$ be a D-feature; (σ, r) is one of the pairs of Definition 3.7. By (U) it is valid: $\bar{t}_r = \bar{t}_\sigma[\bar{u}_0]$. Then $s_i|_r = \beta^i_M[\bar{t}_\sigma[\bar{u}_0]] = (s_i|_\sigma)[\bar{u}_0]$ for all i (Lemma A.1(i)), and by forcing (Lemma 3.2(d)) $\bar{u} = \bar{u}_0$ on $Y_\sigma(D) = Y^*_\sigma$. So $q|_r = \theta(M)[\bar{t}_\sigma[\bar{u}]] = (\theta(M)[\bar{t}_\sigma])[\bar{u}] = (q|_\sigma)[\bar{u}]$. Hence $q \in \mathrm{Feat}(D) = \mathrm{Acc}(D)$ (Theorem 3.4): $\mathrm{inst}(T^*) \subseteq \mathrm{Acc}(D)$, i.e. D is an anchor.

(⇒; under (Rich).) If D is an anchor, every instance of T^* has every D-feature (Theorem 3.4). We exhibit an instance violating a D-feature whenever an event fails; metavariables not mentioned get arbitrary values.

- (R*) fails at $\sigma \in \mathrm{Occ}(T^*)$, carrying M, with common root h. The ancestors of σ lie in $\mathrm{skel}(T^*) \subseteq C(D)$, so $\sigma \in C(D)$ and $\mathsf{Sym}(\sigma)$ is a D-feature. By (Rich) choose $\theta(M)$ with a symbol root $\neq h$ (a closed term, resp. a formula body); by Lemma A.1(iii), $(T^*\theta)|_\sigma$ has that root.
- (R*) holds and (N) fails at (M, m). Take a pattern occurrence $M(\bar{y})$ at σ; σ is a slot. Since the y's are distinct and the β^i_M are closed except for holes, y_m is free in $s_i|_\sigma = \beta^i_M[\bar{y}]$ only if z_m occurs in β^i_M; so $y_m \notin Y_\sigma(D)$, and $\mathsf{Scope}(\sigma)$ is a D-feature. Choose $\theta(M)$ containing z_m: z_m itself, or $R(z_m, \ldots, z_m)$ for a relation symbol R, which exists by (Rich). Then $y_m \in \mathrm{FV}((T^*\theta)|_\sigma)$.
- (R*) and (N) hold and (U) fails at a non-valid pair (σ, r) with witness $\bar{u}$. As in the first part, σ is a slot, $r \in A(D)$ and $Y_\sigma(D) = Y^*_\sigma$, so $\mathsf{Eq}(\sigma, r, \bar{u})$ is a D-feature. If $r \in \mathrm{skel}(T^*)$ has rigid root g, choose $\theta(M)$ with a symbol root $\neq g$; then $((T^*\theta)|_\sigma)[\bar{u}]$ has that root (substituting for free indices keeps a symbol root), while $(T^*\theta)|_r$ has root g. If r carries an occurrence of $M' \neq M$ (of the same sort), choose $\theta(M)$ and $\theta(M')$ with different symbol roots. If r carries $M(\bar{t}_r)$, then $\bar{t}_r \neq \bar{t}_\sigma[\bar{u}]$ because the pair is not valid; pick m with $t_{r,m} \neq t_{\sigma,m}[\bar{u}]$ and put $\theta(M) := \lambda\bar{z}.z_m$ (term-valued) or $\lambda\bar{z}.R(z_m, \ldots, z_m)$ (formula-valued). The two sides then differ in that argument. □

Proof of Corollary 3.9. (a) Every occurrence of a 0-ary metavariable is a pattern occurrence, and $s_i|_\sigma = \beta^i_M$; so (R*) is (R). (N) is vacuous. $Y^*_\sigma = \emptyset$, so (U) at (σ, r) asks whether $s_i|_r = \beta^i_M$ for all i. If $r \in \mathrm{skel}(T^*)$, its root is fixed, and (R) excludes the equation. If r carries $M' \neq M$ (necessarily of the same sort), the equation is the failure of (D) for M, M'. If r carries M, the pair is valid. So (R*)∧(N)∧(U) ⇔ (R)∧(D), and Theorem 3.8 applies.

(b) Formula bodies have symbol roots, and they survive plugging (Lemma A.1(iii)); so the root of $s_i|_\sigma$ is the root of β^i_M at every occurrence: (R*)⇔(R). Assume (R) and (N). For $r \in \mathrm{skel}(T^*)$ (of formula sort) a coincidence would make all roots of the β^i_M equal to the rigid root at r, contradicting (R). For r an occurrence of the same M, $\beta^i_M[\bar{t}_r] = \beta^i_M[\bar{t}_\sigma[\bar{u}]]$ for all i forces $\bar{t}_r = \bar{t}_\sigma[\bar{u}]$ by (N) and Lemma A.1(ii); so either the pair is valid or there is no coincidence. All occurrences have formula sort. What remains of (U) is (D^+).

(c) T_{Ind} and every schema of Table 6 have a single formula metavariable P. In the ZF schemas every occurrence of P is a pattern occurrence (for example $P(x, y, A)$ and $P(x, u, A)$ in $\mathrm{Repl}_{\mathrm{sp}}$ apply P to distinct bound variables), while T_{Ind} has the non-pattern occurrences $P(0)$ and $P(Sx)$; all are in DT°. Part (b) gives (R)∧(N), where (N) splits into one condition per argument place of P; for T_{Ind} it says that some motive has x free. A ground template T^* has $C(\{T^*\})$ = all positions and no slot, so $\mathrm{Feat}(\{T^*\}) = \{T^*\}$ by Theorem 3.4. One instance of a schema never satisfies (R); two instances whose bodies have different main symbols and together use every argument place do (for $\mathrm{Repl}_{\mathrm{sp}}$, for example, the bodies $y = x$ and $\neg(x = A)$). If no datum of T_{Sep} mentions z, the learner stays at the strictly more special template with $P(x, z)$ replaced by $P'(x)$, which is sound. □

Weaker events. The events of Theorem 3.8 cannot be weakened to the candidates suggested by first-order patterns.

Proposition B.2 (Weaker events do not suffice)**.** *[proved; computed]* single Prop D.4, D.5, Ex. D.6

(a) (R)+(N)+(D)*: $T^* = \forall x(0 = f(x))$, $D = \{\forall x(0 = 0), \forall x(0 = x)\}$.* (R*) *and* (N) *hold and* (D) *is vacuous, but $\forall x(f(0) = f(x))$ covers D and misses $\forall x(0 = Sx)$.*

(b) (U) *for pattern occurrences only: $T^* = \forall x\forall y(Sf(x, y) = 0) \wedge \forall x\forall y(f(x, Sy) = 0)$ with the values $\lambda z_1 z_2.z_1$, $\lambda z_1 z_2.z_2$, $\lambda z_1 z_2.Sz_1$.* (R*), (N), (D) *hold, and* (U) *holds for every pair whose first component is the pattern occurrence. But the content at the rigid S above $f(x, y)$ equals the content at the derived occurrence $f(x, Sy)$ under $x \mapsto Sx$, so $\forall x\forall y(f'(y, Sx) = 0) \wedge \forall x\forall y(f'(y, x) = 0)$ covers D and misses $T^*[f := \lambda z_1 z_2.0]$.*

(c) Projections: $T^* = \forall x\forall y(f(x, y) = 0) \wedge \forall x(x = 0)$ with values $\lambda z_1 z_2.z_1$, $\lambda z_1 z_2.z_2$ satisfies (R*), (N)*, but $\forall x\forall y(f(x, y) = 0) \wedge \forall x(f(x, x) = 0)$ covers D. So* (U) *matters even for one metavariable with pattern occurrences only.*

Proof. (a) $\forall x(f(0) = f(x))$ is in DT° ($f(x)$ is a pattern occurrence) and covers $\forall x(0 = 0)$ ($f := \lambda z.0$) and $\forall x(0 = x)$ ($f := \lambda z.z$, $f(0) = 0$). An instance with $f(x) = Sx$ has $f(0) = S0 \neq 0$. The brute-force minimal covering templates are exactly $\forall x(0 = f(x))$ and $\forall x(f(0) = f(x))$.

(b) The data are

$$\forall x\forall y(Sx = 0) \wedge \forall x\forall y(x = 0), \quad \forall x\forall y(Sy = 0) \wedge \forall x\forall y(Sy = 0), \quad \forall x\forall y(SSx = 0) \wedge \forall x\forall y(Sx = 0).$$

Write σ for the left-hand side of the second conjunct (the derived occurrence $f(x, Sy)$) and r for the left-hand side of the first (the rigid S above $f(x, y)$). The data have $(d|_r, d|_\sigma) = (Sx, x), (Sy, Sy), (SSx, Sx)$, and $d|_r = (d|_\sigma)[x \mapsto Sx, y \mapsto y]$ in all three. For the covering template, the pattern occurrence $f'(y, x)$ at σ reads $f' = \lambda z_1 z_2.z_2$, $\lambda z_1 z_2.Sz_1$, $\lambda z_1 z_2.Sz_2$, and then $f'(y, Sx)$ gives Sx, Sy, SSx at r. For $f := \lambda z_1 z_2.0$, T^* gives $\forall x\forall y(S0 = 0) \wedge \forall x\forall y(0 = 0)$, whereas the covering template forces $f' = \lambda z_1 z_2.0$ from the second conjunct and then 0, not $S0$, in the first. (U) holds for the pairs with first component $f(x, y)$ (column (x, y, Sx)): the right-hand zeros have the wrong root, and the only coincidence, with the derived occurrence $f(x, Sy)$ under

$x \mapsto x$, $y \mapsto Sy$, belongs to a valid pair. Brute force (size ≤ 18, arguments ≤ 2, arity ≤ 2) finds exactly two minimal covering templates, T^* (up to argument order) and the displayed one; the referee's code agrees.[single e2, e8; referee t2]

(c) With $d_1 = \forall x \forall y(x = 0) \wedge \forall x(x = 0)$ and $d_2 = \forall x \forall y(y = 0) \wedge \forall x(x = 0)$, the derived occurrence $f(x, x)$ reads x from both values. The rigid x of the second conjunct coincides with the slot's content under $x \mapsto x$, $y \mapsto x$; the instance $f := \lambda z_1 z_2.Sz_1$ of T^* is missed.

On random data (R)∧(N)∧(D) and (R*)∧(N)∧(D) were wrong in 27 and 17 of 248 cases (always "says anchor, is not"; referee: 45, 29 of 540), while (U) restricted to pattern occurrences was never wrong, so (b) needs special structure.[single e4; referee t3]

Example B.3 ((Rich) cannot be dropped)**.** [proved][single Ex. D.7] In pure set theory without parameters (signature $\in, =$; no constants or function symbols) a term over holes is a hole, so the only term bodies are projections $\lambda \bar{z}.z_m$. Hence $T^* = \forall x(f(x) \in x)$ has $\mathrm{inst}(T^*) = \{\forall x(x \in x)\}$, and the single datum $\forall x(x \in x)$ is an anchor (every template covering it contains the whole target), although (R*) fails. So without (Rich) the anchor condition is strictly weaker than (R*)∧(N)∧(U); its exact form is open. With parameters (Rich) holds, since two distinct parameters are closed terms with different roots.

B.5 Rates

Proof of Theorem 3.10. (a) Let $N \geq 1$. By the "if" half of Theorem 3.8, which needs no assumption, "not an anchor" is contained in $\neg(\mathrm{R}^*) \cup \neg(\mathrm{N}) \cup ((\mathrm{R}^*) \wedge (\mathrm{N}) \wedge \neg(\mathrm{U}))$. For a ground T^* all three events are empty; consistently, every nonempty $D \subseteq \mathrm{inst}(T^*)$ equals $\{T^*\}$ and is an anchor. In general:

- $\Pr[\text{all roots at } \sigma \text{ equal}] = \sum_h p_h^N \leq p_{\max}^{N-1} \sum_h p_h = (1 - \rho_\sigma)^{N-1}$, where p_h is the probability of root h at σ.
- $\Pr[z_m \text{ unused by all } \theta_i(M)] = (1 - \nu_{M,m})^N$.
- A coincidence at a non-valid pair (σ, r) on D_N means that some $\bar{u}$ has all $\theta_i \in A_{\sigma,r,\bar{u}}$. Then each of the $\lfloor N/2 \rfloor$ disjoint pairs $(\theta_1, \theta_2), (\theta_3, \theta_4), \ldots$ has a common $\bar{u}$; these pairs are independent, each with probability $1 - \kappa_{\sigma,r}$.

The union bound gives the first inequality. Each term is at most $(1-\lambda)^{\lfloor N/2 \rfloor}$, since $N - 1 \geq \lfloor N/2 \rfloor$ for $N \geq 1$; there are $c(T^*)$ terms. Lower bounds: under (Rich) each single failure event is a non-anchor event (Theorem 3.8, "only if"). All N roots at σ equal the most likely head with probability $(1 - \rho_\sigma)^N$; z_m is unused with probability $(1 - \nu_{M,m})^N$; and a coincidence at (σ, r) has probability $\geq \chi_{\sigma,r}^N$ (Proposition B.4(i)).

(b) $\rho_\sigma > 0$ iff two support elements have different roots at σ; $\nu_{M,m} > 0$ iff some support element uses z_m; and $\kappa_{\sigma,r} > 0$ iff some pair of support elements has no common $\bar{u}$, which by Lemma 3.2(d) (a pairwise condition; finiteness is not used) holds iff the support has no coincidence at (σ, r). So $\lambda > 0$ iff the support satisfies the three events. Then $c(T^*)(1-\lambda)^{\lfloor N/2 \rfloor} \leq \bar{c}\, e^{-\lambda \lfloor N/2 \rfloor} \leq \bar{c}\, e^{-\lambda(N-1)/2}$, which is $\leq \delta$ iff $N \geq 1 + (2/\lambda) \ln(\bar{c}/\delta)$. For a ground T^* the failure probability is 0 for $N \geq 1$.

(c) The version space over tagged calculi factors over tags (Hänni and Claude (Anthropic), 2026), so the verifier is exact iff the tag-i data are an anchor for every i, and

$$\Pr[\text{not exact}] \leq \sum_i \Pr[\text{the tag-}i \text{ data are not an anchor}].$$

Condition on the number $n_i \sim \mathrm{Bin}(N, \pi_i)$ of tag-i data. For all $n_i \geq 0$,

$$\Pr[\text{not an anchor} \mid n_i] \leq \bar{c}_i (1 - \lambda_i)^{\lfloor n_i/2 \rfloor} :$$

for $n_i \geq 1$ this is (a); for $n_i = 0$ the right side is $\bar{c}_i \geq 1$; for a ground tag the left side is $[n_i = 0]$ and the right side is $0^{\lfloor n_i/2 \rfloor} \geq [n_i = 0]$ (with $0^0 = 1$). Next, $(1-\lambda)^{\lfloor n/2 \rfloor} \leq (1-\lambda)^{(n-1)/2} \leq e^{\lambda/2} e^{-\lambda n/2} \leq \sqrt{e}\, e^{-\lambda n/2}$, as $\lambda \leq 1$ (for $\lambda = 1$ check $n = 0, 1$ directly). And $\mathbb{E}[e^{-an_i}] = (1 - \pi_i(1 - e^{-a}))^N \leq \exp(-N\pi_i(1 - e^{-a}))$, with $1 - e^{-a} \geq 3a/4$ for $a = \lambda_i/2 \leq 1/2$ (the left side is concave, and the inequality holds at $a = 1/2$). This gives $\sqrt{e}\, \bar{c}_i \exp(-3N\pi_i\lambda_i/8)$ per tag. Lower bounds: with probability $(1 - \pi_i)^N$ there is no tag-i datum, and then Acc of the tag-i data is empty; this is the exact failure probability of a ground tag. Under (Rich), all tag-i data share the head h at the pattern occurrence, or there are none, with probability $(1 - \pi_i + \pi_i(1 - \rho))^N = (1 - \pi_i\rho)^N$, and then they are no anchor. □

The (U)-terms vanish for a single formula metavariable because, under (R)∧(N), (U) holds automatically (Corollary 3.9(b)), and all occurrence events coincide with (R); so the bound is $(1 - \rho)^{N-1} + \sum_m (1 - \nu_m)^N$.

Proposition B.4 (Coincidence terms)**.** *[proved; computed]* single Prop E.2 *For $(\sigma, r) \in U(T^*)$: (i) the probability of a coincidence at (σ, r) on D_N is $\geq \chi^N_{\sigma,r}$, and under* (Rich) *such a coincidence makes D_N a non-anchor; (ii) $\chi^2_{\sigma,r} \leq 1 - \kappa_{\sigma,r}$; (iii) if Λ has finite support and $\mathcal{K}$ is the family of maximal support sets lying in a common $A_{\sigma,r,\bar{u}}$, the coincidence probability equals $\sum_{\emptyset \neq \mathcal{J} \subseteq \mathcal{K}} (-1)^{|\mathcal{J}|+1} \Lambda(\bigcap \mathcal{J})^N \in [\chi^N, |\mathcal{K}|\chi^N]$. So its exact rate is χ, which can be smaller than the rate $\sqrt{1-\kappa}$ of the upper bound; they agree if $|\mathcal{K}| = 1$.*

Proof of Proposition B.4. (i) For each $\bar{u}$, Pr[all $\theta_i \in A_{\bar{u}}$] $= \Lambda(A_{\bar{u}})^N$; take the supremum. Under (Rich), a coincidence at a non-valid pair is a failure of (U), so D_N is not an anchor (by Theorem 3.8 "only if" if (R*) and (N) hold, and also otherwise). (ii) For every $\bar{u}$, $1-\kappa = \Lambda \otimes \Lambda\{(\theta, \theta') : \exists \bar{u}'\, \theta, \theta' \in A_{\bar{u}'}\} \geq \Lambda(A_{\bar{u}})^2$. (iii) The event says that the set of sampled support points is compatible, i.e. lies in some $A_{\bar{u}}$, i.e. in some $K \in \mathcal{K}$; inclusion–exclusion over $\mathcal{K}$ gives the formula. Each K lies in some $A_{\bar{u}}$, so $\Lambda(K) \leq \chi$ and the probability is at most $|\mathcal{K}|\chi^N$. Conversely $A_{\bar{u}} \cap \operatorname{supp} \Lambda$ is compatible for every $\bar{u}$, hence contained in a maximal compatible set; so $\chi = \max_K \Lambda(K)$, and the probability is at least χ^N. If $\mathcal{K} = \{K\}$ is a singleton, a pair (or a single point) is compatible iff it lies in K, so $1 - \kappa = \Lambda(K)^2 = \chi^2$. □

Two worked laws. *First law* (the referee's example): $T^* = \forall x(0 = f(x))$, with f equal to $\lambda z.0$, $\lambda z.z$, $\lambda z.Sz$ with probabilities 0.4, 0.3, 0.3. The only non-valid pair is (slot of $f(x)$, rigid 0), and its compatible set is $\{\lambda z.0, \lambda z.z\}$ (with $\bar{u} : x \mapsto 0$); $\lambda z.Sz$ is in no $A_{\bar{u}}$. (R*) fails iff all values are equal, (N) fails iff all values are $\lambda z.0$, and (U) fails iff all values lie in $\{0, z\}$. So Pr[not an anchor] $= 0.7^N + 0.3^N$, which at $N = 8$ is 0.057714. It lies between the coincidence lower bound $\chi^8 = 0.7^8 = 0.057648$ and the upper bound $0.4^7 + 0.4^8 + 0.49^4 = 0.059942$ of Theorem 3.10(a); the root and non-vacuity lower bounds alone give $0.4^8 = 0.000655$. *Second law:* $T^* = \forall x \forall y(f(x) = g(y))$ with $f \sim U\{z, Sz\}$ and $g \sim U\{z, 0, Sz\}$ independent; write θ as the pair (value of f, value of g). At the pair $(f(x), g(y))$ the maximal compatible sets are $\{(z, z), (Sz, Sz)\}$ (with $\bar{u} : x \mapsto y$), $\{(z, 0)\}$ $(x \mapsto 0)$ and $\{(z, Sz)\}$ $(x \mapsto Sy)$, of masses $1/3, 1/6, 1/6$; so $|\mathcal{K}| = 3$, $\chi = 1/3$, and $1 - \kappa = 1/9 + 2/36 = 1/6$, so $\sqrt{1-\kappa} = 0.408 > \chi$. At $N = 8$ the exact coincidence probability is $(1/3)^8 + 2(1/6)^8 = 1.54 \cdot 10^{-4}$, against $\chi^N = 1.52 \cdot 10^{-4}$ and $(1 - \kappa)^4 = 7.7 \cdot 10^{-4}$. *Non-circular check of Theorem 3.10(a):* the referee computed exact non-anchor probabilities by inclusion–exclusion over the support, deciding anchor status by feature search rather than by the events of Theorem 3.8, for induction, Separation and the template of Proposition B.2(b); the bound held at every $N \in \{2, 4, 8, 12, 16, 24\}$ (for induction at $N = 24$: exact 0.00100, bound 0.00535).referee t7 The computations of Proposition B.4(iii) were checked on every non-valid pair

of five laws, and exact non-anchor probabilities for two small laws were cross-checked against an independent feature-template test with no disagreement.[single e10]

B.6 Escalations

Let $\Pi(D) := (C(D), (Y_\sigma(D))_\sigma, E(D))$, where $E(D)$ is the set of pairs (σ, r) carrying an Eq-feature. As data grow, C shrinks, scopes grow, and while both are fixed E shrinks; and by Theorem 3.4 a query outside $\mathrm{Acc}(D)$ changes Π (Lemma B.5).

Lemma B.5 (Monotonicity)**.** *[proved]*[single Lemma F.1] *Let $D \subseteq D'$ be nonempty. (i) $C(D') \subseteq C(D)$ and $A(D') \subseteq A(D)$; a slot of D that is still in $A(D')$ is a slot of D'; and $Y_\sigma(D) \subseteq Y_\sigma(D')$ whenever σ is a slot of both. (ii) If $C(D') = C(D)$ and $Y(D') = Y(D)$, then $E(D') \subseteq E(D)$, with the same maps $\bar{u}$. (iii) If $q \notin \mathrm{Acc}(D)$, then $\Pi(D \cup \{q\}) \neq \Pi(D)$.*

Proof. (i) is immediate from the definitions: a slot of D' has its parent in $C(D') \subseteq C(D)$, and scopes are unions over the data. (ii) A feature of D' restricts to D, and its $\bar{u}$ is forced on $Y_\sigma(D) = Y_\sigma(D')$ (Lemma 3.2(d)). (iii) If Π is unchanged, then q has every Sym feature (C unchanged), every Scope feature (Y unchanged) and every Eq feature (E unchanged, with $\bar{u}$ forced by D). So $q \in \mathrm{Feat}(D) = \mathrm{Acc}(D)$ by Theorem 3.4. □

Proof of Theorem 3.11. Let $D_j = \{d, q_2, \ldots, q_j\}$. By Lemma B.5, each step $D_{j-1} \to D_j$ changes Π, and then it does one of the following: (α) shrink C; (β) keep C and enlarge some scope (scopes only grow, and slots are unchanged when C is); (γ) keep C and every scope, and remove at least one pair from E (by (ii), E can only shrink then). (α) happens at most $|C(\{d\})| = n$ times. A position σ is a slot during one interval of steps: it becomes a slot when it leaves C, and it leaves A for ever when its parent leaves C. During that interval Y_σ only grows, inside $\{0, \ldots, \mathrm{b}(\sigma) - 1\}$; so ($\beta$) happens at most $\sum_\sigma \mathrm{b}(\sigma)$ times. A pair (σ, r) can lie in E only while σ is a slot and $r \in A$, which is a single interval. Once removed, it never returns: if it were in E at times $t_1 < t_3$ but not at $t_2 \in (t_1, t_3)$, then σ is a slot and $r \in A$ at t_2, and the feature at t_3 restricts to the data at t_2 (Lemma 3.2(d)), a contradiction. So (γ) happens at most once per ordered pair of incomparable positions of d. The second inequality uses $\mathrm{b}(p) \leq \mathrm{b}_{\max}$ and at most $n(n-1)$ ordered pairs. From no data the first query is escalated, and $\mathrm{b}(p) \leq n - 1$, so $\mathrm{Esc}(\mathrm{DT}^\circ; N) \leq 1 + N + N(N-1) + N(N-1) \leq 2N^2 + 1$.

The linear lower bound. First-order targets already give $\mathrm{Esc}(\mathrm{DT}^\circ; N) \geq N + 1$ for $N \geq 5$. With $k = N - 3$, the chain $S^k 0 = 0$, $S^k a = 0$, $S^{k-1} 0 = 0$, …, $0 = 0$, $0 = S0$, $\neg(0 = 0)$ has $N + 1$ members of size $\leq N$, and the successive acceptance sets are $\{S^k 0 = 0\}$, $\{S^k t = 0\}$, $\{S^{k-1} t = 0\}$, …, $\{t = 0\}$, $\{t = t'\}$ (t, t' closed): each query lies outside the previous set. All are instances of the universal template. Checked for $N = 5, 8, 12, 16$.[single e8] □

The quadratic term is real:

Proposition B.6 (Quadratic escalations)**.** *[proved; computed]*[single Prop F.8] *Let $d_1 = \forall y_1 \ldots \forall y_m (0 = 0 \wedge \cdots \wedge 0 = 0)$ with m atoms, d_2 the sentence d_1 with every left-hand side replaced by a parameter a, and $q_{j,k}$ ($j \leq m$, $k < m$) the sentence d_1 with the j-th left-hand side replaced by the bound variable of index k. (i) In the order $d_1, d_2, q_{1,0}, \ldots, q_{1,m-1}, q_{2,0}, \ldots, q_{m,m-1}$, each sentence lies outside the acceptance set of its predecessors (d_2 violates a Sym feature, each $q_{j,k}$ a Scope feature). All have size $5m-1$. Hence $2 + \lfloor (N+1)/5 \rfloor^2 \leq \mathrm{Esc}(\mathrm{DT}^\circ; N) \leq 2N^2 + 1$ for $N \geq 4$: $\mathrm{Esc}(\mathrm{DT}^\circ; N) = \Theta(N^2)$. (ii) $\mathrm{Ind}(d_1), \mathrm{Ind}(d_2), \mathrm{Ind}(q_{j,k})$, in the same order, are instances of T_{Ind} of size $20m + 1$, and each lies outside the acceptance set of its predecessors.*

The earlier analysis conjectured that escalations are at most *linear* in the size of the first escalated datum[prior induction C11]. By (ii), $1 + m^2$ escalations follow a first datum of size $n = 20m + 1$, which exceeds $c\,n$ once $m > 20c + 1$. So this conjecture is false: for DT°, for the class with nested arguments (whose acceptance sets are smaller), for the universal template and for the induction target. The refutation is asymptotic (for $m = 2, \ldots, 5$ the counts $5, 10, 17, 26$ are below $n = 41, \ldots, 101$). The quadratic bound holds and is tight. Its quadratic part comes from scope growth, which needs deep binder nesting (Proposition B.7).

Proof of Proposition B.6. (i) d_1 and d_2 differ exactly at the m left-hand sides (roots 0 and a). So $C(\{d_1, d_2\})$ is everything else, and the left-hand sides are slots with empty scope. Each $q_{j,k}$ agrees with d_1 off the slots, so C does not change along the chain. At slot j, $q_{j,k}$ has the bound variable with index k free (relative to the slot, which lies under the m binders and no others). Earlier data have at slot j only 0, a, or other bound variables $k' < k$. So $q_{j,k}$ violates $\mathsf{Scope}(\text{slot } j)$ of its predecessors, and $q_{j,k} \notin \text{Feat} = \text{Acc}$. (Without the hard half of Theorem 3.4: T_0 of the predecessors is a covering DT° template that misses $q_{j,k}$.) The size is m binders, $m - 1$ conjunctions and $3m$ symbols in the atoms: $5m - 1$. The chain has $2 + m^2$ members. For $N \geq 4$, $m = \lfloor (N + 1)/5 \rfloor \geq 1$ gives size $5m - 1 \leq N$; the upper bound is Theorem 3.11.

(ii) The motive occurs four times in $\text{Ind}(\varphi)$: as $\varphi(0)$, $\varphi(x)$, $\varphi(Sx)$ and in the conclusion; since x does not occur, all four copies are φ. After $\text{Ind}(d_1)$ and $\text{Ind}(d_2)$ the left-hand sides in all four copies are slots, and their scopes contain no index bound by the motive's own m binders: x does not occur, and d_1, d_2 have no indices at the slots. $\text{Ind}(q_{j,k})$ puts the index k (bound by the motive's binders) at slot j of each copy, which no predecessor has. The size is $4(5m - 1) + 5 = 20m + 1$. For $m > 20c + 1$, $1 + m^2 > c(20m + 1)$. □

Computed. Chains of length $6, 18, 38, 66$ for $m = 2, 4, 6, 8$; for the universal template and $N \in \{4, 5, 9, 14, 19, 23, 24, 29, 34\}$ exactly $2 + \lfloor (N + 1)/5 \rfloor^2$ escalations with sizes $\leq N$; for the induction target, first data of size $41, 61, 81, 101$ followed by $5, 10, 17, 26$ escalations. Every step was certified twice, by the feature verifier and by an explicit covering T_0 that misses the query; the referee's independent code agrees.[single e8, e12; referee t5, t5b]

Proposition B.7 (Binder-free data)**.** *[proved; computed][single Prop F.9] For a chain as in Theorem 3.11, at most $n - 1$ steps of kind (γ) remove a pair whose source has empty scope. If d has no binder, $m - 1 \leq 2n - 1$; on binder-free sentences $N + 1 \leq \text{Esc} \leq 2N$ $(N \geq 5)$.*

Proof of Proposition B.7. (i) For $D \ni d$ define $p \sim_D p'$ iff $p = p'$, or p, p' are positions of every datum and $d'|_p = d'|_{p'}$ for all $d' \in D$. This is an equivalence relation on the positions of d, and it refines as D grows. Consider a (γ)-step $D \to D' = D \cup \{q\}$; it changes neither C nor the scopes, hence neither the slots nor A. Suppose it removes (σ, r) with $Y_\sigma(D) = \emptyset$. Then $\mathsf{Eq}(\sigma, r, \emptyset)$ was a D-feature, i.e. $d'|_r = d'|_\sigma$ for all $d' \in D$, so $\sigma \sim_D r$. It is not a D'-feature, although σ is still a slot, r still admissible and $\text{FV}(q|_\sigma) \subseteq Y_\sigma(D') = \emptyset$; so $q|_r \neq q|_\sigma$ and $\sigma \not\sim_{D'} r$. This is a strict refinement. An equivalence relation on n points refines strictly at most $n - 1$ times.

(ii) If d has no binder, every slot σ has $\text{b}(\sigma) = 0$, since its ancestors lie in C and carry the symbols of d. So all scopes are empty: there are no (β)-steps, and every (γ)-step is of the kind in (i). With at most n (α)-steps, $m - 1 \leq n + (n - 1)$. From no data add 1 for the first escalation; the lower bound $N + 1$ is the binder-free chain above. □

Computed. 40 greedy adversarial chains from binder-free first data of 8–18 positions: no chain exceeded $2n - 1$, and every (γ)-step strictly refined $\sim$.[single e13]

Lemma B.8 (Triangle constraints on equation kills)**.** *[proved; computed]*[single Lemma F.10] *Let $D \subseteq D'$ with $C(D') = C(D)$ and $Y(D') = Y(D)$, and write $\bar{u}_{ab}$ for the forced map of a pair $(a, b) \in E$. (a) If $(k, i), (i, j) \in E(D')$, $k \perp j$, and k, j have the same sort, then $(k, j) \in E(D')$ with $\bar{u}_{kj} = \bar{u}_{ki}[\bar{u}_{ij}]$ on Y_k. (b) If $(k, i), (k, j), (i, j) \in E(D)$ and $(k, i), (k, j) \in E(D')$, then $(i, j) \in E(D')$. (c) If $(i, k), (k, j), (i, j) \in E(D)$ and $(i, k), (k, j) \in E(D')$, then $(i, j) \in E(D')$. Hence a (γ)-step that removes (i, j) also removes one of $(k, i), (k, j)$ for every common in-neighbour k, and one of $(i, k), (k, j)$ for every intermediate k.*

Proof. First, $\bar{u}_{ki}[\bar{u}_{ij}]$ is defined on Y_k: for $y \in Y_k$, $\bar{u}_{ki}(y)$ is read off a datum d with $y \in \mathrm{FV}(d|_k)$, so its free indices lie in $\mathrm{FV}(d|_i) \subseteq Y_i$, the domain of $\bar{u}_{ij}$. (a) k is a slot and $j \in A$. For $d' \in D'$, $d'|_j = (d'|_i)[\bar{u}_{ij}] = ((d'|_k)[\bar{u}_{ki}])[\bar{u}_{ij}] = (d'|_k)[\bar{u}_{ki}[\bar{u}_{ij}]]$, and $\mathrm{FV}(d'|_k) \subseteq Y_k$. (b) By (a) on D and forcing, $\bar{u}_{kj} = \bar{u}_{ki}[\bar{u}_{ij}]$ on Y_k. Let $q \in D' \setminus D$. The maps of (k, i) and (k, j) on D' are the forced maps on D (the domains are unchanged). So $q|_i = (q|_k)[\bar{u}_{ki}]$ and $q|_j = (q|_k)[\bar{u}_{kj}] = ((q|_k)[\bar{u}_{ki}])[\bar{u}_{ij}] = (q|_i)[\bar{u}_{ij}]$; moreover $\mathrm{FV}(q|_i) \subseteq Y_i(D') = Y_i(D)$. So $\mathsf{Eq}(i, j, \bar{u}_{ij})$ holds on D'. (c) Apply (a) to D'; its map $\bar{u}_{ik}[\bar{u}_{kj}]$ equals $\bar{u}_{ij}$ on D by forcing. □

Conjecture B.9 (Equation kills are linear)**.** *[conjecture]*[single Conj. F.7] *The number of (γ)-steps is $O(n)$; hence $\mathrm{Esc}(\mathrm{DT}^\circ; N) = O(N(1 + \mathrm{b}_{\max}))$ for binder depth $\leq \mathrm{b}_{\max}$.*

The evidence is weak: greedy adversarial searches found at most about $1.4m$ kill steps on $m \leq 12$ slots, but they also missed the scope-growth chains of Proposition B.6.[single e6, e6b] Surviving equations need not form an equivalence relation; they compose like a preorder and obey a triangle rule (Lemma B.8), and chains of preorders on m points can have length of order m^2. Proposition B.7 proves the conjecture when all scopes are empty.

Where the difficulty of Conjecture B.9 sits. In a phase with C and the scopes fixed, "same value in every datum" is an equivalence relation on the nodes $A(D) \cup \{(\sigma, \bar{u}_{\sigma r})\}$ (the value of $(\sigma, \bar{u})$ in d being $(d|_\sigma)[\bar{u}]$); it refines with each datum, and (σ, r) survives iff $(\sigma, \bar{u}_{\sigma r}) \equiv r$. So a phase has fewer (γ)-steps than nodes: linearly many if every slot uses $O(1)$ distinct maps, but there can be order n^2 nodes, and up to $n + \sum \mathrm{b}$ phases. *Computed:* 587 composition instances and 935 instances each of (b) and (c), 240 of them in steps that removed some pair, with no failure.[single e13]

B.7 Unions

Proof of Corollary 3.12. (i) If $w_0, w_1, \ldots$ and $L_n = \mathrm{inst}(T_n)$ witness infinite elasticity, then $w_n \notin \mathrm{Acc}(\{w_0, \ldots, w_{n-1}\})$ for $n \geq 1$, so the sequence is bounded by Theorem 3.11 with $d = w_0$. For thickness: $P := \lambda z.(0 = 0)$ puts $\forall x(0 = 0) \wedge (0 = 0)$ in $\mathrm{inst}(\forall x P(x) \wedge P(t))$ for every closed t, and $P := \lambda z.(z = z)$ gives $\forall x(x = x) \wedge (t = t)$, which lies in the t-template only (matching forces $P = \lambda z.(z = z)$ from the pattern occurrence). (ii) is the cited preservation result together with "finite elasticity gives anchors" and the eventual exactness of the cautious verifier once an anchor is in the data (Hänni and Claude (Anthropic), 2026).

(iii) is Proposition B.10. □

Proposition B.10 (Escalations for unions)**.** *[proved; computed]*[single Cor F.4] *With a k-ary relation symbol, a unary function symbol and a constant, $\mathrm{Esc}(H_k(\mathrm{DT}^\circ); N) \geq \lfloor (N-1)/k \rfloor^k$; over $=, +, S, 0$, $\mathrm{Esc}(H_k(\mathrm{DT}^\circ); N) \geq (\lfloor (N-1)/k \rfloor - 1)^k$ for $N \geq 2k + 1$, already for one target.*

Proof. With a k-ary relation symbol P, a unary function symbol g and a constant c, the chain of Hänni and Claude (Anthropic) (2026) for the term signature transfers verbatim: query $P(g^{a_1} c,$

$\dots, g^{a_k}c)$ for $a \in \{0,\dots,n\}^k$, $n+1 = \lfloor (N-1)/k \rfloor$, by non-increasing $\sum_j a_j$. Over $=, +, S, 0$, let $n+1 := \lfloor (N-1)/k \rfloor - 1 \geq 1$ (so $N \geq 2k+1$), and query

$$q_a := \big(S^{a_1}0 + (S^{a_2}0 + \cdots + S^{a_k}0)\big) = 0, \qquad a \in \{0,\dots,n\}^k,$$

ordered by non-increasing $\sum_j a_j$. Each has size $k+1+\sum_j(a_j+1) \leq k+1+k(n+1) \leq N$, and all are instances of the target $(x_1 + \cdots + x_k) = 0$ with 0-ary term metavariables x_j. Every earlier query a' has $a'_j \geq a_j + 1$ for some j; otherwise $a' \leq a$ componentwise with $a' \neq a$, so $\sum a' < \sum a$, contradicting the order. Hence the earlier queries lie in $\bigcup_j \mathrm{inst}(T_j)$ with $T_j := (x_1 + \cdots + S^{a_j+1}(x_j) + \cdots + x_k) = 0 \in \mathrm{DT}^\circ$, and this union misses q_a. So $q_a \notin \mathrm{Acc}_k(\text{earlier queries})$; the first query is escalated because $\mathrm{Acc}_k(\emptyset) = \emptyset$. This gives $(n+1)^k$ escalations. With a k-ary P the same proof works with overhead 1 instead of $k+1$. □

Computed. For $(k, n+1) = (1,6), (2,3), (2,4), (3,2)$: $6, 9, 16, 8$ escalations on sentences of size $\leq N = 8, 9, 11, 10$, each step certified by the explicit k-union and, for the first 12 steps, by the brute-force union verifier.[single e16] The encoded bound over $=, +, S, 0$ was made explicit after the referee round, so it was not checked by the independent referee (Appendix G.3).

Explicit anchors for unions. Let the target be $\bigcup_{i \leq k'} \mathrm{inst}(T_i^*)$, $k' \leq k$, $T_i^* \in \mathrm{DT}^\circ$, and $\Theta_i = \{\theta : T_i^*\theta \in D\}$. Let $\mathrm{Fail}(T^*)$ consist of the failure sets of the three events: $\{\theta : \mathrm{root}((T^*\theta)|_\sigma) = h\}$, $\{\theta : z_m \notin \theta(M)\}$, and $\{\theta : (T^*\theta)|_r = ((T^*\theta)|_\sigma)[\bar{u}]\}$ for non-valid (σ, r).

Proposition B.11 (Explicit untagged anchors)**.** *[proved; computed][single Prop F.5] If for each i the set Θ_i is nonempty and not covered by k members of $\mathrm{Fail}(T_i^*)$, then D is an anchor for the union in $H_k(\mathrm{DT}^\circ)$. For a ground T_i^* (a single axiom) $\mathrm{Fail}(T_i^*) = \emptyset$, and the condition says that the axiom itself is a datum.*

Nonemptiness matters for ZFC's single axioms: with target $\mathrm{inst}(T_{\mathrm{Ind}}) \cup \{\mathrm{Ext}\}$ and $k = 2$, the union verifier rejects Extensionality until it is a datum (computed below; the repair that added this hypothesis is recorded in Appendix G.2). For one formula metavariable, Fail reduces to "all roots equal f" and "argument m unused", the failure family of the earlier untagged PA analysis (Corollary E.4).

Proof of Proposition B.11. Let $\bigcup_{l \leq k} \mathrm{inst}(T_l) \supseteq D$, $T_l \in \mathrm{DT}^\circ$. Assign each $\theta \in \Theta_i$ to some l with $T_i^*\theta \in \mathrm{inst}(T_l)$ (matchers are unique, so data and substitutions correspond). This splits Θ_i into at most k parts, and into at least one nonempty part, since $\Theta_i \neq \emptyset$. Suppose no part's data were an anchor for $\mathrm{inst}(T_i^*)$. By the "if" half of Theorem 3.8 (which needs no (Rich)), each nonempty part then fails (R*), (N) or (U), i.e. lies in a member of $\mathrm{Fail}(T_i^*)$ (all roots at σ equal some h; or z_m unused; or, with (R*) and (N) holding, a coincidence $\bar{u}$ at a non-valid pair). For a ground T_i^* a nonempty part is $\{T_i^*\} = \mathrm{inst}(T_i^*)$, an anchor. So Θ_i would be covered by at most k members of $\mathrm{Fail}(T_i^*)$, contrary to the hypothesis. Hence some part, contained in $\mathrm{inst}(T_l)$, is an anchor, and $\mathrm{inst}(T_l) \supseteq \mathrm{inst}(T_i^*)$. This holds for every i. □

Computed. With target $\mathrm{inst}(T_{\mathrm{Ind}}) \cup \{\mathrm{Ext}\}$ and $k = 2$, three induction data and no Extensionality datum, the union verifier rejects Extensionality; with Extensionality among the data it accepts it; fresh induction instances are accepted in both cases. So the hypothesis $\Theta_i \neq \emptyset$ is needed.[single e15]

Proposition B.12 (The union verifier)**.** *[proved; computed][single Prop F.6] $q \in \mathrm{Acc}_k(D)$ iff every partition of D into at most k nonempty blocks has a block B with $q \in \mathrm{Acc}(B)$. So Acc_k is decidable in time (number of such partitions)$\times$poly.*

This is the case $N = \emptyset$ of Lemma 6.3(b), in the form (1); Theorem 6.16 shows that the test is polynomial for $k \leq 2$ but coNP-complete for each fixed $k \geq 3$.

Proof of Proposition B.12. A k-union $\bigcup_{l \leq k} \mathrm{inst}(T_l)$ contains D iff some partition of D into at most k nonempty blocks $B_1, \ldots, B_j$ has each block covered by its own T_l ($l \leq j$), the remaining templates being unconstrained. Suppose that for some such partition $q \notin \mathrm{Acc}(B_l)$ for every $l \leq j$. Choose T_l covering B_l with $q \notin \mathrm{inst}(T_l)$, and for $l > j$ a ground template other than q (possible since $\mathrm{Acc}(\emptyset) = \emptyset$). This union contains D and misses q, so $q \notin \mathrm{Acc}_k(D)$. Conversely, if every partition has a block B with $q \in \mathrm{Acc}(B)$, let $U = \bigcup_l \mathrm{inst}(T_l) \supseteq D$; assigning each datum to a template that covers it gives a partition, some block B of it has $q \in \mathrm{Acc}(B) \subseteq \mathrm{inst}(T_l) \subseteq U$. Each test $q \in \mathrm{Acc}(B)$ is polynomial by Corollary 3.6. □

Computed. On unlabelled data from PA induction and $\in$-induction (three instances each), $k = 1$ accepts the false $(0 = 0 \wedge \forall x(x = x \to Sx = Sx)) \to \forall x\, x = 0$, while $k = 2, 3$ accept all five fresh instances tried and reject it and two other non-instances (reproduced by the referee).[single e9]

B.8 Complexity of the version space

Proof of Proposition 3.13. Each datum has $4 + (4n - 1) + 1 = 4n + 4$ positions. $C(D_n) = \forall x(\Box = \Box) \wedge C_n$; the two slots have column $(x, 0)$ and scope $\{x\}$. The D_n-features are the Sym and Scope features, the two slot-to-slot equations (identity map), and, for each of the $2n$ zeros of C_n and each slot, "content = slot content with $x := 0$". There are no formula slots, so every metavariable is term-valued with pattern occurrences at the slots, and by Theorem 3.3(b) a saturated template has at each zero either the rigid 0 or a derived occurrence. Every covering template is $\succeq$ one of the templates T_A (A a set of zeros): $f(x)$ at both slots (merging two metavariables at the slots is a specialization that still covers) and $f(0)$ exactly at the zeros in A. For $A \neq B$, the instance $f := \lambda z.Sz$ of T_A has $S0$ exactly at the positions of A, so T_A and T_B have incomparable instance sets. Hence every T_A is minimal ($T \preceq T_A$ covering implies $T \succeq T_B \preceq T_A$ for some B, forcing $B = A$ and $T \equiv T_A$), and $\mathrm{Min}(D_n) = \{T_A\}$, with 4^n elements. By Theorem 3.4, $\mathrm{Acc}(D_n) = \{\forall x(t = t) \wedge C_n : \ \mathrm{FV}(t) \subseteq \{x\},\ t[0/x] = 0\} = D_n$, since a term with root S or $+$ keeps its root under $[0/x]$, so $t \in \{x, 0\}$. □

Computed. $|\mathrm{Min}(D_n)| = 4, 16, 64$ for $n = 1, 2, 3$ from the saturated templates, and $4, 16$ for $n = 1, 2$ by brute force (also by the referee).[single e2; referee t2]

Proposition B.13 (Existence of an lgg)**.** *[proved; computed][single Prop G.3] D has a least covering template iff (i) no Eq-feature of D has $r \in C(D)$, and (ii) in the graph on* $\mathrm{slots}(D)$ *with an edge $\sigma \to r$ for each slot-to-slot Eq-feature, every weakly connected component K has a* source *s with $s \to x$ for all $x \in K \setminus \{s\}$. Then the lgg is the template T_{all}: $C(D)$ with $M_K(Y_s)$ at each source s and $M_K(\bar{u}_{s,x})$ at the other slots x of K, where $\bar{u}_{s,x}$ is the forced map. Both conditions, and T_{all}, are computed in polynomial time from the D-features (Corollary 3.6).*

Proof of Proposition B.13. ($\Leftarrow$) $T_{\mathrm{all}} \in \mathrm{DT}^\circ$, and it covers D with $\theta_d(M_K) := (d|_s)[\bar{z}/Y_s]$, since $d|_x = (d|_s)[\bar{u}_{s,x}]$. Let $q = T_{\mathrm{all}}\theta$. It has every Sym feature, and every Scope feature: $\mathrm{FV}(q|_s) \subseteq Y_s$, and $\mathrm{FV}(q|_x) \subseteq \mathrm{FV}(\bar{u}_{s,x}) \subseteq Y_x$, because $\bar{u}_{s,x}$ is read off data at x. Let $\mathsf{Eq}(\sigma, r, \bar{u})$ be a D-feature. By (i), r is a slot, and σ, r lie in one component K, with source s (put $\bar{u}_{s,s} := \mathrm{id}$). Then $q|_r = \theta(M_K)[\bar{u}_{s,r}]$ and $(q|_\sigma)[\bar{u}] = \theta(M_K)[\bar{u}_{s,\sigma}[\bar{u}]]$. On D both $\bar{u}_{s,r}$ and $\bar{u}_{s,\sigma}[\bar{u}]$ solve the equation of the pair (s, r), so they coincide on Y_s by forcing, and $q|_r = (q|_\sigma)[\bar{u}]$. So $\mathrm{inst}(T_{\mathrm{all}}) \subseteq \mathrm{Feat}(D) = \mathrm{Acc}(D) \subseteq \mathrm{inst}(T_{\mathrm{all}})$. *Syntactic leastness:* let U cover D and take a saturated U' with $U \succeq U'$

(Theorem 3.3(a)). An occurrence of U' in $C(D)$ would be a derived occurrence with an Eq-feature landing in $C(D)$, which (i) excludes; so all occurrences of U' are at slots, every slot is one, and the skeleton of U' is $C(D)$. Map each metavariable L of U', with pattern slot σ_L and pattern arguments $\bar{y}$, to $\lambda\bar{z}.(T_{\text{all}}|_{\sigma_L})[\bar{z}/\bar{y}]$. Each other occurrence $L(\bar{t})$ at r gives the D-feature $(\sigma_L, r, \bar{y} \mapsto \bar{t})$, so r lies in the component of σ_L, and $T_{\text{all}}|_r = (T_{\text{all}}|_{\sigma_L})[\bar{y} \mapsto \bar{t}]$ by forcing. So this substitution maps U' to T_{all}, and $U \succeq U' \succeq T_{\text{all}}$.

($\Rightarrow$) Let T_ℓ be least. $T_\ell \preceq T_0(D)$ makes every position of $C(D)$ rigid in T_ℓ. Suppose an Eq-feature $\varphi = (\sigma, r, \bar{u})$ had $r \in C(D)$. Since T_ℓ covers D, $\text{skel}(T_\ell) \subseteq C(D)$, so the slot σ (whose parent is rigid) carries an occurrence $N(\bar{t})$ in T_ℓ. From $T_\ell = T_\varphi\tau$ we get $T_\ell|_\sigma = \tau(M_\sigma)[Y_\sigma]$, so $\tau(M_\sigma) = \lambda\bar{z}.N(\bar{t}')$, and then $T_\ell|_r = \tau(M_\sigma)[\bar{u}] = N(\bar{t}'[\bar{u}])$ is an occurrence at a rigid position, a contradiction. This proves (i). So the occurrences of T_ℓ are exactly the slots. The same argument shows that a feature $\sigma \to r$ makes σ and r occurrences of the same metavariable. For a metavariable N with pattern occurrence at s, the pair (s, x) is a feature for every other occurrence x of N, because T_ℓ covers D. So the components are the occurrence sets of the metavariables, each with the source s: this is (ii). □

Proof of Corollary 3.14. By Theorem 3.8 (here (Rich) is used), (R*), (N) and (U) hold. So $C(D) = \text{skel}(T^*)$, $\text{slots}(D) = \text{Occ}(T^*)$, $Y_\sigma(D) = Y^*_\sigma$, and every Eq-feature comes from a valid pair: r is an occurrence of the same metavariable. Thus no feature lands in $C(D)$, which is (i). The slot graph has edges only between occurrences of one metavariable, and a pattern occurrence s of M has $s \to x$ for every occurrence x of M (the pair is valid and T^* covers D), which is (ii). T_{all} puts $M_K(Y_s)$ at s and $M_K(\bar{u}_{s,x})$ at x, where by forcing and (N) $\bar{u}_{s,x}$ is the map $\bar{y} \mapsto \bar{t}_x$; so T_{all} is T^* up to renaming and permutation of argument places. The construction of Proposition B.13 is polynomial by Corollary 3.6. □

Computed. The referee's brute force agreed with Proposition B.13 on all 115 decided random cases, and in all 63 anchor cases the lgg was the target.[referee t9]

B.9 Computations

All scripts are in `research/tracks/single/` and are run there with `python3 <script> [args]`; outputs are the `.out` files of the same name. The brute-force enumerator `dtenum.py` does not use the theory: it enumerates prefixes of the common prefix, fillers from a term pool and all sharings, and keeps the determinate covering templates. Agreement with it is therefore an independent check within its bounds. The referee's code in `referee_checks/` is a second, fully independent implementation.

Main agreements. $\text{Min}(D)$ from saturated templates agreed with the brute-force enumerator in all 103 completed random cases, and $\text{Feat}(D)$ with the intersection of all enumerated covering templates on 7621 queries; the referee's independent code, with bounds containing $T_0(D)$ and every T_φ, agreed on 125,109 queries over 519 data sets (Theorems 3.3 and 3.4). Theorem 3.8 agreed with feature-based anchor status on 248/248 random targets with one or two metavariables of all types (referee, independently: 540/540, and 519/519 against brute force), and Corollary 3.9(a), (c) on all of 571 and about 2000 random data sets.

Scripts and what they check: `e1_matching.py` (matching: unique matcher and recovery 1500/1500, generality by freezing 400/400, linear time up to $|s| = 3.3{\cdot}10^6$); `e2_examples.py` (Theorem 3.3, Example B.1, Proposition 3.13, Proposition B.2, Corollary 3.12(i)); `e3_finitary.py 120 7` and `120 8 low` (Theorems 3.3 and 3.4, Proposition B.13 against brute force: 81/81

and 22/22 cases, Acc = Feat on 5804 and 1817 queries); `e4_anchor_random.py 70 S 2`, $S = 11, 12, 13$, and `60 14 2` (Theorem 3.8: prediction = feature truth 248/248; brute force decided 149 and agreed on 144, the other 5 being artifacts of its bounds); `e5_specializations.py` (Corollary 3.9: first-order 571/571; induction 495, Separation 495, Replacement 498, $\in$-induction 491); `e6_escalation.py 60 5` and `e6b_escalation_structured.py 1` (Theorem 3.11, Conjecture B.9); `e7_rates.py` (circular, see above) and `e10_rates_revision.py` (Theorem 3.10, Proposition B.4, ground templates); `e8_counterexamples.py` (Proposition B.2; the chains of Section 3.6); `e9_union.py` and `e15_ground_union.py` (Propositions B.11 and B.12); `e12_c11_refuted.py` (Proposition B.6); `e13_escalation_structure.py 3 40` (Proposition B.7, Lemma B.8); `e16_union_lower_bound.py` (Proposition B.10). The referee's scripts `t1`–`t10` check everything above except the items added in the revision, with full agreement (Feat = brute-force intersection on 125,109 queries; anchor prediction 540/540).
Scripts `e11` and `e14` check Theorem 6.16[single Thm H].

C Proofs for Section 4 (universal axioms)

This appendix gives the proofs for Section 4, in its notation, together with the statements moved out of the main text (Propositions C.1, C.4 and C.5 and Lemma C.3). The default is the λ convention: a 0-ary metavariable takes a term or formula without free bound variables (closed apart from parameters; a formula value may contain its own binders). We write bound variables as names; where indices matter (syntax (dB)) we say so. A *value* is what a substitution assigns to a metavariable.

C.1 Anchors

Proposition C.1 (Instance sets determine schemas)**.** *[proved; computed][cases A1] Assume* ($\mathrm{Rich_c}$)*. Let σ be a first-order schema with 0-ary term metavariables and τ any first-order schema. Then* $\mathrm{inst}(\sigma) \subseteq \mathrm{inst}(\tau)$ *iff every instance of σ at closed terms lies in* $\mathrm{inst}(\tau)$ *iff $\tau \succeq \sigma$, in each syntax.*

Proof. Only the step to $\tau \succeq \sigma$ needs proof. With $\theta(z_i) := t_i$ and $\theta'(z_i) := t'_i$ from ($\mathrm{Rich_c}$), the closed instances $\sigma\theta, \sigma\theta'$ satisfy (R) and (D), so $\mathrm{lgg}(\sigma\theta, \sigma\theta') \equiv \sigma$ (Lemma A.4); τ covers both, so $\tau \succeq \sigma$. □

Proof of Theorem 4.2. (ii)⇔(iii) is the recovery lemma (Lemma A.4) with $\theta_j(z_i) := t_{j,i}$. (ii)⇒(i): a covering τ has $\tau \succeq \mathrm{lgg}(D) \equiv \sigma_\varphi$. (i)⇒(ii): $\mathrm{lgg}(D)$ covers D, hence contains $\mathrm{inst_c}(\varphi)$, hence $\mathrm{lgg}(D) \succeq \sigma_\varphi$ by Proposition C.1; and $\sigma_\varphi \succeq \mathrm{lgg}(D)$ as σ_φ covers D. Only closed instances occur, so the syntax does not matter. □

Further corner cases of Theorem 4.2 [cases §1.3; experiments E1]. Several variables need (D): on diagonal data $\varphi(t, t)$ the lgg for $x + y = y + x$ is the sound specialization $z + z = z + z$. Repeated occurrences of x_i give equal columns and need no further event. Closed subterms of φ and bound occurrences of the name x_i give constant columns, which merge with an x_i-column only if (R) fails: for $x + S0 = Sx$, data terms $S0$ and 0 recover σ_φ, while $S0$ and $SS0$ give $S(z) + S0 = SS(z)$. Numerals have roots 0 and S only, so from numeral data every anchor of $z + 0 = z$ contains $0 + 0 = 0$, which is also an instance of $0 + z = z$. *Evidence* [cases q1_lgg (1), (2); referee r1]: on 141 736 data sets (pairs and triples from 12 closed terms, six formulas) "lgg $\equiv \sigma_\varphi$ iff (R)∧(D)" held throughout; the referee's code agreed on 36 000 data sets from 600 random formulas with binders.

The proof of Proposition 4.3(a) rests on one identity, which generalizes the two-point identity behind Lemma 6.12 [untagged Lemma 5.5, step 1] from one metavariable to several.

Lemma C.2 (Identity lemma). *[proved]*[untagged Lemma 5.5; cases A2′] *Let Z be a set of 0-ary metavariables (of term or formula sort), and let G, K be terms or formulas, possibly with binders, bound variables and parameters, in which members of Z occur only as leaves. Let $\theta_1, \dots, \theta_N$ assign to each $z \in Z$ a value of its sort without free bound variables (closed apart from parameters; a formula value may contain its own binders), such that* (R) *for each $z \in Z$ the roots of $\theta_1(z), \dots, \theta_N(z)$ are not all equal, and* (D) *for distinct $z, z' \in Z$ of the same sort some j has $\theta_j(z) \neq \theta_j(z')$. If $G\theta_j = K\theta_j$ for all j, then $G = K$.*

Proof. Induction on $|G| + |K|$. Substituting values for metavariable leaves changes nothing else; values contain no free bound variables, so no shifting or capture occurs under binders. *Case $G = z \in Z$.* If $K = z$, we are done. If $K = z' \in Z$ with $z' \neq z$, then $\theta_j(z) = \theta_j(z')$ for all j, which is impossible for different sorts and contradicts (D) otherwise. Otherwise the root of K is rigid: a function or relation symbol, a connective, a quantifier, a constant, a parameter or a bound variable. Then every $K\theta_j$ has that root, so all $\theta_j(z) = G\theta_j$ have the same root, contradicting (R). *Case $K \in Z$.* Symmetric. *Otherwise* both roots are rigid. Comparing $G\theta_1 = K\theta_1$ shows that they are the same symbol with the same binding structure, and the children of G and K satisfy the hypothesis. By induction the children are equal, so $G = K$. □

Proof of Proposition 4.3. (a) Let $\sigma \in \mathrm{FO}$ and $D = \{\sigma\theta_1, \dots, \sigma\theta_N\}$ with values without free bound variables satisfying (R) and (D); let $T \in \mathrm{DT}^\circ$ cover D. Write $D_j := \sigma\theta_j$, and for a position c of σ let $\kappa(c) := \sigma|_c$.

Step 1 (skeleton). All data carry σ's symbol at each rigid position of σ. At a metavariable position of σ, carrying z, the roots of the $\theta_j(z)$ are not all equal by (R). So the common prefix $C(D)$ of the data (Definition 3.1) is the set of rigid positions of σ, and its slots are the metavariable positions of σ. By the rigid-prefix lemma (Lemma 3.2(c)), the rigid skeleton of T lies in $C(D)$ with the same symbols, and every occurrence position of T is in $C(D)$ or is a slot. In particular every occurrence position c of T is a position of σ, and $\kappa(c)$ is defined.

Step 2 (bodies). Let M be a metavariable of T of arity n, with a pattern occurrence at c_1, say $M(u_1, \dots, u_n)$ with pairwise distinct bound variables u_m in scope at c_1. By the matching theorem (Theorem 2.5), its value in D_j is $\lambda\bar{h}.\,(D_j|_{c_1})[h_m/u_m]$, and $D_j|_{c_1}$ has no free bound variable outside $\{u_1, \dots, u_n\}$. Since $D_j|_{c_1} = \kappa(c_1)\theta_j$ and the values contain no free bound variables, the free bound variables of $\kappa(c_1)$ are those of $D_j|_{c_1}$, hence among the u_m. Let $B_M := \kappa(c_1)[h_m/u_m]$. It is a template body: it has no free bound variable, and its metavariables are the members of Z in $\kappa(c_1)$, all at leaves. Substituting θ_j commutes with abstracting the u_m, so $\lambda\bar{h}.B_M\theta_j$ is M's value in D_j.

Step 3 (other occurrences). Let $M(\bar{u}^s)$ be any occurrence of M, at c_s. By preservation (Lemma 2.4), $D_j|_{c_s} = (B_M\theta_j)[\bar{u}^s] = (B_M[\bar{u}^s])\theta_j$. The second equality holds because plugging the metavariable-free arguments $\bar{u}^s$ into holes and substituting values for metavariable leaves act on different leaves, and values without free bound variables need no shifting under the binders of B_M. Also $D_j|_{c_s} = \kappa(c_s)\theta_j$. So $G := B_M[\bar{u}^s]$ and $K := \kappa(c_s)$ contain the members of Z only as leaves and agree under every θ_j. By Lemma C.2, $B_M[\bar{u}^s] = \kappa(c_s)$.

Step 4. Let ρ be the template substitution $M \mapsto \lambda\bar{h}.B_M$ for all metavariables M of T. On the rigid skeleton of T, $T\rho$ and σ have the same symbols (Step 1). At every occurrence position c_s, $(T\rho)|_{c_s} = B_M[\bar{u}^s] = \kappa(c_s) = \sigma|_{c_s}$ (Step 3). Every position of T outside argument lists is in the skeleton or is an occurrence, so $T\rho = \sigma$ and $T \succeq \sigma$.

For the instance schema take $\sigma = \sigma_\varphi$ and $D \subseteq \mathrm{inst}_o(\varphi)$: the values are terms over $L \cup \mathrm{Par}$, without bound variables. Every covering $T \in \mathrm{DT}^\circ$ satisfies $T \succeq \sigma_\varphi$, hence $\mathrm{inst}(T) \supseteq \mathrm{inst}(\sigma_\varphi)$,

so D is an anchor for $\mathrm{inst}(\sigma_\varphi)$. Since σ_φ covers D and lies below every covering template, $\mathrm{Min}(D) = \{\sigma_\varphi\}$ up to $\equiv$.

(b) If (R) or (D) fails, $\mathrm{lgg}(D) \not\equiv \sigma_\varphi$ by Plotkin's recovery lemma (Lemma A.4). If $\mathrm{lgg}(D)$ contained $\mathrm{inst}_c(\varphi)$, Proposition C.1 would give $\mathrm{lgg}(D) \succeq \sigma_\varphi$, and $\sigma_\varphi \succeq \mathrm{lgg}(D)$ holds because σ_φ covers D; so $\mathrm{lgg}(D)$ misses a closed instance. It is a legitimate first-order pattern under the λ convention: its metavariables sit at disagreement positions of D, which lie at or below metavariable positions of σ_φ, where every datum has a closed term; so its values on D are closed terms, and it covers D. It lies in $\mathrm{FO} \subseteq \mathrm{DT}^\circ$.

(c) In $T = (f(S0) + f(0) = f(S0))$ no bound variable is in scope, so f has no pattern occurrence and $T \in \mathrm{SO}^\circ \setminus \mathrm{DT}^\circ$. $f := \lambda h.0$ gives $0+0=0$, and $f := \lambda h.h$ gives $S0+0=S0$; both data have different roots at z, so (R) holds. If $f := \lambda h.\beta$ gave $SS0 + 0 = SS0$, then $\beta[0] = 0$ and $\beta[S0] = SS0$. A body whose root is S, $+$, $\cdot$ or a parameter keeps that root after plugging, so $\beta[0] = 0$ forces $\beta \in \{0, h\}$, and then $\beta[S0] \in \{0, S0\}$.

Analogues of (c) with binders cases q1_dt; referee r5: a metavariable without a pattern occurrence can use different arguments on different data, the phenomenon that also makes SO° anchors for induction stronger (Proposition 5.17). (i) $\varphi = \forall y\,(x + y = y + x)$, data at 0 and $S0$, $T_1 = \forall y\,(G(0, S0) + G(y, y) = y + F(y))$: take $G := \lambda h_1 h_2.h_1$, $F := \lambda h.0$, resp. $G := \lambda h_1 h_2.h_2$, $F := \lambda h.S0$. The occurrence $F(y)$ is a pattern occurrence, but G has none. For the instance at $SS0$, $G(y, y) = y$ forces G's body to be a hole, so $G(0, S0) \in \{0, S0\} \neq SS0$. (ii) $\varphi = (x + y = y + x)$, data $(0, S0)$ and $(S0, 0)$, $T_2 = (G(0, S0) = G(S0, 0))$: take $G := \lambda h_1 h_2.(h_1 + h_2)$, resp. $\lambda h_1 h_2.(h_2 + h_1)$. For the instance $0 + SS0 = SS0 + 0$ the body would be $\beta_1 + \beta_2$ with $\beta_1[0, S0] = 0$, forcing $\beta_1 \in \{0, h_1\}$, and $\beta_1[S0, 0] = SS0$, which is impossible. □

Evidence for Proposition 4.3 cases q1_dt, q1_lgg (5); referee r1 (3), r5: a bounded search over covering DT° templates agreed with (R)∧(D) on 99/99 pairs (five formulas, three with binders), the referee's on 60/60, and every formula had an SO° non-anchor satisfying the DT° events. The search is bounded; the statement is proved.

Remarks. (1) With one metavariable, Lemma C.2 is the two-point identity, and Proposition 4.3(a) is Lemma 6.12, including a 0-ary formula metavariable. (2) For closed data, (a) also follows from a direct case analysis of the positions of $\kappa(c_1)$ and $\kappa(c_s)$, including hole positions that receive closed-term arguments at non-pattern occurrences cases Prop A2′. The proof above reorganizes it around Lemma C.2 and allows parameters in the values; it is also the "if" half of Corollary 3.9(a).

C.2 Rates

Theorem 4.4(b) in full. For $k = 2$,

$$P_N = \textstyle\sum_f a_f^N + \sum_g b_g^N + c^N - \sum_{f,g} d_{fg}^N - \sum_f e_f^N,$$

with $a_f = \Lambda(\mathrm{root}(t_1) = f)$, $b_g = \Lambda(\mathrm{root}(t_2) = g)$, $c = \Lambda(t_1 = t_2)$, $d_{fg} = \Lambda(\mathrm{root}(t_1) = f, \mathrm{root}(t_2) = g)$, $e_f = \Lambda(\mathrm{root}(t_1) = f,\, t_1 = t_2)$. For independent components with term law μ and root law p: $P_N = 2S - S^2 + c^N - C$, with $S = \sum_f p_f^N$, $C = \sum_f c_f^N$, $c_f = \sum_{\mathrm{root}(t)=f} \mu(t)^2$.

Numbers cases q1_lgg (4); referee r1 (5). For a Galton–Watson law with root law $0{:}\,.4$, $S{:}\,.3$, $+{:}\,.2$, $\cdot{:}\,.1$ (depth ≤ 4, independent components), the exact values at $N = 2, 4, 6, 10$ are .300, .035, .0049, .00011 ($k = 1$) and .52, .070, .0098, .00022 ($k = 2$), in agreement with 20 000 Monte Carlo runs. Here $\rho_i = 0.5$ (split $\{0, \cdot\}$ against $\{S, +\}$) and $\Lambda(t_1 \neq t_2) > 0.5$, so $\rho = 0.5$; bound (c) is below 0.01 from $N = 11$ ($k = 1$, $2e^{-N/2}$) and $N = 13$ ($k = 2$, $5e^{-N/2}$) in its exponential form, and from $N = 8$ and $N = 9$ in its form $(2k + \binom{k}{2})2^{-N}$.

Proof of Theorem 4.4. The data are i.i.d. $\sim \Lambda$. The failure event is the union of the events $\neg(\mathrm{R}_i)$ and $\neg(\mathrm{D}_{ii'})$.

(a) $\neg(\mathrm{R}_1)$ is the disjoint union over f of A_f: "all N roots are f", with $\Pr[A_f] = p_f^N$.

(b) Let $A = \bigcup_f A_f$ (all roots of t_1 equal), $B = \bigcup_g B_g$ (all roots of t_2 equal, to g), and E: "$t_1 = t_2$ in every datum". The A_f are pairwise disjoint, and so are the B_g. $\Pr[A] = \sum_f a_f^N$, $\Pr[B] = \sum_g b_g^N$, $\Pr[E] = c^N$, $\Pr[A \cap B] = \sum_{f,g} d_{fg}^N$. On E the roots of t_1 and t_2 coincide, so $\Pr[A \cap E] = \sum_f e_f^N = \Pr[B \cap E]$, and $A_f \cap B_g \cap E = \emptyset$ for $f \neq g$ while $A_f \cap B_f \cap E = A_f \cap E$; so $\Pr[A \cap B \cap E] = \sum_f e_f^N$. Inclusion–exclusion gives

$$\Pr[A \cup B \cup E] = \textstyle\sum_f a_f^N + \sum_g b_g^N + c^N - \sum_{f,g} d_{fg}^N - 2\sum_f e_f^N + \sum_f e_f^N,$$

which is the formula. For independent components with term law μ and root law p: $a_f = b_f = p_f$, $d_{fg} = p_f p_g$ (so $\sum_{f,g} d_{fg}^N = S^2$), $c = \sum_t \mu(t)^2$ and $e_f = c_f$.

(c) Union bound: $P_N \leq \sum_i \Pr[\neg(\mathrm{R}_i)] + \sum_{i<i'} \Pr[\neg(\mathrm{D}_{ii'})]$ with $\Pr[\neg(\mathrm{R}_i)] = \sum_f p_{i,f}^N$ and $\Pr[\neg(\mathrm{D}_{ii'})] = c_{ii'}^N$. Let G be a best split for i. The events "all roots of t_i are f", for $f \in G$, are disjoint subsets of "all roots of t_i lie in G", so $\sum_{f \in G} p_{i,f}^N \leq \Lambda(\mathrm{root}(t_i) \in G)^N$; likewise for the complement. Both $\Lambda(\mathrm{root}(t_i) \in G)$ and $\Lambda(\mathrm{root}(t_i) \notin G)$ are at most $1 - \rho_i$, so $\sum_f p_{i,f}^N \leq 2(1-\rho_i)^N \leq 2(1-\rho)^N$, and $c_{ii'}^N = (1 - \Lambda(t_i \neq t_{i'}))^N \leq (1-\rho)^N$. Summing gives $(2k + \binom{k}{2})(1-\rho)^N$, and $1 - \rho \leq e^{-\rho}$. □

C.3 Parameters, guards and capture

Proof of Proposition 4.5. (a) $\varphi(\bar{p})$ substitutes distinct parameters for the free occurrences of the distinct variables $\bar{x}$; as φ has no parameters, the universal closure of $\varphi(\bar{p})$ is $\forall \bar{x} \varphi$ up to the names of bound variables. A member $\varphi(\bar{t})$ of $\mathrm{inst}_o(\varphi)$ follows from $\forall \bar{x} \varphi$ by $\forall$-elimination (the terms contain no bound variable, so the substitution is free), and its closure follows by generalization over its parameters. Conversely $\forall \bar{x} \varphi$ follows from $\varphi(\bar{p})$ by generalization over $\bar{p}$.

(b) Let $\mathcal{H}$ be $H_1(\mathrm{FO})$ or $H_1(\mathrm{DT}^\circ)$ and $\tau \in \mathcal{H}$ with $\mathrm{inst}(\tau) \supseteq \mathrm{inst}_c(\varphi)$. The key point is that $\mathrm{inst}(\tau)$ then contains two closed instances forming an anchor: with $\theta(z_i) := t_i$ and $\theta'(z_i) := t_i'$ from $(\mathrm{Rich_c})$, the closed instances $\sigma_\varphi\theta, \sigma_\varphi\theta'$ satisfy (R) and (D), so $\tau \succeq \sigma_\varphi$ by Theorem 4.2 (for FO) or Proposition 4.3(a) (for DT°). As $\sigma_\varphi \in \mathrm{FO} \subseteq \mathrm{DT}^\circ$ contains $\mathrm{inst}_c(\varphi)$, $\mathrm{cl}_{\mathcal{H}}(\mathrm{inst}_c(\varphi)) = \mathrm{inst}(\sigma_\varphi)$, which is $\mathrm{inst}_o(\varphi)$ under the λ convention with parameters. The inclusion $\mathrm{inst}_c(\varphi) \subseteq \mathrm{inst}_o(\varphi)$ is strict: every z_i occurs in σ_φ, so the text of $\varphi(\bar{p})$ determines the value p_i at z_i, which is not closed. Once the data contain an anchor, the cautious verifier accepts the closure (Lemma 2.8(b)), in particular $\varphi(\bar{p})$. In (Nm) and (dB) the same argument gives $\mathrm{cl}_{H_1(\mathrm{FO})}(\mathrm{inst}_c(\varphi)) = \mathrm{inst}(\sigma_\varphi)$, since Theorem 4.2 holds in every syntax, and $\mathrm{inst}(\sigma_\varphi) \supseteq \mathrm{Cap}(\varphi)$ by Proposition 4.6(a).

(c) An instance of σ_φ whose values are all closed is a closed instance; closed values contain no variable, so there is no capture. Hence $\mathrm{inst}(\sigma_\varphi, \{\mathrm{closed}(z_i)\}_i) = \mathrm{inst}_c(\varphi)$ in every syntax. By the most-specific-guard rule the learner's hypothesis on D is $\mathrm{lgg}(D)$ with every guard of the family that holds on all data. On closed data the values of every metavariable of $\mathrm{lgg}(D)$ are subterms of closed terms, so closed is kept for all of them. Then $\varphi(\bar{p})$ is not accepted: at a metavariable position q of σ_φ the hypothesis either has a metavariable at q (metavariables of $\mathrm{lgg}(D)$ lie at or below the metavariable positions of σ_φ, since all data agree above them), whose value in $\varphi(\bar{p})$ contains a parameter and violates closed, or is rigid at q with the root shared by all data's closed terms, which differs from a parameter. A capture instance is excluded the same way, because its value at some such position contains a bound variable. A datum with a parameter at z_i violates closed for the metavariable of that column and deletes it. □

Computed cases q1_lgg (3): from $0+0=0$, $0+S0=S0$ the guarded learner learns $0+z=z$ with closed(z) and rejects $0+p=p$; after the datum $0+(p+S0)=p+S0$ it accepts $0+q=q$, i.e. $\forall x\,(0+x=x)$. In E1, the guardless DT°_{F} learner on $0+0=0$, $1+0=1$ accepts $w_0+0=w_0$, the axiom in parameter form, but not the string $\forall x\,(x+0=x)$, which is a different datum (Section 2.1).

Lemma C.3 (When capture instances exist)**.** *[proved; computed]*cases Prop A4′(b) $\mathrm{Cap}(\varphi)\neq\emptyset$ *iff: in (Nm) with sentences as data, some name is bound at* every *free occurrence of some x_i; in (Nm) with free names read as parameters, some free occurrence of some x_i is in the scope of a binder; in (dB), every free occurrence of some x_i lies under a binder.*

Proof. (Nm), data are sentences: if a name y is bound at every free occurrence of x_i, then $\theta(z_i):=y$ (other values closed) gives a sentence in $\mathrm{Cap}(\varphi)$. Conversely, if a sentence $\sigma_\varphi\theta\in\mathrm{Cap}(\varphi)$ has the captured name y in $\theta(z_i)$, then every copy of $\theta(z_i)$, one at each free occurrence of x_i, contains y, and since the result has no free names, y is bound at each of them. (Nm), free names read as parameters: $\theta(z_i):=y$ with y bound above some occurrence is an admissible datum (copies of y elsewhere are read as a parameter); conversely capture needs a binder above an occurrence. (dB): if every free occurrence of x_i has binder depth ≥ 1, then $\theta(z_i):=\#0$ is well formed at all of them and captured; conversely a captured value contains some $\#m$, and well-formedness at each occurrence o requires m to be less than the depth of o, so every depth is ≥ 1. □

For example, $x=0\wedge\forall y\,(x=y)$ has no capture sentence in (Nm) and no capture instance in (dB), since its first occurrence of x is under no binder; capture formulas exist in (Nm) when free names are read as parameters.

Proof of Proposition 4.6. (a) Let $\sigma_\varphi\theta$ be an instance (a sentence, or a formula whose free names are parameters). Suppose no value is captured. In (Nm) the substitution is then free for every x_i, and $\sigma_\varphi\theta=\varphi(\bar{\theta})$ is a genuine instance: closed if the values are variable-free, and otherwise at terms whose variable names are bound at no occurrence. Such names are free in the result, so they are parameters (impossible if data must be sentences). In (dB), an index $\#m$ in a value at an occurrence o refers to an enclosing binder if m is less than the binder depth of o (then the value is captured), and is loose otherwise (then the result is not well formed). So a non-captured value has no index, and $\sigma_\varphi\theta$ is a closed or parameter instance. Otherwise $\sigma_\varphi\theta\in\mathrm{Cap}(\varphi)$. The reverse inclusions hold by definition. Disjointness: every z_i occurs in σ_φ, so the text $\sigma_\varphi\theta$ determines each $\theta(z_i)$ (the subtree at an occurrence of z_i), and whether it is captured is a property of that text. The values of a genuine open instance contain no bound variable, so they are not captured.

(b) This is Lemma C.3.

(c) $\exists y\,\neg(y=y)$ is false in every structure, while $\exists y\,\neg(y=t)$ holds in every structure with at least two elements, whatever t denotes. $\forall x\exists y\,(y=Sx)$ is logically valid (take $y:=Sx$), while $\exists y\,(y=Sy)$ is false in $\mathbb{N}$, where S has no fixed point.

(d) By Proposition 4.5(b), $\mathrm{cl}_{H_1(\mathrm{FO})}(\mathrm{inst}_c(\varphi))=\mathrm{inst}(\sigma_\varphi)\supseteq\mathrm{Cap}(\varphi)$; the anchor argument of Theorem 4.2 uses only closed instances, so it holds in (Nm) and (dB). Once the data contain an anchor the cautious verifier accepts the closure (Lemma 2.8(b)). With φ as in (c), the accepted capture instance $\exists y\,\neg(y=y)$ is false in every structure.

(e) Closed values satisfy both guards, so closed data keep both, and $\mathrm{inst}(\sigma_\varphi,\{\mathrm{closed},\mathrm{nocap}\})=\mathrm{inst}_c(\varphi)$ because closedness implies nocap. A genuine parameter instance satisfies nocap, and violates closed(z_i) exactly for the z_i whose value contains a parameter, so only those guards closed(z_i) are deleted; for $k\geq 2$ a datum such as $\varphi(p,0)$ keeps closed(z_2). Once closed(z_i) is deleted for every i, $\mathrm{inst}(\sigma_\varphi,\{\mathrm{nocap}(z_i)\}_i)$ is the set of instances with no captured value, which by

(a) is $\mathrm{inst}_c(\varphi) \cup \mathrm{inst}_o(\varphi) = \mathrm{inst}_o(\varphi)$. Genuine data never violate nocap, so it is never deleted, and capture instances, which violate it, are never accepted. □

Computed cases q1_capture; referee F1, F2: for $\exists y\, \neg(y = x)$, $\mathrm{lgg}(\varphi(0), \varphi(S0)) = \exists y\, \neg(y = z)$ has the instance $\exists y\, \neg(y = y)$, false in all equality structures of sizes 1–5; for seven formulas the decomposition of Proposition 4.6(a) and the criterion of Lemma C.3 were checked on all substituted terms of size ≤ 3; the guards of (e) behave as stated, while the unguarded verifier accepts $\exists y\, \neg(y = y)$. The de Bruijn *first-order* encoding does not prevent capture, which there happens by index: the lgg is $\exists \neg(\#0 = z)$, with capture instance $\exists \neg(\#0 = \#0)$ ($z := \#0$), which the learned guard "no loose index in z" rejects.

C.4 The ω-gap: details

Proof of Proposition 4.7. M_0 is a substructure because L has a constant. (ii)⇒(i): if all $\varphi(\bar{t})$ hold in M, they hold in M_0 by elementarity, and the elements of M_0 are the t^M; so $M_0 \models \forall \bar{x} \varphi$, hence $M \models \forall \bar{x} \varphi$, again by elementarity. The converse direction of (i) is $\forall$-elimination. (i)⇒(ii) is the Tarski–Vaught criterion (Tarski and Vaught, 1957): if $M \models \exists x\, \psi(x, \bar{s})$ with $\bar{s}$ closed but no closed t has $M \models \psi(t, \bar{s})$, then (i) for $\neg\psi(x, \bar{s})$ gives $M \models \forall x\, \neg\psi(x, \bar{s})$, a contradiction. In a model of true arithmetic $M_0 \cong \mathbb{N}$, and M and $\mathbb{N}$ satisfy the same sentences with numerals, so $M_0 \preceq M$. □

Example 4.8(a). Closed terms over $\{0, 1, +, -, \cdot\}$ denote integers, by induction on terms. For an integer n, $n \cdot n \neq 2$, so $\mathbb{R} \models \neg(t \cdot t = 1+1)$ for every closed t; $\sqrt{2}$ refutes $\forall x\, \neg(x \cdot x = 1+1)$. In a real closed field every positive element has a square root, and $1+1 > 0$, so $\mathrm{RCF} \vdash \exists x\, (x \cdot x = 1+1)$. The rationals form an ordered field without $\sqrt{2}$, so $\mathrm{OF} \nvdash \exists x\, (x \cdot x = 1 + 1)$. Here $M_0 \cong \mathbb{Z}$, which is not elementary in $\mathbb{R}$, as Proposition 4.7 requires.

Example 4.8(b). $\emptyset$ is hereditarily finite, and HF is closed under $\{a, b\}$ and $a \cup b$, so every closed term denotes an HF set. Conversely, by $\in$-induction every HF set is denoted: $\{a\} = p(a, a)$ and $\{a_1, \ldots, a_n\} = u(p(a_1, a_1), \{a_2, \ldots, a_n\})$. A set x with $\mathrm{Ind}(x)$ contains $\emptyset$, $\{\emptyset\}$, $\{\emptyset, \{\emptyset\}\}, \ldots$, which are pairwise distinct, so it is infinite; hence $V \models \neg\mathrm{Ind}(t)$ for every closed t. The axiom of Infinity provides an inductive set, so $\forall x\, \neg\mathrm{Ind}(x)$ is false and refuted by ZF. Each closed instance is probably provable in very weak set theories; only its truth is used here.

Example 4.8(c). Assuming PA consistent, $\mathsf{PA} \nvdash \mathrm{Con}(\mathsf{PA})$, so $\mathsf{PA} + \neg\mathrm{Con}(\mathsf{PA})$ has a model M. A closed term t denotes a standard number n, and $\neg\mathrm{Prf}(t, \ulcorner\bot\urcorner)$ is a true Δ_0 sentence because n codes no proof of $\bot$. True Δ_0 sentences in $0, S, +, \cdot$ (bounded quantifiers defined from $+$, since Q says nothing about $<$) are provable in Q (see e.g. Hájek and Pudlák, 1993), so they hold in M, while $M \models \exists x\, \mathrm{Prf}(x, \ulcorner\bot\urcorner)$. So M_0, the standard part, is not elementary in M. (A nonstandard model of true arithmetic, by contrast, satisfies (i) of Proposition 4.7; what fails here is $M \equiv \mathbb{N}$.)

Example 4.9: the model of Q cases Ex. A8; q1_qmodel; referee r1 (4). That $\mathsf{Q} \nvdash \forall x\, (0 + x = x)$ is well known; we give an explicit model. The domain is $\mathbb{N} \cup \{a, b\}$ with $a, b \notin \mathbb{N}$; n, m range over $\mathbb{N}$. The operations are standard on $\mathbb{N}$, and otherwise:

- successor: $Sa = a$, $Sb = b$;
- addition: $n + a = n + b = b$; $a + n = a$, $b + n = b$; $a + a = a + b = a$; $b + a = b + b = b$;
- multiplication: $x \cdot 0 = 0$ for all x; $a \cdot n = b \cdot n = b$ for $n \geq 1$; $n \cdot a = n \cdot b = a$ for $n \geq 1$; $0 \cdot a = 0 \cdot b = 0$; $a \cdot a = a \cdot b = b \cdot a = b \cdot b = a$.

(Q's axioms do not mention $<$, which may be interpreted arbitrarily.) We check Q1–Q7.

- Q1, Q2, Q3: S is injective, 0 is not a successor, and every $x \neq 0$ is a successor ($n+1 = Sn$, $a = Sa$, $b = Sb$).
- Q4, $x + 0 = x$: standard on $\mathbb{N}$; $a + 0 = a$ and $b + 0 = b$.
- Q5, $x + Sy = S(x + y)$. For $y = m$: if $x \in \{a, b\}$ then $x + m = x$ and $Sx = x$; otherwise standard. For $y \in \{a, b\}$: $Sy = y$, and every $x + y$ lies in $\{a, b\}$, which S fixes.
- Q6, $x \cdot 0 = 0$: by definition.
- Q7, $x \cdot Sy = x \cdot y + x$. For $y = m$ and $x = a$: $a \cdot (m + 1) = b$, and $a \cdot m + a$ is $0 + a = b$ if $m = 0$ and $b + a = b$ if $m \geq 1$. For $x = b$: $b \cdot (m + 1) = b$, and $b \cdot m + b$ is $0 + b = b$ or $b + b = b$. For $y \in \{a, b\}$: $Sy = y$, so we need $z + x = z$ for $z = x \cdot y$. If $x = 0$, $z = 0$ and $0 + 0 = 0$. If $x = n \geq 1$, $z = a$ and $a + n = a$. If $x = a$, $z = a$ and $a + a = a$. If $x = b$, $z = a$ and $a + b = a$.

In this model $0 + a = b \neq a$, and every closed instance $0 + t = t$ holds, since closed terms denote standard numbers. So Q together with all closed instances of $0 + x = x$ does not prove $\forall x\,(0 + x = x)$. The script `q1_qmodel.py` checks all seven axioms on all tuples from $\{0, \ldots, 40\} \cup \{a, b\}$, and the referee's independent code on $\{0, \ldots, 60\} \cup \{a, b\}$.

C.5 Consequences for a learner

Proposition C.4 (What is soundly learnable)**.** *[proved]*[cases A10] (1) *From closed instances the soundly learnable object is* $\mathrm{inst}_c(\varphi)$*, with anchors* (R)∧(D) *and the rates of Theorem 4.4. If parameters are admissible values, the cautious single-schema learner over* $H_1(\mathrm{FO})$ *or* $H_1(\mathrm{DT}^\circ)$ *is sound for it only with a closedness guard. In (Nm) and (dB), even without parameters, it must also exclude capture instances whenever* $\mathrm{Cap}(\varphi) \neq \emptyset$ *(Lemma C.3), by* nocap *or by* closed*, which implies it; under the* λ *convention this is automatic.* (2) *The* ω*-step is truth-safe in the intended* M *for every* φ *iff* $M_0 \preceq M$ *(Proposition 4.7).* (3) *In general it is not derivable from the instances in first-order logic, even over* Q *(Example 4.9); it is an* ω*-rule step.* (4) *If the community uses instances at parameters, as proof-assistant practice does when it applies* $\forall x$*,* $\varphi\, x$ *to a variable, the target is* $\mathrm{inst}_o(\varphi)$ *and* $\forall \bar{x} \varphi$ *is obtained soundly (Proposition 4.10); in (Nm) and (dB)* nocap *is still needed when* $\mathrm{Cap}(\varphi) \neq \emptyset$*.*

Proof. (1) Anchors: Theorem 4.2 and Proposition 4.3; rates: Theorem 4.4. Without a closedness guard and with parameters admissible, the closure of $\mathrm{inst}_c(\varphi)$ contains $\varphi(\bar{p})$ (Proposition 4.5(b)); the guard makes $\mathrm{inst}_c(\varphi)$ realizable and is learned from closed data (Proposition 4.5(c)). In (Nm) and (dB), an unguarded learner accepts capture instances after an anchor whenever $\mathrm{Cap}(\varphi) \neq \emptyset$ (Proposition 4.6(d)), and these are not closed instances; closed excludes them (Proposition 4.5(c)), and so does nocap (Proposition 4.6(e)). If $\mathrm{Cap}(\varphi) = \emptyset$, as under the λ convention, no such guard is needed. (2) The statement "for every φ whose closed instances all hold in M, $M \models \forall \bar{x} \varphi$" is condition (i) of Proposition 4.7, hence equivalent to $M_0 \preceq M$; Example 4.8 gives the negative cases. (3) Example 4.9: Q plus all closed instances of $0 + x = x$ does not prove $\forall x\,(0 + x = x)$. (For other φ, e.g. logically valid ones, $\forall \bar{x} \varphi$ may well be derivable.) (4) If the target is $\mathrm{inst}_o(\varphi)$, it equals $\mathrm{inst}(\sigma_\varphi)$ under the λ convention and is realizable, with anchors as in Proposition 4.3(a); in (Nm) and (dB) it is realizable as σ_φ with the guards $\mathrm{nocap}(z_i)$ (Proposition 4.6(e)). It contains $\varphi(\bar{p})$, from which $\forall \bar{x} \varphi$ follows by generalization (Proposition 4.10). □

Proposition C.5 (Refutation and the ω-gap)**.** *[proved]*[cases A11] *Let* $\forall \bar{x} \varphi$ *be false in the intended* M *while all closed instances are true.* (a) *No refutation that ends by evaluating closed instances*

condemns $\forall\bar{x}\varphi$. (b) *A coherence refutation from designated true premises* Γ *exists iff* $\Gamma \cup \{\forall\bar{x}\varphi\}$ *is inconsistent (*RCF *and* ZF *refute the examples of Example 4.8(a),(b);* OF *does not refute (a)).* (c) *In a term-generated* M *such as* $\mathbb{N}$ *this cannot happen (Proposition 4.7); if the channel decides the closed instances of* φ *(as a full* Δ_0 *evaluator does for* Δ_0 *formulas), search finds a false one.* (d) *Capture instances are refuted by pure logic when logically false; when false only in* M*, a designated theory must prove the negation:* $\mathsf{Q} \nvdash \forall y\,\neg(y = Sy)$ *(*$Sa = a$ *in the model of Example 4.9),* PA *proves it. No evaluation of closed instances bears on them.*

Proof. (a) Every closed instance is true in M, so an evaluation of one returns "true" and refutes nothing. (b) A coherence refutation is a derivation of $\bot$ from $\Gamma \cup \{\forall\bar{x}\varphi\}$, which by completeness exists iff this set is inconsistent. RCF proves $\exists x\,(x \cdot x = 1 + 1)$; $\mathbb{Q}$ is a model of OF $+ \forall x\,\neg(x \cdot x = 1 + 1)$; ZF proves $\exists x\,\mathrm{Ind}(x)$ (Appendix C.4). (c) If M is term-generated, $M_0 = M$, so by Proposition 4.7 a false $\forall\bar{x}\varphi$ has a false closed instance. Enumerating closed tuples and deciding each instance finds one at a finite stage. (The same holds whenever $M_0 \preceq M$.) (d) $\neg\exists y\,\neg(y = y)$ is logically valid. In the model of Example 4.9, $Sa = a$, so $\mathsf{Q} \nvdash \forall y\,\neg(y = Sy)$. PA proves it by induction on y: $0 \neq S0$ by Q1, and $y \neq Sy$ implies $Sy \neq SSy$ by Q2. A capture instance is neither a closed nor an open instance (Proposition 4.6(a)), so no evaluation of closed instances concerns it. □

C.6 Computations

The computations cited in Section 4 are the following scripts (directory `research/tracks/cases`): `q1_lgg.py` for Proposition C.1, Theorem 4.2 (141 736 data sets), Proposition 4.5(c), Theorem 4.4 and Proposition 4.3(c), about 85 s; `q1_dt.py` with the bounded search `dt_search.py` for Proposition 4.3, 7 s; `q1_capture.py` for Proposition 4.6 and Lemma C.3; `q1_qmodel.py` for Example 4.9; and the referee's independent `r1_q1.py` and `r5_q1_dt.py` in its subdirectory `referee_code`. Experiment E1 is `experiments/e1_universal.py`, run from `code` (48 s, output `code/results/e1_universal.md`). Each script is run as `python3` *script* from its directory and writes an output file named after it; the numbers quoted are those recorded in the final research notes.

D Proofs for Section 5

This appendix gives the proofs for Section 5, in the order of that section, the theory of pattern templates with several metavariables (Theorem D.3), two results on induction that the main text summarizes (Propositions D.6 and D.7), and the computational evidence (Appendix D.6). Proofs are adapted from the research records named in the source tags and were re-checked in transcription.

D.1 Preliminaries

For a set D of sentences, $C(D)$ is the set of positions at which all data have the same symbol, with all ancestors also in $C(D)$; a *slot* of D is a position outside $C(D)$ whose parent is in $C(D)$. For a template T, $\mathrm{skel}(T)$ is the set of positions outside metavariable occurrences and their arguments.

Lemma D.1 (Rigid prefix)**.** *[proved]*[prior induction C2(a); single Lemma P] *Let* $T \in \mathrm{SO}^\circ$ *and* $D \subseteq \mathrm{inst}(T)$. *Every datum has, at every position of* $\mathrm{skel}(T)$*, the symbol that* T *has there. Hence* $\mathrm{skel}(T) \subseteq C(D)$*, and every occurrence of a metavariable of* T *lies in* $C(D)$ *or is a slot of* D*. (This is the* SO° *form of Lemma 3.2(c).)*

Proof. In SO$^\circ$ all occurrences are rigid and have metavariable-free arguments, so an instance replaces each occurrence by a plugged body and leaves skel(T) unchanged. The parent of an occurrence lies in skel(T). □

Over $L_\in$ with parameters, a metavariable-free argument term is a variable or a parameter, i.e. a leaf. Hence plugging arguments into the holes of a body β does not change its tree shape, and every non-hole position of β keeps its symbol in $\beta[\bar{u}]$. We read bound variables in locally nameless style, so that "v is bound above c" and "v is bound inside the subtree at c" are well defined. $\top := \forall w(w = w)$ and $\bot := \neg\top$ are closed.

D.2 Encodings: proofs for Section 5.2

Proof of Theorem 5.2. (a) Every occurrence in Table 6 applies P to distinct variables bound above it.

(b) "If". Let $N := N_1 = \cdots = N_m$ and $\sigma := F[Z, \ldots, Z]$ with the guards $G = \{\mathrm{fresh}(v, Z) : v$ frame-bound, $v \notin N\}$. Let $\sigma\theta$ satisfy G, with $\psi := \theta(Z)$. The frame-bound names free in ψ lie in N; its other free names are parameters. Let β be ψ with its free occurrences of the names in N replaced by the holes, in argument order. At every occurrence the arguments are the same names, each bound there by the binder that the template intends, so $\beta[\bar{a}_k] = \psi$ for all k and $\sigma\theta = T^*[P := \lambda\bar{z}.\beta]$. Conversely $T^*[P := \lambda\bar{z}.\beta] = F[\psi, \ldots, \psi]$ with $\psi = \beta[\bar{a}_1]$, whose free frame-bound names lie in N, so it satisfies G.

"Only if". Suppose $N_k \neq N_l$ and $\mathrm{inst}(\sigma, G) = \mathrm{inst}(T^*)$ for a first-order schema σ, all of whose metavariables occur, and freshness guards G. Let s_1, s_2 be the instances with bodies $\varphi_1 := \bigwedge_{i\le n}(z_i = z_i)$ and $\varphi_2 := \neg\varphi_1$, and $L := \mathrm{lgg}(s_1, s_2)$. Frame positions are common. At each P-position the roots differ, so L has a metavariable there. The columns at occurrences k and l differ, because φ_1 mentions every argument and the tuples differ at some place. So L has distinct metavariables $Z_k \neq Z_l$ there. Since $s_1, s_2 \in \mathrm{inst}(\sigma)$, $\sigma \succeq L$, say $L = \sigma\rho$. Let s_3 be L with $Z_k := \top$ and every other metavariable $:= \bot$; then $s_3 = \sigma(\rho\theta_3)$. For each metavariable Y of σ, the free names of $\rho\theta_3(Y)$ are those of the rigid part of $\rho(Y)$, which are also free in $\rho\theta_1(Y)$, the value of Y in s_1 (first-order matching is unique). Freshness guards are preserved when free names are removed, and s_1 satisfies G; so $s_3 \in \mathrm{inst}(\sigma, G)$. But $s_3 \notin \mathrm{inst}(T^*)$. If $\beta[\bar{a}_k]$ is the closed $\top$, then β has no hole, since a hole would leave the variable $a_{k,i}$ free in the slot; so $\beta = \top$, and occurrence l would also read $\top$, not $\bot$.

(c) The same proof with indices in place of names. Loose-index guards are also preserved when loose indices are removed. At every occurrence the loose indices of the value must be smaller than its binder depth, so well-formedness implies $\mathrm{loose}(Z) \subseteq \{0, \ldots, d-1\}$ and nothing more. Since $I_1 = I_k$ for all k, $I_1 \subseteq \{0, \ldots, d-1\}$; so the guard $\mathrm{loose}(Z) \subseteq I_1$ is implied iff $I_1 = \{0, \ldots, d-1\}$. □

For Table 11: in de Bruijn notation Sep reads $P(\#0, \#2)$ at depth 3, so the guard "#1 (that is, y) not loose" is needed. Sep$_\mathrm{J}$ reads $P(\#0)$ at depth 3. All three occurrences of EInd read $P(\#0)$, at depths 2, 1, 1. Coll and Repl$_{\exists!}$ read $P(\#1, \#0, \#2)$ in the antecedent and $P(\#1, \#0, \#3)$ in the consequent. Repl$_\mathrm{sp}$ reads $P(\#1, \#0, \#2)$, $P(\#2, \#0, \#3)$, $P(\#1, \#0, \#3)$. Repl$_\mathrm{J}$ reads $P(\#2, \#1)$, $P(\#2, \#0)$, $P(\#0, \#1)$. With $\varphi(x, y)$ only, Coll reads $P(\#1, \#0)$ twice at depths 3 and 4. There #2 is A in the antecedent and Y in the consequent, so the guard $\mathrm{loose}(Z) \subseteq \{\#0, \#1\}$ is needed: without it, $Z := (\#0 = \#2)$ reads $y = A$ in the antecedent and $y = Y$ in the consequent, and the instance forces $Y \in Y$.

schema	named first-order (textbook names)	de Bruijn first-order	named, α-varied
Sep	yes, guard fresh(y, Z)	yes, guard "#1 not loose" (not implied)	yes
Sep_J	yes, guards fresh(X, Z), fresh(Y, Z)	yes, guard loose$(Z) \subseteq \{\#0\}$ (not implied)	yes
EInd	**no** (y vs. x)	**yes**, guard implied by well-formedness	no
Coll, $\text{Repl}_{\exists!}$	yes, guard fresh(Y, Z)	**no** (A is #2 vs. #3)	**no**
Repl_sp, Repl_J	no (y vs. u)	no	no
Coll, $\text{Repl}_{\exists!}$, $\varphi(x, y)$ only	yes, guards fresh(Y, Z), fresh(A, Z)	yes, guard loose$(Z) \subseteq \{\#0, \#1\}$ (not implied)	no

Table 11: Is the schema a guarded first-order pattern? (By Theorem 5.2 and Proposition 5.3; every schema is in PAT.) Without its guard, every guarded entry has a false instance (Proposition 5.5).

Proof of Proposition 5.3. cases Prop B1α The hypothesis class is first-order schemas over the named signature, with formula metavariables and *name metavariables* (at binder-name positions and variable leaves), and guards fresh(v, Z) and distinct(v, w). "If": let $b_1, \ldots, b_K$ be the frame's binders. Let σ be the frame with the name of b_k (at the binder and at every leaf it binds) replaced by a name metavariable v_k, and every occurrence of P replaced by one formula metavariable Z. Its guards are distinct(v_k, v_l) for $k \neq l$, and fresh(v_k, Z) for every b_k that is not an argument binder of P. An instance gives the binders distinct names and Z a formula ψ whose free names are argument names or names the frame does not use. Let β be ψ with the argument names replaced by holes. Every occurrence applies P to the same binders, hence to the same names, so every slot reads $\beta[\text{names}] = \psi$, and the instance is an α-variant of $T^*[P := \lambda\bar{z}.\beta]$. The converse is read off in the same way. "Only if": if occurrences k, l differ at argument place i (binders $b \neq b'$), fix one naming with all frame binders distinct and argue as in Theorem 5.2. The slot texts at k and l differ in s_1 (one mentions the name of b, the other that of b'), and distinct-guards only involve name positions, whose values in s_3 are those in s_1. The ZF classification: Sep and Sep_J have one occurrence. EInd applies P to the binders $\forall y$, $\forall x$ (first) and $\forall x$ (second). In Coll and $\text{Repl}_{\exists!}$ the antecedent's x, y and the consequent's x, y have different binders. Repl_sp and Repl_J differ even in names. □

Proof of Proposition 5.4. Frame positions are common columns. By (R) every P-position is a disagreement column, and the lgg merges equal columns. So positions k, l share a metavariable iff $\varphi_j[\bar{a}_k] = \varphi_j[\bar{a}_l]$ for all j. At a free occurrence of z_i the two texts carry $a_{k,i}$ and $a_{l,i}$, so the texts are equal iff the tuples agree at every place used by φ_j. For the last claim, let U be the set of places used by some body. All positions share one metavariable Z, so for $i \in U$ all occurrences carry the same name (index) a_i. The value of Z in datum j is $\psi_j = \varphi_j[\bar{a}_1]$, whose free frame variables are exactly the a_i with z_i free in φ_j. So the most specific guards say that the value's free frame variables lie in $\{a_i : i \in U\}$. For an instance $F[\psi, \ldots, \psi]$ respecting them, let β be ψ with each free a_i ($i \in U$) replaced by z_i. Then $\beta[\bar{a}_k] = \psi$ for every k, so the instance is $T^*[P := \lambda\bar{z}.\beta]$. □

D.3 First-order failures: proofs for Section 5.3

The false instances of Proposition 5.5, complete list cases §2.4. Each is an instance of the

computed lgg of the two named data, respects its most specific guards, and is not an instance of the schema.

(1) EInd, named; data $\mathrm{EInd}(x = x)$, $\mathrm{EInd}(\neg x \in a)$. The columns are $(y = y, \neg y \in a)$ at occurrence 1 and $(x = x, \neg x \in a)$ at occurrences 2 and 3, so the lgg is $\forall x(\forall y(y \in x \to Z_1) \to Z_2) \to \forall x Z_2$. Take F_1: $Z_1 := \neg y = y$, $Z_2 := \forall w \neg w \in x$. Its antecedent is $\forall x(\mathrm{empty}(x) \to \mathrm{empty}(x))$, which is valid, so F_1 is equivalent to the Π_1 sentence $\forall x \forall w\, \neg w \in x$. The Δ_0 matrix of that sentence fails at the hereditarily finite sets $x = \{\emptyset\}$, $w = \emptyset$; by Δ_0-absoluteness F_1 is false. ZF refutes it from "some set has an element".

(2) $\mathrm{Repl}_{\mathrm{J}}$, named; data $\mathrm{Repl}_{\mathrm{J}}(x \in y)$, $\mathrm{Repl}_{\mathrm{J}}(\neg x = a)$. The lgg is $\forall x \forall y \forall u(Z_1 \wedge Z_2 \to y = u) \to \forall X \exists Y \forall y(y \in Y \leftrightarrow \exists x(x \in X \wedge Z_1))$. Take $Z_1 := y = y$, $Z_2 := \neg u = u$. The antecedent is valid, and at $X = \{\emptyset\}$ the consequent gives a set Y of all sets. Then $Y \in Y$, which contradicts Foundation; without Foundation, Separation yields Russell's set.

(3) $\mathrm{Repl}_{\mathrm{J}}$, de Bruijn: three blocks. $Z_1 := \bot$, $Z_2 := \bot$, $Z_3 := (y = y)$ give the same false consequent.

(4) Coll, de Bruijn; data $\mathrm{Coll}(y = x)$, $\mathrm{Coll}(\neg x = A)$. The two occurrences decouple. $Z_1 := y = y$, $Z_2 := \bot$ give $\forall A(\forall x{\in}A\, \exists y\, y = y \to \exists Y \forall x{\in}A\, \exists y(y \in Y \wedge \bot))$, which is equivalent to $\forall A \neg \exists x\, x \in A$ and is refuted by the HF witness $A = \{\emptyset\}$.

(5) $\mathrm{Repl}_{\exists!}$, de Bruijn; the same data. $Z_1 := y = x$ (and $\exists! y\,(y = x)$ is valid), $Z_2 := \bot$: equivalent to $\forall A \neg \exists x\, x \in A$.

(6) $\mathrm{Repl}_{\mathrm{sp}}$, de Bruijn; the same data, three blocks. $Z_1 := y = x$, $Z_2 := \bot$, $Z_3 := \bot$. The antecedent $\forall x{\in}A\, \exists y(y = x \wedge \forall u(\bot \to u = y))$ is valid, and the instance is equivalent to $\forall A \neg \exists x\, x \in A$.

(7) Guard violations. Sep without $\mathrm{fresh}(y, Z)$: Russell's instance, in both encodings (Proposition 5.8). Coll, $\mathrm{Repl}_{\exists!}$, $\mathrm{Repl}_{\mathrm{sp}}$, named, without $\mathrm{fresh}(Y, Z)$: $Z := (y = Y)$ gives

$$\forall A\,(\forall x{\in}A\, \exists y\,(y = Y) \to \exists Y\, \forall x{\in}A\, \exists y\,(y \in Y \wedge y = Y)).$$

Its antecedent holds, because Y is a parameter there, and its consequent forces $Y \in Y$ for $A \neq \emptyset$, contradicting Foundation. (For $\mathrm{Repl}_{\mathrm{sp}}$ add $\neg u = u$ at the uniqueness occurrence.) Coll with $\varphi(x, y)$ only, de Bruijn, without its guard: $Z := (\#0 = \#2)$, as in Appendix D.2.

(8) Named, α-varied. For Coll, $\mathrm{Repl}_{\exists!}$ the blocks $(y = x)$, $\bot$, and for $\mathrm{Repl}_{\mathrm{sp}}$ the blocks $(y = x)$, $\bot$, $\bot$, each give an instance equivalent to $\forall A \neg \exists x\, x \in A$.

$\mathrm{Repl}_{\mathrm{sp}}$, named, non-anchor data paper review, univzf_repls_pair.py; computed. Proposition 5.7 concerns data with (R) and (N_y). Without (R), false instances arise when freshness guards are available only for formula metavariables, the guard family used throughout Section 5. The bodies $x \in y$ and $y \in A$ both have root $\in$, so the named lgg of $\mathrm{Repl}_{\mathrm{sp}}(x \in y)$ and $\mathrm{Repl}_{\mathrm{sp}}(y \in A)$ is

$$\forall A\,(\forall x\,(x \in A \to \exists y\,(V_0 \in V_1 \wedge \forall u\,(V_2 \in V_3 \to u = y))) \to \exists Y\, \forall x\,(x \in A \to \exists y\,(y \in Y \wedge V_0 \in V_1))),$$

with term metavariables only, so no such guard is learned. Its instance $V_0 := Y$, $V_1 := y$, $V_2 := V_3 := u$ has Y free in the antecedent, i.e. a parameter there, and $y \in Y \wedge Y \in y$ in the consequent. At $A = \{\emptyset\}$ and $Y = \emptyset$ the antecedent holds ($y := \{\emptyset\}$; no set satisfies $u \in u$), while the consequent needs $y \in Y' \in y$, which contradicts Foundation; so the instance is false and refutable in ZF. If freshness guards were also learned for term metavariables ($\mathrm{fresh}(Y, V_0)$ holds on these data), this instance would be excluded; we have not analysed that guard family.

Lemma D.2 (Deep leaf)**.** *[proved]*[cases Lemma C1] *Let s be a sentence of $L_\in$, ℓ a variable leaf of s, and τ a first-order schema with $s \in \mathrm{inst}(\tau)$ and $|\tau| < \mathrm{depth}(\ell)$. Then τ has a formula metavariable Z at a proper prefix q of ℓ with $\mathrm{depth}(q) < |\tau|$. If $s|_q$ occurs in s only at q, then Z occurs in τ only at q, and $s[q \leftarrow v] \in \mathrm{inst}(\tau)$ for every closed formula v. This remains true with freshness guards (named), loose-index guards (de Bruijn), and distinct-guards on name metavariables (α-varied named).*

Proof. Follow the path to ℓ in τ. Since τ has fewer than $\mathrm{depth}(\ell)$ nodes, the path leaves τ at a leaf of τ at depth $< |\tau|$. That leaf is not a constant, because s continues below it, so it is a metavariable Z at a proper prefix q of ℓ. In $L_\in$ every proper prefix of a variable leaf is a formula position (binder-name children are not on the path), so Z is a formula metavariable. If Z also occurred at q', then $s|_{q'} = \theta(Z) = s|_q$, so $q' = q$. Changing $\theta(Z)$ to v changes s only at q. A closed v has no free names and no loose indices, and the other metavariables keep their values, so all guards still hold. □

Proof of Theorem 5.6. Let $\{\tau_1, \dots, \tau_r\}$ cover the schema. Take n with $2n > \max_i |\tau_i|$, a genuine instance s_n and a designated leaf ℓ below; some τ_i covers s_n. By Lemma D.2, once we check that every subterm along the path to ℓ occurs only once in s_n, τ_i contains $s_n[q \leftarrow v]$ for some proper prefix q of ℓ and both $v \in \{\top, \bot\}$. It therefore suffices that, for every proper prefix q of ℓ, one of $\top, \bot$ makes $s_n[q \leftarrow v]$ false. "Make the slot false" means: if q lies inside the slot below e negations, choose v with $\neg^e v \equiv \bot$; the path inside the slot passes only through $\neg$, $\forall$ and $\wedge$, and $\forall w\, v \equiv v$.

(i) EInd, named. $\varphi_n(x) := \neg^{2n} \forall w \neg(w \in x)$ ("x is empty"); $s_n = \mathrm{EInd}(\varphi_n)$; ℓ is the y of occurrence 1. For q the root, take $v := \bot$. For q the antecedent or its $\forall x$-body, take $v := \top$: then s_n is equivalent to $\forall x\, \varphi_n(x)$, "every set is empty", which is false. For q the formula $\forall y(y \in x \to \varphi_n(y))$ or its body, take $v := \bot$: the antecedent becomes $\forall x(\bot \to \varphi_n(x))$, true, and the conclusion is false. For q inside $\varphi_n(y)$, make the slot false: then $\forall y(y \in x \to \bot)$ says that x is empty, the antecedent reads $\forall x(\mathrm{empty}(x) \to \mathrm{empty}(x))$, which is true, and the conclusion is false. Every path subterm inside the slot contains y, which occurs only in occurrence 1 and in the frame atom $y \in x$, so all path subterms are unique.

(ii) Repl_J, both encodings. $\varphi_n(x, y) := \neg^{2n}(x \in y)$; ℓ is the u of occurrence 2, $\varphi_n(x, u)$. The consequent $C := \forall X \exists Y \forall y(y \in Y \leftrightarrow \exists x{\in}X\, x \in y)$ is false: at $X = \{\emptyset\}$, Y would contain every y with $\emptyset \in y$, and $y_0 := \{\emptyset, Y\}$ gives $Y \in y_0 \in Y$, contradicting Foundation. (Without Foundation, $R := \{y \in Y : y \notin y\}$ and $y_1 := R \cup \{\emptyset\}$ give $y_1 \in y_1 \leftrightarrow y_1 \notin y_1$.) For q the root take $\bot$. For q the antecedent, its $\forall x, \forall y, \forall u$ levels or its implication, take $\top$: the antecedent is true and C is false. For q the conjunction, take $\bot$: $(\bot \to y = u)$ is true. Inside $\varphi_n(x, u)$, make the slot false. Uniqueness: the path subterms contain u (named), and in de Bruijn the designated occurrence's arguments $(\#2, \#0)$ differ from $(\#2, \#1)$ and $(\#0, \#1)$.

(iii) Coll, $\mathrm{Repl}_{\exists!}$, $\mathrm{Repl}_\mathrm{sp}$, de Bruijn. $\varphi_n(x, y, A) := \neg^{2n}(y = x \wedge x \in A)$; ℓ is the A in the consequent's occurrence. The antecedent is true for every A, since $y := x$ is the unique witness. Every proper prefix q is the root ($v := \bot$), the $\forall A$-body ($v := \bot$, giving $\forall A\, \bot$), or a subformula of the consequent containing ℓ. Above the slot, $v := \bot$ turns the consequent into $\exists Y \bot$, $\exists Y \forall x \bot$, or $\exists Y \forall x(x \in A \to \bot)$ (that is, "A is empty"); inside the slot (including the conjunction and the atom $x \in A$), make the slot false. In each case the consequent is false for $A \neq \emptyset$, so $\forall A(\text{true} \to \text{false for nonempty } A)$ is false. Uniqueness: in de Bruijn the slot's atom $x \in A$ reads $\#1 \in \#3$, while the frame atoms read $\#0 \in \#1$ (antecedent) and $\#0 \in \#2$ (consequent), and the slot atom $x \in A$ of the other occurrences reads $\#1 \in \#2$ (antecedent occurrence) and, for $\mathrm{Repl}_\mathrm{sp}$, $\#2 \in \#3$ (uniqueness occurrence). Every path subterm inside the slot contains the atom

$\#1 \in \#3$, and every path subterm above it contains the slot; so all are unique. The argument tuple $(\#1, \#0, \#3)$ (resp. that of $\mathrm{Repl}_{\mathrm{sp}}$'s third occurrence) differs from the others.

(iv) Coll, $\mathrm{Repl}_{\exists!}$, $\mathrm{Repl}_{\mathrm{sp}}$, named under α-variation. The same φ_n, with s_n an α-variant of the instance in which all frame binders have distinct names. The designated leaf is the y of the atom $y = x$ in the consequent's occurrence. (The A-leaf of (iii) is not usable: in the named encoding the slot atom $x \in A$ also occurs as a frame atom.) The case analysis is that of (iii); the only new prefix is the atom $y = x$ itself, where $v := \bot$ makes the slot false. Every path subterm inside the slot contains the consequent's own names for x and y, which occur nowhere outside the consequent, and inside the consequent the frame atoms are $x \in A$ and $y \in Y$, never $y = x$. □

Proof of Proposition 5.7. (a) Let $L_S\theta$ satisfy the guards. The antecedent implies $\forall x{\in}A\, \exists y\, Z_1$ (drop the second conjunct). Z_1's value may mention x, y, A and parameters, but not Y (guard). The name u is not bound at either occurrence of Z_1, so a free u would be a parameter at both. Collection, with these parameters, gives $\exists Y \forall x{\in}A\, \exists y{\in}Y\, Z_1$, and ZF proves Collection. Strictness: $Z_1 := (y = x)$, $Z_2 := \bot$ respects the guards. It is not an instance of $\mathrm{Repl}_{\mathrm{sp}}$: a body β with $\beta[x, y, A] = (y = x)$ is $z_2 = z_1$, and then $\beta[x, u, A] = (u = x) \neq \bot$. (b) "If" is the proof of (a), which uses only Collection. "Only if": let $\mathrm{Coll}(\varphi)$ be a Collection instance, with Y not free in φ. Renaming parameters gives an equivalent universal closure, so we may assume $u \notin \mathrm{FV}(\varphi)$. Then $Z_1 := \varphi$, $Z_2 := \bot$ satisfy the guards, and the instance $\forall A(\forall x(x \in A \to \exists y(\varphi \wedge \forall u(\bot \to u = y))) \to \exists Y \forall x(x \in A \to \exists y(y \in Y \wedge \varphi)))$ is logically equivalent to $\mathrm{Coll}(\varphi)$, because $\forall u(\bot \to u = y)$ is valid. □

D.4 Anchors for pattern targets: proofs for Section 5.4

Proof of Theorem 5.9. "If", for $H = \mathrm{SO}^\circ$ (hence for all smaller H). Let $T \in \mathrm{SO}^\circ$ cover D, with $D_j = T\eta_j$.

Step 1 (skeleton). The root of $\varphi_j[\bar{a}_k]$ is $\mathrm{root}(\varphi_j)$. By (R) every slot root of T^* is a disagreement position, while frame positions are common. By Lemma D.1, $\mathrm{skel}(T)$ is a prefix of the frame, and T's occurrences lie at frame positions or at slot roots. For such a position c put $\kappa(c) := T^*|_c$, so that $D_j|_c = \kappa(c)[P := \lambda \bar{z}.\varphi_j]$.

Step 2 (one body per metavariable). Let M have occurrences $M(\bar{u}^1)$ at c_1, …, $M(\bar{u}^r)$ at c_r. The arguments are bound variables in scope or parameters, possibly repeated. Let $\beta_j := \eta_j(M)$, so that $\beta_j[\bar{u}^s] = D_j|_{c_s}$.

(2a) *Shape.* Let p be a slot root of $\kappa(c_1)$. Then $D_j|_{c_1}$ has the formula symbol $\mathrm{root}(\varphi_j)$ at p, so p is a non-hole position of β_j, and $D_j|_{c_s}$ also has $\mathrm{root}(\varphi_j)$ at p. If $\kappa(c_s)$ had a frame symbol at p, all φ_j would share it. If p lay strictly inside a slot of $\kappa(c_s)$ with root p', the non-leaf frame symbol of $\kappa(c_1)$ at p' would be a symbol of β_j, hence of $D_j|_{c_s}$ at p', which is $\mathrm{root}(\varphi_j)$, for all j. Both contradict (R). By symmetry, $\kappa(c_1)$ and $\kappa(c_s)$ have slots at the same positions and the same symbols at non-leaf frame positions. Frame leaves may differ.

(2b) *Frame leaves.* Let q be a frame leaf of $\kappa(c_1)$ carrying the variable v. If v is bound inside $\kappa(c_1)$, then $\beta_1|_q$ is that variable (holes become $\bar{u}^1$, which are bound above c_1 or are parameters), so $\kappa(c_s)|_q$ is the variable bound by the corresponding binder of $\kappa(c_s)$; keep v. If v is bound above c_1, then $\beta_1|_q$ is a hole h_m with $u^1_m = v$ (β_1 has no free bound variable, and a parameter is not v); record m. By (2a), q is a frame leaf of $\kappa(c_s)$ as well, and $\kappa(c_s)|_q = D_1|_{c_s}|_q = u^s_m$.

(2c) *Slot arguments.* Let $\kappa(c_1)$ have the slot $P(\bar{a})$ at p and $\kappa(c_s)$ the slot $P(\bar{b})$ at p. For an argument place i, use (N_i) to pick j_i and a free occurrence q' of z_i in φ_{j_i}. At $p \cdot q'$, $D_{j_i}|_{c_1}$ carries a_i. If a_i is bound inside $\kappa(c_1)$, then $\beta_{j_i}|_{p \cdot q'}$ is that variable, so b_i is the variable bound by the

corresponding binder of $\kappa(c_s)$; keep a_i. Otherwise $\beta_{j_i}|_{p\cdot q'} = h_m$ with $u^1_m = a_i$; record m. Since $D_{j_i}|_{c_s}|_p = \varphi_{j_i}[\bar{b}]$ carries b_i at q', $b_i = u^s_m$.

Let B_M be $\kappa(c_1)$ with every recorded leaf or argument replaced by its recorded hole. Every variable bound above c_1 that occurs in $\kappa(c_1)$ does so at a frame leaf or as a slot argument, so B_M is closed except for holes and the metavariable P. $B_M[\bar{u}^1] = \kappa(c_1)$, and $B_M[\bar{u}^s] = \kappa(c_s)$ for all s by (2a)–(2c). In particular every recorded u^s_m is a bound variable, even when $\bar{u}^s$ contains parameters or repetitions; any valid choice of β_1 and β_{j_i} works. (A term metavariable at a frame leaf is the case where $\kappa(c)$ is a variable.)

Step 3. Let $\sigma(M) := \lambda\bar{h}.B_M$ for every metavariable M of T. Each occurrence $M(\bar{u}^s)$ becomes $B_M[\bar{u}^s] = \kappa(c_s)$, and T's rigid part is F's, so $T\sigma = T^*$. Hence $T \succeq T^*$ and $\mathrm{inst}(T) \supseteq \mathrm{inst}(T^*)$.

"Only if" (witnesses in PAT*).* If all roots are f, replace each $P(\bar{a}_k)$ by $\Phi_f(\bar{a}_k)$, where $\Phi_\in(\bar{a}) = g_1(\bar{a}) \in g_2(\bar{a})$ and $\Phi_=(\bar{a}) = g_1(\bar{a}) = g_2(\bar{a})$ (term metavariables, whose values are projections or parameters), $\Phi_\neg(\bar{a}) = \neg Q(\bar{a})$, $\Phi_\circ(\bar{a}) = Q_1(\bar{a}) \circ Q_2(\bar{a})$ for binary $\circ$, and $\Phi_\mathsf{Q}(\bar{a}) = \mathsf{Q}w.R(\bar{a}, w)$ for $\mathsf{Q} \in \{\forall, \exists, \exists!\}$. All occurrences are pattern occurrences. The template covers D, since atoms of $L_\in$ have variables or parameters as arguments, and it misses $T^*[P := \lambda\bar{z}.\psi]$ for any ψ with another root. If (N_i) fails, replace each $P(\bar{a}_k)$ by $P'(\bar{a}_k$ without its i-th entry). This covers D and misses $T^*[P := \lambda\bar{z}.(z_i = z_i)]$. □

Several metavariables. ZF needs one metavariable, but untagged mixtures behave like templates with several (Section 6). Then a Plotkin-(D)-type event appears, and it depends on the class. Let T^* be a pattern template over $L_\in$ with a metavariable-free, parameter-free frame, metavariables $P, Q, \dots$ (arities $n_P, n_Q, \dots$), every occurrence a pattern occurrence, and data $D_j = T^*[P := \lambda\bar{z}.\varphi_{P,j}, \dots]$. Let (R_P) and $(\mathrm{N}_{P,i})$ be (R) and (N_i) for the bodies of P. For a map $\pi : [n_P] \to [n_Q] \cup \mathrm{Par}$, let $\varphi \circ \pi$ replace each hole z_i by $z_{\pi(i)}$ (or by the parameter $\pi(i)$). The events are: $(\mathrm{D}^{\mathrm{PAT}}_{PQ})$: no bijection π has $\varphi_{Q,j} = \varphi_{P,j} \circ \pi$ for all j; $(\mathrm{D}^{\mathrm{DT}}_{P\to Q})$: no map π has this property; $(\mathrm{D}^{\mathrm{root}}_{PQ})$: some j has $\mathrm{root}(\varphi_{P,j}) \neq \mathrm{root}(\varphi_{Q,j})$. Then $(\mathrm{D}^{\mathrm{root}}_{PQ})$ implies $(\mathrm{D}^{\mathrm{DT}}_{P\to Q})$, which implies $(\mathrm{D}^{\mathrm{PAT}}_{PQ})$. Under $(\mathrm{N}_{P,i})$ for all P, i, $(\mathrm{D}^{\mathrm{DT}}_{P\to Q})$ for all ordered pairs is the event (D^+) of Corollary 3.9(b): a map $\bar{u}$ from the variables at an occurrence $P(\bar{a})$ to terms at an occurrence $Q(\bar{b})$ with $\varphi_{Q,j}[\bar{b}] = (\varphi_{P,j}[\bar{a}])[\bar{u}]$ for all j sends every a_i, which occurs by $(\mathrm{N}_{P,i})$, to some $b_{i'}$ or a parameter, and so is a map π; conversely every such π gives such a $\bar{u}$. So the DT° case of Theorem D.3 is also a case of Theorem 3.8.

Theorem D.3 (Several metavariables)**.** *[proved]*[cases Thm E′] *Assume* (R_P) *and all* $(\mathrm{N}_{P,i})$*, for every metavariable* P*. Then* D *is an anchor in* PAT *iff* $(\mathrm{D}^{\mathrm{PAT}}_{PQ})$ *holds for all* $P \neq Q$*. It is an anchor in* DT° *iff* $(\mathrm{D}^{\mathrm{DT}}_{P\to Q})$ *holds for all ordered pairs. In* SO°*,* $(\mathrm{D}^{\mathrm{root}})$ *for all pairs implies that* D *is an anchor, which implies* $(\mathrm{D}^{\mathrm{DT}})$ *for all ordered pairs; the second implication cannot be reversed. Without* (R_P) *or* $(\mathrm{N}_{P,i})$*,* D *is an anchor in no class between* PAT *and* SO°*.*

The proof of Theorem 5.9 works once each slot of M's first occurrence faces a slot of the *same* metavariable at every other occurrence. If it faces a Q-slot instead, there is a fixed argument map from P's bodies to Q's (Lemma D.5), which the events exclude. Conversely, a violating map lets one rewrite every $Q(\bar{b})$ as an occurrence of P.

The three classes differ. Take $T_B := \forall x\forall y(P(x, y) \to Q(x))$ with bodies $(x \in y, x \in x)$ and $(\neg x = y, \neg x = x)$. Pattern anti-unification returns T_B, but $\forall x\forall y(P(x, y) \to P(x, x)) \in \mathrm{DT}^\circ$ covers the data and misses $\forall x\forall y(x \in y \to \neg x = x)$. Take $T_A := \forall x\forall y(P(x, y) \to Q(x, y))$ with bodies $(x \in y, y \in x)$ and $(\neg x = y, \neg x = x)$. Here D is a DT° anchor, but $\forall x\forall y(G(x, y, x) \to G(x, x, y)) \in \mathrm{SO}^\circ$ covers it and misses the same sentence. So, unlike the one-metavariable case,

SO$^\circ$ and DT$^\circ$ anchors differ even over $L_\in$: a metavariable without a pattern occurrence may choose different arguments on different data. Whether ($\mathrm{D}^{\mathrm{root}}$) is necessary in SO$^\circ$ is open; the bounded searches of Appendix D.6 suggest:

Conjecture D.4. *[conjecture]*[cases §2.9, §6] ($\mathrm{D}^{\mathrm{root}}$) *is not necessary for anchors in* SO$^\circ$*: some data satisfying* (R_P)*, all* ($\mathrm{N}_{P,i}$) *and all* ($\mathrm{D}^{\mathrm{DT}}_{P\to Q}$) *but violating* ($\mathrm{D}^{\mathrm{root}}_{PQ}$) *for some pair are anchors in* SO$^\circ$*.*

Lemma D.5 (Facing)**.** *[proved]*[cases Thm E′] *In the setting of Theorem D.3, let $T \in$ SO$^\circ$ cover D, let M be a metavariable of T with occurrences $M(\bar{u}^s)$ at c_s and bodies β_j, and assume* (R_P) *for all P. (1) $\kappa(c_1)$ and $\kappa(c_s)$ have slots at the same positions and the same symbols at non-leaf frame positions. (2) If $\kappa(c_1)$ has a P-slot $P(\bar{a})$ at p and $\kappa(c_s)$ a Q-slot $Q(\bar{b})$ at p, then $\mathrm{root}(\varphi_{P,j}) = \mathrm{root}(\varphi_{Q,j})$ for all j. If moreover $\bar{u}^1$ is a pattern tuple and all* ($\mathrm{N}_{P,i}$) *hold, there is a map $\pi : [n_P] \to [n_Q] \cup \mathrm{Par}$, independent of j, with $\varphi_{Q,j} = \varphi_{P,j} \circ \pi$ for all j. If also $\bar{u}^s$ is a pattern tuple and all* ($\mathrm{N}_{Q,i}$) *hold, π is a bijection onto $[n_Q]$.*

Proof. (1) is (2a) above, using (R) for the metavariable at each slot; the root claim of (2) is its first sentence. Let $\gamma_j := \beta_j|_p$, so that $\varphi_{P,j}[\bar{a}]$ and $\varphi_{Q,j}[\bar{b}]$ are γ_j with its holes replaced by $\bar{u}^1$, resp. $\bar{u}^s$. Define $\pi(i)$ for $i \le n_P$: by ($\mathrm{N}_{P,i}$) pick j and a free occurrence q' of z_i in $\varphi_{P,j}$; the leaf of $\varphi_{P,j}[\bar{a}]$ at q' is a_i. (α) If a_i is bound above c_1, then $\gamma_j|_{q'} = h_m$ with $u^1_m = a_i$, and m is unique because $\bar{u}^1$ is a pattern tuple. The leaf of $\varphi_{Q,j}[\bar{b}]$ at q' is u^s_m, which is free relative to p; so it is some $b_{i'}$ or a parameter, and we put $\pi(i) := i'$, resp. that parameter. (β) If a_i is bound by a binder at relative position r between c_1 and p, then $\gamma_j|_{q'}$ is the variable bound at r, and the leaf of $\varphi_{Q,j}[\bar{b}]$ at q' is the variable bound at r in $\kappa(c_s)$, which lies above p; so it is some $b_{i'}$, and $\pi(i) := i'$. In both cases $\pi(i)$ depends only on a_i (through m or r), not on j or q'. Now compare the two plugged texts leaf by leaf through γ_j. Constants, parameters and variables bound inside γ_j agree. A hole h_m gives u^1_m, a free leaf of the slot and hence some a_i, and $u^s_m = b_{\pi(i)}$ (or the parameter $\pi(i)$). A variable bound between c_1 and p gives a_i and $b_{\pi(i)}$. Hence $\varphi_{Q,j} = \varphi_{P,j} \circ \pi$. If $\bar{u}^s$ is also a pattern tuple, every u^s_m is a bound variable, so π maps into $[n_Q]$. It is injective: distinct a_i give distinct m or distinct r, hence distinct variables, and the two cases give variables bound above, resp. below, c_s. Every $b_{i'}$ free in some $\varphi_{Q,j}[\bar{b}]$ arises in this way, so under ($\mathrm{N}_{Q,i}$) π is onto. □

Proof of Theorem D.3. "If" parts. Let T in the class cover D. Step 1 of Theorem 5.9 holds verbatim, using (R_P) for every P. For each metavariable M of T choose c_1 to be a pattern occurrence in PAT and DT$^\circ$, and any occurrence in SO$^\circ$. Then every slot of $\kappa(c_1)$ faces, in each $\kappa(c_s)$, a slot of the *same* metavariable. In SO$^\circ$ this follows from ($\mathrm{D}^{\mathrm{root}}$) and the root claim of Lemma D.5; in DT$^\circ$ from (D^{DT}) and the map; in PAT (where $\bar{u}^s$ is a pattern tuple too) from ($\mathrm{D}^{\mathrm{PAT}}$) and the bijection. Steps 2b, 2c and 3 of Theorem 5.9 then apply slot by slot, with ($\mathrm{N}_{P,i}$) for the metavariable of each slot, and give $T \succeq T^*$.

"Only if". If ($\mathrm{D}^{\mathrm{PAT}}_{PQ}$) fails via a bijection π, let T' be T^* with every $Q(\bar{b})$ replaced by $P(b_{\pi(1)}, \dots, b_{\pi(n_P)})$, a pattern occurrence. Then $T'[P := \lambda \bar{z}.\varphi_{P,j}] = D_j$, because $\varphi_{P,j} \circ \pi$ plugged with $\bar{b}$ is $\varphi_{Q,j}[\bar{b}]$. T' misses every instance of T^* in which the P- and Q-bodies have different roots. So D is not an anchor in PAT or in any larger class. If ($\mathrm{D}^{\mathrm{DT}}_{P\to Q}$) fails via π, the same construction with arguments $b_{\pi(i)}$ or the parameter $\pi(i)$ gives $T' \in$ DT$^\circ$ (P keeps its pattern occurrences, and the new occurrences have metavariable-free arguments), which covers D and misses the same instances. In (c), an SO$^\circ$ anchor is a DT$^\circ$ anchor, hence satisfies (D^{DT}) by (b). Non-reversibility is the example T_A above. There ($\mathrm{D}^{\mathrm{PAT}}$) and ($\mathrm{D}^{\mathrm{DT}}$) hold: the only map fitting the first datum is the swap, which fails on the second. The template $\forall x \forall y (G(x,y,x) \to G(x,x,y))$ covers the data with $G := \lambda h_1 h_2 h_3.\, h_3 \in h_2$, resp. $\lambda h_1 h_2 h_3. \neg h_1 = h_2$. It misses $\forall x \forall y (x \in y \to \neg x = x)$, because

a body with $\beta[x, y, x] = (x \in y)$ is $h_1 \in h_2$ or $h_3 \in h_2$, so that $\beta[x, x, y] \in \{x \in x, y \in x\}$. Without ($\mathrm{R}_P$) or ($\mathrm{N}_{P,i}$), the witnesses of Theorem 5.9 applied to P work in every class. □

Proof of Theorem 5.11. By Theorem 5.9, "not an anchor" is $\bigcup_f A_f \cup \bigcup_i B_i$, where A_f says that all roots are f and B_i that z_i is never free. In inclusion–exclusion over these events, every intersection containing two distinct A_f is empty. The intersection of A_F (or of no A, $F = *$) with B_i, $i \in I$, is the event that all N samples lie in $C_{F,I}$, of probability $\mu(C_{F,I})^N$, and its sign is $(-1)^{[F\neq *]+|I|+1}$. The upper bound is the union bound. □

D.5 PA induction: proofs for Section 5.5

Write $\mathrm{Ind}(\varphi) = \mathrm{imp}(U, V)$ with $U = \mathrm{and}(a, s)$, $s = \forall x\, W$, $W = \mathrm{imp}(b, c)$ and $V = \forall x\, d$. The *motive slots* a, b, c, d hold $\varphi[0/x]$, φ, $\varphi[Sx/x]$, φ, and the *five-node frame* consists of the positions of $\mathrm{imp}, \mathrm{and}, s, W, V$. Two facts are used throughout [prior induction C3, Lemma A]: (i) substituting terms for x changes neither the root of a formula nor its *logical size* ℓ (the number of atoms, connectives and quantifiers); (ii) if x is free in φ and $\varphi[u/x] = \varphi[v/x]$, then $u = v$. Also $St \neq x$, $St \neq 0$ and $St \neq t$ for every term t. With $L = \ell(\varphi) \geq 1$, the slots have logical sizes a, b, c, d: L; W: $2L + 1$; s: $2L + 2$; V: $L + 1$; U: $3L + 3$; and the whole sentence $4L + 5$. These six values are pairwise distinct.

Proof of Proposition 5.12. [prior induction B1, B3] (a) For $\varphi_1 = (x + 0 = x)$ and $\varphi_2 = (0 + x = x)$ the a-column is $(0 + 0 = 0, 0 + 0 = 0)$ and stays. The b-column $(x + 0 = x, 0 + x = x)$ gives $z_{(x,0)} + z_{(0,x)} = x$. The c-column $(Sx + 0 = Sx, 0 + Sx = Sx)$ gives $z_{(Sx,0)} + z_{(0,Sx)} = Sx$. The d-column equals the b-column. The four index tuples are distinct. The displayed instance ($z_1 = z_2 = z_3 := 0$, $z_4 := S0$) is false: $0 + 0 = x$ forces $x = 0$, and then $0 + S0 = 1 = Sx$, so the premise holds, while $\forall x(0 + 0 = x)$ fails at $x = 1$. A single instance I is covered by the ground schema I. For $\mathrm{Ind}(x = x), \mathrm{Ind}(Sx = Sx)$ the lgg is $(t = t) \wedge \forall x(u = u \to Su = Su) \to \forall x(u = u)$, and every sentence instance has the valid conclusion $\forall x(u = u)$. (b) By fact (i) the a, b, c columns all have mixed roots, so each becomes a metavariable indexed by its column. If some φ_j has x free, then $\varphi_j[0/x]$, φ_j, $\varphi_j[Sx/x]$ are pairwise distinct, so the three columns and their metavariables are distinct; the d-column equals the b-column. The displayed instance has a true premise and the false conclusion $\forall x(0 = S0)$. The lgg differs from L_∞ only if all roots are equal (probability $\sum_f p_f^N \leq p_{\max}^{N-1}$) or no motive has x free (probability $(1 - q)^N$). □

Proof of Theorem 5.13. [prior induction B2] Let $\varphi_n := (g_n(x) = x)$ and $n < m$, $d := m - n$, so that $g_m(t) = g_n(g_d(t))$ as terms. In the b-slot both left sides begin with n copies of $0 + \cdot$, so the lgg descends n times and meets the column $(x, g_d(x))$. Its roots are a variable and $+$, giving $b := z_{(x,g_d(x))}$, so the slot is $g_n(b) = x$. Likewise the a-slot meets $(0, g_d(0))$ and gives $g_n(a) = 0$, and the c-slot meets $(Sx, g_d(Sx))$ and gives $g_n(c) = Sx$. The d-slot repeats b. The tuples indexing a, b, c have first entries $0, x, Sx$, so the metavariables are distinct, and the lgg is L_n. (For any subset of F' with least index n the same descent gives L_n.) For $M \geq n$, $I_M^* \in \mathrm{inst}(L_n)$: put $a, b := g_{M-n}(0)$ and $c := g_{M-n}(S0)$, and use $g_n(g_{M-n}(t)) = g_M(t)$. I_M^* is false: $g_M(0)$ denotes 0 and $g_M(S0)$ denotes 1; the base holds, $g_M(0) = x$ forces $x = 0$ and then $g_M(S0) = Sx$, while $\forall x(g_M(0) = x)$ fails at 1. It is refuted in Q from the closed facts $g_M(0) = 0$ and $g_M(S0) = S(g_M(0))$, which Q proves by Σ_1-completeness, and $\neg(g_M(0) = S0)$. Assume $g_M(0) = x$; by congruence and transitivity $g_M(S0) = Sx$. Discharging and generalizing gives the step clause, modus ponens gives $\forall x(g_M(0) = x)$, and instantiating at $S0$ contradicts $\neg(g_M(0) = S0)$. Now a schema covering $\mathrm{Ind}(\varphi_n)$ and $\mathrm{Ind}(\varphi_m)$ is $\succeq L_n$ and contains I_n^*. A union of k truth-sound schemas covering $k + 1$

members of F' would put two of them into one schema. If a finite union $\bigcup \mathrm{inst}(\tau_l)$ contained F', some τ_l would cover two members of F' and hence contain the false, Q-refutable sentence I_n^*; so no such union lies between F' and $\mathrm{Th}(\mathbb{N})$, or between F' and a consistent extension of Q. The motives of F' are atomic, so the same holds for open induction. □

Guards prior induction B2 remark. Data from F' instantiate b and c by terms containing x, and a by closed terms, so the strongest occurrence and freshness guards learnable from F' are $x \in \mathrm{FV}(b)$, $x \in \mathrm{FV}(c)$ and "a closed". The padded instance $a := 0$, $b := 0 \cdot x$, $c := S0 + 0 \cdot x$ satisfies them and has the truth values of I_n^*.

Proof of Theorem 5.14. prior induction referee R1 Idea: for each instance that is not trivially true, a convergent rewrite system over the numerals and a new constant g gives an algebra containing $\mathbb{N}$ in which the step clause fails at g; amalgamating these algebras over $\mathbb{N}$ gives a model of all instances and all true closed quantifier-free sentences. In detail: In a sentence instance, $z_1 := t$ is closed (it occurs outside the binder) and $z_2 := u$ has $\mathrm{FV}(u) \subseteq \{x\}$. Let k be the value of t, and let $\hat{u}(g)$ be u with x replaced by a new constant g and each maximal closed subterm replaced by its numeral.

Case T: $\hat{u}(x)$ is literally $\underline{k}{+}x$. In every L_A-algebra M containing $\mathbb{N}$ as a subalgebra, $t{+}a = u(a)$ for every a, so the conclusion and hence the instance hold.

Case W: otherwise. Let $s := \underline{k} + g$ and $r := \hat{u}(g)$, so $s \neq r$. Consider ground terms over the numerals and g, with the rules $R_{\mathbb{N}}$ that evaluate $+$ and $\cdot$ on numerals, and one more rule ρ: $\rho := r \to s$ if s is a proper subterm of r, and $\rho := s \to r$ otherwise. *Termination*: if r is a numeral, use the lexicographic measure (number of g's, number of closed non-numeral positions). If $\rho = s \to r$ with $|r| \geq |s|$, use (closed non-numeral positions, occurrences of s); no ρ-step creates a new occurrence of s, since an occurrence straddling the rewritten position would force $r = g$ or r closed. Otherwise use (closed non-numeral positions, size). In the first case a ρ-step removes a g and an $R_{\mathbb{N}}$-step lowers the second component. In the other two cases an $R_{\mathbb{N}}$-step lowers the first component, and a ρ-step keeps it (both sides contain g and have no closed non-numeral subterm, and the context above still contains g) and lowers the second. *Confluence*: there are no critical pairs. The left sides of $R_{\mathbb{N}}$ are closed, with only numerals as proper subterms; the left side of ρ contains g, is $R_{\mathbb{N}}$-normal, and is ground. So $R_{\mathbb{N}} \cup \{\rho\}$ is convergent (critical pair lemma and Newman's lemma). Let $M_{t,u}$ be the ground terms modulo convertibility. Distinct numerals are distinct normal forms, so $\mathbb{N}$ is a subalgebra of $M_{t,u}$, and $M_{t,u} \models s = r$, i.e. $t + g = u(g)$. The step clause fails at g. If $\rho = s \to r$, the term $\underline{k} + Sg$ is normal, with root $+$, while the normal form of $Su(g)$ is Sr or a numeral. If $\rho = r \to s$, the normal form of $Su(g)$ is Ss or s, and $\underline{k} + Sg$ is normal and differs from both. So the instance holds in $M_{t,u}$, its antecedent being false.

Amalgamation. Take one $M_{t,u}$ for each Case-W instance, rename them apart over $\mathbb{N}$, and let M be the free completion of their union: an operation whose arguments all lie in one M_i is computed there (this is well defined on $\mathbb{N}$, a subalgebra of each), and any other application is a new formal element. Each M_i is a subalgebra of M. Closed terms evaluate in $\mathbb{N}$, so M satisfies every true closed quantifier-free sentence. Each Case-W instance keeps its quantifier-free witness g, and each Case-T instance holds in every algebra containing $\mathbb{N}$. A refutation by sound rules would make some false closed quantifier-free sentence true in M, but such sentences have the same truth value in M as in $\mathbb{N}$. For Δ_0 premises with primitive $<$, interpret $a < b$ as "$a, b \in \mathbb{N}$ and $a < b$"; bounded quantifiers with standard bounds then range over standard elements, so Δ_0 truth is preserved. □

Proof of Theorem 5.16. prior induction C3 We write the proof for parameter-free motives, as in the

earlier analysis. (With parameters, the statement for DT° is Corollary 3.9(c) single Cor D.3; we also checked that the steps below go through unchanged.) *"If".* Fix $\varphi^\circ \in D$ with x free (N) and let $T \in \mathrm{DT}^\circ$ cover D. *Step 1.* By (R) and fact (i) the four motive slots have different roots in different data, so by Lemma D.1 skel(T) is a prefix of the five-node frame. The occurrences of T sit at some of the positions of Ind, U, V, s, a, W, b, c, d, at most one per position. *Step 2.* Every metavariable has a pattern occurrence at one of these formula positions, so all metavariables are formula-valued, and all arguments are metavariable-free terms. Only x is in scope at W, b, c, d, and nothing at the others. So every metavariable is 0-ary, or unary with a pattern occurrence $M(x)$ at W, b, c or d. *Step 3.* Let $\kappa(\cdot)$ denote slot contents in $\mathrm{Ind}(\varphi^\circ)$. A 0-ary metavariable takes a closed value, so it cannot occur at W, b, c, d, whose contents have x free. It cannot occur at two of Ind, U, V, s, a, whose contents have pairwise different logical sizes; so it occurs exactly once. A unary M with a pattern occurrence at π has the value $\lambda x.\kappa(\pi)$, and each occurrence $M(t)$ at σ needs $\kappa(\pi)[t/x] = \kappa(\sigma)$; by fact (i), σ lies in π's logical-size class, so $\pi = \sigma = W$ or $\pi, \sigma \in \{a, b, c, d\}$. If M has a pattern occurrence at b or d, take π there, so $\kappa(\pi) = \varphi^\circ$; fact (ii) forces $t = x$ at b, d, $t = Sx$ at c, and $t = 0$ at a. Otherwise M's pattern occurrences are at W (and then M occurs only there) or at c; in the latter case $\kappa(c)[t/x] = \varphi^\circ[St/x]$, and occurrences at b, d or a would need $St = x$ or $St = 0$, which is impossible. So M occurs only at c, as $M(x)$. *Step 4.* Map a 0-ary metavariable at slot ρ to $T_{\mathrm{Ind}}|_\rho$; M with a pattern occurrence at b or d to $\lambda y.P(y)$; M with pattern occurrences only at c to $\lambda y.P(Sy)$; and M_W to $\lambda y.(P(y) \to P(Sy))$. By Step 3 every occurrence of T becomes T_{Ind}'s subformula at its position, so $T\sigma = T_{\mathrm{Ind}}$ and $T \succeq T_{\mathrm{Ind}}$. *"Only if".* If all motives are closed, $Z_1 \to \forall x\, Z_2$ (size 4) covers D and misses $\mathrm{Ind}(x = x)$, whose conclusion is not $\forall x\, Z_2$ with Z_2 closed. If all roots are f, use $(\Phi_f \wedge Z_1) \to Z_2$, where the filler of the a-slot is $\Phi_f = z_1\, R\, z_2$ for an atomic root $R \in \{=, <\}$ (z_i 0-ary term metavariables), $\neg Z$, $Z' \circ Z''$, or $\mathsf{Q}y\, K(y)$ with $K(y)$ a pattern occurrence. Each has size ≤ 7, covers every $\mathrm{Ind}(\varphi)$ with root f (because $\varphi[0/x]$ has root f), and misses $\mathrm{Ind}(\neg x = 0)$ (if f is atomic) or $\mathrm{Ind}(x = x)$. These witnesses lie in DT°_s for every $s \geq 7$. □

The following result is summarized after Theorem 5.16; sizes are as defined there.

Proposition D.6 (Generalizations, lgg, size threshold)**.** *[proved]* prior induction C4, C6.2 *(a) Up to renaming, exactly 26 templates of* DT° *contain* $\mathrm{inst}(T_{\mathrm{Ind}})$*. Each takes a prefix of the frame, puts fresh metavariables at the cut slots, and partitions the motive slots present, where* $P(0)$ *and* $P(Sx)$ *join a class only together with a* $P(x)$ *slot. For every anchor these 26 are exactly the covering templates, so* $\mathrm{lgg}_{\mathrm{DT}^\circ}(D) = T_{\mathrm{Ind}}$*. (b)* $\mathrm{inst}(T_{\mathrm{Ind}})$ *is the intersection of the members of* DT°_s *that contain it iff* $s \geq 12$*. For* $7 \leq s \leq 11$*, the cautious verifier over* DT°_s *accepts on every anchor the false sentence* $(0 = 0 \wedge \forall x(x = x \to Sx = Sx)) \to \forall x(x = 0)$*.*

Proposition 5.15 reads T_{pat} off the list in (a): it is the least generalization without non-pattern occurrences. Part (b) is Lemma 2.8 at work.

Proof of Proposition D.6. prior induction C4, C6.2; referee C (a) Every frame generalization is $\succeq T_{\mathrm{Ind}}$ by the substitution of Step 4 above, so it contains $\mathrm{inst}(T_{\mathrm{Ind}})$. Conversely, $\mathrm{inst}(T_{\mathrm{Ind}})$ contains anchors, and Steps 1–3 show that every template of DT° covering an anchor is a frame generalization. The count: the prefixes of the five-node frame are $\emptyset$, $\{\mathrm{imp}\}$, $\{\mathrm{imp}, U\}$, $\{\mathrm{imp}, V\}$, $\{\mathrm{imp}, U, V\}$, $\{\mathrm{imp}, U, s\}$, $\{\mathrm{imp}, U, s, V\}$, $\{\mathrm{imp}, U, s, W\}$ and the full frame. The motive slots present are none, none, a, d, $\{a, d\}$, a, $\{a, d\}$, $\{a, b, c\}$ and $\{a, b, c, d\}$. The admissible partitions number $1, 1, 1, 1, 2, 1, 2, 4$ and 13 (for the full frame: 4 with b, d together, 9 with b, d apart), 26 in all. For an anchor, the covering templates are these 26, and T_{Ind} (full frame, one class) is below all of them. (b) Sizes count symbols, an occurrence $M(\bar{t})$ counting $1 + \sum |t_i|$. "If":

$T_8 = (P(0) \wedge Z) \to \forall x P(x)$ (size 8), $T_{11} = (Z \wedge \forall x(P(x) \to P(Sx))) \to Z'$ (size 11) and $T_{12} = T_{\text{pat}}$ (size 12) contain $\text{inst}(T_{\text{Ind}})$. A sentence in $\text{inst}(T_{12})$ has the form $\alpha \wedge \forall x(\beta \to \gamma) \to \forall x \beta$ with β a motive; T_{11} adds $\gamma = \beta[Sx/x]$ and T_8 adds $\alpha = \beta[0/x]$. So the intersection is $\text{inst}(T_{\text{Ind}})$. "Only if" ($s \leq 11$): sharing a metavariable between d and b needs W and V rigid (five rigid nodes) and fillers of sizes $\geq 1, 2, \geq 2, 2$, so size ≥ 12; sharing between d and c needs ≥ 13. So templates of size ≤ 11 impose at most $a = d[0/x]$, $a = b[0/x]$, $c = b[Sx/x]$ and a frame prefix. The sentence $s^* = (0 = 0 \wedge \forall x(x = x \to Sx = Sx)) \to \forall x(x = 0)$ satisfies all three equations, but $b \neq d$; so it is no induction instance, and it is false. Combine with Lemma 2.8. □

Proof of Proposition 5.15. prior induction C9 $\text{PAT} \subseteq \text{DT}^\circ$, and D is an anchor in DT° (Theorem 5.16). So the covering pattern templates are the frame generalizations of Proposition D.6(a) without non-pattern occurrences: the a-slot gets a 0-ary metavariable or is cut, and the c-slot gets its own $M_c(x)$ or is cut. The most specific of them is the full frame with b, d sharing $P(x)$, which is T_{pat}; all others are above it. The displayed instance ($Z := 0 = 0$, $P, Q := \lambda x.\, x = 0$) has a true premise and a false conclusion, and its c-slot $x = 0$ is not $P(Sx)$. □

Proof of Proposition 5.17. prior induction C7 "Only if". If (R) fails, use the root-specialized templates above. If (B) fails, let $\Theta = \{\theta_1, \ldots, \theta_k\}$ be the parent terms of the free occurrences of x in the motives of D; each is $\neq x$ with $\text{FV} \subseteq \{x\}$. $T_\Theta := Z \to \forall x\, P(\theta_1, \ldots, \theta_k)$ covers D: in a motive, replace each maximal occurrence of some θ_i in which x is free by the hole z_i (closed motives are their own bodies). It misses $\text{Ind}(x = x)$, because in its instances every free x of the conclusion's body lies inside some plugged $\theta_i \neq x$. "If". Let $\varphi^\circ \in D$ have an unshielded free occurrence of x at position q, and let $T \in \text{SO}^\circ$ cover D. By (R) and Lemma D.1 the skeleton is a frame prefix, and all metavariables are formula-valued. For each M fix a body β°_M solving all its occurrences in $\text{Ind}(\varphi^\circ)$. Plugging terms preserves logical size, so the occurrences of M lie in one class: $\{a, b, c, d\}$, $\{W\}$, or a single closed depth-0 slot. *Key fact.* If M occurs at b or d with arguments $\bar{t}$, then β°_M has a hole z_m exactly at q, with $t_m = x$. The x at q must come from a plugged argument at some position $p \leq q$; if $p < q$, the subterm at p would be a proper term $\neq x$ with $\text{FV} \subseteq \{x\}$ containing the occurrence, contradicting unshieldedness. The same holds at W, at position $1 \cdot q$. At c, position q holds Sx, so $\beta^\circ_M|_q$ is z_m with $t_m = Sx$ or $S(z_m)$ with $t_m = x$; the hole cannot sit higher, because q's parent is a predicate (a formula position) or a function symbol whose subterm contains a variable bound inside φ°, which no argument contains. At a, position q holds 0: either a hole with argument 0 or the constant 0 of the body. *Substitution.* (i) M occurring at b or d, with hole z_m at q: $M \mapsto \lambda \bar{z}.P(z_m)$. Every occurrence $M(\bar{u})$ at a motive slot has $u_m = \kappa(\cdot)|_q \in \{0, x, Sx\}$, which is T_{Ind}'s argument there. (ii) M at c but not at b, d: with z_m at q, $M \mapsto \lambda \bar{z}.P(z_m)$ (arguments 0 at a, Sx at c); with $S(z_m)$ at q, $M \mapsto \lambda \bar{z}.P(Sz_m)$, and then M does not occur at a, since $S(u_m) = 0$ is impossible. (iii) M only at W: $M \mapsto \lambda \bar{z}.(P(z_m) \to P(Sz_m))$. (iv) M only at a: $M \mapsto \lambda \bar{z}.P(0)$. (v) M only at a depth-0 slot ρ: $M \mapsto \lambda \bar{z}.T_{\text{Ind}}|_\rho$. Then $T\sigma = T_{\text{Ind}}$. □

The Sub encoding summarized in Section 5.5:

Proposition D.7 (Sub encoding)**.** *[proved]*prior induction C5, P1, P2, P6 *Record each induction as the step from* $\text{Sub}(\varphi, x, 0, \alpha)$ *and* $\text{Sub}(\varphi, x, Sx, \beta)$ *to* $\alpha \wedge \forall x(\varphi \to \beta) \to \forall x \varphi$*, where the decidable judgment* $\text{Sub}(\varphi, x, t, \psi)$ *says that* $\psi = \varphi[t/x]$ *(Hänni and Claude (Anthropic), 2026). With* x *fixed, this is a first-order pattern in* P, α, β*. The lgg of genuine data recovers it iff* (R) *and* (N)*, so the failure probability is that of Theorem 5.11 for one argument place:* $\sum_f p_f^N + (1-q)^N - \sum_f r_f^N$*, where* r_f *is the probability of root* f *with* x *not free. If the induction variable varies, the variables must also not all be equal (or must be* α*-normalized). With the decidable guard "all Sub premises are true", the cautious verifier accepts only genuine inductions.*

Proof of Proposition D.7. prior induction C5, P1, P2, P6 The metavariables P, α, β receive φ, $\varphi[0/x]$ and $\varphi[Sx/x]$. By fact (i) the three (R) events of Plotkin's lemma coincide with (R). The three (D) events coincide with (N), because $\varphi \neq \varphi[0/x]$ iff $\varphi \neq \varphi[Sx/x]$ iff $\varphi[0/x] \neq \varphi[Sx/x]$ iff x is free in φ: at a free occurrence the three formulas carry x, 0 and Sx. The rate is inclusion–exclusion over "all roots are f" (disjoint) and "all motives closed". If the induction variable is a metavariable X, its column consists of the induction variables, which adds the event that they are not all equal. Genuine data satisfy the guard "all Sub premises true", so by the most specific guard rule the verifier accepts inst(lgg$D$) intersected with the Sub-true steps. A Sub-true instance of σ_{Sub} has $\alpha = P[0/x]$ and $\beta = P[Sx/x]$ with P a formula, i.e. it is a genuine induction. □

Proof of Proposition 5.18. prior induction B6 (a) In first-order logic with equality, $\forall x(x = t \to \varphi) \leftrightarrow \varphi[t/x]$ when $x \notin \mathrm{FV}(t)$. Apply this with $t = 0$, y, Sy; since $\mathrm{FV}(\varphi) \subseteq \{x\}$, y is not free in φ, and the step clause becomes $\forall y(\varphi[y/x] \to \varphi[Sy/x])$, an α-variant of $\forall x(\varphi \to \varphi[Sx/x])$. A sentence instance IndEq(ψ) forces $\mathrm{FV}(\psi) \subseteq \{x\}$, since the first conjunct would leave any other free variable free. Under universal closure, $P := (x = 0 \vee \neg x = y)$ gives a false instance: at $y = 1$ the premises hold (the step's hypothesis holds only at $y = 0$, and its conclusion always holds), while $\forall x(x = 0 \vee \neg x = 1)$ fails at $x = 1$. (b) The four occurrences of P carry the same ψ_j in each datum and all other positions agree, so the lgg is IndEq applied to the anti-unifier of the ψ-column, computed once (equal columns get equal results). It is IndEq(P) iff the ψ_j do not all have the same root; (D) is vacuous with one metavariable. (c) By (a) the instances give every parameter-free induction instance. For $\varphi(x, \bar{p})$ apply parameter-free induction to $\psi(x) := \forall \bar{p}[\varphi(0, \bar{p}) \wedge \forall z(\varphi(z, \bar{p}) \to \varphi(Sz, \bar{p})) \to \varphi(x, \bar{p})]$. Here $\psi(0)$ is trivial, $\psi(x) \to \psi(Sx)$ uses the step clause, and $\forall x\, \psi(x)$ gives induction for φ at every $\bar{p}$. Conversely, PA proves every instance. The trick raises quantifier complexity, which is why parameter-free $\mathrm{I}\Sigma_n^-$ is weaker than $\mathrm{I}\Sigma_n$ (Kaye et al., 1988). □

Recorded motives in Lean and Metamath prior induction C12, P8, P9; referees A, C. In Lean 4 (version 4.32.0, checked locally in the earlier analysis and by its referee), `Nat.rec` takes the motive {*motive* : `Nat` → `Sort` u} as an *implicit* argument. The `induction` tactic elaborates to `Nat.recAux` (`fun` $n \Rightarrow \dots$), so the kernel term records the motive as a λ. For eliminators marked `elab_as_elim` the elaborator infers the motive by abstracting the discriminants, using `kabstract`. For closed ∀-form steps the recorded motive is the unique matcher of T_{Ind}, so recording it changes neither anchors nor rates. In CIC proof terms, though, it is recoverable only up to conversion. Writing the motive as a λ-redex app($\lambda x.P, t$), with trusted β, gives a first-order pattern in one metavariable whose lgg is exact iff the motives' roots are not all equal. But a *named* outer binder $\forall n$ captures a parameter n in P: $P := (x = 0 \vee \neg x = n)$ yields, after β, a sentence with a true antecedent and the false consequent $\forall n(n = 0 \vee \neg n = n)$. So the binder must be nameless or guarded by $n \notin \mathrm{FV}(P)$.

Metamath's set.mm states induction (`finds`, `nnind`) with implicit-substitution hypotheses and distinct-variable conditions (Megill and Wheeler, 2019). Transcribed to PA, the premises are H1 $x = 0 \to (\varphi \leftrightarrow \psi)$, H2 $x = y \to (\varphi \leftrightarrow \chi)$, H3 $x = Sy \to (\varphi \leftrightarrow \theta)$, H4 ψ and H5 $\chi \to \theta$, and the conclusion is φ. This is a first-order pattern in $\varphi, \psi, \chi, \theta$. A set of guards from $\{x \notin \mathrm{FV}(\psi), x \notin \mathrm{FV}(\chi), x \notin \mathrm{FV}(\theta), y \notin \mathrm{FV}(\varphi), x \neq y\}$ makes every instance a derived rule of PA iff it contains $y \notin \mathrm{FV}(\varphi)$ and one of $x \notin \mathrm{FV}(\chi)$, $x \notin \mathrm{FV}(\theta)$ [proved]. *Sufficiency*: fix a model M of PA in which the universal closures of the premises hold, a valuation v in M, and let $X = \{a \in M : M \models \varphi(v[x{:=}a])\}$. H1 and H4 give $0 \in X$. For $a \in X$ put $w = v[x{:=}a, y{:=}a]$ and $w' = v[x{:=}a{+}1, y{:=}a]$. Then $\varphi(w)$ holds since $y \notin \mathrm{FV}(\varphi)$, and H2 gives $\chi(w)$. Then H5 gives

$\theta(w')$, either at w' (if $x \notin \mathrm{FV}(\chi)$) or at w (if $x \notin \mathrm{FV}(\theta)$). H3 at w' gives $\varphi(w')$, so $a+1 \in X$. X is definable in M from the parameters of v, so induction in M gives $X = M$. Hence in every model of PA the universal closures of the premises imply that of the conclusion, i.e. the rule is a derived rule of PA. *Necessity*: $\varphi = \neg x = SSy$, $\psi = \neg 0 = SSy$, $\chi = \neg y = SSy$, $\theta = \neg Sy = SSy$ violates only $y \notin \mathrm{FV}(\varphi)$, has PA-provable premises, and concludes the false $\neg x = SSy$. The instance $\varphi = (x = 0)$, $\psi = (0 = 0)$, $\chi = \theta = (x = 0)$ violates only the two x-guards. So set.mm's conditions, read as "not free", are sound but not minimal. Learned from canonical uses, the most specific guards include the spurious $y \notin \mathrm{FV}(\psi)$, which only a non-canonical use removes.

D.6 Computational evidence

Everything above is proved. The computations corroborate it and found no counterexample. Scripts and outputs are in `research/tracks/cases/` and `research/prior/induction/`. Each record also has an independent referee implementation that imports none of the researcher's code. Bounded searches are evidence, not proof.

- *Encodings and first-order failures* cases z1, z4, z5, z8; referee r2. The criterion of Theorem 5.2 was computed from the frames. The witness s_3 was checked for all 7 non-first-order cases. The block structure of Proposition 5.4 matched on all pool pairs with (R) and on 2417 random pairs. Every exhibited false sentence is an instance of the computed lgg and not of the schema. For $n \leq 10$ the path subterms in Theorem 5.6(i)–(iii) are unique, and 2136 random small generalizations τ of s_n contain both $s_n[q \leftarrow \top]$ and $s_n[q \leftarrow \bot]$. The pure-logic steps of the case analysis hold on all $\in$-structures of size ≤ 3. The guard criterion of Proposition 5.8(b) matched on all pool pairs and on 1537 random pairs.

- *Anchors, pattern lgg and rates* cases z2, z3, z6, z7, z9; referee r3, r4. For each of the seven schemas, all 120 pairs of a 16-body pool were checked with a reduced DT° enumeration, which is complete for anchor-hood: anchor $\iff$ (R)$\wedge$(N) on 120/120. Full DT° and SO° enumerations on the smaller frames, and bounded DT°/SO° searches on all frames (7×20 pairs), agree, with identical SO° and DT° verdicts. The referee's SO° search, with repeated and parameter arguments, found 0 disagreements in 460 runs. An n-ary pattern anti-unifier with a permutation merge is exact iff (R)$\wedge$(N) on all pairs and 400 random triples per schema; the referee's implementation agrees on 3500 data sets, and 10 500 random instances of its lggs are genuine. We did not run the published implementation of Baumgartner et al. (2017). For several metavariables, the pattern-lgg criterion held on 3×1500 random data sets and the DT° criterion on 2×150 pairs. Every SO° pair with ($\mathrm{D}^{\mathrm{root}}$) was an anchor, and none violating (D^{DT}) was. The few pairs with (D^{DT}) but not ($\mathrm{D}^{\mathrm{root}}$) were SO° anchors under the bounded search, which suggests Conjecture D.4. The exact rates agree with Monte Carlo.

- *Induction* prior induction b1–b4, c0–c5; referees B, C. An exactly false lgg instance was found for 2357 of the 2556 pairs of quantifier-free motives of size ≤ 5; the smallest unsound pair has motive sizes $3+3$. lgg $= L_n$ for all $0 \leq n < m \leq 10$. For K_0, a congruence-closure test on 2224 instances finds the step clause refutable over $\mathbb{N}$ exactly in the 12 Case-T instances. For all 351 pairs of a 27-motive pool, all SO° templates of size ≤ 14 (arity ≤ 2, arguments of size ≤ 3; 4.6 million covering templates) were classified: DT° anchor $\iff$ (R)$\wedge$(N), and SO° anchor $\iff$ (R)$\wedge$(B), on 351/351. The referee found no disagreement on 176 further data sets. With bounded arity, though, some non-(B) sets are SO° anchors, since Proposition 5.17 needs unbounded arity. The 26 generalizations and the threshold 12 were reproduced. The Sub-lgg and DT° verdicts coincide on all pairs and 2925 triples. IndEq

and Ind agree on 30 000 random pairs of a motive and a finite structure.

- *Experiment E2* experiments E2: see Table 9 and Section 7.

E Proofs and details for Section 6

Notation is that of Section 6.1. "Template" means a template in DT°; "covering X" means having every element of X as an instance. The finitariness facts of Proposition 6.2 are used throughout.

E.1 The setting

Proof of Lemma 6.3. (a) $R^* = \bigcup_i \mathrm{inst}(\sigma_i) \in H_k$ because $k \geq k'$; it contains D and avoids N. Larger D or N shrink the version space, so they enlarge its intersection (Theorem 2.7(a), Lemma 2.10(b)). (b) Each displayed union is in H_k (at most k blocks), contains D and avoids N, so it lies in $\mathrm{VS}_k(D, N)$; hence $\mathrm{Acc}_k(D, N)$ is contained in the displayed intersection. Conversely let $h = \bigcup_l \mathrm{inst}(\tau_l) \in \mathrm{VS}_k(D, N)$. Assign each datum to the first slot covering it; this partitions D into at most k nonempty blocks $B \subseteq \mathrm{inst}(\tau_l)$. By Proposition 6.2(b), $\tau_l \succeq T_B$ for some $T_B \in \mathrm{Min}(B)$, and $\mathrm{inst}(T_B) \subseteq \mathrm{inst}(\tau_l) \subseteq h$ avoids N, so $T_B \in \mathrm{Min}_N(B)$ and $h \supseteq \bigcup_B \mathrm{inst}(T_B)$. So every member of the version space contains a displayed union, and the displayed intersection is contained in $\mathrm{Acc}_k(D, N)$. Decidability: there are finitely many partitions, each $\mathrm{Min}(B)$ is finite and computable, and avoidance of a finite N and membership of a query are decided by matching (Theorem 2.5). □

With $N = \emptyset$ and $\mathrm{Acc}(B) := \bigcap \mathrm{Cov}(B) = \bigcap_{T \in \mathrm{Min}(B)} \mathrm{inst}(T)$ (and $\mathrm{Acc}(\emptyset) := \emptyset$), part (b) says, since the choices T_B are independent across blocks:

$$q \in \mathrm{Acc}_k(D) \iff \forall \pi \; \exists B \in \pi : \; q \in \mathrm{Acc}(B), \tag{1}$$

where π ranges over the partitions of D into at most k nonempty blocks.

Lemma E.1 (monotonicity of coherence)**.** *[proved]*untagged Lemma 5.1 *X is d-coherent iff some $T \in \mathrm{Cov}(X)$ is not d-refuted. d-coherence is downward closed ($\emptyset \neq Y \subseteq X$ and X d-coherent imply Y d-coherent) and antitone in d.*

Proof. If $T \in \mathrm{Cov}(X)$ is not d-refuted, then $T \succeq T_0 \in \mathrm{Min}(X)$, and $\mathrm{inst}(T_0) \subseteq \mathrm{inst}(T)$ is not d-refuted either; the converse holds as $\mathrm{Min}(X) \subseteq \mathrm{Cov}(X)$. A covering template of X covers Y. And $\mathrm{Ref}_d \subseteq \mathrm{Ref}_{d+1}$. □

E.2 Positive data: bounds, spare slots and head classes

Proof of Proposition 6.4. D is a finite union of ground templates and avoids N. For the second claim take a true non-ground σ: the class contains $\mathrm{inst}(\sigma)$ and all its finite subsets, all inside Tr, on which the oracle answers identically. So the learner is a function of the text, and Gold's locking-sequence argument for superfinite classes applies (Gold, 1967); it needs no computability. □

Proof of Theorem 6.5. (a) $h \subseteq R^*$ and $N \cap R^* = \emptyset$ give $h \cap N = \emptyset$, and $D \subseteq h \in H_k$. (b) Let $h = \bigcup_l \mathrm{inst}(\tau_l) \in \mathrm{VS}_k(D, N)$ and $L_j := \{l : \tau_l \text{ covers a datum of } D_j\}$. Each L_j is nonempty because D_j is. By cross-separation a slot covering data of two targets would meet N, so the L_j are pairwise disjoint and $|L_i| \leq k - (k' - 1) = m$. Every datum of D_i is covered by a slot in L_i, and these slots avoid N, so by hypothesis $\bigcup_{l \in L_i} \mathrm{inst}(\tau_l) \supseteq \mathrm{inst}(\sigma_i)$; hence $h \supseteq \mathrm{inst}(\sigma_i)$. For the

second claim, $h := \bigcup \mathrm{inst}(F) \cup \bigcup_{l \neq i} \mathrm{inst}(\sigma_l)$ uses at most $m + k' - 1 = k$ slots, contains D, avoids N (the σ_l lie in R^*) and misses q. If the instance sets are pairwise disjoint and the first condition fails, some admissible family misses some $q \in \mathrm{inst}(\sigma_i)$, which lies in no other instance set, so the second claim applies: the condition is necessary and sufficient. □

In general a target's slots may be shared with others, provided every shared slot contains the whole target.

Theorem E.2 (slot accounting)**.** *[proved]*[untagged Thm 4.1] *Fix a target i, a bound k and a sound N, and let $\bar{D}_i := D \setminus D_i$. Assume (Abs): every N-avoiding template covering a datum of D_i and one of $\bar{D}_i$ contains* $\mathrm{inst}(\sigma_i)$*. Let c_i be the least number of N-avoiding templates not containing* $\mathrm{inst}(\sigma_i)$ *that cover $\bar{D}_i$. (a) If every family of at most $k - c_i$ N-avoiding templates covering D_i has union $\supseteq \mathrm{inst}(\sigma_i)$, then $\mathrm{inst}(\sigma_i) \subseteq \mathrm{Acc}_k(D, N)$. (b) If some such family and some optimal cover of $\bar{D}_i$ both miss $q \in \mathrm{inst}(\sigma_i)$, then $q \notin \mathrm{Acc}_k(D, N)$.*

Corollary E.3 (slot accounting for PA)**.** *[proved; computed]*[untagged Cor 4.2] *Let the targets be* Q*'s axioms Q1–Q7 (numbered as in Section 2.6) as ground sentences in closure-normal form ($\neg Sp = 0$; $Sp = Sp' \to p = p'$; $\neg p = 0 \to \exists y\, p = Sy$; $p{+}0 = p$; $p{+}Sp' = S(p{+}p')$; $p{\cdot}0 = 0$; $p{\cdot}Sp' = p{\cdot}p'{+}p$) and T_{Ind}, so $k' = 8$. (Abs) holds for induction for every motive: a template covering a* Q*-axiom and an induction instance lies above F_0, or above $F_0 \to F_1$ for Q2 and Q3 (F_0, F_1 0-ary formula metavariables), and both contain* $\mathrm{inst}(T_{\mathrm{Ind}})$*. On positive data $c = 4$ (Q4–Q7 share $z_0 = z_1$; Q1, Q2, Q3 stay alone), so at $k = 8$ induction gets four slots: an induction instance whose motive has x free and a main connective unseen in the data is accepted iff the induction data cannot be split into four blocks each root-homogeneous or all-vacuous (without vacuous motives: iff at least five main connectives occur). With the three negatives $0 = S0$, $(p + 0) = 0$, $(p \cdot 0) = S0$, $c = 7$, induction gets one slot, and the tagged anchor condition* (R)+(N) *suffices. (In the Sub-encoded presentation of induction of the earlier analysis of untagged PA, $c = 1$* [prior pa-untagged]*.)*

The computed check of $c = 4$ and $c = 7$ (24/24 cases)[untagged u2] holds by construction given (Abs) and the anchor theorem for T_{Ind} (Theorem 5.16), so it is a consistency check of the implementation, not independent evidence.

Proof of Theorem E.2. (a) Let $h \in \mathrm{VS}_k(D, N)$ with slots τ_l. If some slot contains $\mathrm{inst}(\sigma_i)$ we are done. Otherwise, by (Abs), no slot covers both a datum of D_i and a datum of $\bar{D}_i$. The slots covering $\bar{D}_i$-data form an N-avoiding cover of $\bar{D}_i$ by templates not containing $\mathrm{inst}(\sigma_i)$, so there are at least c_i of them, and they are disjoint from the slots covering D_i. So at most $k - c_i$ slots cover D_i, and by hypothesis their union contains $\mathrm{inst}(\sigma_i)$. (If $k \leq c_i$, this shows that some slot of every h contains $\mathrm{inst}(\sigma_i)$.) (b) $\bigcup \mathrm{inst}(F \cup G)$ has at most k slots, contains D, avoids N and misses q. □

Proof of Corollary E.3. (Abs). At the top level of a sentence no bound variable is in scope, so a metavariable whose pattern occurrence is not under a binder is 0-ary. A Q-axiom with root $\neg$ or $=$ and an induction instance (root $\to$) have no common root, so their minimal covering template is the bare F_0. Q2 and Q3 have root $\to$; their antecedents ($=$, resp. $\neg$) differ in head from the induction antecedent ($\wedge$), and their consequents ($=$, resp. $\exists$) from the induction consequent ($\forall$); the two slots are closed and not equal in either datum, so the minimal covering template is $F_0 \to F_1$. This holds for every motive. Every covering template lies above one of these (Proposition 6.2), and both contain $\mathrm{inst}(T_{\mathrm{Ind}})$. $c = 4$ *with* $N = \emptyset$. Any template covering two of Q1, Q2, Q3, or one of them and one of Q4–Q7, lies above F_0 or $F_0 \to F_1$ (compare the roots and,

for Q2 and Q3, the heads of antecedents and consequents), hence contains inst(T_{Ind}). So Q1, Q2 and Q3 need a slot each and Q4–Q7 at least one; $z_0 = z_1$ covers Q4–Q7 and contains no induction instance. $c = 7$ *with* $N = \{0 = S0, (p + 0) = 0, (p \cdot 0) = S0\}$. Q4 and Q5 have the lgg $p + z_0 = z_1$ (common prefix $=, +, p$; closed slots), which has the instance $(p + 0) = 0 \in N$; Q6 and Q7 have $p \cdot z_0 = z_1 \ni (p \cdot 0) = S0$; the other pairs among Q4–Q7 have $z_0 = z_1 \ni 0 = S0$. All covering templates of these pairs lie above the lggs, so Q4–Q7 need a slot each. (F_0 is now refuted, $F_0 \to F_1$ is not, so (Abs) still holds.) Then induction gets $k - 7 = 1$ slot, and by Theorem E.2 it is learned exactly iff its data contain an anchor, which for T_{Ind} is (R)+(N) (Theorem 5.16). *The four-slot criterion.* Let q be an induction instance whose motive has x free and has a main connective that occurs in no datum. By Theorem 5.16, a block of induction data is a non-anchor iff its motives all have the same root f or are all vacuous; then $T_{\mathrm{Ind}}[P := \lambda x.f(\dots)]$, resp. $T_{\mathrm{Ind}}[P := \lambda x.A]$, covers the block and misses q. Let χ be the least number of blocks in a partition of the induction data into blocks that are root-homogeneous or all-vacuous (without vacuous motives: the number of main connectives occurring). If $\chi \leq 4$, these templates for an optimal partition, together with the cover {Q1, Q2, Q3, $z_0 = z_1$} of the Q-data, form a hypothesis with 8 slots missing q (Theorem E.2(b)). If $\chi > 4$, every family of at most four templates covering the induction data has, by partition into first covering members, an anchor block, whose template contains inst(T_{Ind}) (Theorem E.2(a)). □

Corollary E.4 (thresholds in terms of failure families)**.** *[proved]*[untagged Cor 3.3] *Call T a* proper specialization *of σ if* $\mathrm{inst}(T) \subsetneq \mathrm{inst}(\sigma)$. *A* failure family $\mathrm{Fail}(\sigma)$ *is a set of proper specializations such that every finite non-anchor $X \subseteq \mathrm{inst}(\sigma)$ lies in* $\mathrm{inst}(F)$ *for some $F \in \mathrm{Fail}(\sigma)$. (Examples: for first-order patterns the head specializations $\sigma[x \mapsto f(\bar{z})]$ and the identifications $\sigma[y \mapsto x]$, by Plotkin's* (R)+(D)*; for T_{Ind} the nine root specializations $T_{\mathrm{Ind}}[P := \lambda x.f(\bar{P}(x))]$, $f \in \{=, <, \neg, \wedge, \vee, \to, \leftrightarrow, \forall, \exists\}$ (eight in the untagged prototype, whose arithmetic has no $<$), and the vacuous one $T_{\mathrm{Ind}}[P := \lambda x.A]$.) Let $m = k - k' + 1$. (a) If $D_i \subseteq \bigcup_{j \leq m} \mathrm{inst}(F_j)$ with $F_j \in \mathrm{Fail}(\sigma_i)$, and some $q \in \mathrm{inst}(\sigma_i) \setminus \bigcup_j \mathrm{inst}(F_j)$ lies in no other* $\mathrm{inst}(\sigma_l)$*, then $q \notin \mathrm{Acc}_k(D, N)$ for every sound N. (b) Suppose N cross-separates D, N* target-separates *D_i (it meets every template that covers a datum of D_i and whose instance set neither contains nor is contained in* $\mathrm{inst}(\sigma_i)$*), and the instance sets are pairwise disjoint. Then* $\mathrm{inst}(\sigma_i) \subseteq \mathrm{Acc}_k(D, N)$ *iff D_i is not covered by m proper specializations of σ_i whose union misses an instance of σ_i. If moreover no m members of* $\mathrm{Fail}(\sigma_i)$ *cover* $\mathrm{inst}(\sigma_i)$*, this holds iff D_i is not covered by m members of* $\mathrm{Fail}(\sigma_i)$.

Proof. (a) Apply Theorem 6.5(a) to $h = \bigcup_j \mathrm{inst}(F_j) \cup \bigcup_{l \neq i} \mathrm{inst}(\sigma_l)$, which has k slots and satisfies $D \subseteq h \subseteq R^*$. (b) If exactness for i fails, then by Theorem 6.5(b) (which uses cross-separation) some N-avoiding family of at most m templates covers D_i and misses some $q \in \mathrm{inst}(\sigma_i)$. No member contains inst(σ_i), so by target-separation each member covering a D_i-datum is a proper specialization. Each block $X_j = D_i \cap \mathrm{inst}(T_j)$ is covered by a template not containing inst(σ_i), so it is not an anchor and lies in some inst(F_j), $F_j \in \mathrm{Fail}(\sigma_i)$. Conversely, a cover by proper specializations that misses q gives $q \notin \mathrm{Acc}_k(D, N)$ by Theorem 6.5(b) and disjointness, and a cover by m failure templates misses some q under the extra hypothesis, so (a) applies. □

On positive data ($N = \emptyset$) target i is learned exactly if every partition of D_i into at most k blocks (at most $k - c_i$ under (Abs)) has an anchor block, and not if D_i is covered by $k - k' + 1$ proper specializations missing an instance of σ_i that lies in no other target [untagged Prop 6.1]. For PA the gap between $k - c$ and $k - k' + 1$ is the whole story: $c = 1$ for Sub-encoded steps, $c = 4$ for raw sentences, and $c = 7 = k' - 1$ with three negatives. The classical sufficient condition "not covered by k failure sets" is vacuous for T_{Ind} at $k \geq 9$, because the nine root specializations

cover every induction instance. Plotkin's failure families are not m-noncovering for every m once data are read under universal closure (Proposition E.6; the earlier contrary claim is withdrawn, Appendix G.2).

Head classes. In the rest of this subsection sentences are identified up to bijective renaming of parameters, as the closure reading demands and as the implementation does (parameters are renamed canonically by first occurrence). Remark E.5 summarizes the results; Propositions E.6 to E.8 state and prove them.

Remark E.5 (head classes and numerals). [proved; computed]untagged U17 Because data are read under universal closure, a one-variable instance schema $\varphi(z)$ is covered by its $h = |F|+p+1$ head specializations (F the function symbols, p the number of parameters of φ; $5+p$ in arithmetic), one per head class of the value of z, a parameter outside φ being one class (Proposition E.6). So once $m \geq h$ (i.e. at least $h-1$ spare slots) the "failure family" covers everything. For quantifier-free, parameter-free φ, under the hypotheses of Proposition E.7, the exact threshold is known for $m \leq h$ (Proposition E.7) and, over a unary signature, in closed form (Proposition E.8). In particular *numeral-only data never make the cautious k-union learner accept $\forall x\varphi$ with one spare slot* when $\varphi(a)$ lies in no other target's instance set: the sound split $\{\varphi(0)\} \cup \mathrm{inst}(\varphi(Su))$ misses $\varphi(a)$ (Theorem 6.5(a)). Brute-force checks agree with all three propositions (2000/2000, 60/60, 254/254, 2509/2509 cases). Beyond this the spare-slot threshold is open (binary symbols with $m > h$, several metavariables, templates with parameters, T_{Ind} with $m \geq 9$, where the nine root specializations already cover $\mathrm{inst}(T_{\mathrm{Ind}})$).

Proposition E.6 (finitely many head classes). *[proved; computed]untagged Prop 3.4 Let $\sigma = \varphi(z)$ have a single metavariable, a 0-ary term metavariable z occurring in φ; let F be the function symbols (constants included) and $c_1, \ldots, c_p$ the parameters of φ. Every closed term t has one head class* $\mathrm{head}(t) \in F \cup \{c_1, \ldots, c_p\} \cup \{\nu\}$*, where ν means "a parameter other than the c_j". Put $\sigma_f := \varphi(f(u_1, \ldots, u_r))$ (fresh 0-ary u's), $\sigma_{c_j} := \varphi(c_j)$, $\sigma_\nu := \varphi(b)$ for a parameter b not in φ. Then (a)* $\mathrm{inst}(\sigma) = \bigcup_g \mathrm{inst}(\sigma_g)$ *over the $h = |F| + p + 1$ classes, and (b) no σ_g is redundant.*

Proof. (a) For an instance $\varphi(t)$: if $t = f(\bar{t})$ it is $\sigma_f[\bar{u} := \bar{t}]$; if $t = c_j$ it is σ_{c_j}; if t is a parameter $b' \notin \{c_j\}$, the renaming swapping b and b' maps $\varphi(b)$ to $\varphi(b')$, since φ contains neither, so the two are the same datum. (b) Suppose a renaming π maps $\varphi(t)$ to $\varphi(t')$. At the rigid occurrences of c_j in φ it fixes c_j, so it maps other parameters to other parameters; at the position of z it maps t to t'. So $\mathrm{head}(t) = \mathrm{head}(t')$, and σ_g covers only instances whose z-value has class g. □

Computed[untagged u12(a)]: the 5 head-class templates of $z + 0 = z$ and the 6 of $c_1 + z = z + c_1$ (one parameter) cover 2000/2000 random canonical instances each, and each template is needed.

Proposition E.7 (threshold for one-variable instance schemas). *[proved; computed]untagged Prop 3.5 Assume (H1) $\sigma_i = \varphi(z)$ with φ quantifier-free and parameter-free and z a 0-ary term metavariable occurring in it; (H2) every datum of D_i has a z-value with at most one parameter, and T is the set of these values with the parameter renamed a; (H3) N cross-separates D, the instance sets are pairwise disjoint, every D_j is nonempty, and $m = k - k' + 1$. Let* $\mathrm{cl}(T)$ *be the set of head classes occurring in T and $h = |F| + 1$. (a) If $m < |\mathrm{cl}(T)|$, then* $\mathrm{inst}(\sigma_i) \subseteq \mathrm{Acc}_k(D, N)$*. (b) If some class is missing from T, then* $\mathrm{inst}(\sigma_i) \subseteq \mathrm{Acc}_k(D, N)$ *iff $m < |\mathrm{cl}(T)|$. (c) If all h classes occur and $m = h$, then* $\mathrm{inst}(\sigma_i) \subseteq \mathrm{Acc}_k(D, N)$ *iff for each $f \in F$ of arity $r \geq 1$ the block T_f of terms with head f has* $\mathrm{lgg} T_f = f(x_1, \ldots, x_r)$ *with distinct variables, i.e. Plotkin's* (R)+(D) *holds for T_f (each argument column has two different heads, the parameter counting as one head, and no two columns are equal).*

Proof of Proposition E.7. Step 0 (least covering template of a block). Let $\emptyset \neq B \subseteq D_i$ have z-values $\{t_b\}$. Every $T \in \mathrm{Cov}(B)$ satisfies $\mathrm{inst}(T) \supseteq \mathrm{inst}(\varphi(\mathrm{lgg}\{t_b\}))$, the lgg (Plotkin's) taken with a as a constant. Indeed: the data are quantifier-free, so T has no binder in its rigid part, no bound variable is in scope, and every metavariable is 0-ary: T is a first-order pattern. It covers each datum through a representative $\varphi(t_b[n_b/a])$. By Plotkin's theorem, $T \succeq \mathrm{lgg}\{\varphi(t_b[n_b/a])\} = \varphi(G)$ with $G := \mathrm{lgg}\{t_b[n_b/a]\}$ (lgg commutes with the parameter-free context φ: rigid positions carry constant tuples, and the z-positions carry the same tuple, hence the same variable). If all n_b equal some n, $G = \mathrm{lgg}\{t_b\}[n/a]$, the same up to renaming. Otherwise G has no rigid parameter (a rigid n at position q would force $t_b|_q = a$ and $n_b = n$ for all b), and $G \succeq \mathrm{lgg}\{t_b\}$: the rigid symbols of G are function symbols at positions where all t_b agree, and two variable positions of G with the same tuple $(t_b[n_b/a]|_q)_b$ also have the same tuple $(t_b|_q)_b$ (undo the renamings), so mapping each variable of G to the corresponding subterm of $\mathrm{lgg}\{t_b\}$ is well defined and turns G into $\mathrm{lgg}\{t_b\}$. So $\mathrm{inst}(T) \supseteq \mathrm{inst}(\varphi(G)) \supseteq \mathrm{inst}(\varphi(\mathrm{lgg}\{t_b\}))$. The template $\varphi(\mathrm{lgg}\{t_b\})$ covers B, lies below σ_i, and so avoids every sound N.

Step 1 (reduction). By Theorem 6.5(b), exactness for i holds iff every N-avoiding family of at most m templates covering D_i has union $\supseteq \mathrm{inst}(\sigma_i)$. Partitioning D_i by first covering member and using Step 0, this holds iff for every partition π of D_i (equivalently of T) into at most m blocks, $\bigcup_{B\in\pi} \mathrm{inst}(\varphi(\mathrm{lgg}T_B)) = \mathrm{inst}(\sigma_i)$. Since z occurs in the parameter-free φ, $\varphi(t)$ determines t up to renaming, so this says that every closed term is, up to renaming, an instance of some $\mathrm{lgg}T_B$: $\bigcup_{B\in\pi} \mathrm{inst}(\mathrm{lgg}T_B) = G_{\mathrm{cl}}$, the closed terms modulo renaming.

Step 2 (mixed blocks). A block with two head classes has a variable as lgg, which covers G_{cl}. So failing partitions are head-pure, and a head-pure partition has at least $|\mathrm{cl}(T)|$ blocks. This proves (a).

Step 3 (a missing class). If a class g is missing from T, the head partition ($|\mathrm{cl}(T)|$ blocks) has lggs that are constants, the parameter a, or terms with heads in $\mathrm{cl}(T)$, so no term of class g is covered and it fails. With (a), this proves (b).

Step 4 ($m = h$, all classes present). The only head-pure partition with at most h blocks is the head partition, and it covers G_{cl} iff each block covers its class. A constant class (a 0-ary f, or ν) has one term, which its block covers. For f of arity $r \geq 1$, $\mathrm{inst}(\mathrm{lgg}T_f) \supseteq f(G_{\mathrm{cl}}^r)$ iff $\mathrm{lgg}T_f = f(x_1, \ldots, x_r)$ with distinct variables: a non-variable argument misses terms with another head there, and a shared variable misses terms with different subterms in the two arguments. Plotkin's construction gives a variable in argument j iff column j has two different heads, and the same variable in two arguments iff the columns are equal. □

Proposition E.8 (closed form for unary signatures)**.** *[proved; computed]*[untagged Prop 3.6] *Assume (H1)–(H3) with function symbols a finite set C of constants and one unary S (so (H2) is automatic and $h = |C| + 2$). Put $T_0 := T$, $T_{j+1} := \{u : Su \in T_j\}$, and let r be the least j with $|\mathrm{cl}(T_j)| < h$. Then $\mathrm{inst}(\sigma_i) \subseteq \mathrm{Acc}_k(D, N)$ iff $m < \kappa(T) := (h-1)r + |\mathrm{cl}(T_r)|$.*

Proof of Proposition E.8. Let $\kappa(T)$ be the least number of blocks of a partition of T whose lggs fail to cover G_{cl}; by Steps 1–3, exactness holds iff $m < \kappa(T)$, and failing partitions are head-pure. If $r = 0$ (a class is missing), $\kappa(T) = |\mathrm{cl}(T_0)|$ by Step 3. If all h classes are present, each of the $h - 1$ constant classes (C and ν) is a single term and forms a block of a head-pure partition, and the other blocks partition the S-class. Since lgg of an S-block is S applied to the lgg of the arguments and $\mathrm{inst}(Ss) = S(\mathrm{inst}s)$, the S-blocks cover $S(G_{\mathrm{cl}})$ iff the corresponding blocks of T_1 cover G_{cl}. So $\kappa(T) = (h-1) + \kappa(T_1)$, and induction on r gives $\kappa(T) = (h-1)r + |\mathrm{cl}(T_r)|$. □

Computed[untagged u12]: for $z + 0 = z$ and 60 random data sets, exactness at $m = h = 5$, decided by brute force over all partitions with coverage of all 608 canonical instances of depth

≤ 2 (decisive, since every failure in Step 4 is witnessed at depth ≤ 2), agreed with (c) in 60/60 cases (18 exact), and with $m = |\mathrm{cl}(T)| - 1$ the target was exact in 60/60; the closed form agreed with brute force on all 254 term sets of size ≤ 7 over $\{S^j b : j \leq 3, b \in \{0, \nu\}\}$ and all 2509 sets of size ≤ 6 over $\{S^j b : j \leq 3, b \in \{0, c, \nu\}\}$.

E.3 Self-generated negatives

Corollary E.9 (self-generated negatives)**.** *[proved]*[untagged Cor 2.2] *Assume (Dec) and fix an effective enumeration of $\mathcal{S}$. (a) If every $T \in \mathrm{Min}(\{a, b\})$ with $a \in A_i$, $b \in A_j$, $i < j$, is d-refuted, then one refuted instance n_T per such T gives a set $N_\times$ satisfying (X) of Theorem 6.6, with $|N_\times| \leq \mu \sum_{i<j} |A_i||A_j|$, where $\mu = \max |\mathrm{Min}(\{a, b\})|$ over cross anchor pairs. (b) The set $N(D, d)$ of first d-refuted instances of all d-refuted $T \in \mathrm{Min}(\{a, b\})$, $a, b \in D$, is computable from D without labels, is contained in Ref_d, and contains a choice of $N_\times$. Hence $\mathrm{Acc}_{k'}(D, N(D, d))$ is sound at all times, decidable, and equal to R^* as soon as D contains anchors A_i satisfying the hypothesis of (a).*

Proof. (a) If T' covers $a \in A_i$ and $b \in A_j$, then $T' \in \mathrm{Cov}(\{a, b\})$, so $T' \succeq T$ for some $T \in \mathrm{Min}(\{a, b\})$ and $n_T \in \mathrm{inst}(T) \subseteq \mathrm{inst}(T')$. (b) $\mathrm{Min}(\{a, b\})$ is computable and "d-refuted" decidable; the first refuted instance is found by enumeration. Within-target pairs may contribute further refuted sentences, which are false and harmless. Apply Lemma 6.3 and Theorem 6.6. □

Remark E.10 (combining with the k-union learner). [proved][untagged Rem 5.3] Let $k \geq k'$ be known and accept q iff $q \in \mathrm{Acc}_d \cap \mathrm{Acc}_k(D, N(D, d))$. This is target-sound at all times, by Lemma 6.3, with or without separation. Under $\mathrm{RS}_d(D)$ with $k = k'$, once every D_i contains an anchor, it equals R^*: DTRC is exact by Theorem 6.8(iii), and $\mathrm{RS}_d(D)$ makes every minimal template of every cross pair d-refuted, so $N(D, d)$ satisfies (X) and Theorem 6.6 applies. The anchor hypothesis is needed for this argument: DTRC can be exact without anchors, because refuted templates of whole clusters drop out, while $N(D, d)$ contains only refutations of pair templates (Appendix G.2).

For PA (Table 13) the four sentences $0 = S0$, $0 = 0 \to 0 = S0$, $p + 0 = 0$, $p \cdot 0 = S0$ refute every cross template; since the Q/induction templates F_0 and $F_0 \to F_1$ contain all of $\mathrm{inst}(T_{\mathrm{Ind}})$, three suffice for the k-union learner (Corollary E.3).

E.4 dtrc: proofs

Proof of Theorem 6.8. (i) By Proposition 6.2(c) some $T_0 \in \mathrm{Min}(X)$ has $\mathrm{inst}(T_0) \subseteq \mathrm{inst}(\sigma_i) \subseteq R^* \subseteq \mathrm{Tr}$, so $\mathrm{inst}(T_0) \cap \mathrm{Ref}_d = \emptyset$ by (OS). (ii) Call a cluster pure if it lies inside one D_i; initially all are. If $C \subseteq D_i$ and $C' \subseteq D_j$ are pure with $i \neq j$, every $T \in \mathrm{Cov}(C \cup C')$ covers a datum of D_i and one of D_j, so it is d-refuted by $\mathrm{RS}_d(D)$: the union is not coherent and is never formed. So all clusters stay pure. The process stops after finitely many merges. At termination, two final clusters inside one D_i would have a coherent union by (i), contradicting termination. So each nonempty D_i is exactly one final cluster, whatever the policy. (iii) For the final cluster D_i, the T_0 of (i) lies in $\mathrm{Min}^+_d(D_i)$, so $\mathrm{Acc}_d(D_i) \subseteq \mathrm{inst}(T_0) \subseteq \mathrm{inst}(\sigma_i)$ and $\mathrm{Min}^+_d(D_i) \neq \emptyset$. If $A_i \subseteq D_i$ is an anchor, every $T \in \mathrm{Min}(D_i)$ covers A_i, so $\mathrm{inst}(T) \supseteq \mathrm{inst}(\sigma_i)$ and $\mathrm{Acc}_d(D_i) \supseteq \mathrm{inst}(\sigma_i)$. (iv) Under global separation, (ii) and (iii) hold for every $D \subseteq R^*$. So DTRC fails to be exact only if some D_i lacks an anchor, which is the failure event of the tagged learner given the labels; the union bound over i gives the sum, and averaging the tagged bound over the binomial number of target-i data gives the rate. (v) A pair answer depends only on the two clusters. There are $n(n-1)/2$ initial pairs; there are at most $n - 1$ merges, and after each only pairs with the new cluster (at most n)

need retesting. For the online policy under $\mathrm{RS}_d(D)$, by induction on the data there is at most one cluster per target at all times: a datum of target i joins the first coherent cluster, cross unions are incoherent, and the cluster of target i is coherent with it by (i). So each datum is tested against at most k' clusters, and the final pass tests the $k'(k'-1)/2$ pairs of pure clusters. □

Proof of Theorem 6.10. (a) A pure final cluster $C \subseteq D_i$ contributes $\mathrm{Acc}_d(C) \subseteq \mathrm{inst}(\sigma_i)$, as in the proof of Theorem 6.8(iii). A mixed final cluster C is d-coherent or an incoherent singleton contributing $\emptyset$ (Proposition E.11), and it contributes $\mathrm{Acc}_d(C) \subseteq \mathrm{inst}(T)$ for each $T \in \mathrm{Min}_d^+(C)$, which is part of $\mathrm{Res}_d(D)$ (take $X = C$). (b) $\mathrm{Acc}_d(C) \subseteq \mathrm{inst}(T) \subseteq \mathrm{Tr}$. (c) By definition; $\mathrm{Res}_d(D)$ decreases in d because coherence and non-refutation are antitone in d (Lemma E.1). (d) In the example of Section 6.5 all data are quantifier-free, so every minimal template is a Plotkin lgg (Proposition E.14(b)). $\{g, 0+0=0\}$ has the lgg $0+z=z$ (the columns $(S0, 0)$ at the second argument and at the right-hand side are equal), which is true and never refuted; $\{g, 0+0 = 0, S0+0 = S0\}$ has $z_0+z_1 = z_2$ and $\{S0+0 = S0, g\}$ has $z_0+z_1 = S0$, both refuted by $0+0 = S0$. So in the order $g, 0+0=0, S0+0=S0$ the final clusters are $\{g, 0+0=0\}$ and $\{S0+0=S0\}$; $p+0=p$ is in neither acceptance set, while $0+p=p$ is accepted. Computed[untagged u9, Example B], with the referee's variant with overlapping instance sets ($\sigma_2 = 0+z=z$) behaving the same way in 2 of 3 orders. □

Proof of Lemma 6.12. Two-point identity. Let $G(z), K(z)$ be expressions in which z occurs only as a leaf, with $G(t_j) = K(t_j)$ for $j = 1, 2$. Then $G = K$. Induction on G and K: if $G = z$ and $K \neq z$, then K has a rigid head h (a symbol or an index) that survives substitution, and $K(t_j) = t_j$ forces $h = \mathrm{head}(t_1)$ and $h = \mathrm{head}(t_2)$, a contradiction; symmetrically for $K = z$. If neither is z, their heads agree (compare at t_1), and we recurse into the arguments; binders are harmless, because z is not a bound variable and the t_j are closed. *Skeleton.* The two data agree everywhere except at and below the z-positions of φ, where their heads differ. By Lemma 3.2(c) the rigid skeleton of a covering template T lies in the common prefix of the data, so it avoids the z-positions and everything below them; every occurrence of T is at a position of φ. *Bodies.* For each metavariable M of T choose a pattern occurrence $M(\bar{y})$ at position p. For any closed t put $\theta_t(M) := \lambda\bar{y}.(\varphi(t)|_p)$. This is a legal body: t is closed, so the bound variables free in $\varphi(t)|_p$ are those free in $\varphi(t_1)|_p$, which are among $\bar{y}$ because T covers $\varphi(t_1)$. *Every occurrence fits.* For an occurrence $M(\bar{u})$ at position p' put $G(z) := (\lambda\bar{y}.\varphi(z)|_p)(\bar{u})$ (plugging) and $K(z) := \varphi(z)|_{p'}$. G contains z only as a leaf, since $\bar{u}$ is metavariable-free. T covers both data with the unique matchers $\theta_{t_1}, \theta_{t_2}$, and substituting a closed t_j for z commutes with abstraction and plugging, so $G(t_j) = K(t_j)$; hence $G = K$ and $G(t) = K(t)$. With the skeleton, $T\theta_t = \varphi(t)$, so $\mathrm{inst}(T) \supseteq \mathrm{inst}(\varphi(z))$. □

Proof of Proposition 6.13. (a) $\varphi(z)$ covers D, and by Lemma 6.12 every covering template contains $\mathrm{inst}(\varphi(z))$; so some covering template is unrefuted iff $\varphi(z)$ is (Lemma E.1). (b) In closure-normal form a parameter is read universally; it must be fresh, so that $\varphi(a)$ is not an instance at a special parameter. If all instances are true, so is the closure of $\varphi(a)$, i.e. $\forall x\varphi$; conversely every instance follows from $\forall x\varphi$. (c) If $\forall x\varphi$ is true, every instance is true, so every subset of D is coherent (Theorem 6.8(i) with the single target $\varphi(z)$), and no two clusters can remain at termination. In the single final cluster every unrefuted minimal template contains $\mathrm{inst}(\varphi(z))$ (Lemma 6.12), and one of them is contained in it (Proposition 6.2(c)), so $\mathrm{Acc}_d = \mathrm{inst}(\varphi(z))$. The oracle is a fixed function of the query, so the practices "n ground instances" and "$\varphi(z)$" produce the same run. (d) By (a); a subset whose terms all have the same head, e.g. S, has other minimal templates such as $\varphi(Sz)$, which need not be refuted. (e) The first claim is the definition of completeness for false instances plus (b), and Proposition 6.14 applied to $T = \varphi(z)$ gives it for R-Δ_0 and quantifier-free

φ. R-Δ_0 verifies universal statements only by equality reasoning from true closed equations, so a false sentence whose refutation needs a true universal premise that this reasoning cannot verify is never refuted; the noisy-PA computation (Example E.13) contains such a sentence, the wrong-base induction variant $(\neg S0 = 0 \wedge \forall x(\neg x = 0 \to \neg Sx = 0)) \to \forall x\, \neg x = 0$, whose refutation needs the universal premise $\forall x(\neg x = 0 \to \neg Sx = 0)$, which follows from Q1 untagged referee U7; u8. At a finite budget, a merge through a false universal whose least counterexample lies beyond the budget is part of Res_d (Theorem 6.10). In $\mathbb{R}$ as an ordered field every closed term denotes an integer (Example 4.8(a)), and no integer squares to 2, so every closed instance of $\neg(z \cdot z = 1 + 1)$ is true, while $\neg(p \cdot p = 1 + 1)$, read universally, is false; a decision procedure for real closed fields would refute it. $\mathsf{Q} \vdash 0 + \underline{n} = \underline{n}$ for each n but $\mathsf{Q} \nvdash \forall x(0 + x = x)$ is Example 4.9. □

Computed untagged u6: numeral instances of Q4–Q6, as 12 ground targets, form three clusters with templates $z_0 + 0 = z_0$, $z_0 + Sz_1 = S(z_0 + z_1)$, $z_0 \cdot 0 = 0$, and DTRC accepts $p + 0 = p$; instances of the false $n \cdot n = n$ ($n = 0, 1$) stay apart (refuted at 2); $\neg(n \cdot n = 2)$, $n \leq 2$, merge and the true $\neg(p \cdot p = 2)$ is accepted.

Proof of Proposition 6.14. Let $s = T\theta$ be false. Its universal closure is false in $\mathbb{N}$, so some assignment $\bar{n}$ of numerals to the parameters of s makes $s[\bar{n}]$ false. Let θ' agree with θ on term metavariables, and for a formula metavariable F let $\theta'(F) := (0 = 0)$ if $\theta(F)[\bar{n}]$ is true in $\mathbb{N}$ and $(0 = S0)$ otherwise. Then $s' := T\theta'$ is quantifier-free, since T's rigid part, the term values and the new formula values are. Moreover $s'[\bar{n}]$ has the truth value of $s[\bar{n}]$: the rigid part has no binders, so it is a Boolean combination of equations between terms built from rigid symbols and unchanged term values, and of occurrences of formula metavariables, whose replacements have the same truth values at $\bar{n}$. So s' is false. At depth at least $\max(\bar{n}, |s'|)$, R-Δ_0 instantiates the parameters of s' (which are among those of s) by $\bar{n}$ and evaluates the resulting closed quantifier-free sentence, so $s' \in \mathrm{Ref}_\infty$. If T is in Res_∞, no instance is refuted, so T has no false instance. □

Proof of Theorem 6.15. Construction. By the MRDP theorem (Davis et al., 1961; Matiyasevich, 1993) there are polynomials $P(e, \bar{x})$, $Q(e, \bar{x})$ with natural coefficients, $\bar{x} = x_1 \dots x_k$, such that $e \in K \iff \exists \bar{x} \in \mathbb{N}^k\, P(e, \bar{x}) = Q(e, \bar{x})$, K the halting set. Let P_e, Q_e be the terms with the numeral $\underline{e}$ substituted, and put

$$H_e(z) := \exists x_1 \dots \exists x_k\, (z = SS(x_1 + \dots + x_k) \wedge P_e(\bar{x}) = Q_e(\bar{x})),$$

with a 0-ary term metavariable z under k binders (z is closed, so no capture). Let $T_e := \neg H_e(z)$, $A_e := \neg H_e(0)$, $B_e := \neg H_e(S0)$, $q_e := \neg H_e(SS0)$. Practice $\mathcal{P}_e$ has the single target T_e; practice $\mathcal{P}'_e$ has the ground targets A_e, B_e; both have the data law uniform on $\{A_e, B_e\}$. *Facts.* (1) A_e, B_e are true, since $0 \neq SSu$ and $S0 \neq SSu$; so $\mathcal{P}'_e$ is valid. (2) $\mathcal{P}_e$ is valid iff $e \notin K$: its instances, including the parameter instance read universally, are all true iff no solution exists. (3) $\{A_e, B_e\}$ is an anchor for T_e (Lemma 6.12: heads 0 and S). (4) If $e \in K$ with solution $\bar{x}^*$, let $\bar{c}$ be the numeral of $2 + \sum x_i^*$. The instance $\neg H_e(\bar{c})$ is false, and R-Δ_0 refutes it at depth $d_e := \max(|\neg H_e(\bar{c})|, \max_i x_i^*)$: refuting $\neg\psi$ means verifying $\psi = \exists \bar{x}(\dots)$, done by the witnesses $\bar{x}^*$ and closed evaluation of two atoms; no universal has to be verified. By (3) every template covering A_e and B_e contains $\neg H_e(\bar{c})$, so $\mathcal{P}'_e$ is separated at depth d_e and $\mathrm{Res}_\infty(\mathcal{P}'_e) = \emptyset$. (5) If $e \notin K$, T_e is truth-sound and never refuted, and $\mathrm{Res}_\infty(\mathcal{P}'_e) \supseteq \mathrm{inst}(T_e) \ni q_e$; $\mathcal{P}'_e$ is separated at no depth, so (b) asks nothing of it. (6) If $e \notin K$, $\mathcal{P}_e$ has a single target, is separated vacuously, and its data contain the anchor $\{A_e, B_e\}$ with positive probability. *Reduction.* Let $S := \{e : \exists t\ \Pr[q_e \in \mathrm{Acc}_{t'} \text{ for some } t' \leq t] > \frac{1}{2}\}$. If $e \notin K$, (b) for $\mathcal{P}_e$ gives eventual acceptance of $q_e \in R^*_{\mathcal{P}_e}$ with probability $\geq 1 - \delta > \frac{1}{2}$; by continuity along the increasing events "accepted by time t", some single t has probability $> \frac{1}{2}$, so $e \in S$. If $e \in K$, (a)

for $\mathcal{P}'_e$ (with $R^* = \{A_e, B_e\}$, $\mathrm{Res}_\infty = \emptyset$, $q_e \notin R^*$) gives $\Pr[q_e \text{ ever accepted}] \le \delta < \frac{1}{2}$, so $e \notin S$. Hence $S = \omega \setminus K$. The learner, the data law and the oracle are computable uniformly in e (Ref_d consists of the sentences of size $\le d$ refuted with numeral bounds d, decidable uniformly in d), and $\mathcal{P}_e$, $\mathcal{P}'_e$ have the same data law and oracle; so the probability that q_e is accepted by time t is computable from (e, t) (for randomized learners, approximable from below), and S is Σ_1. But $\omega \setminus K$ is not Σ_1. □

R-Δ_0 cannot verify the bounded universal quantifiers of Kleene's T predicate, which is why the construction uses MRDP (Appendix G.2). Alternatively, keep the Kleene-T form and take as oracle the full evaluation of closed Δ_0 sentences with bounded quantifiers as primitives, which is also sound and decidable per depth.

E.5 Whole-cluster tests and noise

Proposition E.11 (whole-cluster versus pairwise tests)**.** *[proved; computed]*[untagged Prop 5.8] *Coherence is downward closed but not transitive. Every cluster built by* DTRC *is coherent at all times (except a refutable singleton, which contributes nothing), whereas single linkage on the pairwise-coherence graph can build clusters with no unrefuted minimal template. Under* $\mathrm{RS}_d(D)$*, the coherent subsets of* D *are exactly the subsets of the* D_i*, and single linkage, complete linkage and* DTRC *agree. Without separation,* DTRC *returns some maximal partition into coherent blocks, which may depend on the policy.*

Example: $a = (0+0=0)$, $b = (S0+0=S0)$, $c = (0+S0=S0)$. The pairs $\{a, b\}$, $\{a, c\}$ have the unrefuted templates $z_0+0=z_0$, $0+z_0=z_0$, while $\{b, c\}$ and $\{a, b, c\}$ have only $z_0+z_1=S0$, resp. $z_0+z_1=z_2$, refuted by $0+0=S0$. Single linkage lumps all three; DTRC returns $\{\{a, b\}, \{c\}\}$ or $\{\{a, c\}, \{b\}\}$.

Noise. Let $D = D_{\text{clean}} \cup E$ with mistakes $E \cap R^* = \emptyset$.

Proposition E.12 (noise)**.** *[proved; computed]*[untagged Prop 5.9] *(a) A refutable mistake stays an incoherent singleton. (b) Under* $\mathrm{RS}_d(D_{\text{clean}})$ *no coherent set contains clean data of two targets, so mistakes never bridge targets. (c) An unrefutable mistake can be absorbed into a target's cluster and make it unsound, and (d) it can* fragment *the target:* $C \cup \{m\} \cup C'$ *can be incoherent although* $C \cup C'$ *is coherent. (e) For* $|C| > e$ *the* trimmed verifier $\mathrm{Acc}^e_d(C) := \bigcap\{\mathrm{inst}(T) : T \text{ not } d\text{-refuted}, |C \setminus \mathrm{inst}(T)| \le e\}$ *is contained in* $\mathrm{inst}(\sigma_i)$ *when* C *has at most* e *mistakes and its clean part lies in* D_i*, and equals* $\mathrm{inst}(\sigma_i)$ *when moreover every subset of* C *of size* $\ge |C| - e$ *contains an anchor.*

Robust DTRC asserts only clusters of multiplicity $\ge s > e$, verified with Acc^e_d. If $\mathrm{RS}_d(D_{\text{clean}})$ holds, every mistake is refutable or *noise-separated* (every template covering it and a clean datum is refuted), and every coherent set of mistakes has multiplicity $< s$, then the asserted clusters are exactly the D_i of multiplicity $\ge s$, verified soundly, and exactly once every subset of D_i of size $\ge |D_i| - e$ contains an anchor. Fragmentation is not prevented, and a frequent coherent family of mistakes is a systematic error that no positive-data method can tell from a rule. A computed example on PA (Example E.13) shows absorption of the false instance $J^*_0 = (0+0=0 \wedge \forall x(0+x = 0 \to 0+Sx = S0)) \to \forall x(0+x=0)$ into a non-diverse induction cluster, its rejection by the trimmed verifier, and order-dependent fragmentation.

Proof of Proposition E.11. Non-transitivity and policy dependence: the example after the proposition (all data quantifier-free, so the minimal templates are Plotkin lggs). DTRC merges only coherent unions, so its clusters stay coherent, apart from a singleton $\{m\}$ with $m \in \mathrm{Ref}_d$, which no merge involves since every covering template of a set containing m has m as an instance. Under $\mathrm{RS}_d(D)$ the subsets of the D_i are coherent (Theorem 6.8(i)) and every other set is not, so the pairwise-coherence graph is a disjoint union of cliques D_i, and all three procedures return $\{D_i\}$. □

Proof of Proposition E.12. (a) $m \in \mathrm{inst}(T)$ for every $T \in \mathrm{Cov}(X)$ with $m \in X$. (b) A template covering clean $a \in D_i$ and $b \in D_j$ covers $\{a, b\}$ and is d-refuted. (c) By definition; Example E.13 gives an instance. (d) Example E.13. (e) First, the definition of $\mathrm{Acc}^e_d(C)$ agrees with the finite form

$$\mathrm{Acc}^e_d(C) = \bigcap_{E' \subseteq C,\, |E'| \le e} \; \bigcap_{T \in \mathrm{Min}^+_d(C \setminus E')} \mathrm{inst}(T) :$$

every unrefuted T with $|C \setminus \mathrm{inst}(T)| \le e$ covers $C \setminus E'$ for $E' := C \setminus \mathrm{inst}(T)$ and lies above some $T_0 \in \mathrm{Min}(C \setminus E')$, which is unrefuted; conversely each such T_0 is an unrefuted template missing at most e data ($C \setminus E' \neq \emptyset$ since $|C| > e$). Soundness: take E' to be the mistakes; $C \setminus E' \subseteq D_i$, so Proposition 6.2(c) gives an unrefuted $T_0 \in \mathrm{Min}(C \setminus E')$ with $\mathrm{inst}(T_0) \subseteq \mathrm{inst}(\sigma_i)$. Exactness: for each E', $C \setminus E'$ contains an anchor, so every member of $\mathrm{Min}(C \setminus E')$ contains $\mathrm{inst}(\sigma_i)$. *Robust* DTRC. Refutable and noise-separated mistakes never join a cluster with clean data, so by (b) and Theorem 6.8(ii) the clean data form the clusters D_i, and clusters of mistakes have multiplicity $< s$ and are not asserted. Each asserted D_i has no mistakes, so (e) applies with zero mistakes. If noise-separation is dropped, each asserted cluster still has clean data of one target (by (b)) and, by assumption, at most e mistakes, so (e) gives soundness and the stated exactness per cluster. □

Example E.13 (noisy PA)**.** [computed]untagged u8; reproduced by the referee To 60 clean PA data add three refutable mistakes (e.g. $p + 0 = Sp$), two unrefutable false ones (J_0^*, and the wrong-base variant $(\neg S0 = 0 \wedge \forall x(\neg x = 0 \to \neg Sx = 0)) \to \forall x\, \neg x = 0$, whose refutation needs Q1), and two true non-targets ($0 + p = p$, $\neg Sp = p$). The refutable mistakes stay incoherent singletons; the other mistakes stay singletons of multiplicity 1, since their merges with the diverse induction cluster are refuted, so with $s = 2$ none is asserted. *Absorption*: J_0^* together with the non-diverse data $\mathrm{Ind}(S^n 0 + x = S^n x)$, $n < 3$, is coherent, with the single unrefuted minimal template $(f(0) + 0 = f(0) \wedge \forall x(f(0) + x = f(x) \to f(0) + Sx = Sf(x))) \to \forall x(f(0) + x = f(x))$, which is K_0-shaped and not below T_{Ind} (unique among 999 covering templates of size ≤ 20, by the referee's enumeration). The plain verifier accepts the false J_0^* and $J_1^* = (S0 + 0 = S0 \wedge \forall x(S0 + x = S0 \to S0 + Sx = SS0)) \to \forall x(S0 + x = S0)$; the trimmed verifier with $e = 1$ rejects both and accepts the genuine $\mathrm{Ind}(S^3 0 + x = S^3 x)$. *Fragmentation*: in the order $\mathrm{Ind}(0 + x = x)$, J_0^*, $\mathrm{Ind}(S0 + x = Sx)$, then three diverse inductions, DTRC returns the unsound cluster $\{\mathrm{Ind}(0 + x = x), J_0^*, \mathrm{Ind}(S0 + x = Sx)\}$ and a cluster of the diverse inductions; with the diverse data first it puts all five inductions together and leaves J_0^* alone. Merging most specific or largest clusters first is a heuristic remedy, not a theorem.

E.6 Complexity of untagged verification and of the refutation test

Proposition E.14 (cost of a test)**.** *[proved; computed]*untagged Prop 5.10 *(a) Given* $\mathrm{Min}(X)$*, a d-coherence test needs at most* $|\mathrm{Min}(X)|$ *refutation checks. (b) If X is quantifier-free, every covering template is a first-order pattern and* $\mathrm{Min}(X) = \{\mathrm{lgg}(X)\}$ *(Plotkin's lgg with 0-ary term and formula variables), computable in polynomial time. (c) If $X \subseteq \mathrm{inst}(\sigma)$ contains an anchor of*

σ, *then, under* (Rich), $\mathrm{Min}(X) = \{\sigma\}$, *computable from* X *in polynomial time. (d) For one negative* n*: some* $T \in \mathrm{Min}(X)$ *avoids* n *iff* $n \notin \mathrm{Acc}(X)$*, decidable in polynomial time. (e) In every coherence test of the PA runs (*$63 + 102 + 98$ *tests) and ZF runs (*$164 + 131 + 147$ *tests) of Section 6.11,* $|\mathrm{Min}(X)| = 1$*; the worst case met was in low-diversity induction pairs (*$|\mathrm{Min}|$ *up to 27).*

Proof. (a) By definition. (b) The data have no binders, so neither has the rigid part of a covering template; no bound variable is in scope, so pattern occurrences, and hence all metavariables, are 0-ary. Plotkin's lgg is least among first-order patterns (Plotkin, 1970; Reynolds, 1970). (c) X is itself an anchor (its covering templates cover the anchor), so Corollary 3.14 makes σ the least covering template, computed by the polynomial lgg construction of Proposition B.13. (d) If $n \notin \mathrm{Acc}(X) = \bigcap \mathrm{Cov}(X)$, some covering T misses n, and so does a minimal T_0 below it; the converse is trivial. $\mathrm{Acc}(X)$ is decided in polynomial time by the feature verifier (Theorem 3.4 and Corollary 3.6). (e) Computed[untagged u3, u5, u0]. □

Features. Common prefix $C(B)$, slots, $A(B)$, scopes $Y_\sigma(B)$, the features $\mathrm{Sym}(p)$, $\mathrm{Scope}(\sigma)$, $\mathrm{Eq}(\sigma, r, \bar{u})$ and $\mathrm{Feat}(B)$ are as in Definition 3.1, and $\mathrm{Acc}(B) = \mathrm{Feat}(B)$, decidable in polynomial time (Theorem 3.4 and Corollary 3.6). For positions $\sigma \perp r$ of the same sort and a sentence d, the *matcher* e_d, if it exists, is the unique partial map on $\mathrm{FV}(d|_\sigma)$ with $d|_r = (d|_\sigma)[e_d]$ (forcing, Lemma 3.2(d)). With (1), "$q \in \mathrm{Acc}_k(D)$" is in coNP for every k: a partition into at most k blocks B with $q \notin \mathrm{Acc}(B)$ is a certificate of non-membership.

Lemma E.15 (failure types)**.** *[proved; computed]*[single Lemma H.1] *Fix* q*. Say that* d agrees with q above p *if every strict ancestor of* p *is a position of* d *with the same root symbol as in* q *(then* p *is a position of* d*). For nonempty finite* B*,* $q \notin \mathrm{Acc}(B)$ *iff* B *fits one of the following* types*, indexed by positions of* q*:*

- $\mathrm{Sym}(p, h)$*,* $h \neq \mathrm{root}(q|_p)$*: every* $d \in B$ *agrees with* q *above* p *and has root* h *at* p*;*
- $\mathrm{Scope}(\sigma, y)$*,* $y \in \mathrm{FV}(q|_\sigma)$*: every* $d \in B$ *agrees with* q *above* σ*, and* $y \notin \mathrm{FV}(d|_\sigma)$*;*
- $\mathrm{Eq}(\sigma, r)$*,* $\sigma \perp r$ *of the same sort: every* $d \in B$ *agrees with* q *above* σ *and above* r *and has a matcher* e_d*; the* e_d *are pairwise compatible; and either* e_q *does not exist or it is incompatible with some* e_d*,* $d \in B$*.*

There are at most $|q|(|\Sigma_D| + 2b_{\max}(q)) + |q|^2$ *types (*Σ_D *the non-index symbols of the data,* $b_{\max}(q)$ *the binder depth of* q*). For each type, "*B *fits" is a conjunction of:* $B \subseteq$ *a fixed* allowed *set;* B *a clique of a fixed compatibility graph (Eq-types);* B *meets a fixed* witness *set (only for Eq-types where* e_q *exists).*

Proof. ($\Rightarrow$) q lacks some B-feature; take the first applicable case. If q lacks $\mathrm{Sym}(p)$ for some $p \in C(B)$, let p' be the topmost position on the path to p where q's root differs from B's common root h'; its strict ancestors agree with q, so p' is a position of q, and B fits $\mathrm{Sym}(p', h')$. If q has every Sym feature but lacks $\mathrm{Scope}(\sigma)$, the strict ancestors of σ lie in $C(B)$ with q's symbols; any $y \in \mathrm{FV}(q|_\sigma) \setminus Y_\sigma(B)$ gives the type $\mathrm{Scope}(\sigma, y)$. If q has every Sym and Scope feature but lacks $\mathrm{Eq}(\sigma, r, \bar{u})$, each e_d exists and, by forcing, is the restriction of $\bar{u}$ to $\mathrm{FV}(d|_\sigma)$, so the e_d are pairwise compatible; also $\mathrm{FV}(q|_\sigma) \subseteq Y_\sigma(B) = \mathrm{dom}\,\bar{u}$ and $q|_r \neq (q|_\sigma)[\bar{u}]$. So either e_q does not exist, or $e_q(y) \neq \bar{u}(y) = e_d(y)$ for some y and some d with $y \in \mathrm{FV}(d|_\sigma)$.

($\Leftarrow$) By induction on the size of $q|_\sigma$ (resp. $q|_p$). The named positions lie in $A(B)$, since their strict ancestors lie in $C(B)$ with q's symbols. $\mathrm{Sym}(p, h)$: $p \in C(B)$, and q lacks $\mathrm{Sym}(p)$. $\mathrm{Scope}(\sigma, y)$: if σ is a slot, q lacks $\mathrm{Scope}(\sigma)$. If $\sigma \in C(B)$ with common root g and q's root differs,

$\mathrm{Sym}(\sigma)$ fails. Otherwise g is not an index ($g = y$ would put y into every $\mathrm{FV}(d|_\sigma)$; another index would give $\mathrm{FV}(q|_\sigma) = \{g\} \not\ni y$), so y occurs, shifted if g binds, in some child $q|_{\sigma\cdot i}$, and B fits $\mathrm{Scope}(\sigma \cdot i, y')$; apply induction. $\mathrm{Eq}(\sigma, r)$: if σ is a slot, $\bar{u} := \bigcup_d e_d$ is well defined on $Y_\sigma(B)$ by compatibility, $\mathrm{Eq}(\sigma, r, \bar{u})$ is a B-feature, and q has it only if e_q exists and $e_q \subseteq \bar{u}$, which the type excludes. If $\sigma \in C(B)$ with common root g and q's root differs, $\mathrm{Sym}(\sigma)$ fails. If g is an index (free at σ), then $d|_r = e_d(g)$ for each d, and compatibility makes all $d|_r$ equal to one term u, so r and the positions of u below it lie in $C(B)$; e_q exists with $e_q(g) = q|_r$, and the type forces $q|_r \neq u$, so a Sym feature fails inside u. If g is a non-index symbol, every $d|_r = (d|_\sigma)[e_d]$ has root g, so $r \in C(B)$, and if $q|_r$ has another root, $\mathrm{Sym}(r)$ fails. Otherwise pass to the children: the restricted matchers $e_d^{(i)}$ exist and are pairwise compatible (a local index bound by g is mapped to itself). The failure of e_q shows at some child i and some index z free in $q|_{\sigma\cdot i}$: q's child matcher does not exist, maps z to a wrong value, or two children of q disagree on z. If some $d \in B$ has z free in $d|_{\sigma\cdot i}$, compatibility fixes the correct value and B fits $\mathrm{Eq}(\sigma \cdot i, r \cdot i)$ (in the disagreement case, at the child with the wrong value); otherwise B fits $\mathrm{Scope}(\sigma \cdot i, z)$. Apply induction.

The count: Sym-types are indexed by a position of q and a symbol of the data or one of fewer than $b(p)$ indices, Scope-types by a position and an index, Eq-types by a pair of positions. The conditions are read off the definitions. □

Computed[single e11]: on 600 random cases (mixtures of 1–3 random DT° templates with low-diversity bodies, 3–6 data), "$q \notin \mathrm{Acc}(B)$" agreed with "B fits a type" on all 118,818 pairs (B, q), with 15,220 Eq-types having non-trivial compatibility graphs.

Proof of Theorem 6.16(a). $k = 1$ is Theorem 3.4. For $k = 2$: by (1), $q \notin \mathrm{Acc}_2(D)$ iff $q \notin \mathrm{Acc}(D)$ or there are types τ_1, τ_2 and a partition $D = B_1 \sqcup B_2$ with B_l fitting τ_l. Fix (τ_1, τ_2) and, for Eq-types that need one, witnesses $w_l \in B_l$ (polynomially many tuples). With variables x_d ("$d \in B_1$"), "B_l fits τ_l" is a conjunction of clauses with at most two literals: $\neg x_d$ for d not allowed by τ_1; x_d for d not allowed by τ_2; $\neg x_d \vee \neg x_{d'}$ for d, d' incompatible for τ_1; $x_d \vee x_{d'}$ for d, d' incompatible for τ_2; and x_{w_1}, $\neg x_{w_2}$. 2-SAT is solvable in linear time (Aspvall et al., 1979). The algorithm first checks $q \in \mathrm{Acc}(D)$; after that, a solution with an empty block would make D fit a type, i.e. $q \notin \mathrm{Acc}(D)$, so every solution is a genuine 2-partition. □

Proof of Theorem 6.16(b). Membership in coNP was noted above. Hardness: reduction from graph k-colourability, NP-complete for each fixed $k \geq 3$ (Karp, 1972; Garey and Johnson, 1979). Let $G = (V, E)$ with $V = \{1, \ldots, n\}$, $E = \{e_1, \ldots, e_m\}$, each edge oriented (tail, head). Under m binders $\forall y_1 \ldots \forall y_m$ put $\mathrm{tup}_u := t_{u,1} + (t_{u,2} + (\cdots + (t_{u,m} + 0)))$ with $t_{u,i} = y_i$ if $u \in e_i$ and 0 otherwise; let c_u map y_i to 0 if u is the tail of e_i and to $S0$ if it is its head; and put

$$d_u := \forall \bar{y}\,\big(S^u(\mathrm{tup}_u) = 0 \wedge S^u(\mathrm{tup}_u[c_u]) = 0\big), \qquad q := \forall \bar{y}\,\big(S^n(0) = 0 \wedge S^{n+1}(0) = 0\big),$$

of total size $O(n(n+m))$. Let σ_w, r_w be the positions at S-depth w in the left-hand sides of the two conjuncts. *Claim: for nonempty $B \subseteq D$, $q \notin \mathrm{Acc}(B)$ iff the vertices of B form an independent set.* Then by (1), $q \notin \mathrm{Acc}_k(D)$ iff V splits into at most k independent sets, iff G is k-colourable. *Independent $\Rightarrow$ $q \notin \mathrm{Acc}(B)$.* If $|B| = 1$, $\mathrm{Acc}(\{d_u\}) = \{d_u\} \not\ni q$. If $|B| \geq 2$, let w be the least vertex of B; all $d_u \in B$ have S above depth w on both sides, and at depth w the datum d_w has root + and the others S, so σ_w, r_w are slots. For $u \in B$, $d_u|_{r_w} = (d_u|_{\sigma_w})[c_u]$. As no y_i is free in two members (independence), the restrictions of the c_u are compatible, and their union $\bar{u}$ makes $\mathrm{Eq}(\sigma_w, r_w, \bar{u})$ a B-feature. But $q|_{\sigma_w} = S^{n-w}(0)$ and $q|_{r_w} = S^{n+1-w}(0)$ are closed and different. *An edge $\{u, v\} \subseteq B \Rightarrow q \in \mathrm{Acc}(B)$.* Acc is monotone in the data, so it suffices that $q \in \mathrm{Feat}(\{d_u, d_v\})$. With $w = \min(u, v)$, $C(\{d_u, d_v\}) = \forall^m(S^w(\Box) = 0 \wedge S^w(\Box) = 0)$ with

slots σ_w, r_w; q has every Sym feature ($n > w$) and every Scope feature (its contents there are closed). Eq-features with source σ_w: for (σ_w, r_w), the edge's index y_e is free at σ_w in both data and $c_u(y_e) \neq c_v(y_e)$, so by forcing there is no common $\bar{u}$; for the other admissible term positions $p \perp \sigma_w$ (the right-hand 0's and the S-positions of the second conjunct above r_w), the datum with vertex w has root $+$ at σ_w, and after any substitution, but 0 or S at p. Eq-features with source r_w have empty scope (the c_u are closed), so they are equalities; the datum with vertex w has root $+$ at r_w, the candidates carry 0 or S, except σ_w, which contains a free index while r_w is closed. So the only features are Sym and Scope features, which q has. □

Proof of Theorem 6.16(c). (i) Reduction from Set Cover (Karp, 1972): universe $\{1, \ldots, n\}$, sets $F_1, \ldots, F_t$, bound k. Let $d_u := \bigwedge_{j \leq t}(c_{uj} = 0)$ with $c_{uj} = 0$ if $u \in F_j$ and $S0$ otherwise, and $q := \bigwedge_{j \leq t}(S0 = 0)$. If some u lies in no F_j, then $d_u = q \in D$, there is no cover, and $q \in \mathrm{Acc}_k(D)$, consistently. Otherwise read off the types of q (Lemma E.15); let ℓ_j be the left-hand side of conjunct j, where data have 0 or S and q has S. The type $\mathrm{Sym}(\ell_j, 0)$ has allowed set exactly $\{d_u : u \in F_j\}$. There are no other Sym-types (below an S at ℓ_j data and q both have 0) and no Scope-types (no indices). All matchers are empty maps, so an Eq-type needs $q|_\sigma \neq q|_r$; the pairs with $q|_\sigma \neq q|_r$ that data can satisfy are (ℓ_j, a 0-position), its converse, and $(\ell_j \cdot 1, \ell_{j'})$ with $u \notin F_j$ (and its converse); each forces $d|_{\ell_{j'}} = 0$ for some j', so its allowed set lies inside some $\{d_u : u \in F_{j'}\}$. Formula-sort pairs are void. So a block fails iff it lies inside some $\{d_u : u \in F_j\}$, and $q \notin \mathrm{Acc}_k(D)$ iff k of the sets cover the universe. All data are quantifier-free, so all covering templates are first-order patterns. (ii) For binder-free D every matcher is the empty map, so Eq-types have no incompatible pairs and need no witness unless e_q exists, in which case it is compatible with every e_d and the type is void. So every type has the form "$B \subseteq$ allowed set", there are polynomially many, and $q \notin \mathrm{Acc}_k(D)$ iff D is covered by at most k allowed sets, which for fixed k is decided by trying all k-tuples. □

Proof of Theorem 6.16(d). In NP. Some $T \in \mathrm{Min}(X)$ avoids R iff some covering template does (it lies above a minimal one), iff some saturated covering template does (minimal templates are saturated, Theorem 3.3(c)). A saturated template is determined by an occurrence set inside $A(X)$ and a pattern slot per metavariable, a certificate of polynomial size, checked by matching. *NP-hard* (from CNF-SAT; Garey and Johnson, 1979). Take variables $x_1, \ldots, x_n$, distinct parameters $a_1, \ldots, a_n$, and $X = \{d_1, d_2\}$ with $d_1 := \forall x \bigwedge_j (x = a_j)$, $d_2 := \forall x \bigwedge_j (a_j = a_j)$. The common prefix is everything except the left-hand sides σ_j (column (x, a_j), scope $\{x\}$); the right-hand sides z_j carry the rigid a_j. Besides Sym and Scope features, the X-features are exactly $\mathrm{Eq}(\sigma_j, z_j, x \mapsto a_j)$ ($\sigma_i \to \sigma_j$ fails on d_2 as $a_i \neq a_j$; $\sigma_i \to z_j$ would need $a_i = a_j$; there are no formula slots). So the saturated covering templates are $T_A := \forall x \bigwedge_j (f_j(x) = \zeta_j)$, $A \subseteq [n]$, with $\zeta_j = f_j(a_j)$ for $j \in A$ and $\zeta_j = a_j$ otherwise; they are pairwise incomparable ($f_j := \lambda z.Sz$ separates them), so $\mathrm{Min}(X) = \{T_A\}$. For a clause C (w.l.o.g. not containing both x_j and $\neg x_j$) put $n_C := \forall x \bigwedge_j \gamma_j$ with $\gamma_j = (x = a_j)$ if x_j does not occur in C, $\gamma_j = (Sx = a_j)$ if $x_j \in C$, and $\gamma_j = (Sx = Sa_j)$ if $\neg x_j \in C$. By matching, where $\gamma_j = (x = a_j)$ the matcher takes $f_j := \lambda z.z$ and both choices of ζ_j fit; where the left side is Sx it takes $f_j := \lambda z.Sz$, and then $\zeta_j = a_j$ fits only $Sx = a_j$ while $\zeta_j = Sa_j$ fits only $Sx = Sa_j$. So $n_C \in \mathrm{inst}(T_A)$ iff the assignment "x_j true iff $j \in A$" falsifies C, and some $T \in \mathrm{Min}(X)$ avoids $R := \{n_C : C \in \varphi\}$ iff φ is satisfiable. The size is $O(n|\varphi|)$. *Polynomial cases.* Whether X has a least covering template is decidable in polynomial time, and then it is computed (Proposition B.13), and $\mathrm{Min}(X)$ is that template; under (Rich) an anchor gives one (Corollary 3.14). If $r \in R \cap \mathrm{Acc}(X)$, every covering template has r as an instance. □

In this reduction d_1 is false, so the hardness does not transfer to DTRC's use (true data, false

negatives in a fixed world); that case is open. Computed[single e11, e14]: the 2-SAT verifier equals the brute-force partition verifier on all 5,918 queries; for 12 graphs (including K_3, C_5, K_4, the Petersen graph) $q_G \in \mathrm{Acc}_k(D_G)$ iff $\chi(G) > k$ for $k = 1, \ldots, 4$; on 25 random set systems $q \in \mathrm{Acc}_k(D)$ iff no k sets cover; for $n \leq 6$, $|\mathrm{Min}(X)| = 2^n$ in the SAT reduction, and the test equals satisfiability on 360/360 random CNFs. These results were not refereed independently (Appendix G.3).

E.7 MDL

MDL with likelihood errs in two directions. For schemas it errs towards incompleteness: a split is a sound but incomplete hypothesis, which never accepts induction on an unused connective. For rarely used ground axioms it errs towards unsoundness: two ground axioms used m times each are merged by the naive code into their lgg once m times the body cost is below the template saving, and for $0 + 0 = 0$ and $0 \cdot 0 = 0$ that lgg, $z = 0$, is false. The refutation test of DTRC catches the second error, and the anchor condition (data diversity) controls the first. This is the size principle (Tenenbaum and Griffiths, 2001) and MDL (Rissanen, 1978) meeting a hypothesis space whose boundaries are logical rather than statistical. The definitions and results follow.

For a finite set H of templates and data D, each datum d is assigned a covering template $T_d \in H$ (the cheapest if several), with unique matcher θ_d, and $L(H, D) = \sum_{T \in H} L_{\mathrm{tmpl}}(T) + \sum_{d \in D}[L_{\mathrm{idx}}(T_d) + L_{\mathrm{body}}(\theta_d \mid T_d)]$. Without the body term this is the parent report's observation that MDL without likelihood picks the bare metavariable (Hänni and Claude (Anthropic), 2026). The split hypothesis is $H_F = \{Q1, \ldots, Q7\} \cup \{T_f : f \in F\}$ with F the main connectives of the induction data and $T_f := T_{\mathrm{Ind}}[P := \lambda x.f(P_1(x), \ldots)]$ (for "=": $\lambda x.t_L(x) = t_R(x)$; for $\forall, \exists$: $\lambda x.\forall y\, P_1(x, y)$). For a datum $\mathrm{Ind}(\varphi)$ with root f the matcher of T_f has bodies of total size $|\varphi| - 1$; under T_{Ind} the body is $\lambda x.\varphi$.

Proposition E.16 (a naive code over-splits)**.** *[proved]*[untagged Prop 6.2] *Let $L_{\mathrm{tmpl}}(T) = \lambda|T|$ and $L_{\mathrm{body}}(\theta) = \lambda \sum_M |\theta(M)|$ with $\lambda = \log_2 A$, A the number of symbols; let L_{idx} be the two-part maximum-likelihood code of the label sequence (empirical entropy plus $\frac{|H|-1}{2} \log_2 n + O(1)$, e.g. the KT code (Krichevsky and Trofimov, 1981)); let n_{Ind} be the number of induction data and $\hat{H}$ the empirical entropy of their main connectives. Then*

$$L(H_F, D) - L(H_{\mathrm{true}}, D) = \lambda\Big(\sum_{f \in F} |T_f| - |T_{\mathrm{Ind}}|\Big) + \frac{|F| - 1}{2} \log_2 n - n_{\mathrm{Ind}}(\lambda - \hat{H}) + O(1).$$

Since $\hat{H} \leq \log_2 |F| < \lambda$, the difference tends to $-\infty$ linearly in n_{Ind}. With a uniform index code ($\log_2 |H|$ bits per datum) the same holds once the induction frequency exceeds $\log_2((7 + |F|)/8)/\lambda$. Applied recursively, naive MDL keeps splitting each class by its next symbol while the class has enough data.

Proof. Template part: as displayed. Body part: each induction datum costs λ fewer bits under its T_f, for every root, $\forall$ and $\exists$ included. Index part: the split labels refine the true labels by the root of the induction data, so their empirical entropy is $n\hat{H}(\text{true labels}) + n_{\mathrm{Ind}}\hat{H}$ (chain rule), and the parametric term grows by $\frac{|F|-1}{2} \log_2 n$. Ground axioms cost the same under both hypotheses. For the uniform code, the index cost grows by $n \log_2((7 + |F|)/8)$ and the body cost falls by $n_{\mathrm{Ind}}\lambda$. □

Proposition E.17 (well-specified codes gain at most $O(\log n)$)**.** *[proved]*[untagged Prop 6.3] *Let the data be i.i.d. from a law P_0 (template index and instance law), and let the code for H_{true} be a Bayesian mixture over an r-parameter family containing P_0 with regret at most $\frac{r}{2} \log_2 n + C$*

(Rissanen, 1986; Clarke and Barron, 1990) (a standard fact, cited from memory). Then for every fixed split hypothesis and $K > 0$, for a constant C',

$$P\big[L(H_F, D) \le L(H_{\text{true}}, D) - \tfrac{r}{2}\log_2 n - K - C'\big] \le 2^{-K}.$$

Proof. The data part of $L(H_F, \cdot)$ is a prefix code for the data sequence given the template list, i.e. a sub-probability Q; by the no-hypercompression inequality (see Grünwald, 2007), $P[-\log_2 Q(D) \le -\log_2 P_0(D) - K] \le 2^{-K}$. The data part of $L(H_{\text{true}}, \cdot)$ is at most $-\log_2 P_0(D) + \frac{r}{2}\log_2 n + C$, and the template parts are constants; take C' to absorb C and the template parts. □

The regret bound and the no-hypercompression inequality are standard, but we cite them from memory and did not re-check them. With a well-specified code, splitting cannot win linearly; the decision falls to lower-order terms, which favour fewer templates. This does not prove that MDL selects the target partition. Computed (Table 12): the naive code splits from $n \approx 10^3$; under the natural law G1 even the depth-aware code DPC splits for $n \ge 1.6 \cdot 10^4$, because usage really differs across connectives (terms under a quantifier see a bound variable); under G2, for which DPC is well specified, DPC keeps T_{Ind} whole with a margin growing like $\log n$, while the misspecified PC eventually splits.

Table 12: $L(H_{\text{root}}) - L(H_{\text{true}})$ in bits on PA data (Q and induction, induction frequency 0.5); negative means that MDL splits induction by main connective. NAIVE: Proposition E.16 with $A = 23$; PC: sequential KT per template with context (parent symbol, child index); DPC: KT with context (depth below the motive root, parent, child index). G1: a natural usage law (random motives with quantifiers; scope-dependent terms); G2: a law for which DPC is well specified (symbols depend only on depth, parent and index). Computed[untagged u7], reproduced exactly by the referee.

usage law	n	NAIVE	PC	DPC
G1 natural	1000	−223	+2325	+2701
	16000	−16466	−1199	−3036
	64000	−68243	−19783	−34412
G2 well specified for DPC	1000	−915	+1141	+1428
	16000	−21467	+1053	+2822
	64000	−87156	−1522	+3564

E.8 Natural examples: details

Proposition E.18 (a passing run is DTRC with a fixed oracle)**.** *[proved][untagged Prop 5.12] The implementation keeps a global set N of refuted sentences; a template is reported refuted iff it covers some $n \in N$ or its own budgeted search finds a refuted instance, which is then added to N. Every coherence decision is logged. A run on data D (fixed order and policy) ending with N_{final}* passes the audit *if every "coherent" decision had a minimal template covering no element of N_{final}, and every asserted template avoids N_{final}. If the sentence refuters are sound, then $\text{Ref}_d := N_{\text{final}}$ satisfies (OS) and (Dec), and the run's final clusters and asserted templates coincide with those of DTRC_d with this fixed oracle on the same data, order and policy.*

Proof. (OS) holds because N_{final} consists of sentences refuted by sound procedures; (Dec) holds by matching. The merge policy is a deterministic function of the sequence of coherence answers, so

it suffices that each answer agrees with the fixed oracle: "incoherent" answers agree because $N \subseteq N_{\text{final}}$ at every time (each refuted minimal template covers its refuting sentence), and "coherent" answers agree by the audit. The asserted templates are the members of $\text{Min}(C)$ not reported refuted: every excluded member covers an element of N_{final}, and every included one avoids N_{final} by the audit. □

Refuter soundness is supported by code reading and by the referee's fuzzing (0 of 3000 true arithmetic and 1206 true set-theoretic sentences refuted). Every DTRC run passed the audit on the first pass (24 runs checked), and the re-audit after the separation check found no inconsistent decision. (The prototype's first version decided refutation by template-specific searches, which is not monotone in $\succeq$; the global refuted set and the audit replaced it, and no result changed, Appendix G.2.) Caveats: completeness of the computed minimal templates was checked against brute-force enumeration (by the researcher on induction pairs; by the referee's independent enumerator on all PA and ZF cross pairs, several within-schema sets, the K_0 pair and the J_0^* clusters), always with no covering template outside the computed ones; internal caps never changed a result; templates are matched literally, not modulo renaming of their parameters, which can only under-accept.

Table 13: PA cross merges: the unique minimal covering template of each cross pair and an R-Δ_0 refutation (computed[untagged u1]; the referee's independent enumeration of all 68 covering templates of size ≤ 14 of the 49 tested cross pairs, 21 Q–Q and 7×4 Q–induction, agrees). Numbering as in Section 2.6, in parameter form. Q_i^s is the instance schema of Q_i (closed terms for the parameters), an alternative presentation of Q.

cross pair	minimal template	refuting instance
different roots (Q1–Q4 etc.); Q_i–Ind, root $\neq\!\!\rightarrow$	F_0	$0 = S0$
Q2–Q3, Q2–Ind, Q3–Ind	$F_0 \rightarrow F_1$	$0 = 0 \rightarrow 0 = S0$
Q4–Q6, Q4–Q7, Q5–Q6, Q5–Q7	$z_0 = z_1$	$0 = S0$
Q4–Q5	$p + z_0 = z_1$	$p + 0 = 0$ at $p := 1$
Q6–Q7	$p \cdot z_0 = z_1$	$p \cdot 0 = S0$ at $p := 0$
Q_i^s–Q_j^s (21 pairs, 4 samples each)	various	all refuted

In the instance-schema presentation, one Q2^s sample lacked an S-free instance and gave the sound specialization $SSz_0 = Sz_1 \rightarrow Sz_0 = z_1$. The PA near-miss set had 104 sentences (50 step-by-two and 50 wrong-base induction variants, 4 mutated axioms); 13 variants are genuine inductions (vacuous motives) and were accepted, the other 91 rejected. Held-out data come from other seeds; the referee found no leakage. Separation on the data was verified for all 147, 201 and 175 cross pairs of the three PA runs.

Example E.19 (ZF cross merges)**.** [computed][untagged u4, u13; confirmed by the referee] The ZF targets in closure-normal form: Ext $\forall x(x \in p \leftrightarrow x \in p') \rightarrow p = p'$; Pair $\exists c(p \in c \wedge p' \in c)$; Union $\exists b \forall c(c \in b \leftrightarrow \exists d(d \in p \wedge c \in d))$; Power $\exists b \forall c(c \in b \leftrightarrow \forall d(d \in c \rightarrow d \in p))$; Inf (empty set and successor written out with $\in$); Found $\exists y(y \in p) \rightarrow \exists y(y \in p \wedge \forall z(z \in y \rightarrow \neg z \in p))$; Separation $\exists b \forall c(c \in b \leftrightarrow c \in p \wedge P(c))$; Replacement $\forall x(x \in p \rightarrow \exists y(P(x,y) \wedge \forall z(P(x,z) \rightarrow z = y))) \rightarrow \exists b \forall y(y \in b \leftrightarrow \exists x(x \in p \wedge P(x,y)))$; $\in$-induction $\forall x(\forall y(y \in x \rightarrow P(y)) \rightarrow P(x)) \rightarrow \forall x P(x)$. All 36 cross pairs (one instance per schema) have a unique minimal covering template: F_0 or $F_0 \rightarrow F_1$ in 22 pairs, refuted by an HF counterexample ($\neg p = p$, resp. $p = p \rightarrow \neg p = p$, at $p := \emptyset$); in 14 pairs a template keeping quantifier structure, refuted by logic: $\exists x F_0(x)$ (Pair–Union, Pair–Power,

Pair–Sep, Union–Inf, Power–Inf, Inf–Sep; instance $\exists x \neg p = p$), $\exists x(F_0(x) \wedge F_1(x))$ (Pair–Inf), $(\forall x F_0(x)) \to F_1$ (Ext–Rep, Ext–EInd), $F_0 \to \exists x F_1(x)$ (Found–Rep), $(\forall x(F_0(x) \to F_1(x))) \to F_2$ (Rep–EInd), and naive comprehension $\exists x \forall y(y \in x \leftrightarrow F_0(y))$ (Union–Power, Union–Sep, Power–Sep), refuted through Russell's instance (Herbrand: $y := x$). The referee's independent enumerator listed all 598 covering templates of size ≤ 14 of 63 ZF cross pairs (two samples per schema): none outside the computed minimal ones, all refuted with the global-negative-set oracle (the prototype's first version had left 120 of them unrefuted, all above refuted naive comprehension; Appendix G.2). Within each schema, three random instances have the schema as unique minimal template (confirmed by the referee for Separation, $\in$-induction and Replacement).

Proof of Proposition 6.17. Every instance holds in M_0 with $x := u$; HF is a transitive part of M_0, so R-HF's steps (Δ_0 facts about HF, HF witnesses and counterexamples) are sound in M_0; R-logic refutes only logical falsehoods. A witness b of $\exists x \forall y(y = y \to y \in x)$ gives $b \in b$; or $r = \{y \in b : y \notin y\}$ gives $r \in r \leftrightarrow r \notin r$. □

Example E.20 (DTRC on ZF and ZFC)**.** [computed]untagged u5, u10 *ZF* (three seeds of 45 data): besides the numbers in Section 6.11, 60/60 held-out instances of each schema were accepted, separation on the data was verified on all 467, 371 and 348 cross pairs, and $|\mathrm{Min}| = 1$ in all 442 tests. *ZFC*: in closure-normal form, with p the family,

$$\begin{aligned}\mathrm{AC}(p) = [\forall y(y \in p \to \exists z\, z \in y) \wedge \forall y \forall w((y \in p \wedge w \in p \wedge \neg y = w) \to \neg \exists z(z \in y \wedge z \in w))] \\ \to \exists c \forall y(y \in p \to \exists z(z \in y \wedge z \in c \wedge \forall u((u \in y \wedge u \in c) \to u = z))).\end{aligned}$$

AC is true in V and not refuted. Its 12 cross pairs (two samples per schema) each have one minimal template: $F_0 \to F_1$ (with Ext and EInd) and F_0 (with Pair, Union, Power, Inf, Sep), refuted by an HF counterexample; $F_0 \to \exists x F_1(x)$ (with Found) and $F_0 \to \exists x \forall y F_1(y, x)$ (with Rep), refuted by logic. On ZFC data (three seeds of 50): 10 pure clusters, schema clusters exact, AC accepted, 180/180 held-out schema instances, separation verified on 614, 294 and 428 cross pairs. *Weak Union and Power* (part B): with logic and HF only, the two weak forms merge into the universal-set cluster and $\exists x \forall y(y = y \to y \in x)$ is accepted (8 clusters); with the closure of Russell's Separation instance designated (bootstrapped coherence), 9 pure clusters and the universal set rejected. Over the UnionW–PowerW pair, the referee's enumerator found 81 of 91 covering templates unrefuted, all above the universal-set schema, as Proposition 6.17 predicts.

K_0. $K_0 = (z_1{+}0 = z_1) \wedge \forall x(z_1{+}x = z_2 \to z_1{+}Sx = Sz_2) \to \forall x(z_1{+}x = z_2)$ is the first-order lgg (named encoding) of $\mathrm{Ind}(0{+}x = x)$ and $\mathrm{Ind}(S0{+}x = Sx)$. By Theorem 5.14, no sound refutation from quantifier-free truths exists, so R-Δ_0 and logic refute none of its instances, among them the false J_0^*. In DT$^\circ$ the pair has two minimal covering templates (computed; the referee's enumerator found no other among 6258 covering templates of size ≤ 24): $T_{\mathrm{Ind}}[P := \lambda x.(f(0) + x = f(x))]$ and the K_0-shaped template of Example E.13. The intersection verifier excludes J_0^* (it is not an instance of the first). The K_0-shaped template has J_0^* as an instance ($f := \lambda h.0$) and is refuted by coherence with designated Q: the refutation of J_0^* through the recursion axiom Q5, $\forall x \forall y\,(x + Sy = S(x + y))$, goes through once Q5 is designated prior induction B5(d) (see the text after Theorem 5.14).

Computations. The scripts are in `research/tracks/untagged/code` (Python 3.11, no external packages; `shrun_all.sh`, about 15 minutes), with outputs `<script>.out`: u0 (minimal templates vs brute force), u1 (Table 13), u2 ($c = 4/7/7$), u3 (DTRC on PA), u4, u5, u5b (ZF cross

merges, DTRC on ZF, weak forms), u6 ($\forall x\varphi$ from numerals; non-transitivity), u7 (Table 12), u8 (Example E.13), u9 (clean fragmentation), u10 (ZFC), u12 (head classes), u13, u14 (monotone oracle, audit). The checks of Theorem 6.16 are `research/tracks/single/e11_union_complexity.py` and `e14_dtrc_step2.py`; the referee's independent code is in `research/tracks/untagged/referee_code`.

Credits. Gold's theorem (Gold, 1967); lgg and witness events (Plotkin, 1970; Reynolds, 1970); finite elasticity of bounded unions (Wright, 1989; Motoki et al., 1991); minimal multiple generalizations for unions of pattern languages (Arimura et al., 1994) (overlap with Theorem 6.16 not checked); MRDP (Davis et al., 1961; Matiyasevich, 1993); Δ_0 absoluteness (Kunen, 1980; Jech, 2003); MDL and coding (Rissanen, 1978, 1986; Clarke and Barron, 1990; Krichevsky and Trofimov, 1981; Grünwald, 2007); the size principle (Tenenbaum and Griffiths, 2001); conceptual clustering (Michalski and Stepp, 1983), in whose spirit the clustering is. Covering problems for first-order patterns, relevant to the open spare-slot thresholds, are decidable by results on complement problems and ground reducibility (Lassez and Marriott, 1987; Comon and Jacquemard, 2003) (cited from memory, not re-checked). As far as we know, the refutation test, the separation theorem, the depth impossibility, the head-class thresholds and the implementation audit are new.

F Implementation and experiments: proofs and details

This appendix states and proves the results of Section 7 that the main text only cites: the normal form for minimal covering templates (Appendix F.1), the refuter and the soundness of the oracles (Appendix F.2), the guarantees for DTRC, the sharing pass and the separation of the mixtures (Appendix F.3), and the stress sets, hard pairs and false merges (Appendix F.4). It ends with further numbers, tables and the learning curves (Appendix F.5) and the commands that reproduce them (Appendix F.6). We use the de Bruijn conventions of Appendix A and the substitution facts of Lemma A.1: plugging is compositional, injective in every hole that occurs in the body, and preserves the root symbol of a body whose root is not a hole; formula bodies never have a hole as root.

F.1 The normal form

Let D be a finite nonempty set of data, with parameters compared by name. Let $A(D)$ be the common-prefix tree of D with its slots, as in Definition 3.1: the positions p such that all data carry the same label and arity at p and above it, together with the *slots*, the maximal positions at which they disagree. The *column* at p is $(d|_p)_{d\in D}$; it is *open* if some $d|_p$ contains a variable bound above p. Let $\mathcal{T}(D)$ be $A(D)$ with each common atom node whose subtree contains a slot with an open column cut to a leaf slot (an *F-slot*); so every term slot of $\mathcal{T}(D)$ has a closed column.

Definition F.1 (Normal configurations)**.** experiments §1.3 For a slot σ of $\mathcal{T}(D)$ let $\bar{y}_\sigma$ list the variables free in its column, M_σ a fresh metavariable of the slot's sort and arity $|\bar{y}_\sigma|$, and $\beta_d(\sigma)$ the abstraction of $d|_\sigma$ over $\bar{y}_\sigma$. A node π is *derivable* from σ, $\sigma \to \pi$, if metavariable-free terms $\bar{t} =: \bar{t}(\sigma,\pi)$ satisfy $\beta_d(\sigma)[\bar{t}] = d|_\pi$ for all $d \in D$ ($\bar{t}$ is then unique). A *normal configuration* is a set S' of slots, none derivable from another, an antichain $A \supseteq S'$ of nodes of $\mathcal{T}(D)$ with every slot at or below a member of A, and a choice of $c(\pi) \in S'$ with $c(\pi) \to \pi$ for each $\pi \in A \setminus S'$. Its template has $M_\sigma(\bar{y}_\sigma)$ at $\sigma \in S'$, $M_{c(\pi)}(\bar{t}(c(\pi),\pi))$ at $\pi \in A \setminus S'$, and the common labels elsewhere.

Proposition F.2 (Normal form)**.** *[proved; computed]*[experiments Props E1-lit, E1-al; E0, E0b] *Let D be finite and nonempty, and compare parameters by name.* (a) *Every normal configuration lies in DT°_F and covers D.* (b) *Every $T \in \mathrm{DT}^\circ_F$ covering D satisfies $T \succeq N$ for a normal configuration N.* (c) *Hence the minimal normal configurations form a finite set $\mathrm{Min}^{\mathrm{lit}}_F(D)$; every covering $T \in \mathrm{DT}^\circ_F$ lies above a member; members are minimal among all covering templates in DT°_F; and if $D \subseteq \mathrm{inst}(T^*)$ with $T^* \in \mathrm{DT}^\circ_F$, some member lies below T^*.*

In outline: rigid symbols survive instantiation, so the skeleton of a covering template lies in $A(D)$, and no occurrence sits strictly below an F-slot. Pushing pattern occurrences down through common labels, dropping argument places that no datum uses, and replacing an own slot derivable from another by the derived occurrence specializes the template, keeps it covering, and ends in a normal configuration.

Proof. "Covers" means literal covering throughout.

(a) Let N be a normal configuration. An own occurrence $M_\sigma(\bar{y}_\sigma)$ is a pattern occurrence: its arguments are distinct variables bound above σ. Put $\theta_d(M_\sigma) := \lambda\bar{z}.\beta_d(\sigma)$. This is a body, since $\bar{y}_\sigma$ lists every index free in $d|_\sigma$, and abstraction inverts plugging, so the occurrence produces $d|_\sigma$. A derived occurrence $M_{c(\pi)}(\bar{t})$ at π produces $\beta_d(c(\pi))[\bar{t}] = d|_\pi$ by the definition of $\to$; its arguments are subterms of the data with no metavariables, and their free indices are free in $d|_\pi$ (every hole of $\beta(c(\pi))$ occurs in some $\beta_d(c(\pi))$), so they refer to binders above π. Positions not at or below A carry labels common to all data. Hence $N\theta_d = d$ for every d. Every metavariable has its pattern occurrence at its own slot, so $N \in \mathrm{DT}^\circ$; term metavariables are own only at term slots of $\mathcal{T}(D)$, whose columns are closed, so they are 0-ary and $N \in \mathrm{DT}^\circ_F$.

(b) Let $T \in \mathrm{DT}^\circ_F$ and $d = T\theta_d$ for all $d \in D$. Each of the following steps replaces T by $T\sigma$ for a template substitution σ (so the new template is $\preceq T$), keeps it in DT°_F, and keeps it covering D.

Step 0 (where occurrences sit). By Lemma 2.4, every position of the skeleton of T carries the same label in T and in every datum. So the skeleton lies in the common part of $A(D)$, and every occurrence position, whose parent is a skeleton position or which is the root, is a common position or a slot of $A(D)$. No occurrence lies strictly below an F-slot ν. Otherwise ν is a skeleton position. Some slot σ below ν has an open column; it is not a skeleton position, because the data disagree there, so some occurrence sits at a position π with $\nu < \pi \le \sigma$. Then π lies inside the atom at ν, so the occurrence is a 0-ary term metavariable, and $d|_\pi = \theta_d(f)$ is closed for every d. But for some d the subterm $d|_\sigma$ of $d|_\pi$ contains an index bound above σ, hence above ν (atoms contain no binders), hence above π; so $d|_\pi$ is not closed. Consequently every occurrence of T sits at a node of $\mathcal{T}(D)$; and every slot of $\mathcal{T}(D)$, F-slots included, lies at or below an occurrence (an F-slot in the skeleton would have an occurrence strictly below it).

Step 1 (push down). Let M have a pattern occurrence $M(\bar{z})$ at a node p of $\mathcal{T}(D)$ that is not a slot. Then all data carry the same label h at p, and $\theta_d(M) = \lambda\bar{z}.\beta_d$ with β_d the abstraction of $d|_p$ over $\bar{z}$. Specialize $M := \lambda\bar{z}.B$:

- h a connective or quantifier: $B = h(N_1(\bar{z}_1), \dots)$ with fresh formula metavariables; $\bar{z}_i = \bar{z}$, or under a quantifier the newly bound variable followed by $\bar{z}$;
- h a relation symbol, p not an F-slot: every slot below p has a closed column; B is the common part of the β_d with a fresh 0-ary term metavariable at each such slot;
- h a function symbol (M is then a 0-ary term metavariable): $B = h(f_1, \dots, f_m)$ with fresh 0-ary f_i;

- h a leaf: B is h if h is 0 or a parameter (the same name in all data), and the hole z_k if h is a bound variable, which must then be among the arguments $\bar{z}$ of M; M disappears.

The new metavariables take the bodies read off the β_d (children of β_d, or the closed columns at the slots), so the result covers D; at M's other occurrences $M(\bar{t})$ the plugged body is $B[\bar{t}]$, which agrees with $\beta_d[\bar{t}]$ on the common part, and on the closed parts the values coincide. The new metavariables have pattern occurrences at p's children or at the slots below p, and new term metavariables are 0-ary. Each step adds the node p of $\mathcal{T}(D)$ to the skeleton, and the skeleton stays inside $\mathcal{T}(D)$ by Step 0, so the process terminates. Afterwards every pattern occurrence sits at a slot of $\mathcal{T}(D)$.

Step 2 (own slots and arguments). For each metavariable M choose a pattern occurrence $M(\bar{y})$, at a slot σ. Every index free in the column at σ is among $\bar{y}$, since the matcher reads $\theta_d(M)$ off this occurrence and bodies have no free index. If some y_m is free in no $d|_\sigma$, the hole z_m occurs in no $\theta_d(M)$; specialize $M := \lambda\bar{z}.M'(\bar{z} \setminus z_m)$. Repeating, the arguments are exactly the indices of $\bar{y}_\sigma$; reordering them gives an equivalent template. So M is M_σ up to renaming, with $\theta_d(M_\sigma) = \beta_d(\sigma)$. Distinct metavariables get distinct own slots, since a position carries one occurrence. Other occurrences of M, pattern or not, are treated as derived.

Step 3 (forced arguments). Let $M_\sigma(\bar{u})$ be an occurrence at π. Then $\beta_d(\sigma)[\bar{u}] = d|_\pi$ for all d. Every hole of M_σ occurs in some $\beta_d(\sigma)$ (Step 2), and plugging is injective in a hole that occurs, so $\bar{u}$ is determined: $\sigma \to \pi$ and $\bar{u} = \bar{t}(\sigma, \pi)$. This also shows the uniqueness of $\bar{t}(\sigma, \pi)$ claimed in Definition F.1. Now T is a configuration: S' is the set of own slots, $A = \mathrm{Occ}(T)$ is an antichain of nodes of $\mathcal{T}(D)$ that covers every slot (Step 0), and each derived occurrence names its source. Only the independence of S' may fail.

Step 4 (independence). Let $\sigma \in S'$ be derivable from $\rho \in S' \setminus \{\sigma\}$, with $\bar{t} = \bar{t}(\rho, \sigma)$. The free indices of $\bar{t}$ are free in some $d|_\sigma$ (every hole of $\beta(\rho)$ occurs in some $\beta_d(\rho)$), so they are among $\bar{y}_\sigma$, and $\lambda\bar{z}.M_\rho(\bar{t}')$, with $\bar{t}'$ the abstraction of $\bar{t}$ over $\bar{y}_\sigma$, is a template body. Specialize M_σ to it. An occurrence $M_\sigma(\bar{u})$ at π becomes $M_\rho(\bar{t}'[\bar{u}])$, and $\beta_d(\rho)[\bar{t}'[\bar{u}]] = (\beta_d(\rho)[\bar{t}'])[\bar{u}] = \beta_d(\sigma)[\bar{u}] = d|_\pi$; so by Step 3 it is the derived occurrence $M_\rho(\bar{t}(\rho, \pi))$. M_ρ keeps its own occurrence, and $|S'|$ drops by one. Iterating gives a normal configuration $N \preceq T$.

(c) There are finitely many own sets, antichains and choices, and $\bar{t}(\sigma, \pi)$ is unique, so there are finitely many normal configurations; let $\mathrm{Min}_F^{\mathrm{lit}}(D)$ be the minimal ones, one per $\equiv$-class. A covering T is $\succeq N$ for a normal N by (b), and N lies above a minimal normal configuration. If a covering T' satisfied $M \succ T'$ for some $M \in \mathrm{Min}_F^{\mathrm{lit}}(D)$, then by (b) $T' \succeq N_1$ for a normal N_1, and $M \succ N_1$ (if $N_1 \succeq M$ then $T' \succeq M$), contradicting the minimality of M among normal configurations. The last clause is (b) applied to T^*. □

Alignment. For parameter-free T^* literal covering and covering up to renaming coincide, so $D \subseteq \mathrm{inst}(T^*)$ gives a member of $\mathrm{Min}_F^{\mathrm{lit}}(D)$ below T^* experiments Cor E1-pf. For templates with rigid parameters this fails (Example A.6), and $\mathrm{Min}_F(D)$ is computed over alignments. Proposition A.5, whose finiteness hypothesis (F) for DT_F° is Proposition F.2(c), gives the properties of $\mathrm{Min}(D)$ for $\mathrm{Min}_F(D)$ with respect to templates all of whose parameters occur in their skeleton; a parameter that occurs only inside an argument of a non-pattern occurrence is not covered. On the experiment data the aligned and literal sets differed on none of the 720 data sets of E2 and in one call of all DTRC runs of E3, and no cap on alignments or normal configurations was hit.

Computed checks experiments E0, E0b; referee §3. E0 compares the code with a bounded enumerator of covering templates written by a referee of the earlier analysis (de Bruijn levels, imported read-only) on 48 arithmetic data sets: 40 pairs of induction instances and 8 others. It enumerates 3703 covering DT_F° templates of size ≤ 14 and 2515 DT° templates of size ≤ 13. Every enumerated

template lies above a computed minimum, every computed minimum within the size bounds is enumerated, and none lies strictly below one. The coverage is narrow: no $<$, $\in$, $\leftrightarrow$ or parameters, and 41 of the 48 sets have $|\mathrm{Min}| = 1$. E0b draws a random generating template T^* with the referee's generator and 2–3 of its instances, 200 data sets per language and with or without a rigid parameter in T^*, and checks: (a) the computed minima cover D, lie in DT°_F and are pairwise incomparable; (b) some minimum is $\preceq T^*$; (c) along random specialization walks from T^* (≤ 8 steps), every covering template reached lies above a minimum and none strictly below one.

language	rigid param. in T^*	Min	data sets	minima	(a) fail	(b) fail	walk templates	(c) fail
PA	no	literal	200	200	0	0	380	0
PA	no	aligned	200	200	0	0	380	0
PA	yes	literal	200	200	0	**17**	317	**6**
PA	yes	aligned	200	203	0	0	320	0
ZF	no	literal	200	200	0	0	408	0
ZF	no	aligned	200	206	0	0	408	0
ZF	yes	literal	200	204	0	**36**	381	**27**
ZF	yes	aligned	200	220	0	0	419	0

The literal set fails only for generating templates with a rigid parameter, as Example A.6 predicts. The aligned set also differs from the literal one on 38/200 (PA) and 72/200 (ZF) data sets generated from a parameter-free T^*: the data's bodies carry parameters, and alignment finds more specific templates with rigid parameters.

F.2 The refuter and the soundness of the oracles

The refuter tries, within its budget: all $\top/\bot$ assignments to the formula metavariables; one metavariable running through a pool of small bodies while the others are $\top$ (or $\bot$); enumerations of small bodies; and bodies read off the data.

The evaluators. PA (truth in $\mathbb{N}$) evaluates numerals exactly, decides atoms with symbolic variables by polynomial normal forms, enumerates bounded quantifiers with numeral bounds, searches numeral counterexamples and witnesses, and case-splits an unbounded quantifier into $x \in \{0, 1\}$ and $x = x' + 2$. *ZF* (truth in V) uses hereditarily finite sets of rank ≤ 3 and *generic objects*: an unbounded quantifier is split into x in the transitive closure E of the concrete values in scope, x an in-scope generic, or x a fresh generic. Both use Kleene's strong connectives (Kleene, 1952) with supervaluation over unknown leaves (van Fraassen, 1966). Both first simplify by *constant propagation*: propositional laws valid in every nonempty structure, after replacing by $\top$ or $\bot$ the atoms whose value does not depend on the variables ($t = t$; closed arithmetic atoms; $t < t$; $x \in x$, false by Foundation).

Proof of Proposition 7.1. Both evaluators work on a formula together with an environment. The universal closure of a datum is evaluated by treating its parameters as universally quantified. We prove, by induction on the formula: a verdict True (False) is correct for every admissible interpretation of the environment.

PA. The environment maps variables to numerals or to *symbols*, which stand for arbitrary natural numbers; "admissible" means any values of the symbols. Atoms: at numerals the evaluation is exact. With symbols, terms built from $0, S, +, \cdot$ denote polynomials with nonnegative integer coefficients; $s = t$ holds for all values iff the polynomials p_s, p_t are equal, since polynomials that agree on $\mathbb{N}^k$ are equal; if $p_s - p_t$ has all coefficients ≥ 0 and a positive constant term

then $s > t$ for all values; the other classifications are symmetric, and everything else is unknown. Connectives: Kleene's strong tables are monotone in the information order, so known values stay correct. Supervaluation: for fixed values of the symbols, the actual truth values of the unknown leaves form one of the enumerated assignments, and the known leaves have their computed values by induction; so a verdict common to all assignments is correct. Quantifiers: bounded quantifiers with numeral bounds are enumerated exactly. For an unbounded $\forall x$, a verdict for the body with x a fresh symbol holds for all values of x; a numeral counterexample is a counterexample; and in the case split every number is one of $0, \ldots, K-1$ or $x' + K$ with $x' \in \mathbb{N}$, so if all cases are True then $\forall x$ is true and if one case is False there is a counterexample. One-point forms ($\forall x(x = t \to \varphi)$ with x not in t) are equivalent to $\varphi[t/x]$. $\exists$ is dual. Stripping a vacuous quantifier is valid in a nonempty domain. A local budget overrun returns unknown, which is always sound.

ZF. The environment maps variables to HF sets or to generics. A *realization* ν maps each generic G to a set $\nu(G) \notin E_G$, where E_G is the transitive closure of the concrete values in scope when G was introduced, with $\nu(G) \neq \nu(H)$ for every generic H in scope at that time. Two generics in one environment were introduced in nested scopes, so the later one was introduced with the earlier in scope. Atoms: between HF sets, membership and equality are exact (HF sets are standard; equality by Extensionality). $G \in c$ is False if $c \in E_G$ or every element of c lies in E_G: in both cases $c \subseteq E_G$ (E_G is transitive) and $\nu(G) \notin E_G$. $G = c$ is False if $c \in E_G$. $G \in G$ is False by Foundation. $G = H$ is False for distinct generics, by the freshness condition. Every other mixed atom is unknown. Connectives and supervaluation: as for PA, for each fixed realization. Unbounded $\forall x$: the cases are $x \in E$ (the transitive closure of the concrete values in scope), x equal to an in-scope generic, and x a fresh generic H with $E_H = E$. Fix a realization ν and a set a. Either $a \in E$, or $a = \nu(G)$ for an in-scope G, or ν extended by $H \mapsto a$ is a realization. So if every case is True under every realization, the body holds for every a. If some case is False under every realization, there is a counterexample; for the fresh case this needs some set outside E and different from the finitely many values of the in-scope generics, and such a set exists because no set contains all sets. Extra HF search values are concrete counterexamples or witnesses. $\exists$ is dual. Memoization is keyed on the formula and the environment's values, and generic identifiers are unique, so a verdict is reused only for the same environment.

Constant propagation (both). The rewrite rules ($A \wedge \top \equiv A$, $A \to \bot \equiv \neg A$, $\neg\neg A \equiv A$, $Qx\,\top \equiv \top$, ...) are logical equivalences in every structure with a nonempty domain; $t = t$ is logically true; closed arithmetic atoms are evaluated exactly; $t < t$ is false in $\mathbb{N}$; $x \in x$ is false in V by Foundation. So the simplified formula has the same truth value as the original in $\mathbb{N}$ (resp. V) under every environment.

The ZF argument uses Foundation, Extensionality on HF sets, the existence of HF sets, and that no set contains all sets; not Infinity. □

The implementation makes a new generic distinct from the generics created before it; the proof needs this only for the generics in scope, and only generics of one environment are ever compared.

Independent checks experiments §3.2; recheck R3, R4, R11. The referee's evaluators were copied unchanged and re-run against the revised oracles. The independent PA evaluator is three-valued and decides only through exact bounded quantifiers, counterexamples and witnesses. The independent ZF evaluator evaluates exactly in finite structures $V_4 \cup \{\text{extra objects}\}$ in which the HF sets are standard, the extras are never elements of HF sets, membership into and among the extras is arbitrary but irreflexive, and equality is identity. These structures satisfy exactly what the proof uses, so every True or False verdict must hold in them. R3 compared the verdicts logged during DTRC runs on the mixtures with mistakes and on the near-miss sets: 2197 PA verdicts and

413 ZF verdicts of quantifier depth ≤ 4 plus 179 of depth 5–6, with 0 contradictions. R4 compared 3000 random PA and 600 random ZF sentences, with 0 contradictions. R11 is a mutation test of the ZF check: dropping the in-scope-generic case is detected 39/600 times and dropping the fresh-generic case 292/600 times; mutations of the generic-versus-HF atom rules are not detected, because those rules rarely decide a verdict, and the referee verified them by hand.

F.3 Guarantees for dtrc, the sharing pass, and the separation of the mixtures

Throughout, the world is $\mathbb{N}$ or V and the refuter is sound; (H1)–(H4) are the hypotheses of Theorem 7.2.

Proposition F.3 (Merge monotonicity; within-target merges succeed)**.** *[proved]* experiments Prop E3 (a) *Let $D \subseteq D'$ and the refuter be ideal. If every member of $\mathrm{Min}_F(D)$ is refuted, so is every member of $\mathrm{Min}_F(D')$.* (b) *Let $\emptyset \neq X \subseteq \mathrm{inst}(T^*)$, where $T^* \in \mathrm{DT}^\circ_F$ has all its parameters in its skeleton and all its instances true. Then some $T_0 \in \mathrm{Min}_F(X)$ with $T^* \succeq T_0$ is never refuted, so the merge test on X succeeds for every sound refuter and budget.*

Proof. (a) Let $M' \in \mathrm{Min}_F(D')$; it is a member of $\mathrm{Min}^{\mathrm{lit}}_F(\alpha'(D'))$ for an alignment α' of D'. Each global name of α' names exactly one parameter in every datum of D', hence of D. So there is an alignment α of D (for G take the parameters of the reference datum of D that carry a global name of α') and a bijective renaming π of parameter names with $\pi(\alpha'_d(d)) = \alpha_d(d)$ for all $d \in D$; then $\pi M'$ covers $\alpha(D)$ literally. By Proposition F.2(c), $\pi M' \succeq T_0$ for some $T_0 \in \mathrm{Min}^{\mathrm{lit}}_F(\alpha(D))$, and $T_0 \succeq^\sim M$ for some $M \in \mathrm{Min}_F(D)$, the minimal elements of a finite union. So $M' \succeq^\sim M$, and the false instance of M lies in $\mathrm{inst}^\sim(M')$; an ideal refuter refutes M'. (This argument avoids Proposition A.5(b), which needs all parameters of M' in its skeleton.) (b) By Proposition A.5(c), which applies because every parameter of T^* occurs in its skeleton, some $T_0 \in \mathrm{Min}_F(X)$ satisfies $T^* \succeq^\sim T_0$. Its instances are, up to renaming of parameters, instances of T^*, hence true, and no sound refuter refutes it. □

The literal set. When some datum of X is parameter-free there is a single alignment, and the implementation computes $\mathrm{Min}^{\mathrm{lit}}_F(X)$ instead of $\mathrm{Min}_F(X)$. Then every T^* as in (b) is parameter-free (a parameter in the skeleton of T^* occurs in every instance, by Lemma 2.4), and Proposition F.2(c) gives a literal member below T^*. So (b), and every consequence of (b) below (Theorem 7.2, Propositions F.4 and F.5), holds for the implementation as well.

Proof of Theorem 7.2. No datum is discarded: genuine data are true and the oracle is sound.

Invariant: every cluster is contained in some D_i. It holds for singletons. When clusters $A \subseteq D_i$ and $B \subseteq D_j$ are tested, the test fails if $i \neq j$, by (H4), and succeeds if $i = j$, by Proposition F.3(b) applied to $A \cup B \subseteq \mathrm{inst}(T_i)$. So merges preserve the invariant. A pair is marked failed only if it is cross-label, and every later pair of clusters containing its two parts is cross-label as well; so the inheritance of failures never blocks a within-label merge.

The final clusters are the nonempty D_i. Each merge reduces the number of clusters, so the agglomeration terminates. Every pair of current clusters stays in the implementation's queue (with minimum score 0) until it is tested or inherits a failure. At the end, therefore, every pair of final clusters has failed, directly or by inheritance, and is cross-label. With the invariant, the final clusters are exactly the nonempty D_i.

Soundness. By Proposition F.3(b) there is $T_0 \in \mathrm{Min}_F(D_i)$ with $T_i \succeq T_0$, and no sound refuter refutes it. So T_0 is among the intersected templates, and $\mathrm{Acc}(D_i) \subseteq \mathrm{inst}(T_0) \subseteq \mathrm{inst}(T_i)$.

Exactness. If D_i contains an anchor of T_i, every member of $\mathrm{Min}_F(D_i)$ covers the anchor, so its instance set contains $\mathrm{inst}(T_i)$. The set of unrefuted members is nonempty (it contains T_0), so $\mathrm{Acc}(D_i) \supseteq \mathrm{inst}(T_i)$.

(H-ref). If verdicts do not depend on the call history, the final verification of D_i sees the same verdicts as the tagged refutation-filtered verifier on D_i, so the acceptance sets coincide. □

Proposition F.4 (Per-cluster soundness does not depend on the budget)**.** *[proved]* experiments Prop E5 *If $C \subseteq \mathrm{inst}(T^*)$, where $T^* \in \mathrm{DT}^\circ_F$ has all its parameters in its skeleton and all its instances true, then $\mathrm{Acc}(C) \subseteq \mathrm{inst}(T^*)$ for every sound refuter; more refutations only enlarge $\mathrm{Acc}(C)$.*

Proof. T_0 of Proposition F.3(b) is intersected, so $\mathrm{Acc}(C) \subseteq \mathrm{inst}(T_0) \subseteq \mathrm{inst}(T^*)$; fewer intersected templates give a larger set. □

So a budgeted refuter harms the soundness of DTRC only through clustering: an unrefuted merge across targets or with a mistake. Such a mixed cluster is truth-sound if one of its unrefuted minimal templates is valid, and may accept false sentences otherwise (Proposition 7.4). Fewer refutations also cost exactness inside a pure cluster: in E2, T_{Ind} at $N = 2$ is exact in 27/30 runs with refutation and 25/30 without.

Proposition F.5 (The sharing pass)**.** *[proved]* experiments Prop E4-share, corrected *Assume* (H1), (H3), *data $D \subseteq \bigcup_l \mathrm{inst}(T_l)$ (a datum may instantiate several targets),* ($\mathrm{H4}^+$) *every merge or sharing test on a set contained in no $\mathrm{inst}(T_l)$ fails, and* (H5) *each* phase-one cluster*, i.e. final cluster of the agglomeration, lies in exactly one $\mathrm{inst}(T_l)$, written $T_{l(C)}$. Then distinct phase-one clusters have distinct labels. After the sharing pass each phase-one cluster C, with $l := l(C)$, has become $C^+ = D \cap \mathrm{inst}(T_l)$, in any order; and $\mathrm{Acc}(C^+) \subseteq \mathrm{inst}(T_l)$, with equality whenever C^+ contains an anchor of T_l. Under* (H-ref)*, $\mathrm{Acc}(C^+)$ is the acceptance set of the tagged learner with membership labels for T_l. A target that labels no phase-one cluster gets no cluster of its own, and may be learned incompletely even when $D \cap \mathrm{inst}(T_l)$ contains an anchor of T_l (Example F.6).*

Proof. Call a cluster *pure* if it is contained in some $\mathrm{inst}(T_l)$. All clusters are pure at all times: a test whose union lies in no $\mathrm{inst}(T_l)$ fails by ($\mathrm{H4}^+$), so a successful merge or addition has a pure union. Inheritance of failures is correct: if a test on $A \cup X$ failed, then $A \cup X$ lies in no $\mathrm{inst}(T_l)$ (otherwise it passes, by Proposition F.3(b)), and neither does any superset.

Distinct labels. If two phase-one clusters $C \neq C'$ had $l(C) = l(C') = l$, then $C \cup C' \subseteq \mathrm{inst}(T_l)$ would pass the merge test, and the pair is never marked failed (by the inheritance remark); so the phase would not have ended.

The sharing pass. For each phase-one cluster C, with $l := l(C)$, let C' be the growing cluster, initially C. For $d \in D \cap \mathrm{inst}(T_l)$, $C' \cup \{d\} \subseteq \mathrm{inst}(T_l)$ passes by Proposition F.3(b). For $d \notin \mathrm{inst}(T_l)$, $C' \cup \{d\}$ lies in no $\mathrm{inst}(T_j)$: not in $\mathrm{inst}(T_l)$ because of d, and not in $\mathrm{inst}(T_j)$ for $j \neq l$ because $C \subseteq C'$ lies in exactly one target's instance set, by (H5); so the test fails by ($\mathrm{H4}^+$). By induction C' stays inside $\mathrm{inst}(T_l)$, and it ends as $C \cup (D \cap \mathrm{inst}(T_l)) = D \cap \mathrm{inst}(T_l)$, since genuine data are never discarded.

Acceptance. $\mathrm{Acc}(C^+) \subseteq \mathrm{inst}(T_l)$ by Proposition F.4; with an anchor in C^+, equality holds as in the proof of Theorem 7.2; and under (H-ref) the tagged learner with membership labels for T_l sees the same data C^+ and the same verdicts.

Nothing in the argument gives every target a phase-one cluster: the agglomeration may assign all data of T_l to clusters labelled by other targets, and the sharing pass only enlarges existing clusters. □

Without (H-ref) the proposition gives exactness only where the membership-labelled learner's exactness comes from an anchor: that learner's exactness can also come from refutations that DTRC's refuter, with another call history, need not find.

Example F.6 (A target without a cluster)**.** [proved; computed]experiments E4-share, corrected Let the targets be $U_{\text{add0}} = (z + 0 = z)$, $U_{0\text{add}} = (0 + z = z)$ and $T_3 = (1 + z = Sz)$, all valid, and $D = \{0+0=0,\ 0+2=2,\ 1+0=1,\ 1+1=2\}$. A subset of D lies in no single instance set iff it contains one of the cross pairs $\{0+0=0, 1+1=2\}$, $\{0+2=2, 1+0=1\}$, $\{0+2=2, 1+1=2\}$, whose unique minimal templates $z+z=w$, $z_0+z_1=Sz_2$ and $z_0+Sz_1=2$ are false; so (H4$^+$) holds for an ideal refuter, by Proposition F.3(a). In the implemented merge order (largest common prefix first, ties in data order) the agglomeration builds $\{1+0=1, 1+1=2\}$, which lies only in $\text{inst}(T_3)$, and $\{0+0=0, 0+2=2\}$, which lies only in $\text{inst}(U_{0\text{add}})$; so (H5) holds. The sharing pass adds nothing, U_{add0} labels no cluster, and DTRC+share rejects $2+0=2$ and $3+0=3$. The tagged learner with membership labels sees $\{0+0=0, 1+0=1\}$ for U_{add0}, an anchor ($\text{Min}_F = \{z+0=z\}$), and is exact. The run was computed with the implementation at budget 80. Nested targets, such as $z+0+0=z+0$ inside $z+0=z$ in the near-miss set of E4(a), also own no cluster; there the union stays exact.

In every run of E3 (both PA regimes and ZF, seeds 0–4) each target with data labels a phase-one cluster and (H5) holds (computed for this paper by re-running DTRC without sharing).

Lemma F.7 (Root symbols under derivation)**.** *[proved]*experiments Lemma E8.1 *In $\mathcal{T}(D)$, if $\sigma \to \pi$ then for every $d \in D$: $d|_\pi = d|_\sigma$ if σ is a term slot, and $d|_\pi$ has the root symbol of $d|_\sigma$ if σ is a formula slot. Consequently a* clash slot *(a slot at which two data a, b have different root symbols $r_a \neq r_b$) derives only slots at which a and b have the root symbols r_a and r_b; it derives no interior node.*

Proof. Term slots of $\mathcal{T}(D)$ have closed columns, so $\beta_d(\sigma) = d|_\sigma$ has no hole, and $\beta_d(\sigma)[\bar{t}] = d|_\sigma$. A formula body's root is not a hole, and plugging preserves the root of such a body. Interior nodes of $\mathcal{T}(D)$ carry a label common to all data. □

Proposition F.8 (One-variable instance schemas)**.** *[proved; computed]*experiments Prop E7; referee R2 *Let T^* have one 0-ary term metavariable z, at rigid positions, no other metavariable and no parameter, and let $D \subseteq \text{inst}(T^*)$ contain a parameter-free datum. Then $\text{Min}_F(D) = \{T^*\}$ iff the terms substituted for z do not all have the same root symbol (0 and each parameter count as root symbols); otherwise $\text{Acc}(D) \neq \text{inst}(T^*)$. So for i.i.d. terms with root law (p_h), $\Pr[\text{not exact after } N] = \sum_h p_h^N$, and numerals drawn uniformly from 0..7 form an anchor iff they include 0 and a nonzero numeral.*

This is Plotkin's (R) for the implemented class (Theorem 4.2); the referee found no violation in 3000 random data sets over 9 templates.

Proof. Write $d = T^*[z := t_d]$. Some datum has no parameter, so there is a single alignment α, which renames all parameters apart, and $\text{Min}_F(D)$ consists of the $\succeq^\sim$-minimal elements of $\text{Min}_F^{\text{lit}}(\alpha(D))$. As T^* has no parameter, $\alpha(D) \subseteq \text{inst}(T^*)$, with substituted terms $\alpha_d(t_d)$, and these have pairwise equal roots iff the t_d do: the root of the parameter-free t_d is not a parameter, and renaming changes no other root. So it suffices to prove the claims for $\text{Min}_F^{\text{lit}}(D')$, for every $D' \subseteq \text{inst}(T^*)$ with a parameter-free datum; this also covers the literal set that the implementation uses here. All positions of D' outside the z-positions are common.

($\Leftarrow$) If the roots of the t_d are not all equal, every z-position is a slot, with the same closed column $(t_d)_d$, and there is no other slot. An own set contains at most one of these slots, since each derives the others. By Lemma F.7 a derived occurrence at π needs $d|_\pi = t_d$ for all d, impossible at an interior node, where all data share a root. So the only normal configuration has one own z-slot and derived occurrences at the others: it is T^*.

($\Rightarrow$) If all t_d have root h, the z-positions are interior nodes with label h. Own slots lie strictly below z-positions and are term slots with closed columns; by Lemma F.7 such a slot derives only nodes π with $d|_\pi = d|_\sigma$, which excludes the z-positions and the nodes above them (a term never equals a proper subterm). So every normal configuration, hence every member, carries h rigidly at the z-positions, and rigid symbols survive instantiation. An instance of T^* whose substituted term has another root (0 if $h \neq 0$, $S0$ if $h = 0$) is therefore an instance of no member, and $\mathrm{Acc}(D) \neq \mathrm{inst}(T^*)$.

For i.i.d. terms the probability that all N roots coincide is $\sum_h p_h^N$. For numerals uniform in 0..7 the roots are 0 (probability 1/8) and S, so N draws fail to be an anchor with probability $(7/8)^N + (1/8)^N$. In E1 every parameter instance canonicalizes to the same datum, so among $N \geq 2$ distinct data some datum is parameter-free. □

The mixture targets. In the implementation the ZF-mix targets are $\mathrm{Ext} = \forall x \forall y(\forall z(z \in x \leftrightarrow z \in y) \rightarrow x = y)$, $\mathrm{Pair} = \forall x \forall y \exists z \forall u(u \in z \leftrightarrow u = x \vee u = y)$, $\mathrm{Union} = \forall x \exists y \forall z(z \in y \leftrightarrow \exists u(u \in x \wedge z \in u))$, $\mathrm{Pow} = \forall x \exists y \forall z(z \in y \leftrightarrow \forall u(u \in z \rightarrow u \in x))$, $\mathrm{Inf} = \exists x(\exists y(y \in x \wedge \forall z \neg z \in y) \wedge \forall y(y \in x \rightarrow \exists z(z \in x \wedge \forall u(u \in z \leftrightarrow u \in y \vee u = y))))$, $\mathrm{Found} = \forall x(\exists y\, y \in x \rightarrow \exists y(y \in x \wedge \forall z(z \in y \rightarrow \neg z \in x)))$, $T_{\mathrm{Sep}} = \forall x \exists y \forall z(z \in y \leftrightarrow z \in x \wedge P(z,x))$, $\mathrm{Repl_E} = \forall x(\forall y(y \in x \rightarrow \exists z(P(y,z,x) \wedge \forall u(P(y,u,x) \rightarrow u = z))) \rightarrow \exists y \forall z(z \in y \leftrightarrow \exists u(u \in x \wedge P(u,z,x))))$, and T_{EInd}. The PA-mix targets are Q1–Q7 as in Section 2.6, T_{Ind}, and the three instance schemas.

Proposition F.9 (Separating the mixture targets)**.** *[proved; computed]*experiments Prop E8; E10 *Let $T_i \neq T_j$ both be PA-mix targets (Q1–Q7, T_{Ind}, U_{add0}, U_{mul0}, $U_{0\mathrm{add}}$) or both ZF-mix targets (*Ext, Pair, Union, Pow, Inf, Found, T_{Sep}, $\mathrm{Repl_E}$, T_{EInd}*), and $a \in \mathrm{inst}(T_i) \setminus \mathrm{inst}(T_j)$, $b \in \mathrm{inst}(T_j) \setminus \mathrm{inst}(T_i)$. Then every member of $\mathrm{Min}_F(\{a,b\})$ has a form in Table 14 and the false instance shown there.*

case	target pairs	members of $\mathrm{Min}_F(\{a,b\})$	false instance
A	root clash. PA: Q_i–T_{Ind}, Q_i–U, T_{Ind}–U (31 pairs); ZF: Inf–any, T_{EInd}–any (15)	P	$\bot$
B	clash below a common $\forall$-prefix. PA: Q_i–Q_j (21); ZF: the other 17 pairs among Ext, Pair, Union, Pow, Found, T_{Sep}, $\mathrm{Repl_E}$	$\forall x\, P(x)$ or $\forall x \forall y\, P(x,y)$	$\forall \bar{x}\, \bot$
C	comprehension clash. ZF: Union–Pow, Union–T_{Sep}, Pow–T_{Sep}	$\forall x \exists y \forall z\, (z \in y \leftrightarrow P(z,x))$	$P := \neg z \in z$
D	ZF: Found–$\mathrm{Repl_E}$	$\forall x\, (A(x) \rightarrow \exists y\, B(y,x))$	$A := \top$, $B := \bot$
E	PA: two of $U_{\mathrm{add0}}, U_{\mathrm{mul0}}, U_{0\mathrm{add}}$ (3)	$z_0 = z_1$, $z_0 = 0$, or $z_0 + z_1 = R$	$1 = 0$; $S(r) + 0 = r$

Table 14: Minimal covering templates of cross pairs of mixture targets (Proposition F.9). In case E, z_0, z_1 occur only where shown, R is built from the common prefix of the two substituted terms, and r is R with all metavariables set to 0. Russell's instance in C fails at $z := y$.

Proof of Proposition F.9. Cases A–D. Inspection of the targets shows, for every listed pair, a position p above all metavariable positions of both targets at which the two targets differ, while

all positions not below p carry the same rigid symbols in both (in case D there are two such positions). In case A, p is the root: Q1–Q7 have root $\forall$, T_{Ind} root $\to$, the instance schemas root $=$; Inf has root $\exists$, T_{EInd} root $\to$, and the other ZF targets root $\forall$. In case B, p is the node right below the common prefix $\forall x$ or $\forall x \forall y$. There the roots differ, except for Q4–Q6 and Q5–Q7, where both data carry an equation whose term disagreement contains a bound variable, so p is an F-slot. In case C, p is the right side of $\leftrightarrow$ in the comprehension frame $\forall x \exists y \forall z(z \in y \leftrightarrow \cdot)$, where the targets read $\exists u(\dots)$, $\forall u(\dots)$, and $z \in x \wedge \varphi$. In case D the two positions are the antecedent under $\forall x$ ($\exists$ against $\forall$) and the body under the consequent's $\exists y$ ($\wedge$ against $\forall$).

Hence for instances a, b, $\mathcal{T}(\{a, b\})$ is the shared prefix with one slot (two in D), and its interior nodes carry the shared labels. Each slot is a clash slot, or in case B possibly an F-slot, whose root pair is $(=, =)$ while the interior nodes are quantifiers. By Lemma F.7 no slot derives an interior node, and in case D neither slot derives the other, since their root pairs $(\exists, \forall)$ and $(\wedge, \forall)$ differ. So the only normal configuration puts an own metavariable at each slot, applied to the variables free in its column. In C these are z and x (each of the three right sides mentions both, and a Separation body cannot mention y); in D, x for the antecedent and y, x for the consequent body. Parameters occur only inside the slots' columns, so every alignment gives the same template, and $\mathrm{Min}_F(\{a, b\})$ is this single template. The instances are false: $\bot$; $\forall \bar{x} \bot$ (the domain is nonempty); Russell's $\forall x \exists y \forall z(z \in y \leftrightarrow \neg z \in z)$, which fails at $z := y$; and $\forall x(\top \to \exists y \bot)$.

Case E. All slots have closed term columns, so by Lemma F.7 $\sigma \to \pi$ requires identical columns. (i) U_{add0}–U_{mul0}: $a = (t + 0 = t)$, $b = (s \cdot 0 = 0)$. The left sides clash ($+$ against $\cdot$); the right sides t and 0 form a slot unless $t = 0$. No two of these columns are identical, and no column can be derived at a common node, so $\mathrm{Min}_F = \{z_0 = z_1\}$ or $\{z_0 = 0\}$, with the false instance $1 = 0$. (ii) U_{mul0}–$U_{0\mathrm{add}}$: $a = (t \cdot 0 = 0)$, $b = (0 + u = u)$; symmetric to (i). (iii) U_{add0}–$U_{0\mathrm{add}}$: $a = (t + 0 = t)$ with $t \neq 0$ (otherwise $a \in \mathrm{inst}(U_{0\mathrm{add}})$) and $b = (0 + u = u)$ with $u \neq 0$. The left sides share $+$, with slots σ_1 (column $(t, 0)$) and σ_2 (column $(0, u)$). A node π with $a|_\pi = t$ is σ_1 or the root of the right side, where b has $u \neq 0$; a node with $b|_\pi = u$ is σ_2 or the root of the right side, where a has $t \neq 0$; slots inside the right side have columns of proper subterms. So σ_1, σ_2 derive no other node and are derived from none, and every normal configuration has the form $z_0 + z_1 = R$ with z_0, z_1 occurring only there and R built from the common prefix of t and u. Setting every metavariable of R to 0 gives a term r; with $z_0 := S(r)$ and $z_1 := 0$ the instance $S(r) + 0 = r$ is false in $\mathbb{N}$ for every value of its parameters. □

Proof of Corollary 7.3. "Only if" is the case $|A| = |B| = 1$; "if" follows from Proposition F.3(a) with $\{a, b\} \subseteq A \cup B$. Then apply Proposition F.9; disjointness is read off its proof. □

F.4 Hard pairs and false merges

The stress sets. The *PA near-miss* set has the targets $z+0 = z$, $z+1 = Sz$, $z \cdot 1 = z$, $0+z = z$, the nested $(z+0)+0 = z+0$, T_{Ind} and two reorderings of induction (conjuncts swapped; curried). The *ZF near-miss* set has Separation, its conjunct-swapped form, $\mathrm{Repl}_{\mathrm{E}}$, T_{EInd}, Found and the minimal-element schema. The *hard ZF* set has T_{Sep}, Separation from $\bigcup a$, $\mathrm{Repl}_{\mathrm{E}}$, Collection, T_{EInd}, $\in$-induction along $z \in y \in x$ (EInd2), the minimal-element schema, and the bounding forms Union_W, Power_W of Section 6.11, with the leading $\forall a$ kept.

Proposition F.10 (Two hard ZF pairs)**.** *[proved; computed]* experiments Prop E9; E9, E9b (a) *The unique minimal covering template of* $\mathrm{Union}_W = \forall a \exists b \forall x\,(\exists y(y \in a \wedge x \in y) \to x \in b)$ *and* $\mathrm{Power}_W = \forall a \exists b \forall x\,(\forall y(y \in x \to y \in a) \to x \in b)$ *is the universal-set schema* $U = \forall a \exists b \forall x\,(P(x, a) \to x \in b)$ *(here with the leading* $\forall a$ *kept, so* P *is binary, unlike the parameter form of Proposition 6.17). The*

ZF oracle refutes its instance $P := \top$, whereas no oracle sound for HF *plus a universal element refutes any instance (Proposition 6.17).* (b) *Let* $\mathrm{EInd2}(\psi) = \forall x\,(\forall y{\in}x\,\forall z{\in}y\,\psi(z) \to \psi(x)) \to \forall x\,\psi(x)$. *When the motives' root symbols differ, the $\in$-induction motive does not begin with $\forall$, and some motive uses its variable (the most frequent case), the minimal template of an* EInd *datum and an* EInd2 *datum is $T_{EE} = (\forall x(\forall y(y \in x \to P_0(y)) \to P_1(x))) \to \forall x\,P_1(x)$, which is false. Some other merges of this pair are valid (e.g.* $\mathrm{EInd}(x = x)$ *with* $\mathrm{EInd2}(x = x)$*), and some false ones are refuted only at budget* 400.

Proof. (a) Union_W and Power_W share the prefix $\forall a \exists b \forall x(\cdot \to x \in b)$ and clash at the antecedent ($\exists$ against $\forall$), whose column mentions x and a but not b. By Lemma F.7 this slot derives no other node, so the universal-set schema U is the unique normal configuration and the unique minimal covering template. Its instance with $P := \top$ simplifies to $\exists b \forall x\,(x \in b)$ (the quantifier $\forall a$ is vacuous). With no concrete value in scope, $E = \emptyset$ and there is no generic, so the only case for b is a fresh generic G; then $\forall x\,(x \in G)$ is False in its case $x := G$, because $G \in G$ is False: generics obey Foundation. So the oracle returns False, soundly by Proposition 7.1: in V, a set b with $\forall x\,(x \in b)$ would satisfy $b \in b$. The impossibility for oracles sound in $\mathrm{HF} \cup \{u\}$ is Proposition 6.17.

(b) Let $e = \mathrm{EInd}(\varphi)$ and $e' = \mathrm{EInd2}(\psi)$, where the roots of φ and ψ differ, φ does not begin with $\forall$, and φ or ψ uses its variable. The common prefix is $(\forall x(\forall y(y \in x \to \square_1) \to \square_2)) \to \forall x\,\square_3$, with columns $(\varphi(y), \forall z(z \in y \to \psi(z)))$ at $\square_1$ and $(\varphi(x), \psi(x))$ at $\square_2$ and $\square_3$. All three are clash slots. The columns at $\square_2$ and $\square_3$ are identical (each sits under one binder), so each derives the other. The root pairs of $\square_1$ and $\square_2$ are $(\mathrm{root}\,\varphi, \forall)$ and $(\mathrm{root}\,\varphi, \mathrm{root}\,\psi)$. If they differ, Lemma F.7 excludes derivations between $\square_1$ and $\square_2, \square_3$. If they coincide (ψ begins with $\forall$), a size argument excludes them: terms of $L_\in$ are variables or parameters, so plugging does not change the size of a body, while $\forall z(z \in y \to \psi(z))$ is larger than $\psi(x)$ and $\psi(t)$. No slot derives an interior node (Lemma F.7). So the only normal configuration, up to renaming, has own slots $\square_1, \square_2$ and $\square_3$ derived from $\square_2$; it is T_{EE}: P_0 is unary, since only y occurs in the column at $\square_1$ and $z \in y$ mentions it, and P_1 is unary, since some motive uses its variable. Its instance with $P_0 := \bot$, $P_1(x) := \forall z\,\neg z \in x$ has the premise $\forall x(\forall y(y \in x \to \bot) \to \forall z\,\neg z \in x)$, i.e. $\forall x(x = \emptyset \to x = \emptyset)$, which is true, and the conclusion $\forall x\,\forall z\,\neg z \in x$, which fails at $x = \{\emptyset\}$.

The remaining claims are computed (E9, E9b); the valid merges are covered by Lemma F.11(5). □

If neither motive uses its variable, P_1 is 0-ary and the template is valid: when P_1 is false its premise $\forall x\,\neg\forall y(y \in x \to P_0(y))$ fails at $x = \emptyset$. This is why (b) assumes that some motive uses its variable.

Constant propagation is needed for (b): without it $\forall y\,(y \in x \to \bot)$ and $\forall z\,\neg z \in x$ are different unknown leaves, and 0 of 12 random EInd/EInd2 merges are refuted, against 10 of 12 with it (E9b). The surviving templates of E9 are $(\forall x(\forall y{\in}x\,\forall z{\in}y\,P_0(z,y) \to P_1(x))) \to \forall x\,P_1(x)$ (pairs and sets protocol) and the same with unary $P_0(z)$ (sets protocol). They are false: with $P_0 := \bot$ and $P_1(x) := \alpha(x) := \forall y{\in}x\,\forall z{\in}y\,\bot$ the premise is $\forall x(\alpha(x) \to \alpha(x))$, and $\forall x\,\alpha(x)$ fails at $\{\{\emptyset\}\}$. The refuter finds this at budget 400, not at 80 (one-hot pass, witness $(\forall x(\forall y{\in}x\,\forall z{\in}y\,\bot \to \neg x = w_0)) \to \forall x\,\neg x = w_0$, false at $w_0 = \{\{\emptyset\}\}$). Valid merges also occur: $\mathrm{EInd}(x = x)$ with $\mathrm{EInd2}(x = x)$, and Separation with Separation from $\bigcup a$ on the body $x \in a$ (Separation with swapped conjuncts is equivalent to Separation).

Lemma F.11 (Valid residual templates)**.** *[proved]*[experiments Lemma V] *Every instance of the following is true in $\mathbb{N}$ (1–4) or V (5): (1) $S^n 0 + z = S^n z$; (2) $z + 1 = Sz$, $z + 0 = z$, $0 + z = z$; (3) $(\alpha \wedge \dots) \to \dots$ with first conjunct α one of $z < 0$, $0 = Sz$, $\exists x(x < t \wedge \dots)$ for $t \in$*

$\{0, 0+0, 0\cdot 0\}$; *(4)* $(P(0) \wedge \forall x(P(x) \to P(Sx))) \to \forall x\, P(Sx)$; *(5)* $\forall x(P(x) \to x = x) \to \forall x(x = x)$ *and* $\forall x(\forall y(y \in x \to P(y)) \to x = x) \to \forall x(x = x)$.

Proof. (1) $n + t = S^n t$. (2) True schemas. (3) α is false. (4) Induction gives $\forall x\, P(x)$. (5) The conclusion is true. □

The false templates met in E4(b). Each was found refuted by the post-hoc refuter; the false instances were checked by hand experiments §3.6.

- PA, numerals, seed 0, budget 80: $(\forall x(P_0(x,0) \to P_1(x,0)) \wedge \forall x(\forall y(P_0(y,x) \to P_1(y,x)) \to \forall y(P_0(y,Sx) \to P_2(y,x)))) \to \forall x \forall y(P_0(y,x) \to P_1(y,x))$. With $P_0 := \top$, $P_1(y,x) := (x = 0)$, $P_2 := \top$ the premises are $\forall x(\top \to 0 = 0)$ and $\forall x(\cdots \to \forall y(\top \to \top))$, both true, and the conclusion $\forall x\,(x = 0)$ is false.
- PA, numerals, seed 1, budget 80: $(\forall x(P_0(x,0) \to P_1(x,1)) \wedge \forall x(\forall y(P_0(y,x) \to P_1(y,x)) \to \forall y(P_0(y,Sx) \to P_1(y,Sx)))) \to \forall x \forall y(P_0(y,x) \to P_1(y,x))$. With $P_0 := \top$, $P_1(y,x) := \neg x = 0$ the premises $\neg 1 = 0$ and $\forall x(\neg x = 0 \to \neg Sx = 0)$ are true, and the conclusion $\forall x\, \neg x = 0$ fails at 0.
- ZF, budget 80, seeds 0–2: five merges of a Separation-capture mistake with Separation instances, e.g. $\forall a \exists b \forall x\,(x \in b \leftrightarrow x \in a \wedge \forall u\, P(u,x,b,a))$. With $P := \neg x \in b$ the body becomes $x \in b \leftrightarrow x \in a \wedge \neg x \in b$, contradictory for $x \in a$, so the sentence fails at $a = \{\emptyset\}$. The post-hoc witnesses of the other four reduce in the same way to $x \in b \leftrightarrow x \in a \wedge \neg x \in b$, or to $x \in b \leftrightarrow x \in a \wedge \neg w_0 \in b$, which fails at $a = \{\emptyset\}$, $w_0 = \emptyset$ (if $\emptyset \in b$ the right side is false at $x = \emptyset$; if not, it is true).
- PA, both regimes, seed 0, budgets 80 and 400: T_F of Proposition 7.4, with the false instance

$$q_F = (\exists x(0 = 0 \wedge \neg\, 0 = 2) \wedge \forall x(\exists y(x = 0 \wedge \neg\, x = 2) \to \exists y(y < Sx \wedge \neg\, Sx = 2))) \\ \to \forall x\, \exists y\, (y < Sx \wedge \neg\, Sx = 2).$$

F.5 Further numbers

Leading quantifiers stripped. With leading universal quantifiers stripped, Q4 becomes an instance of $z + 0 = z$ and Q6 of $z \cdot 0 = 0$, and stripped targets have rigid parameters. The referee re-ran DTRC (first version) with all data and targets stripped: ARI 1 on unambiguous data in PA-mix seeds 0–4 and ZF-mix seeds 0–2, Q4 and Q6 absorbed into the clusters of these schemas (semantically right, scored "not exact"), all other targets exact, and no non-target probe referee R13.

E1, further numbers. For the two-variable axiom $x + y = y + x$ the mean N to exactness is 4.18 (mixed regime) and 11.85 (numerals). From $0 + 0 = 0$ and $1 + 0 = 1$ the learned $z + 0 = z$ accepts $w_0 + 0 = w_0$, the axiom in parameter form, but not the different datum $\forall x\,(x + 0 = x)$ experiments E1.

E2, further exhibits. Accepted sentences certified false by the oracle ($N = 2$): Separation, both encodings, $\forall x \exists y \forall z\,(z \in y \leftrightarrow z \in x \wedge \neg w_0 \in y)$; Replacement, first-order de Bruijn, $\forall x(\forall y(y \in x \to \exists z(y \in x \wedge \forall u(\neg \forall v\,(v = v) \to u = z))) \to \exists y \forall z(z \in y \leftrightarrow \exists u(u \in x \wedge \forall v\,(v = v))))$; $\in$-induction, named, $(\forall x(\forall y(y \in x \to x = y) \to \neg x \in w_0)) \to \forall x\, \neg x \in w_0$, false with $w_0 = \{\{\emptyset\}\}$ (the premise only says $\emptyset \notin w_0$); induction, first-order de Bruijn, $(0 = w_0 \wedge \forall x(x = 0 \to 0 = x)) \to \forall x\,(x = 0)$; induction, genuine pattern lgg, $(0 = 0 \wedge \forall x(w_0 = Sx + (0 + x) \to 0 = 1 \cdot Sx)) \to \forall x\,(w_0 = Sx + (0 + x))$. Per-probe counts are in `code/results/e2_schemas.md`.

E3, mixed regime and timing. DTRC is exact on 19/20 schema targets, DTRC+share and both tagged learners on 20/20, and skeleton clustering on 17/20, with 197 non-target probes. A DTRC run took at most 0.8 s (PA) or 6 s (ZF) experiments E3.

E4(b), outcome of the mixed clusters. Table 15 lists, per data regime and refuter budget, the seeds in which a false template was accepted and the residual templates, checked with a post-hoc refuter (budget 4000 for PA, 1500 for ZF) and by hand (Lemma F.11, Appendix F.4).

data	budget	seeds with a false accepted template	false templates	valid residual templates
PA, mixed	80, 400	0	T_F	$(z < 0 \wedge \dots) \to \dots$, $(0 = Sz \wedge \dots) \to \dots$
PA, mixed	2000	none	–	the same, and $(P(0) \wedge \forall x(P(x) \to P(Sx))) \to \forall x\, P(Sx)$
PA, numerals	80	0, 1	T_F; two induction-step templates	false-antecedent forms with $x < 0 + 0$, $x < 0 \cdot 0$, $0 = Sz$; the $\forall x\, P(Sx)$ form; $0 + z = z$
PA, numerals	400	0	T_F	the same
PA, numerals	2000	none	–	the same, and the form with $x < 0$
ZF	80	0, 1, 2	five Separation-capture templates	$\forall x(P(x) \to x = x) \to \forall x\,(x = x)$
ZF	400	none	–	the same

Table 15: E4(b): mixed clusters with injected mistakes experiments E4; code/results/e4_stress.md.

The fates of the mistakes per run (discarded as refuted, singleton, or merged), the ARI on clean data, held-out acceptance and the oracle-call counts are in `code/results/e4_stress.md`. Refuted mistakes number 2–3 per run.

E4(c), unequal frequencies. PA: induction 60, U_{add0} 20, U_{mul0} 3, U_{0add} 1; ZF: Separation 30, Replacement 4, $\in$-induction 1; 3 seeds. Induction and Separation are exact in 3/3 seeds, Replacement and U_{mul0} in 2/3, U_{add0} in 1/3 (2/3 with sharing; $0 + 0 = 0$ is its only anchor). Targets with one instance are never generalized. No non-target probe was found experiments E4.

E5, learning curves. Streams are drawn i.i.d. from a mixture: PA, each Q axiom 0.02, induction 0.50, U_{add0} 0.16, U_{mul0} 0.12, U_{0add} 0.08; ZF, each axiom 0.03, Separation 0.30, Replacement 0.27, $\in$-induction 0.25. Figure 2 shows the curves summarized in Section 7.3; they are dominated by the first appearance of each ground axiom; first-N tables per target and seed are in `code/results/e5_curves.md`.

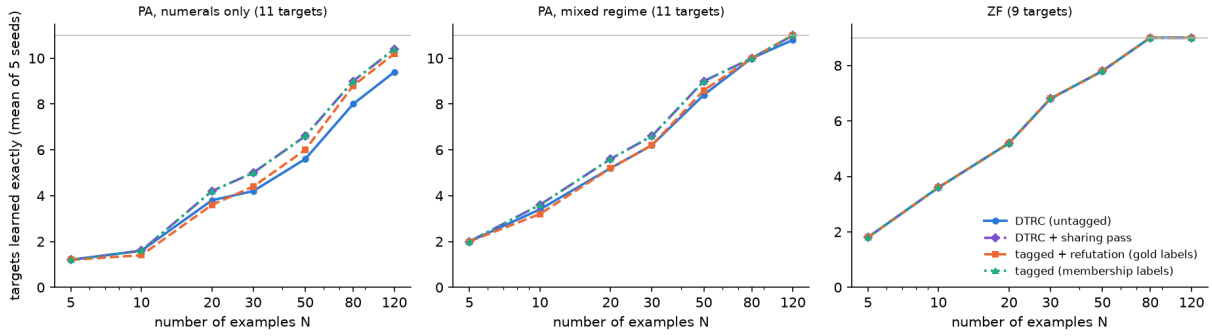


Figure 2: E5: targets learned exactly after N examples, mean over seeds 0–4 (rows of one stream share seeds). Left: PA, numerals only (11 targets); middle: PA, mixed regime; right: ZF (9 targets) experiments E5.

E6, slot counts. With leading quantifiers kept, all of Q's axioms are instances of $\forall x\, P(x)$, so Q takes one slot on positive data (four in the parameter form of Corollary E.3) and seven with refutation negatives. Without negatives, $k = 8$ and $k = 9$ accept none of 8 held-out induction instances, since each induction datum gets its own slot. With negatives, $k = 8$ accepts all 8 from the anchor pair $\mathrm{Ind}(x = x), \mathrm{Ind}(\neg x = 0)$; $k = 9$ accepts none from these two (the spare slot keeps them apart) and all 8 from three. With third motive $\exists y\, x < y$ every split of the three induction data into two blocks has an anchor block, as the slot accounting requires; with third motive $x + 0 = x$ the block $\{\mathrm{Ind}(x = x), \mathrm{Ind}(x + 0 = x)\}$ is root-homogeneous, and acceptance relies on DT°_{F} (Section 7.3) experiments E6.

E9, all counts. Table 16 gives the cross-merge counts summarized in Section 7.3.

targets	merges	refuted	distinct minimal templates	distinct unrefuted
PA-mix	2200 / 2200	2200 / 2200	15 / 13	0 / 0
ZF-mix	540 / 540	540 / 540	5 / 5	0 / 0
PA near-miss	1080 / 1080	1073 / 1080	61 / 36	5 / 0
ZF near-miss	225 / 225	225 / 225	5 / 5	0 / 0
ZF hard	432 / 432	431 / 430	14 / 15	1 / 2

Table 16: E9: cross-target merges, pairs of single data / sets of one or two data per target; data instantiating both targets excluded; budget 80 experiments E9.

F.6 Reproduction

The package `dtrc` (pure Python) is independent of the prototype of Section 6.11. All commands run from `axiom-schemas/code/`; the experiment scripts are in `experiments/`. `sh run_all.sh` runs the tests and E0–E9 (593 s wall time on 4 shared cores); E10 and E9b take a few seconds. Outputs other than timings reproduce exactly. The results of the first version are kept in `code/results_v1/`.

The scripts are `e0_crossval.py` (run as `14 40 F` and `13 40 full`), `e0b_property.py 200`, `e1_universal.py`, `e2_schemas.py`, `e3_mix.py`, `e4_stress.py`, `e5_curves.py`, `e6_kunion.py`, `e7_blowup.py`, `e9_separation.py`, `e9b_nosimplify.py` and `e10_pairtable.py 100`, each

writing `results/eN_*.md`; `python3 -m pytest -q tests` runs the 43 unit tests, and `sh experiments/recheck/run_recheck.sh` re-runs the referee's oracle checks (outputs in `experiments/recheck/*.out`). Seeds are listed in each results file.

G How the results were established

The results come from a process with built-in adversarial checking. This appendix describes the process, lists what it corrected, and says what it did not cover. All roles below (researchers, referees, section writers, reviewers and editors) were played by separate instances of Claude (Anthropic). No human has checked the mathematics or the code, and nothing has been verified in a proof assistant. "Independent" means a separate session that wrote its own code and imported none of the researcher's code; it does not mean an independent person. Researcher and referee were instances of the same model, so an error that the model tends to make could survive both. The brute-force checks, which do not depend on the theory under test, are the part of the verification least exposed to this.

G.1 Process

The earlier analysis. The questions followed the earlier analysis of PA induction (Section 2), whose record is in `research/prior/`; gray tags and the tables below call it "prior". It supplied the determinate-template learner for induction, the first-order impossibility results and the Sub and IndEq encodings (Section 5.5). It had three tracks: motive-annotated instances, raw instances with first-order patterns, and second-order templates. A referee attacked each track once with independent code (`research/prior/induction/*-referee.md`). The notes were not revised afterwards, so this paper uses those results as the referee reports correct them (Table 18, rows 1–5). Three reviewers checked a short analysis of untagged PA (`research/prior/pa-untagged/`) by proof and with brute-force version spaces (row 6).

Brief and tracks. A brief (`research/00-brief.md`) fixed the commissioner's questions (Section 1.1), the conventions (closure-normal form, de Bruijn indices, the class DT$^\circ$), what the prior work had established, and a proposed method (DTRC) with intended theorems. Tracks could change the method if they argued for the change. Four tracks then worked in parallel, each developed by one researcher: "single" (one template, Section 3), "cases" (Questions 1 and 2, Sections 4 and 5), "untagged" (many schemas without labels, Section 6) and "experiments" (the package `dtrc`, Section 7). Each researcher wrote a record (`notes.md`) with complete proofs, a status for every claim, and scripts with saved outputs. When a track needed another track's result that had not yet been refereed, it stated that result as a hypothesis. For example, the untagged track assumed the finiteness of minimal covering templates, which is now Theorem 3.3. Each record went through one round of refereeing; results added afterwards were not refereed again (Appendix G.3).

Adversarial refereeing. A referee then received each record with the task of finding errors. The referee checked every proof line by line and re-ran the researcher's scripts against their saved outputs. The referee also tested the claims with code written from scratch: its own term representation, matcher, enumerator of covering templates, anti-unifiers and evaluators. Bounded brute-force enumeration provided ground truth independent of the theory. Where the researcher's evidence was circular (anchor status decided by the anchor theorem's own events), the referee

replaced it with exact computation. Each claim received one of four verdicts: holds, holds with fix, false, or unclear. A claim that could not be confirmed was marked unclear (Table 17).

Revision. Each researcher then wrote a final record (`notes-final.md`). It supersedes the first version, which is kept unchanged. It restates every result, corrected where needed, and ends with a verification log that answers every referee point. Withdrawn claims stay in the record, labelled as withdrawn, and refuted conjectures are labelled refuted. The experiments track re-ran the referee's oracle checks against its revised code and found no contradiction. The outputs of the first versions of both implementations are kept.

Writing. Each section was written from the final records only. Its writer re-checked the proofs while transcribing them, and recorded every deviation from the records in `paper/writer-notes.txt`. Where records disagree, the paper follows the statement that a referee checked. For example, the experiments record says that the size-24 template of Example B.1 lies below two of the eight earlier templates. The single record and its referee say instead that five of the eight lie strictly above it, and the paper follows them. Each theorem-like statement carries a status and a gray tag naming its source, which leads to its proof, its scripts and its referee verdict.

Review of the paper. The draft of Sections 2 to 7 and their appendices was then reviewed by five reviewers, each with one lens: mathematical correctness (three reviewers, for Sections 2 and 3, for Sections 4 and 5 and for Sections 6 and 7), consistency across sections, and readability. The mathematical reviewers checked every statement against the final records and every appendix proof line by line, compared every number of Section 7 with the result files, and wrote their own scripts, among them an implementation of de Bruijn terms, DT° matching and features that imports no project code. They found two false propositions, both among the results that no referee had attacked: an alignment result for parameters and the guarantee of the sharing pass. They also found several overclaims and missing hypotheses (Table 18, rows 31–37). One editor per section then applied the fixes under common decisions, re-ran the counterexamples, and reported every change (`research/paper-review/`). The introduction, the discussion and this appendix were not reviewed in that round; they were revised afterwards to match the edited sections.

Table 17: The records and their refereeing ("row" refers to Table 18).

Record	Examples of the referee's independent checks	Outcome
Prior: annotated instances	Own anti-unifier, matcher and motive generator on a richer law; 40 000 data sets per anchor theorem; Lean, set.mm	Core confirmed; row 5; one numerical error
Prior: raw instances	Own anti-unifier; exact evaluator in $\mathbb{N}$; ground congruence closure	Row 4; new Theorem R1 (Theorem 5.14)
Prior: second-order	Own code with de Bruijn levels; the track's enumeration counts reproduced; own pattern anti-unifier	Rows 1, 2, 5; C11 unclear (row 3)
Prior: untagged PA	Brute-force version spaces	Row 6
single	Own matcher, enumerator, features, events: Theorem 3.4 on 125 109 queries, Theorem 3.8 on 540 targets; exact rates	No error in the main theorems; rows 7, 8
cases	Own lgg, encodings, model of Q; SO° search with repeated and parameter arguments; pattern anti-unifier	All hold; row 9 and five minor points

Record	Examples of the referee's independent checks	Outcome
untagged	Own DT° enumerator on 112 cross pairs; refuter fuzzing with 4206 true sentences; merge orders	15 hold, 5 with fix; rows 12–17
experiments	Full re-run (identical up to timings); own PA and ZF evaluators, 0 contradictions in 6283 comparisons; mutation test	Rows 18–22; no reported number wrong

G.2 Corrections

Table 18 lists the substantive corrections, with the finder named in the group heading unless the row says otherwise. The referee reports, the verification logs, `paper/writer-notes.txt` and the review reports of `research/paper-review/` list the minor ones: wording, constants, citations, and hypotheses made explicit.

Table 18: Substantive corrections.

#	Original claim (record)	What was wrong	Now
The earlier analysis (prior), found by its referees A, B, C and reviewers			
1	Minimal covering templates are finitely many, also with metavariables inside arguments (C8.3).	False: $\{\mathrm{Ind}(x = x), \mathrm{Ind}(0 = x)\}$ has an infinite antichain of minimal covering templates; the enumeration-based learner was unsound for a target with nested arguments.	Class DT°, where C8.3 holds (Theorem 3.3 and Remark A.3).
2	$\{\mathrm{Ind}(x = x), \mathrm{Ind}(0 = x)\}$ has 8 minimal covering templates (C8.2).	An artifact of a size bound; exactly 4 (counted by the single track, confirmed by its referee).	Example B.1.
3	Escalations are linear in the size of the first escalated datum (conjecture C11).	Refuted by the single track and its referee: $1 + m^2$ escalations follow an induction datum of size $20m + 1$; $\Theta(N^2)$ in general.	Proposition B.6 and Theorem 3.11.
4	Every over-general lgg of raw induction data is refuted through one of two named instances.	False: K_0 is consistent with all true closed quantifier-free sentences, so it is never refuted (Theorem R1).	Theorem 5.14; Propositions 5.18 and D.7.
5	The Lean-style β-redex encoding is sound (P8); `Nat.rec`'s motive is explicit (C12).	A named outer binder $\forall n$ captures a parameter of the motive; in Lean 4 the motive is implicit.	Guard $n \notin \mathrm{FV}(P)$ (Appendix D.5).
6	Untagged PA: the general sufficient conditions apply, the exact threshold is open, and world refutation plays no role.	The conditions are vacuous for PA. The threshold is $k - 1$ failure sets, since Q's axioms can share one slot. Two world refutations forbid this, and at $k = k'$ the anchor has tagged size.	Theorem E.2 and Corollary E.3.
Research tracks, found by the track's referee			
7	Rates (single Thm E) and untagged anchors (Prop. F.5) hold for every template.	False for ground templates such as ZFC's single axioms: $c = 0$, $\lambda = \min \emptyset$, and F.5 holds vacuously.	Theorem 3.10 and Proposition B.11.
8	C11 "proved in quadratic form"; "matching lower bounds"; an anchor's lgg is the target.	C11 is refuted; the lower bounds cover only two of three terms; the lgg claim needs (Rich).	Proposition B.4 and Corollary 3.14.
9	Question 1: first-order anti-unification learns the closed-instance schema; without a guard its only error is the ω-step.	Capture: $\mathrm{lgg}(\exists y \neg(y = 0), \exists y \neg(y = S0))$ has the instance $\exists y \neg(y = y)$, false in every structure. The unguarded verifier is unsound outright.	Propositions 4.6 and C.4.

#	Original claim (record)	What was wrong	Now
10	Proof of the ZF anchor theorem: "occurrence arguments are bound variables".	False with parameter or repeated arguments; the theorem survives.	Appendix D.4.
11	The named lgg of spelled-out Replacement is truth-sound, via Collection (cases C3).	Correct for anchor data; sharpened in revision: truth-sound over B iff B proves Collection. Non-anchor data: row 33.	Proposition 5.7.
12	Spare-slot threshold under target-separation (untagged Cor. 3.3(b)).	The proof needs cross-separation.	Corollary E.4.
13	Plotkin's failure families are m-noncovering for every m (caveat).	False under canonical parameter names: five head specializations cover all instances of $z + 0 = z$.	Withdrawn; Proposition E.6.
14	DTRC costs $O(n^2)$.	This counts tests; one test may need $4^{\Theta(n)}$ refutation checks.	Proposition E.14 and Theorem 6.16.
15	The prototype runs DTRC with a fixed oracle.	Its refutation was not monotone: 120 of 598 ZF cross templates stayed unrefuted above a refuted one.	Audit (Proposition E.18); no result changed.
16	If no instance of $\varphi(z)$ is refuted, $\forall x\varphi$ is true (untagged Prop. 5.6(e)).	This needs an oracle that refutes all false closed instances. R-Δ_0 is one for quantifier-free φ, not for every quantified φ, and then only at unbounded depth (at a fixed depth, relative to Res_d).	Propositions 6.13 and 6.14.
17	Depth relativization is necessary (untagged Thm 5.7), via Kleene's T.	R-Δ_0 cannot verify the bounded universals of T.	Rebuilt on MRDP (Theorem 6.15).
18	The computed Min(D) has a member below every target (experiments Prop. E1, v1).	False with rigid parameters: names were compared literally, while matching is up to renaming. 21 of 300 random ZF data sets failed.	Proposition F.2 and Proposition A.5, the latter only for templates whose parameters all occur in their skeleton (row 31).
19	"PA mistakes merge only into valid templates."	False: DTRC accepted the false sentence q_F at refuter budgets 80 and 400.	Withdrawn (Proposition 7.4).
20	The pattern lgg equals the DT°_F learner on the ZF schemas.	The baseline generalized at formula level, not by genuine pattern anti-unification.	Withdrawn (Table 9).
21	Headlines: 54/55 and 45/45 targets exact; 2190/2190 cross merges refuted.	Not the numerals-only regime; mostly ground axioms; held-out sets overlapped training; only 12 and 5 distinct templates; a valid near-miss merge missed.	Redone (Tables 10 and 16); Proposition F.9.
22	DTRC is exact exactly where the tagged learner is, with the same verdicts.	"Only if" fails for overlapping instance sets; verdicts depend on the call history.	Theorem 7.2.
Writing stage, found by the section writers			
23	The all-tied first-order lgg is a sound specialization (cases B2).	Only with the most specific freshness guards.	Proposition 5.4.
24	Every non-first-order ZF case has a false lgg instance from two data (cases §2.4).	Not $\mathrm{Repl}_{\mathrm{sp}}$ in the named encoding with textbook names, on anchor data (row 33).	Proposition 5.5.
25	DTRC with the k-union learner is exact when DTRC is and $k = k'$ (untagged Rem. 5.3).	Needs an anchor in every target's data.	Remark E.10.
26	PA: an induction on an unseen connective is accepted iff $\chi > k - 4$.	Needs a non-vacuous motive.	Corollary E.3.

#	Original claim (record)	What was wrong	Now
27	The refutation test with several negatives is open (untagged Prop. 5.10).	NP-complete by the later single Thm H, whose proof uses a false datum.	Theorem 6.16(d).
28	The merged $\in$-induction template is false (experiments Prop. E9(b)).	Valid if no motive uses its variable.	Proposition F.10.
29	DTRC with sharing is exact when the membership-tagged learner is (experiments E4-share).	Only if that exactness comes from an anchor, and only for targets that label a phase-one cluster (row 32).	Proposition F.5.
30	Termination of the rewriting in the proof of Theorem R1.	Imprecise in the first case.	Appendix D.5.
Review of the paper, found by its reviewers			
31	The aligned minimal templates have the properties of $\mathrm{Min}(D)$ for every template; if some datum is parameter-free, the aligned and the literal set agree (experiments Prop. E1-al; extended to DT° in writing).	False if a parameter of the template occurs only inside an argument of a non-pattern occurrence. For a target with only logically valid instances the unique aligned (and literal) minimal template has false instances, so the cautious learner is unsound; the second claim fails too.	Restricted to templates whose parameters all occur in their skeleton (Proposition A.5, Remark A.7); experiment targets are parameter-free.
32	With sharing, each target's cluster grows to all its data, so DTRC+share is exact where the membership-labelled learner is through an anchor (experiments E4-share; row 29).	A target may label no phase-one cluster. With three valid targets ($z + 0 = z$, $0 + z = z$, $1 + z = Sz$) and four data meeting all hypotheses, DTRC+share rejects $2 + 0 = 2$, while the membership-labelled learner has an anchor and is exact.	Proposition F.5 restricted, Example F.6; every E3 target labels one (computed).
33	$\mathrm{Repl}_{\mathrm{sp}}$, named, textbook names: no two instances give a false lgg instance (writing; row 24).	Only for anchor data. From $\mathrm{Repl}_{\mathrm{sp}}(x \in y)$ and $\mathrm{Repl}_{\mathrm{sp}}(y \in A)$ the lgg has only term metavariables, so no freshness guard is learned, and one of its instances is refutable in ZF.	Restricted to anchor data (Propositions 5.5 and 5.7; Appendix D.3).
34	The ω-step fails in nonstandard models of PA, and is derivable only by an ω-rule (cases A10).	It is truth-safe in every model of true arithmetic, nonstandard ones included (Proposition 4.7); and $\forall \bar{x} \varphi$ is often first-order derivable, Example 4.9 shows only that it need not be.	"In a model of $\mathsf{PA} + \neg\mathrm{Con}(\mathsf{PA})$" (Example 4.8); "in general not derivable, even over Q" (Proposition C.4).
35	In (Nm) and (dB) the learner needs nocap in addition whenever a free x_i lies under a binder; one parameter datum makes it accept $\mathrm{inst}_o(\varphi)$ (cases A10, A4′).	closed already excludes capture; capture must be excluded exactly when $\mathrm{Cap}(\varphi) \neq \emptyset$, which can fail with x_i under a binder; for $k \geq 2$ every z_i needs a parameter datum.	Lemma C.3, Proposition 4.6(e), Proposition C.4, Table 5.
36	E6: the $k = 9$ union verifier accepts from three induction instances, which reproduces the earlier untagged result (experiments E6).	With third motive $x + 0 = x$ this is an artifact of DT°_F: a DT° template with $f(x)$ covers the root-homogeneous block, has only true instances and misses a held-out instance.	Caveat in Section 7.3 (E6), limitation (6) of Section 7.4.
37	T_{Ind} has eight root specializations, so the open threshold case is $m \geq 8$ (untagged §§3, 10).	L_A contains $<$: there are nine (eight in the prototype's language).	Remark E.5, Appendix E.2.

G.3 What was not independently refereed

The referees saw the first versions of the records. Results added afterwards were checked by the Claude instances that added them, by proof and where possible by computation, often with the referee's code. The section writers re-read them, but no independent referee attacked them. They are:

- single: Propositions B.4 and B.7, Lemma B.8 (suggested by the referee), Example B.3 ((Rich) is needed), Theorem 6.16 and Lemma E.15 (single Thm H), the encoded lower bound over $=, +, S, 0$ in Proposition B.10 (single Cor. F.4, made explicit after the referee round), and the repairs in rows 7 and 8;
- cases: Proposition 4.6 with its capture criterion (now Lemma C.3), Proposition 5.3 with case (iv) of Theorem 5.6, part (b) of Proposition 5.7, Theorem D.3 with Lemma D.5, and the several-metavariable part of Corollary 5.10;
- untagged: Propositions E.6 to E.8, Proposition E.14, Proposition 6.11, Proposition E.18, Proposition 6.14, the ZFC computation of Example E.20, and the MRDP construction for Theorem 6.15;
- experiments (version 2): Proposition A.5 (Prop. E1-al) and Proposition F.5, both false as first stated (rows 31, 32); Lemma F.7; Proposition F.9 with Corollary 7.3; Proposition F.10; Lemma F.11; and experiment E0b. The extended oracles of Proposition 7.1 passed the referee's re-run evaluators;
- the earlier analysis: Theorem R1 (Theorem 5.14) was proved by a referee, and no second referee checked it; the writer of Section 5 re-checked it (row 30). The other prior results used here were confirmed by the prior referees, and several are also special cases of refereed track results (Theorem 2.5, Corollary 3.9, Theorem 3.3);
- writing: the extension of Proposition A.5 from DT°_F to DT° (restricted as in row 31), the extension of Proposition 4.3(a) to parameter data and 0-ary formula metavariables (via Lemma C.2), rows 23–30, and the direct proofs in Appendix D.5;
- review of the paper (written or changed by reviewers and section editors, and checked by them by hand and by computation): the restricted statements and proofs of Propositions A.5 and F.5, with Remark A.7 and Example F.6; the new proof of Proposition F.3(a); the computed check that every E3 target labels a phase-one cluster; the DT°-versus-DT°_F caveat of E6; the non-anchor $\mathrm{Repl}_{\mathrm{sp}}$ pair in Appendix D.3; the corrected Proposition 4.6(e) and Proposition C.4; Conjecture D.4 as a formal statement, and the remark in Appendix D.4 that the DT° case of Theorem D.3 is a case of Theorem 3.8; Example 3.5, the example for (U) after Definition 3.7, and Example 6.9; the count of nine root specializations of T_{Ind} (row 37); and the renumbering of Q's axioms in Corollary E.3 and Table 13, which follows Section 2.6.

Some things were not checked at all. Bounded searches are evidence, not proof. The textbook wordings of the ZF axioms were recalled, not checked against the books. Citations marked as unverified remain so, and the experiments use small synthetic data.

G.4 Conjectures and open problems

The paper states two formal conjectures. The evidence for Conjecture B.9 is weak: the greedy searches behind it also missed the quadratic chains of Proposition B.6. Conjecture D.4, that root-

level distinctness is not necessary for SO° anchors with several metavariables, rests on bounded searches (Appendix D.6). Two hypotheses are assumed rather than proved in general. The first is refutation separation. It is proved for Q's axioms with T_{Ind} and, for an ideal refuter, for the experimental mixtures (Corollary 7.3); for ZF and ZFC it was verified on every sampled cross pair (one or two sampled instances per schema) and on every run's data (Section 6.11); and it fails for some similar targets (Lemma F.11 and Proposition 6.17). The second is (H-ref), which the implemented refuter violates (Theorem 7.2); the sharing pass needs, besides, (H4$^+$) and (H5) and helps only targets that label a phase-one cluster (Proposition F.5). The question whether ZF without Foundation proves Collection is open to us (row 11). Two problems that one record left open are settled in another: finiteness (Theorem 3.3) and the complexity of the refutation test (Theorem 6.16). Section 8.3 collects all open problems of the records.

G.5 Reproducing the results

Everything is in `axiom-schemas/` in the accompanying repository.

- *Code* (`code/`, with `README.md`). There, `python3 -m pytest -q tests` runs 43 unit tests in about 10 s, and `sh run_all.sh` regenerates every table, data file and figure of Section 7 in `results/`. It took 593 s on 4 shared cores, and 674 s in the referee's re-run. In `experiments/recheck/`, `sh run_recheck.sh` re-runs the referee's oracle checks against the final code (about 6 min). Outputs other than timings reproduce exactly; the first version's results are in `results_v1/` (Appendix F.6).
- *Research records* (`research/`). The brief is `00-brief.md`. Each `tracks/<track>/` holds `notes.md` (first version), `referee.md` and `notes-final.md` (ending with the verification log), the researcher's scripts with their `.out` files (run there as `python3 <script> [args]`; the untagged prototype is in `code/` with `run_all.sh`), and the referee's code in `referee_checks/` (single; experiments, `sh run_checks.sh`, about 10 min) or `referee_code/` (cases, untagged). The earlier analysis is in `prior/induction/`, with the referees' code in `referee_A/`, `referee_C/` and `rb*.py` and the Lean checks in `lean/`, and in `prior/pa-untagged/`. Some cases and prior scripts import the first-order lgg from the parent report's code, in the directory `research/theory/T1-code/` of `inferential-learning/`, which lies next to `axiom-schemas/`. Each section's computations are listed in Appendices B.9, C.6, D.6, E.8 and F.6.
- *Review of the paper* (`research/paper-review/`): the review reports `review-*.md`, the per-section issue lists, decisions and editors' reports in `edits/`, and the reviewers' scripts in `scratch/` (run as `python3 -I <script>`; some import the package `dtrc` from `code/` through a path set at their top).
- *Paper* (`paper/`): the section files, `writer-notes.txt` and `build.sh`.